

Question ID d1a81099

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: d1a81099

$y > 4x + 8$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
2	19
4	30
6	41

B.

x	y
2	8
4	16
6	24

C.

x	y
2	13
4	18
6	23

D.

x	y
2	13
4	21
6	29

ID: d1a81099 Answer

Correct Answer: A

Rationale

Choice A is correct. In each choice, the values of x are 2, 4, and 6. Substituting the first value of x , 2, for x in the given inequality yields $y > 4(2) + 8$, or $y > 16$. Therefore, when $x = 2$, the corresponding value of y must be greater than 16. Of the given choices, only choice A is a table where the value of y corresponding to $x = 2$ is greater than 16. To confirm that the other values of x in this table and their corresponding values of y are also solutions to the given inequality, the values of x and y in the table can be substituted for x and y in the given inequality. Substituting 4 for x and 30 for y in the given inequality yields $30 > 4(4) + 8$, or $30 > 24$, which is true. Substituting 6 for x and 41 for y in the given inequality yields $41 > 4(6) + 8$, or

41 > **32**, which is true. It follows that for choice A, all the values of ***x*** and their corresponding values of ***y*** are solutions to the given inequality.

Choice B is incorrect. Substituting **2** for ***x*** and **8** for ***y*** in the given inequality yields **8** > **4(2) + 8**, or **8** > **16**, which is false.

Choice C is incorrect. Substituting **2** for ***x*** and **13** for ***y*** in the given inequality yields **13** > **4(2) + 8**, or **13** > **16**, which is false.

Choice D is incorrect. Substituting **2** for ***x*** and **13** for ***y*** in the given inequality yields **13** > **4(2) + 8**, or **13** > **16**, which is false.

Question Difficulty: Medium

Question ID c8f6ca7c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: c8f6ca7c

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of **12** and the fourth odd integer is at most **26** less than the sum of the first and third odd integers. Which inequality represents this situation?

- A. $12(x + 6) \leq x + (x + 4) - 26$
- B. $12(x + 6) \geq 26 - (x + (x + 4))$
- C. $12(x + 4) \leq x + (x + 3) - 26$
- D. $12(x + 4) \geq 26 - (x + (x + 3))$

ID: c8f6ca7c Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the four odd integers are consecutive, ordered from least to greatest, and that the first odd integer is represented by x . It follows that the second odd integer is represented by $x + 2$, the third odd integer is represented by $x + 4$, and the fourth odd integer is represented by $x + 6$. Therefore, the product of **12** and the fourth odd integer is represented by $12(x + 6)$, and **26** less than the sum of the first and third odd integers is represented by $x + (x + 4) - 26$. Since the product of **12** and the fourth odd integer is at most **26** less than the sum of the first and third odd integers, it follows that $12(x + 6) \leq x + (x + 4) - 26$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID dcce85c2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: dcce85c2

What is the y-coordinate of the y-intercept of the graph of $\frac{3x}{7} = -\frac{5y}{9} + 21$ in the xy-plane?

ID: dcce85c2 Answer

Correct Answer: 189/5, 37.8

Rationale

The correct answer is $\frac{189}{5}$. A y-intercept of a graph in the xy-plane is a point where the graph intersects the y-axis, which is a point with an x-coordinate of **0**. Substituting **0** for x in the given equation yields $\frac{3(0)}{7} = -\frac{5y}{9} + 21$, or $0 = -\frac{5y}{9} + 21$. Subtracting **21** from both sides of this equation yields $-21 = -\frac{5y}{9}$. Multiplying both sides of this equation by -9 yields $189 = 5y$. Dividing both sides of this equation by **5** yields $\frac{189}{5} = y$. Therefore, the y-coordinate of the y-intercept of the graph of the given equation in the xy-plane is $\frac{189}{5}$. Note that 189/5 and 37.8 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 1f7d5800

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 1f7d5800

For a literature class, a student's essay needs to have no more than **5,000** words. If a student's essay already has **3,026** words, what is the maximum number of additional words the student can include in the essay?

- A. **1,974**
- B. **3,026**
- C. **5,000**
- D. **8,026**

ID: 1f7d5800 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a student's essay needs to have no more than **5,000** words. It's also given that the student's essay already has **3,026** words. Let x represent the number of additional words the student includes in the essay. After including x additional words, the essay will have a total of $3,026 + x$ words. Since the essay needs to have no more than **5,000** words, it follows that $3,026 + x \leq 5,000$. Subtracting **3,026** from both sides of this inequality yields $x \leq 1,974$. Thus, the maximum number of additional words the student can include in the essay is **1,974**.

Choice B is incorrect. This is the number of words already in the student's essay, rather than the maximum number of additional words the student can include.

Choice C is incorrect. This is the total number of words the essay can contain, rather than the maximum number of additional words the student can include.

Choice D is incorrect. This is the number of words the essay would contain if the student added **5,000** more words, rather than the maximum number of additional words the student can include.

Question Difficulty: Easy

Question ID 8bb5aaf8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 8bb5aaf8

If $3x - 27 = 24$, what is the value of $x - 9$?

- A. 1
- B. 8
- C. 24
- D. 35

ID: 8bb5aaf8 Answer

Correct Answer: B

Rationale

Choice B is correct. Dividing each side of the given equation by 3 yields $x - 9 = 8$. Therefore, the value of $x - 9$ is 8.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $3x - 27$, not $x - 9$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID ec0d8652

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: ec0d8652

What is the y-intercept of the graph of $y = 34x + 81$ in the xy-plane?

- A. $(0, 81)$
- B. $(0, 34)$
- C. $(0, -34)$
- D. $(0, -81)$

ID: ec0d8652 Answer

Correct Answer: A

Rationale

Choice A is correct. In the xy-plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a y-intercept of $(0, b)$. Therefore, the y-intercept of the graph of $y = 34x + 81$ is $(0, 81)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 1d9a6078

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 1d9a6078

$$\begin{aligned} 48x - 72y &= 30y + 24 \\ ry &= \frac{1}{6} - 16x \end{aligned}$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

ID: 1d9a6078 Answer

Correct Answer: -34

Rationale

The correct answer is -34 . A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system can be written in standard form by subtracting $30y$ from both sides of the equation to yield $48x - 102y = 24$. The second equation in the given system can be written in standard form by adding $16x$ to both sides of the equation to yield $16x + ry = \frac{1}{6}$. The coefficient of x in this second equation, 16 , is $\frac{1}{3}$ times the coefficient of x in the first equation, 48 . For the lines to be parallel the coefficient of y in the second equation, r , must also be $\frac{1}{3}$ times the coefficient of y in the first equation, -102 . Thus, $r = \frac{1}{3}(-102)$, or $r = -34$. Therefore, if the given system has no solution, the value of r is -34 .

Question Difficulty: Hard

Question ID 2226c9a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 2226c9a5

Line p is defined by $2y + 18x = 9$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

- A. -9
- B. $-\frac{1}{9}$
- C. $\frac{1}{9}$
- D. 9

ID: 2226c9a5 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that line r is perpendicular to line p in the xy -plane. This means that the slope of line r is the negative reciprocal of the slope of line p . If the equation for line p is rewritten in slope-intercept form $y = mx + b$, where m and b are constants, then m is the slope of the line and $(0, b)$ is its y -intercept. Subtracting $18x$ from both sides of the equation $2y + 18x = 9$ yields $2y = -18x + 9$. Dividing both sides of this equation by 2 yields $y = -9x + \frac{9}{2}$. It follows that the slope of line p is -9 . The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of -9 is $\frac{-1}{-9}$, or $\frac{1}{9}$. Thus, the slope of line r is $\frac{1}{9}$.

Choice A is incorrect. This is the slope of line p , not line r .

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p .

Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p .

Question Difficulty: Hard

Question ID 1cc17260

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 1cc17260

$$2x + 16 = a(x + 8)$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a ?

ID: 1cc17260 Answer

Correct Answer: 2

Rationale

The correct answer is **2**. An equation with one variable, x , has infinitely many solutions only when both sides of the equation are equal for any defined value of x . It's given that $2x + 16 = a(x + 8)$, where a is a constant. This equation can be rewritten as $2(x + 8) = a(x + 8)$. If this equation has infinitely many solutions, then both sides of this equation are equal for any defined value of x . Both sides of this equation are equal for any defined value of x when $2 = a$. Therefore, if the equation has infinitely many solutions, the value of a is **2**.

Alternate approach: If the given equation, $2x + 16 = a(x + 8)$, has infinitely many solutions, then both sides of this equation are equal for any value of x . If $x = 0$, then substituting **0** for x in $2x + 16 = a(x + 8)$ yields $2(0) + 16 = a(0 + 8)$, or $16 = 8a$. Dividing both sides of this equation by **8** yields $2 = a$.

Question Difficulty: Medium

Question ID a229cabe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a229cabe

$$\begin{aligned}8x + y &= 5 \\ y &= 9x + 1\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -6
- B. $\frac{4}{17}$
- C. $\frac{6}{17}$
- D. 4

ID: a229cabe Answer

Correct Answer: B

Rationale

Choice B is correct. The second equation in the given system is $y = 9x + 1$. Substituting $9x + 1$ for y in the first equation in the given system yields $8x + 9x + 1 = 5$, which is equivalent to $17x + 1 = 5$. Subtracting 1 from both sides of this equation yields $17x = 4$. Dividing both sides of this equation by 17 yields $x = \frac{4}{17}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID cd2646d2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: cd2646d2

A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 2% salt by weight, with y kilograms of a high-salt mixture, which is 96% salt by weight, to create 24 kilograms of a mixture that is 4% salt by weight. Which equation represents this situation?

- A. $0.96x + 0.02y = (0.04)(24)$
- B. $0.02x + 0.96y = (0.04)(24)$
- C. $0.96x + 0.02y = 24$
- D. $0.02x + 0.96y = 24$

ID: cd2646d2 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that a chemist mixes x kilograms of a low-salt mixture, which is 2% salt by weight. Multiplying 0.02 by the amount of the low-salt mixture, x kilograms, yields $0.02x$ kilograms of salt in the low-salt mixture. It's also given that the chemist mixes y kilograms of a high-salt mixture, which is 96% salt by weight. Multiplying 0.96 by the amount of the high-salt mixture, y kilograms, yields $0.96y$ kilograms of salt in the high-salt mixture. Therefore, the total amount of salt in the combined mixture is $0.02x + 0.96y$ kilograms. It's given that the low-salt mixture and the high-salt mixture together create 24 kilograms of a combined mixture that is 4% salt by weight. Thus, the amount of salt in the combined mixture is $0.04(24)$ kilograms. Since the total amount of salt in the combined mixture equals the amount of salt in the low-salt mixture and the amount of salt in the high-salt mixture, the equation $0.02x + 0.96y = (0.04)(24)$ represents this situation.

Choice A is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight.

Choice C is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight, and where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Choice D is incorrect. This equation represents a situation where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Question Difficulty: Easy

Question ID 92921647

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 92921647

For the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 and passes through the point $(0, -8)$. Which equation defines f ?

- A. $f(x) = 3x$
- B. $f(x) = 3x - 8$
- C. $f(x) = 3x + 5$
- D. $f(x) = 3x + 11$

ID: 92921647 Answer

Correct Answer: B

Rationale

Choice B is correct. An equation defining a linear function can be written in the form $f(x) = mx + b$, where m and b are constants, m is the slope of the graph of $y = f(x)$ in the xy -plane, and $(0, b)$ is the y -intercept of the graph. It's given that for the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 . Therefore, $m = 3$. It's also given that this graph passes through the point $(0, -8)$. Therefore, the y -intercept of the graph is $(0, -8)$, and it follows that $b = -8$. Substituting 3 for m and -8 for b in the equation $f(x) = mx + b$ yields $f(x) = 3x - 8$. Thus, the equation that defines f is $f(x) = 3x - 8$.

Choice A is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 0)$, not $(0, -8)$.

Choice C is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$, not $(0, -8)$.

Choice D is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 11)$, not $(0, -8)$.

Question Difficulty: Easy

Question ID 9868b7fc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 9868b7fc

A model estimates that whales from the genus *Eschrichtius* travel **72** to **77** miles in the ocean each day during their migration. Based on this model, which inequality represents the estimated total number of miles, x , a whale from the genus *Eschrichtius* could travel in **16** days of its migration?

- A. $72 + 16 \leq x \leq 77 + 16$
- B. $(72)(16) \leq x \leq (77)(16)$
- C. $72 \leq 16 + x \leq 77$
- D. $72 \leq 16x \leq 77$

ID: 9868b7fc Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the model estimates that whales from the genus *Eschrichtius* travel **72** to **77** miles in the ocean each day during their migration. If one of these whales travels **72** miles each day for **16** days, then the whale travels **72(16)** miles total. If one of these whales travels **77** miles each day for **16** days, then the whale travels **77(16)** miles total. Therefore, the model estimates that in **16** days of its migration, a whale from the genus *Eschrichtius* could travel at least **72(16)** and at most **77(16)** miles total. Thus, the inequality $(72)(16) \leq x \leq (77)(16)$ represents the estimated total number of miles, x , a whale from the genus *Eschrichtius* could travel in **16** days of its migration.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

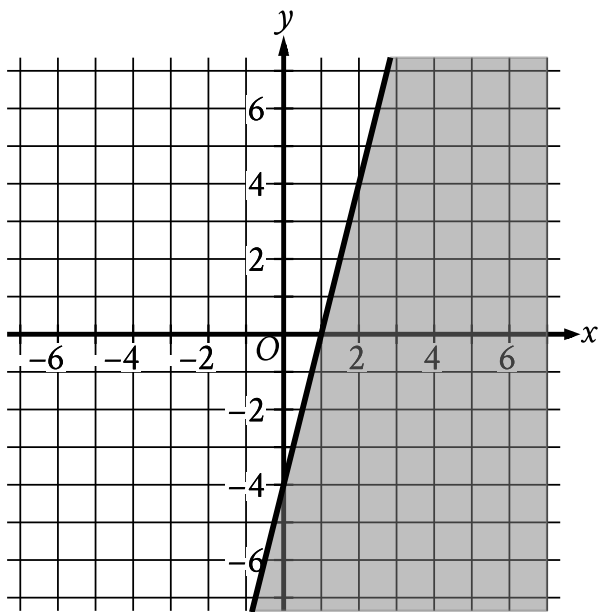
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 9a1cd96b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 9a1cd96b



The shaded region shown represents the solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A. $(-5, -6)$
- B. $(-2, 5)$
- C. $(1, 4)$
- D. $(6, -2)$

ID: 9a1cd96b Answer

Correct Answer: D

Rationale

Choice D is correct. Since the shaded region shown represents the solutions to an inequality, an ordered pair (x, y) is a solution to the inequality if it's represented by a point in the shaded region. Of the given choices, only $(6, -2)$ is represented by a point in the shaded region. Therefore, the ordered pair $(6, -2)$ is a solution to this inequality.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID e1f981b8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: e1f981b8

The function g is defined by $g(x) = 6x$. For what value of x is $g(x) = 54$?

ID: e1f981b8 Answer

Correct Answer: 9

Rationale

The correct answer is **9**. It's given that $g(x) = 6x$. Substituting **54** for $g(x)$ in the given function yields $54 = 6x$. Dividing both sides of this equation by **6** yields $x = 9$. Therefore, the value of x when $g(x) = 54$ is **9**.

Question Difficulty: Easy

Question ID 3b188791

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 3b188791

$$3x + 21 = 3x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k ?

ID: 3b188791 Answer

Correct Answer: 21

Rationale

The correct answer is **21**. It's given that the equation $3x + 21 = 3x + k$ has infinitely many solutions. If an equation in one variable has infinitely many solutions, then the equation is true for any value of the variable. Subtracting $3x$ from both sides of the given equation yields $k = 21$. Since this equation must be true for any value of x , the value of k is **21**.

Question Difficulty: Medium

Question ID 0c5cc8c7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 0c5cc8c7

In August, a car dealer completed **15** more than **3** times the number of sales the car dealer completed in September. In August and September, the car dealer completed **363** sales. How many sales did the car dealer complete in September?

ID: 0c5cc8c7 Answer

Correct Answer: 87

Rationale

The correct answer is **87**. It's given that in August, the car dealer completed **15** more than **3** times the number of sales the car dealer completed in September. Let x represent the number of sales the car dealer completed in September. It follows that $3x + 15$ represents the number of sales the car dealer completed in August. It's also given that in August and September, the car dealer completed **363** sales. It follows that $x + (3x + 15) = 363$, or $4x + 15 = 363$. Subtracting **15** from each side of this equation yields $4x = 348$. Dividing each side of this equation by **4** yields $x = 87$. Therefore, the car dealer completed **87** sales in September.

Question Difficulty: Hard

Question ID 8e21bc65

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 8e21bc65

$f(x) = 39$

For the given linear function f , which table gives three values of x and their corresponding values of $f(x)$?

- A.

x	$f(x)$
0	0
1	0
2	0
- B.

x	$f(x)$
0	39
1	39
2	39
- C.

x	$f(x)$
0	0
1	39
2	78
- D.

x	$f(x)$
0	39
1	0
2	−39

ID: 8e21bc65 Answer

Correct Answer: B

Rationale

Choice B is correct. For the given linear function f , $f(x)$ must equal **39** for all values of x . Of the given choices, only choice B gives three values of x and their corresponding values of $f(x)$ for the given linear function f .

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID d73f1c33

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: d73f1c33

If $x = 40$, what is the value of $x + 6$?

- A. 34
- B. 40
- C. 46
- D. 64

ID: d73f1c33 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $x = 40$. Adding 6 to both sides of this equation yields $x + 6 = 40 + 6$, or $x + 6 = 46$. Therefore, the value of $x + 6$ is 46.

Choice A is incorrect. This is the value of $x - 6$, not $x + 6$.

Choice B is incorrect. This is the value of x , not $x + 6$.

Choice D is incorrect. This is the value of $x + 24$, not $x + 6$.

Question Difficulty: Easy

Question ID 73e31b5e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 73e31b5e

$$\begin{aligned}y &= 6x + 16 \\ -7x - y &= 36\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(-4, -8)$
- B. $(-\frac{20}{13}, -\frac{80}{13})$
- C. $(4, 40)$
- D. $(20, 136)$

ID: 73e31b5e Answer

Correct Answer: A

Rationale

Choice A is correct. The given system of linear equations can be solved by the substitution method. The first equation in the given system of equations defines y as $6x + 16$. Substituting $6x + 16$ for y in the second equation of the given system of equations yields $-7x - (6x + 16) = 36$. Applying the distributive property on the left-hand side of this equation yields $-7x - 6x - 16 = 36$, or $-13x - 16 = 36$. Adding 16 to both sides of this equation yields $-13x = 52$. Dividing both sides of this equation by -13 yields $x = -4$. Substituting -4 for x in the first equation of the given system of equations, $y = 6x + 16$, yields $y = 6(-4) + 16$, or $y = -8$. Therefore, the solution (x, y) to the given system of equations is $(-4, -8)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 7714fd2b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 7714fd2b

The function f is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

ID: 7714fd2b Answer

Correct Answer: 67

Rationale

The correct answer is **67**. It's given that $f(5) = 32$. Therefore, for the given function f , when $x = 5$, $f(x) = 32$. Substituting **5** for x and **32** for $f(x)$ in the given function $f(x) = 4x + k(x - 1)$ yields $32 = 4(5) + k(5 - 1)$, or $32 = 20 + 4k$. Subtracting **20** from each side of this equation yields $12 = 4k$. Dividing each side of this equation by **4** yields $k = 3$. Substituting **3** for k in the given function $f(x) = 4x + k(x - 1)$ yields $f(x) = 4x + 3(x - 1)$, which is equivalent to $f(x) = 4x + 3x - 3$, or $f(x) = 7x - 3$. Substituting **10** for x into this equation yields $f(10) = 7(10) - 3$, or $f(10) = 67$. Therefore, the value of $f(10)$ is **67**.

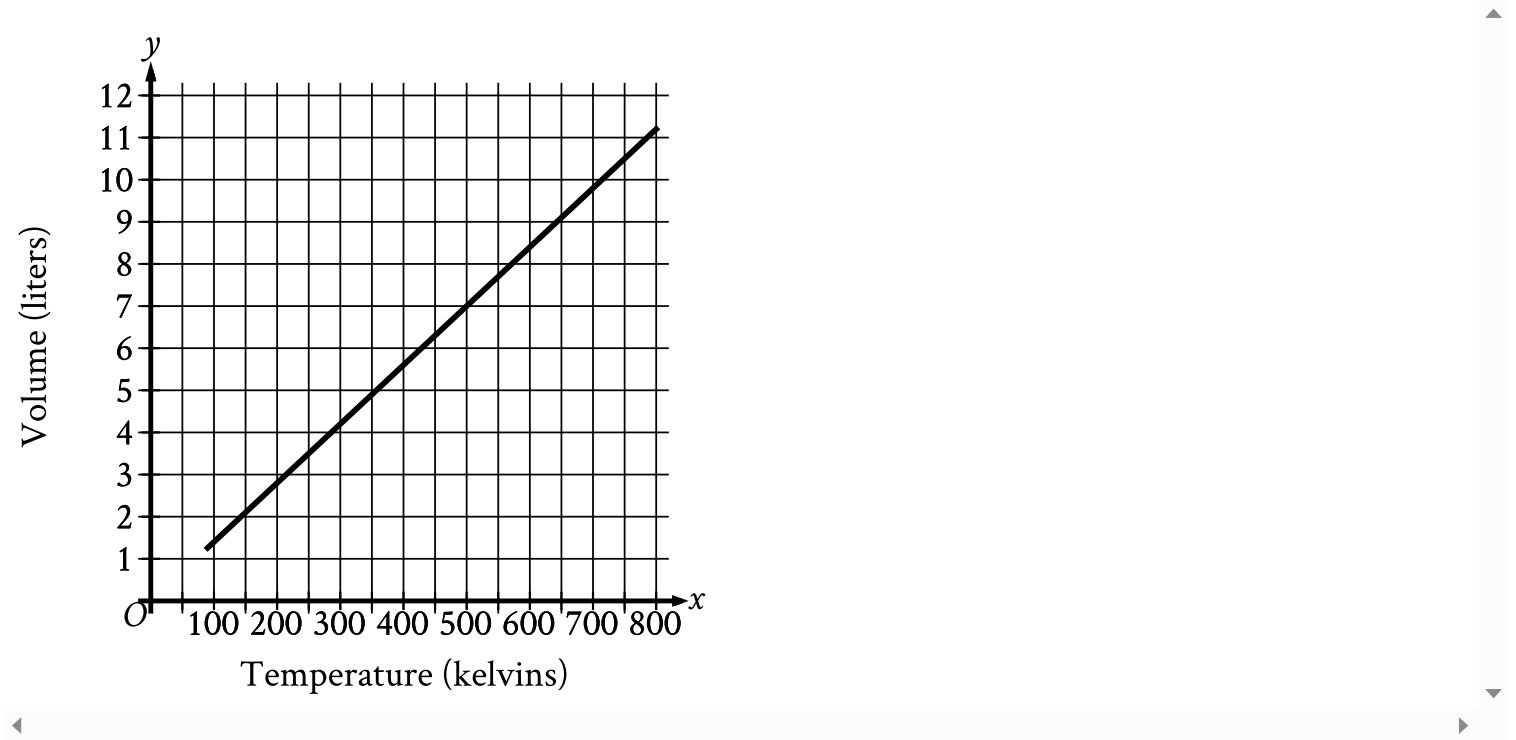
Question Difficulty: Hard

Question ID 3eef8a08

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 3eef8a08

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y , in liters, of the hydrogen when its temperature is x kelvins.



What is the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins?

- A. **0**
- B. $\frac{7}{500}$
- C. **7**
- D. $\frac{500}{7}$

ID: 3eef8a08 Answer

Correct Answer: C

Rationale

Choice C is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents volume, in liters. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins is represented by the y -coordinate of the point on the graph that has an x -coordinate of **500**. The point on the graph with an x -coordinate of **500** has a y -coordinate of **7**. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is **500** kelvins is **7**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID deed83a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: deed83a1

$$38(x - n) = 38y + 38n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

- A. $4x - 4y = 8n$
- B. $4x + 4y = 4n$
- C. $4x + 4y = 8n$
- D. $4x - 4y = 4n$

ID: deed83a1 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel and distinct if the coefficients of x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$, and the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$. It's given that one of the equations in a system of two linear equations is $38(x - n) = 38y + 38n$, where n is a positive constant. Distributing 38 on the left-hand side of this equation yields $38x - 38n = 38y + 38n$. Adding $38n$ to both sides of this equation yields $38x = 38y + 76n$. Subtracting $38y$ from both sides of this equation yields $38x - 38y = 76n$. Therefore, the given equation can be written in the form $Ax + By = C$, where $A = 38$, $B = -38$, and $C = 76n$. The equation in choice D, $4x - 4y = 4n$, is written in the form $Dx + Ey = F$, where $D = 4$, $E = -4$, and $F = 4n$. Using these values, it follows that the value of $\frac{D}{A}$ is $\frac{4}{38}$, or $\frac{2}{19}$; the value of $\frac{E}{B}$ is $\frac{-4}{-38}$, or $\frac{2}{19}$; and the value of $\frac{F}{C}$ is $\frac{4n}{76n}$, or $\frac{1}{19}$, since n is a positive constant. Since $\frac{D}{A} = \frac{E}{B}$ and since $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$, it follows that the lines represented by these two equations are parallel and distinct. Therefore, the second equation in this system of equations with no solution could be $4x - 4y = 4n$.

Choice A is incorrect. The system consisting of this equation and the given equation has infinitely many solutions rather than no solution.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Choice C is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Question Difficulty: Hard

Question ID 9d50428c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 9d50428c

Which of the following systems of linear equations has no solution?

- A. $x = 3$
 $y = 5$
- B. $y = 6x + 6$
 $y = 5x + 6$
- C. $y = 16x + 3$
 $y = 16x + 19$
- D. $y = 5$
 $y = 5x + 5$

ID: 9d50428c Answer

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m , are the same and are distinct if their y -coordinates of the y -intercepts, b , are different. In the equations $y = 16x + 3$ and $y = 16x + 19$, the values of m are each 16 , and the values of b are 3 and 19 , respectively. Since the slopes of these lines are the same, and the y -coordinates of the y -intercepts are different, it follows that the system of linear equations in choice C has no solution.

Choice A is incorrect. The lines represented by the equations in this system are a vertical line and a horizontal line. Therefore, this system has a solution, $(3, 5)$, rather than no solution.

Choice B is incorrect. The two lines represented by these equations have different slopes and the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 6)$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 5 that have the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 5)$, rather than no solution.

Question Difficulty: Hard

Question ID b62ca837

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: b62ca837

Ellen bought two types of snack bags for a school event. Each box of chips Ellen bought contained **24** bags and cost **\$11**. Each box of crackers Ellen bought contained **16** bags and cost **\$9**. It cost Ellen **\$497** before tax to buy **968** bags. If x is the number of boxes of chips and y is the number of boxes of crackers that Ellen bought, which of the following systems of equations represents this situation?

- A. $11x + 16y = 968$
 $24x + 9y = 497$
- B. $11x + 9y = 968$
 $24x + 16y = 497$
- C. $11x + 9y = 497$
 $24x + 16y = 968$
- D. $9x + 11y = 497$
 $16x + 24y = 968$

ID: b62ca837 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that Ellen bought two types of snack bags: chips and crackers. It's also given that each box of chips contains **24** bags. Thus, if Ellen bought x boxes of chips, the number of bags of chips is **24x**. Additionally, it's given that each box of chips costs **\$11**. Thus, if Ellen bought x boxes of chips, the cost for all the bags of chips is **11x**. It's also given that each box of crackers contains **16** bags. Thus, if Ellen bought y boxes of crackers, the number of bags of crackers is **16y**. Additionally, it's given that each box of crackers costs **\$9**. Thus, if Ellen bought y boxes of crackers, the cost for all the bags of crackers is **9y**. It's given that it cost Ellen **\$497** to buy **968** bags. Since Ellen bought **24x** bags of chips and **16y** bags of crackers, the total number of snack bags that Ellen bought can be represented by the equation **24x + 16y = 968**. Additionally, since the cost for all the bags of chips is **11x** and the cost for all the bags of crackers is **9y**, the total cost for all the snack bags that Ellen bought can be represented by the equation **11x + 9y = 497**. Therefore, the system consisting of the equations **11x + 9y = 497** and **24x + 16y = 968** represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This system of equations represents a situation where the total cost is **\$968**, not **\$497**, and the total number of bags is **497**, not **968**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 47cc6093

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 47cc6093

In the linear function h , $h(28) = 15$ and $h(26) = 22$. Which equation defines h ?

- A. $h(x) = -\frac{2}{7}x + 23$
- B. $h(x) = -\frac{2}{7}x + 113$
- C. $h(x) = -\frac{7}{2}x + 23$
- D. $h(x) = -\frac{7}{2}x + 113$

ID: 47cc6093 Answer

Correct Answer: D

Rationale

Choice D is correct. An equation defining h can be written in the form $y = mx + b$, where $y = h(x)$, m represents the slope of the graph of $y = h(x)$ in the xy -plane, and b represents the y -coordinate of the y -intercept of the graph. It's given that $h(28) = 15$ and $h(26) = 22$. It follows that the points $(28, 15)$ and $(26, 22)$ are on the graph of $y = h(x)$ in the xy -plane. The slope can be found by using any two points, (x_1, y_1) and (x_2, y_2) , and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(28, 15)$ and $(26, 22)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{22 - 15}{26 - 28}$, or $m = -\frac{7}{2}$. Substituting $-\frac{7}{2}$ for m and $(28, 15)$ for (x, y) in the equation $y = mx + b$ yields $15 = (-\frac{7}{2})(28) + b$, or $15 = -98 + b$. Adding 98 to both sides of this equation yields $113 = b$. Substituting $-\frac{7}{2}$ for m and 113 for b in the equation $y = mx + b$ yields $y = -\frac{7}{2}x + 113$. Since $y = h(x)$, it follows that the equation that defines h is $h(x) = -\frac{7}{2}x + 113$.

Choice A is incorrect. For this function, $h(26) = \frac{109}{7}$, not $h(26) = 22$.

Choice B is incorrect. For this function, $h(28) = 105$, not $h(28) = 15$, and $h(26) = \frac{739}{7}$, not $h(26) = 22$.

Choice C is incorrect. For this function, $h(28) = -75$, not $h(28) = 15$, and $h(26) = -68$, not $h(26) = 22$.

Question Difficulty: Hard

Question ID e9a2206b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: e9a2206b

What value of t is the solution to the equation $0.8t - 0.46 = 8(t - 0.001) + 1.9$?

ID: e9a2206b Answer

Correct Answer: -.3266, -.3267, -49/150

Rationale

The correct answer is $-.3267$. Applying the distributive property to the right-hand side of the given equation yields $0.8t - 0.46 = 8t - 0.008 + 1.9$, or $0.8t - 0.46 = 8t + 1.892$. Subtracting $0.8t$ from both sides of this equation yields $-0.46 = 7.2t + 1.892$. Subtracting 1.892 from both sides of this equation yields $-2.352 = 7.2t$. Dividing both sides of this equation by 7.2 yields $\frac{-2.352}{7.2} = t$. Therefore, the value of t is approximately -0.32667 . Note that -.3267, -.3266, -0.326, and -0.327 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 9f5ff8c2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 9f5ff8c2

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. Which of the following is the best interpretation of 9 in this context?

- A. Tavon will keep the vine plant for 9 months.
- B. The vine plant is expected to grow 9 inches each month.
- C. The vine plant is expected to grow to a maximum length of 9 inches.
- D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

ID: 9f5ff8c2 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. For a function defined by an equation of the form $f(t) = mt + b$, where m and b are constants, b represents the value of $f(0)$, or the value of $f(t)$ when the value of t is 0 . Therefore, for the function defined by $f(t) = 14t + 9$, 9 represents the value of $f(t)$ when the value of t is 0 . This means that 0 months after the vine plant was purchased, the estimated length of the vine plant was 9 inches. Therefore, the best interpretation of 9 in this context is the estimated length of the vine plant was 9 inches when Tavon purchased it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. The vine plant is expected to grow 14 inches, not 9 inches, each month.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 26e3a069

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 26e3a069

The function g is defined by $g(x) = 10x + 8$. What is the value of $g(x)$ when $x = 8$?

- A. 0
- B. 8
- C. 10
- D. 88

ID: 26e3a069 Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $g(x)$ when $x = 8$ can be found by substituting 8 for x in the given equation $g(x) = 10x + 8$. This yields $g(8) = 10(8) + 8$, or $g(8) = 88$. Therefore, when $x = 8$, the value of $g(x)$ is 88.

Choice A is incorrect. This is the value of x when $g(x) = 8$, rather than the value of $g(x)$ when $x = 8$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a08e1d3e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: a08e1d3e

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

- A. 21
- B. 28
- C. 32
- D. 35

ID: a08e1d3e Answer

Correct Answer: A

Rationale

Choice A is correct. The x -intercept of a graph in the xy -plane is the point on the graph where $y = 0$. It's given that function h is defined by $h(x) = 4x + 28$. Therefore, the equation representing the graph of $y = h(x)$ is $y = 4x + 28$. Substituting 0 for y in the equation $y = 4x + 28$ yields $0 = 4x + 28$. Subtracting 28 from both sides of this equation yields $-28 = 4x$. Dividing both sides of this equation by 4 yields $-7 = x$. Therefore, the x -intercept of the graph of $y = h(x)$ in the xy -plane is $(-7, 0)$. It's given that the x -intercept of the graph of $y = h(x)$ is $(a, 0)$. Therefore, $a = -7$. The y -intercept of a graph in the xy -plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $y = 4x + 28$ yields $y = 4(0) + 28$, or $y = 28$. Therefore, the y -intercept of the graph of $y = h(x)$ in the xy -plane is $(0, 28)$. It's given that the y -intercept of the graph of $y = h(x)$ is $(0, b)$. Therefore, $b = 28$. If $a = -7$ and $b = 28$, then the value of $a + b$ is $-7 + 28$, or 21.

Choice B is incorrect. This is the value of b , not $a + b$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $-a + b$, not $a + b$.

Question Difficulty: Hard

Question ID edd6232c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: edd6232c

$5x + 14y = 45$
 $10x + 7y = 27$

The solution to the given system of equations is (x, y) . What is the value of xy ?

ID: edd6232c Answer

Correct Answer: 1.8, 9/5

Rationale

The correct answer is $\frac{9}{5}$. Multiplying the first equation in the given system by **2** yields $10x + 28y = 90$. Subtracting the second equation in the given system, $10x + 7y = 27$, from $10x + 28y = 90$ yields $(10x + 28y) - (10x + 7y) = 90 - 27$, which is equivalent to $10x + 28y - 10x - 7y = 63$, or $21y = 63$. Dividing both sides of this equation by **21** yields $y = 3$. The value of x can be found by substituting **3** for y in either of the two given equations. Substituting **3** for y in the equation $10x + 7y = 27$ yields $10x + 7(3) = 27$, or $10x + 21 = 27$. Subtracting **21** from both sides of this equation yields $10x = 6$. Dividing both sides of this equation by **10** yields $x = \frac{6}{10}$, or $x = \frac{3}{5}$. Therefore, the value of xy is $(\frac{3}{5})(3)$, or $\frac{9}{5}$. Note that 9/5 and 1.8 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 76d7f04a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 76d7f04a

What value of p satisfies the equation $2p + 275 = 325$?

- A. 5
- B. 25
- C. 48
- D. 300

ID: 76d7f04a Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 275 from both sides of the given equation yields $2p = 50$. Dividing both sides of this equation by 2 yields $p = 25$. Therefore, the value of p that satisfies the given equation is 25 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of p that satisfies the equation $(2 + p) + 275 = 325$, not $2p + 275 = 325$.

Choice D is incorrect. This is the value of p that satisfies the equation $2p - 275 = 325$, not $2p + 275 = 325$.

Question Difficulty: Easy

Question ID 697c5748

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 697c5748

Number of cars	Maximum number of passengers and crew
3	174
5	284
10	559

The table shows the linear relationship between the number of cars, c , on a commuter train and the maximum number of passengers and crew, p , that the train can carry. Which equation represents the linear relationship between c and p ?

- A. $55c - p = -9$
- B. $55c - p = 9$
- C. $55p - c = -9$
- D. $55p - c = 9$

ID: 697c5748 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that there is a linear relationship between the number of cars, c , on a commuter train and the maximum number of passengers and crew, p , that the train can carry. It follows that this relationship can be represented by an equation of the form $p = mc + b$, where m is the rate of change of p in this relationship and b is a constant. The rate of change of p in this relationship can be calculated by dividing the difference in any two values of p by the difference in the corresponding values of c . Using two pairs of values given in the table, the rate of change of p in this relationship is $\frac{284-174}{5-3}$, or 55 . Substituting 55 for m in the equation $p = mc + b$ yields $p = 55c + b$. The value of b can be found by substituting any value of c and its corresponding value of p for c and p , respectively, in this equation. Substituting 10 for c and 559 for p yields $559 = 55(10) + b$, or $559 = 550 + b$. Subtracting 550 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $p = 55c + b$ yields $p = 55c + 9$. Subtracting 9 from both sides of this equation yields $p - 9 = 55c$. Subtracting p from both sides of this equation yields $-9 = 55c - p$, or $55c - p = -9$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b1145a50

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: b1145a50

$2.6 + x = 2.8$

What value of x is the solution to the given equation?

ID: b1145a50 Answer

Correct Answer: 0.2, 1/5

Rationale

The correct answer is **.2**. Subtracting **2.6** from each side of the given equation yields $x = \mathbf{0.2}$. Therefore, the value of x that's the solution to the given equation is **0.2**. Note that .2 and 1/5 are examples of ways to enter a correct answer.

Question Difficulty: Easy

Question ID d31c0757

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: d31c0757

If $\frac{x}{8} = 5$, what is the value of $\frac{8}{x}$?

ID: d31c0757 Answer

Correct Answer: .2, 1/5

Rationale

The correct answer is $\frac{1}{5}$. Since the number 5 can also be written as $\frac{5}{1}$, the given equation can also be written as $\frac{x}{8} = \frac{5}{1}$. This equation is equivalent to $\frac{8}{x} = \frac{1}{5}$. Therefore, the value of $\frac{8}{x}$ is $\frac{1}{5}$. Note that 1/5 and .2 are examples of ways to enter a correct answer.

Alternate approach: Multiplying both sides of the equation $\frac{x}{8} = 5$ by 8 yields $x = 40$. Substituting 40 for x into the expression $\frac{8}{x}$ yields $\frac{8}{40}$, or $\frac{1}{5}$.

Question Difficulty: Medium

Question ID b59c4af9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: b59c4af9

If $46 = 16 + 2(x - 8)$, what is the value of $2(x - 8)$?

- A. 16
- B. 23
- C. 30
- D. 38

ID: b59c4af9 Answer

Correct Answer: C

Rationale

Choice C is correct. Subtracting **16** from both sides of the given equation yields $30 = 2(x - 8)$. Therefore, the value of $2(x - 8)$ is **30**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 0eb2c6a2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 0eb2c6a2

Students and chaperones from a middle school are going on a field trip to a museum. Admission to the museum will cost **\$8.50** for each student and **\$12** for each chaperone. It will cost a total of **\$730** for admission to the museum for all **83** people on the field trip. How many students are going on the field trip?

- A. **31**
- B. **70**
- C. **76**
- D. **86**

ID: 0eb2c6a2 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that admission to the museum will cost **\$8.50** for each student and **\$12** for each chaperone. It's also given that it will cost a total of **\$730** for admission to the museum for all **83** people on the field trip. Let ***x*** represent the number of students and ***y*** represent the number of chaperones going on the field trip. Since there are a total of **83** people going on the field trip, it follows that ***x* + *y* = 83**. Since admission to the museum for each student costs **\$8.50** and each chaperone costs **\$12**, the total cost can be represented by the equation **8.50*x* + 12*y* = 730**. The equations ***x* + *y* = 83** and **8.50*x* + 12*y* = 730** form a system of equations. Subtracting ***x*** from both sides of the equation ***x* + *y* = 83** yields ***y* = 83 - *x***. Substituting **83 - *x*** for ***y*** in the equation **8.50*x* + 12*y* = 730** yields **8.50*x* + 12(83 - *x*) = 730**, or **-3.50*x* + 996 = 730**. Subtracting **996** from both sides of this equation yields **-3.50*x* = -266**. Dividing both sides of this equation by **-3.50** yields ***x* = 76**. Therefore, **76** students are going on the field trip.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2a58b08a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 2a58b08a

For a snowstorm in a certain town, the minimum rate of snowfall recorded was **0.6** inches per hour, and the maximum rate of snowfall recorded was **1.8** inches per hour. Which inequality is true for all values of s , where s represents a rate of snowfall, in inches per hour, recorded for this snowstorm?

- A. $s \geq 2.4$
- B. $s \geq 1.8$
- C. $0 \leq s \leq 0.6$
- D. $0.6 \leq s \leq 1.8$

ID: 2a58b08a Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that for a snowstorm in a certain town, the minimum rate of snowfall recorded was **0.6** inches per hour, the maximum rate of snowfall recorded was **1.8** inches per hour, and s represents a rate of snowfall, in inches per hour, recorded for this snowstorm. It follows that the inequality $0.6 \leq s \leq 1.8$ is true for all values of s .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 389ff1fb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 389ff1fb

$10x = 86$

What value of x is the solution to the given equation?

ID: 389ff1fb Answer

Correct Answer: 8.6, 43/5

Rationale

The correct answer is **8.6**. Dividing both sides of the given equation by **10** yields $x = 8.6$. Therefore, **8.6** is the solution to the given equation. Note that 8.6 and 43/5 are examples of ways to enter a correct answer.

Question Difficulty: Easy

Question ID a6ab27d1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a6ab27d1

$$\begin{aligned}y &= 3x \\ 2x + y &= 12\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $5x$?

- A. 24
- B. 15
- C. 12
- D. 5

ID: a6ab27d1 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given by the first equation in the system that $y = 3x$. Substituting $3x$ for y in the equation $2x + y = 12$ yields $2x + 3x = 12$, or $5x = 12$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID a4b53a50

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: a4b53a50

$66x = 66x$

How many solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: a4b53a50 Answer

Correct Answer: C

Rationale

Choice C is correct. If the two sides of a linear equation are equivalent, then the equation is true for any value. If an equation is true for any value, it has infinitely many solutions. Since the two sides of the given linear equation $66x = 66x$ are equivalent, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

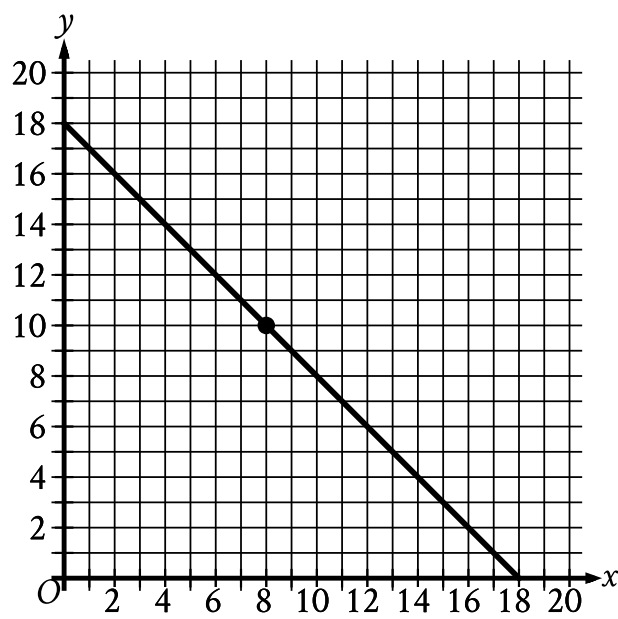
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID f084b2c3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: f084b2c3



The graph in the xy -plane models the possible combinations of length x , in meters (**m**), and width y , in meters, for a rectangle with a perimeter of **36 m**. Which statement is the best interpretation of the point **(8, 10)** in this context?

- A. The length is **10 m** less than the perimeter, and the width is **8 m** less than the perimeter.
- B. The length is **10 m**, and the width is **8 m**.
- C. The length is **8 m**, and the width is **10 m**.
- D. The length is **8 m** less than the perimeter, and the width is **10 m** less than the perimeter.

ID: f084b2c3 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the graph in the xy -plane models the possible combinations of length x , in meters (m), and width y , in meters, for a rectangle with a perimeter of **36 m**. Since x represents the length, in meters, and y represents the width, in meters, the point **(8, 10)** in the xy -plane represents a rectangle whose length is **8 m** and whose width is **10 m**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is an interpretation of the point **(10, 8)**, not **(8, 10)**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 62a13f1d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 62a13f1d

$$\begin{aligned} -x - wy &= -337 \\ 2x - wy &= 47 \end{aligned}$$

In the given system of equations, w is a constant. In the xy -plane, the graphs of these equations intersect at the point $(q, 19)$, where q is a constant. What is the value of w ?

ID: 62a13f1d Answer

Correct Answer: 11

Rationale

The correct answer is **11**. It's given that the graphs of the equations in the given system intersect at the point $(q, 19)$, where q is a constant. Therefore, the coordinates of this point must satisfy both equations. Substituting the point $(q, 19)$ into the first equation, $-x - wy = -337$, yields $-q - w(19) = -337$. Adding $19w$ to both sides of this equation yields $-q = -337 + 19w$, which is equivalent to $q = 337 - 19w$. Substituting the point $(q, 19)$ into the second equation yields $2q - w(19) = 47$. Substituting $337 - 19w$ in place of q in the equation $2q - w(19) = 47$ yields $2(337 - 19w) - 19w = 47$. Applying the distributive property to the left-hand side of this equation yields $674 - 38w - 19w = 47$. Combining like terms on the left-hand side of this equation yields $674 - 57w = 47$. Subtracting 674 from both sides of this equation yields $-57w = -627$. Dividing both sides of this equation by -57 yields $w = 11$.

Question Difficulty: Hard

Question ID a762cee2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: a762cee2

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of **33** in this context?

- A. The estimated height, in cm, of the liquid at the start of the project
- B. The estimated height, in cm, of the liquid at the end of the project
- C. The estimated change in the height, in cm, of the liquid each day
- D. The estimated number of days for all of the liquid to evaporate

ID: a762cee2 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. For a linear function in the form $f(x) = a + bx$, where a and b are constants, a represents the value of $f(0)$ and b represents the rate of change of the function. It follows that in the given function, **33** represents the value of $f(0)$. Therefore, the best interpretation of **33** in this context is the estimated height, in cm, of the liquid at the start of the project.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The estimated change in the height, in cm, of the liquid each day is **0.18**, not **33**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 710119ea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 710119ea

John paid a total of **\$165** for a microscope by making a down payment of **\$37** plus *p* monthly payments of **\$16** each. Which of the following equations represents this situation?

- A. $16p - 37 = 165$
- B. $37p - 16 = 165$
- C. $16p + 37 = 165$
- D. $37p + 16 = 165$

ID: 710119ea Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that John made a **\$16** payment each month for *p* months. The total amount of these payments can be represented by the expression **16*p***. The down payment can be added to that amount to find the total amount John paid, yielding the expression **16*p* + 37**. It's given that John paid a total of **\$165**. Therefore, the expression for the total amount John paid can be set equal to that amount, yielding the equation **16*p* + 37 = 165**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 5e8699ad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 5e8699ad

Jay walks at a speed of **3** miles per hour and runs at a speed of **5** miles per hour. He walks for *w* hours and runs for *r* hours for a combined total of **14** miles. Which equation represents this situation?

- A. $3w + 5r = 14$
- B. $\frac{1}{3}w + \frac{1}{5}r = 14$
- C. $\frac{1}{3}w + \frac{1}{5}r = 112$
- D. $3w + 5r = 112$

ID: 5e8699ad Answer

Correct Answer: A

Rationale

Choice A is correct. Since Jay walks at a speed of **3** miles per hour for *w* hours, Jay walks a total of **3w** miles. Since Jay runs at a speed of **5** miles per hour for *r* hours, Jay runs a total of **5r** miles. Therefore, the total number of miles Jay travels can be represented by **3w + 5r**. Since the combined total number of miles is **14**, the equation **3w + 5r = 14** represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 815ba8ee

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 815ba8ee

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of $1,120$ points. The equation $80S + 90C = 1,120$ represents this situation. How many more points is a circle token worth than a square token?

- A. 950
- B. 90
- C. 80
- D. 10

ID: 815ba8ee Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the equation $80S + 90C = 1,120$ represents this situation, where S is the number of square tokens won, C is the number of circle tokens won, and $1,120$ is the total number of points the tokens are worth. It follows that $80S$ represents the total number of points the square tokens are worth. Therefore, each square token is worth 80 points. It also follows that $90C$ represents the total number of points the circle tokens are worth. Therefore, each circle token is worth 90 points. Since a circle token is worth 90 points and a square token is worth 80 points, then a circle token is worth $90 - 80$, or 10 , more points than a square token.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of points a circle token is worth.

Choice C is incorrect. This is the number of points a square token is worth.

Question Difficulty: Medium

Question ID 1ff7bc88

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 1ff7bc88

For the linear function f , $f(0) = 17$ and $f(1) = 17$. Which equation defines f ?

- A. $f(x) = \frac{1}{17}$
- B. $f(x) = 1$
- C. $f(x) = 17$
- D. $f(x) = 34$

ID: 1ff7bc88 Answer

Correct Answer: C

Rationale

Choice C is correct. An equation defining the linear function f can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y-intercept of the graph of $y = f(x)$ in the xy-plane. The slope of the graph of $y = f(x)$ can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are any two points that the graph passes through. If $f(0) = 17$, it follows that the graph of $y = f(x)$ passes through the point $(0, 17)$. If $f(1) = 17$, it follows that the graph of $y = f(x)$ passes through the point $(1, 17)$. Substituting $(0, 17)$ and $(1, 17)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{17 - 17}{1 - 0}$, which is equivalent to $m = \frac{0}{1}$, or $m = 0$. Since the graph of $y = f(x)$ passes through $(0, 17)$, it follows that $b = 17$. Substituting 0 for m and 17 for b in the equation $f(x) = mx + b$ yields $f(x) = 0x + 17$, or $f(x) = 17$. Thus, the equation that defines f is $f(x) = 17$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b176c98f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b176c98f

- The function $f(x)$ is defined as 19 more than 4 times a number x . If $y = f(x)$ is graphed in the xy -plane, what is the best interpretation of the x -intercept?
- A. When $f(x) = 0$, the number is $-\frac{19}{4}$.
 - B. When the number is 0, $f(x) = 19$.
 - C. The value of $f(x)$ increases by 1 for each increase of 4 in the value of the number.
 - D. For each increase of 1 in the value of the number, $f(x)$ increases by 4.

ID: b176c98f Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function $f(x)$ is defined as 19 more than 4 times a number x . This can be represented by the equation $f(x) = 4x + 19$. The x -intercept of the graph of $y = f(x)$ in the xy -plane is the point where the graph intersects the x -axis, or the point on the graph where the value of $f(x)$ is equal to 0. Substituting 0 for $f(x)$ in the equation $f(x) = 4x + 19$ yields $0 = 4x + 19$. Subtracting 19 from each side of this equation yields $-19 = 4x$. Dividing each side of this equation by 4 yields $x = -\frac{19}{4}$. Therefore, when $f(x) = 0$, the number is $-\frac{19}{4}$.

Choice B is incorrect. This is the best interpretation of the y -intercept, not the x -intercept, of the graph of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the best interpretation of the slope, not the x -intercept, of the graph of the function.

Question Difficulty: Hard

Question ID 6e850bbb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 6e850bbb

$4x + 1 = 33$

Which equation has the same solution as the given equation?

- A. $4x = 32$
- B. $4x = 5$
- C. $4x = 1$
- D. $4x = -32$

ID: 6e850bbb Answer

Correct Answer: A

Rationale

Choice A is correct. Subtracting 1 from both sides of the given equation, $4x + 1 = 33$, yields $4x = 32$. Therefore, $4x = 32$ has the same solution as the given equation.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e7743fef

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: e7743fef

If $2 + x = 60$, what is the value of $16 + 8x$?

ID: e7743fef Answer

Correct Answer: 480

Rationale

The correct answer is **480**. Multiplying both sides of the given equation by 8 yields $8(2 + x) = 8(60)$, or $16 + 8x = 480$. Therefore, if $2 + x = 60$, the value of $16 + 8x$ is **480**.

Question Difficulty: Easy

Question ID 4d170b67

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 4d170b67

x	$f(x)$
1	−64
2	0
3	64

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Function f is defined by $f(x) = ax + b$, where a and b are constants. What is the value of $a - b$?

- A. −64
- B. 62
- C. 128
- D. 192

ID: 4d170b67 Answer

Correct Answer: D

Rationale

Choice D is correct. The table gives that $f(x) = 0$ when $x = 2$. Substituting 0 for $f(x)$ and 2 for x into the equation $f(x) = ax + b$ yields $0 = 2a + b$. Subtracting $2a$ from both sides of this equation yields $b = -2a$. The table gives that $f(x) = -64$ when $x = 1$. Substituting $-2a$ for b , -64 for $f(x)$, and 1 for x into the equation $f(x) = ax + b$ yields $-64 = a(1) + (-2a)$. Combining like terms yields $-64 = -a$, or $a = 64$. Since $b = -2a$, substituting 64 for a into this equation gives $b = (-2)(64)$, which yields $b = -128$. Thus, the value of $a - b$ can be written as $64 - (-128)$, which is 192.

Choice A is incorrect. This is the value of $a + b$, not $a - b$.

Choice B is incorrect. This is the value of $a - 2$, not $a - b$.

Choice C is incorrect. This is the value of $2a$, not $a - b$.

Question Difficulty: Hard

Question ID 828fbdfc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 828fbdfc

A moving truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than **4,600** pounds. What is the maximum number of boxes this truck can tow in a trailer with a weight of **500** pounds if each box weighs **120** pounds?

- A. **34**
- B. **35**
- C. **38**
- D. **39**

ID: 828fbdfc Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than **4,600** pounds. If the trailer has a weight of **500** pounds and each box weighs **120** pounds, the expression **$500 + 120b$** , where **b** is the number of boxes, gives the combined weight of the trailer and the boxes. Since the combined weight must be no more than **4,600** pounds, the possible numbers of boxes the truck can tow are given by the inequality **$500 + 120b \leq 4,600$** . Subtracting **500** from both sides of this inequality yields **$120b \leq 4,100$** . Dividing both sides of this inequality by **120** yields **$b \leq \frac{205}{6}$** , or **b** is less than or equal to approximately **34.17**. Since the number of boxes, **b** , must be a whole number, the maximum number of boxes the truck can tow is the greatest whole number less than **34.17**, which is **34**.

Choice B is incorrect. Towing the trailer and **35** boxes would yield a combined weight of **4,700** pounds, which is greater than **4,600** pounds.

Choice C is incorrect. Towing the trailer and **38** boxes would yield a combined weight of **5,060** pounds, which is greater than **4,600** pounds.

Choice D is incorrect. Towing the trailer and **39** boxes would yield a combined weight of **5,180** pounds, which is greater than **4,600** pounds.

Question Difficulty: Medium

Question ID 5c60c35f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 5c60c35f

$4x + 5 = 165$

What is the solution to the given equation?

ID: 5c60c35f Answer

Correct Answer: 40

Rationale

The correct answer is **40**. Subtracting **5** from both sides of the given equation yields $4x = 160$. Dividing both sides of this equation by **4** yields $x = 40$. Therefore, the solution to the given equation is **40**.

Question Difficulty: Easy

Question ID fcc4a886

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: fcc4a886

A factory makes **9**-inch, **7**-inch, and **4**-inch concrete screws. During a certain day, the number of **9**-inch concrete screws that the factory makes is **5** times the number *n* of **7**-inch concrete screws, and the number of **4**-inch concrete screws is **22**. During this day, the factory makes **100** concrete screws total. Which equation represents this situation?

- A. $9(5n) + 7n + 4(22) = 100$
- B. $9n + 7n + 4n = 100$
- C. $5n + 22 = 100$
- D. $6n + 22 = 100$

ID: fcc4a886 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day at a factory, the number of **7**-inch concrete screws the factory makes is *n* and the number of **4**-inch concrete screws the factory makes is **22**. It's also given that during this day the number of **9**-inch concrete screws the factory makes is **5** times the number of **7**-inch concrete screws, or **5n**. Therefore, the total number of **7**-inch, **9**-inch, and **4**-inch concrete screws is *n* + **5n** + **22**, or **6n** + **22**. It's given that during this day, the factory makes **100** concrete screws total. Thus, the equation **6n** + **22** = **100** represents this situation.

Choice A is incorrect. This equation represents a situation where the total length, in inches, of all the concrete screws, rather than the total number of concrete screws, is **100**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the total number of **9**-inch concrete screws and **4**-inch concrete screws, not including the **7**-inch concrete screws, is **100**.

Question Difficulty: Hard

Question ID fabe653f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: fabe653f

$$\begin{aligned}y &= 9x + 12 \\ x + 7y &= 20\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: fabe653f Answer

Correct Answer: 3

Rationale

The correct answer is **3**. It's given that $y = 9x + 12$. Substituting $9x + 12$ for y in the second equation in the system, $x + 7y = 20$, yields $x + 7(9x + 12) = 20$, which gives $x + 63x + 84 = 20$, or $64x + 84 = 20$. Subtracting **84** from each side of this equation yields $64x = -64$. Dividing each side of this equation by **64** yields $x = -1$. Substituting -1 for x in the first equation in the system, $y = 9x + 12$, yields $y = 9(-1) + 12$, or $y = 3$. Therefore, the value of y is **3**.

Question Difficulty: Hard

Question ID 772b3070

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 772b3070

Line k is defined by $y = 7x + \frac{1}{8}$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

- A. -8
- B. $-\frac{1}{7}$
- C. $\frac{1}{8}$
- D. 7

ID: 772b3070 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that line k is defined by $y = 7x + \frac{1}{8}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line defined by this equation in the xy -plane and b represents the y -coordinate of the y -intercept of this line. Therefore, the slope of line k is 7 . It's also given that line j is perpendicular to line k in the xy -plane. Therefore, the slope of line j is the opposite reciprocal of the slope of line k . The opposite reciprocal of 7 is $-\frac{1}{7}$. Therefore, the slope of line j is $-\frac{1}{7}$.

Choice A is incorrect. This is the opposite reciprocal of the y -coordinate of the y -intercept, not the slope, of line k .

Choice C is incorrect. This is the y -coordinate of the y -intercept of line k , not the slope of line j .

Choice D is incorrect. This is the slope of a line that is parallel, not perpendicular, to line k .

Question Difficulty: Medium

Question ID 6d1cc26a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 6d1cc26a

Line h is defined by $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$. Line j is perpendicular to line h in the xy -plane. What is the slope of line j ?

- A. $-\frac{7}{5}$
- B. $-\frac{5}{7}$
- C. $\frac{7}{5}$
- D. $\frac{5}{7}$

ID: 6d1cc26a Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that line h is defined by $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$. This equation can be written in slope-intercept form $y = mx + b$, where m is the slope of line h and b is the y -coordinate of the y -intercept of line h . Adding 70 to both sides of $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$ yields $\frac{1}{5}x + \frac{1}{7}y = 70$. Subtracting $\frac{1}{5}x$ from both sides of this equation yields $\frac{1}{7}y = -\frac{1}{5}x + 70$. Multiplying both sides of this equation by 7 yields $y = -\frac{7}{5}x + 490$. Therefore, the slope of line h is $-\frac{7}{5}$. It's given that line j is perpendicular to line h in the xy -plane. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Therefore, the slope of line j is the negative reciprocal of the slope of line h . The negative reciprocal of $-\frac{7}{5}$ is $\frac{-1}{(-\frac{7}{5})}$, or $\frac{5}{7}$. Therefore, the slope of line j is $\frac{5}{7}$.

Choice A is incorrect. This is the slope of a line in the xy -plane that is parallel, not perpendicular, to line h .

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of $-\frac{7}{5}$.

Choice C is incorrect. This is the negative, not the negative reciprocal, of $-\frac{7}{5}$.

Question Difficulty: Hard

Question ID da7cded0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: da7cded0

A gym charges its members a onetime **\$36** enrollment fee and a membership fee of **\$19** per month. If there are no charges other than the enrollment fee and the membership fee, after how many months will a member have been charged a total of **\$188** at the gym?

- A. **4**
- B. **5**
- C. **8**
- D. **10**

ID: da7cded0 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that a gym charges its members a onetime **\$36** enrollment fee and a membership fee of **\$19** per month. Let x represent the number of months at the gym after which a member will have been charged a total of **\$188**. If there are no charges other than the enrollment fee and the membership fee, the equation $36 + 19x = 188$ can be used to represent this situation. Subtracting **36** from both sides of this equation yields $19x = 152$. Dividing both sides of this equation by **19** yields $x = 8$. Therefore, a member will have been charged a total of **\$188** after **8** months at the gym.

Choice A is incorrect. A member will have been charged a total of $\$(36 + 19 \times 4)$, or **\$112**, after **4** months at the gym.

Choice B is incorrect. A member will have been charged a total of $\$(36 + 19 \times 5)$, or **\$131**, after **5** months at the gym.

Choice D is incorrect. A member will have been charged a total of $\$(36 + 19 \times 10)$, or **\$226**, after **10** months at the gym.

Question Difficulty: Easy

Question ID 3ac93939

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 3ac93939

$\frac{1}{3}(x + 6) - \frac{1}{2}(x + 6) = -8$

What value of x is the solution to the given equation?

ID: 3ac93939 Answer

Correct Answer: 42

Rationale

The correct answer is **42**. The expression $(x + 6)$ is a factor of both terms on the left-hand side of the given equation. Therefore, the given equation can be written as $(x + 6)(\frac{1}{3} - \frac{1}{2}) = -8$, or $(x + 6)(-\frac{1}{6}) = -8$. Multiplying each side of this equation by -6 yields $x + 6 = 48$. Subtracting **6** from each side of this equation yields $x = 42$. Therefore, the value of x that is the solution to the given equation is **42**.

Question Difficulty: Hard

Question ID ac02dad2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: ac02dad2

$$2x + y = 37$$

In triangle QRS , sides QR and RS each have a length of x centimeters and side SQ has a length of y centimeters. The given equation represents this situation. Which of the following is the best interpretation of 37 in this context?

- A. The difference, in centimeters, between the lengths of sides QR and SQ
- B. The difference, in centimeters, between the lengths of sides QR and RS
- C. The sum of the lengths, in centimeters, of the three sides of the triangle
- D. The length, in centimeters, of one of the two sides of equal length

ID: ac02dad2 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that in triangle QRS , sides QR and RS each have a length of x centimeters. Therefore, the expression $2x$ represents the sum of the lengths, in centimeters, of sides QR and RS . It's also given that side SQ has a length of y centimeters. Therefore, the expression $2x + y$ represents the sum of the lengths, in centimeters, of sides QR , RS , and SQ . Since $2x + y$ is the sum of the lengths, in centimeters, of the three sides of the triangle and $2x + y = 37$, it follows that 37 is the sum of the lengths, in centimeters, of the three sides of the triangle.

Choice A is incorrect. The difference, in centimeters, between the lengths of sides QR and SQ is $x - y$, not 37 .

Choice B is incorrect. The difference, in centimeters, between the lengths of sides QR and RS is $x - x$, or 0 , not 37 .

Choice D is incorrect. The length, in centimeters, of one of the two sides of equal length is x , not 37 .

Question Difficulty: Medium

Question ID 2573a53e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 2573a53e

x	10	15	20	25
$f(x)$	82	137	192	247

The table shows four values of x and their corresponding values of $f(x)$. There is a linear relationship between x and $f(x)$ that is defined by the equation $f(x) = mx - 28$, where m is a constant. What is the value of m ?

ID: 2573a53e Answer

Correct Answer: 11

Rationale

The correct answer is **11**. It's given that $f(x)$ is defined by the equation $f(x) = mx - 28$, where m is a constant. It's also given in the table that when $x = 10$, $f(x) = 82$. Substituting **10** for x and **82** for $f(x)$ in the equation $f(x) = mx - 28$ yields, $82 = m(10) - 28$. Adding **28** to both sides of this equation yields $110 = 10m$. Dividing both sides of this equation by **10** yields $11 = m$. Therefore, the value of m is **11**.

Question Difficulty: Medium

Question ID 7c24f272

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 7c24f272

A business owner plans to purchase the same model of chair for each of the **81** employees. The total budget to spend on these chairs is **\$14,000**, which includes a **7%** sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

- A. **\$148.15**
- B. **\$161.53**
- C. **\$172.84**
- D. **\$184.94**

ID: 7c24f272 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that a business owner plans to purchase **81** chairs. If ***p*** is the price per chair, the total price of purchasing **81** chairs is **81*p***. It’s also given that **7%** sales tax is included, which is equivalent to **81*p*** multiplied by **1.07**, or **81(1.07)*p***. Since the total budget is **\$14,000**, the inequality representing the situation is given by **81(1.07)*p* ≤ 14,000**. Dividing both sides of this inequality by **81(1.07)** and rounding the result to two decimal places gives ***p* ≤ 161.53**. To not exceed the budget, the maximum possible price per chair is **\$161.53**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the maximum possible price per chair including sales tax, not the maximum possible price per chair before sales tax.

Choice D is incorrect. This is the maximum possible price if the sales tax is added to the total budget, not the maximum possible price per chair before sales tax.

Question Difficulty: Hard

Question ID 35ade89d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 35ade89d

x	$g(x)$
1	32
2	28
3	24
4	20

For the linear function g , the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

- A. 4
- B. 16
- C. 32
- D. 36

ID: 35ade89d Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that for the linear function g , the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 32. Substituting 1 for x and 32 for $g(x)$ in $g(x) = mx + b$ yields $32 = m(1) + b$, or $32 = m + b$. Subtracting b from both sides of this equation yields $32 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 28. Substituting 2 for x and 28 for $g(x)$ in $g(x) = mx + b$ yields $28 = m(2) + b$, or $28 = 2m + b$. Substituting $32 - b$ for m in this equation yields $28 = 2(32 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $28 = 64 - 2b + b$, or $28 = 64 - b$. Subtracting 64 from both sides of this equation yields $-36 = -b$. Dividing both sides of this equation by -1 yields $36 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 11436130

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 11436130

The perimeter of an isosceles triangle is **36** feet. Each of the two congruent sides of the triangle has a length of **10** feet. What is the length, in feet, of the third side?

- A. **10**
- B. **12**
- C. **16**
- D. **18**

ID: 11436130 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the perimeter of an isosceles triangle is **36** feet and that each of the two congruent sides has a length of **10** feet. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $10 + 10 + x = 36$ can be used to represent this situation, where x is the length, in feet, of the third side. Combining like terms on the left-hand side of this equation yields $20 + x = 36$. Subtracting **20** from each side of this equation yields $x = 16$. Therefore, the length, in feet, of the third side is **16**.

Choice A is incorrect. This would be the length, in feet, of the third side if the perimeter was **30** feet, not **36** feet.

Choice B is incorrect. This would be the length, in feet, of the third side if the perimeter was **32** feet, not **36** feet.

Choice D is incorrect. This would be the length, in feet, of the third side if the perimeter was **38** feet, not **36** feet.

Question Difficulty: Medium

Question ID 25f7665c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 25f7665c

$$\begin{aligned}\frac{3}{2}y - \frac{1}{4}x &= \frac{2}{3} - \frac{3}{2}y \\ \frac{1}{2}x + \frac{3}{2} &= py + \frac{9}{2}\end{aligned}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

ID: 25f7665c Answer

Correct Answer: 6

Rationale

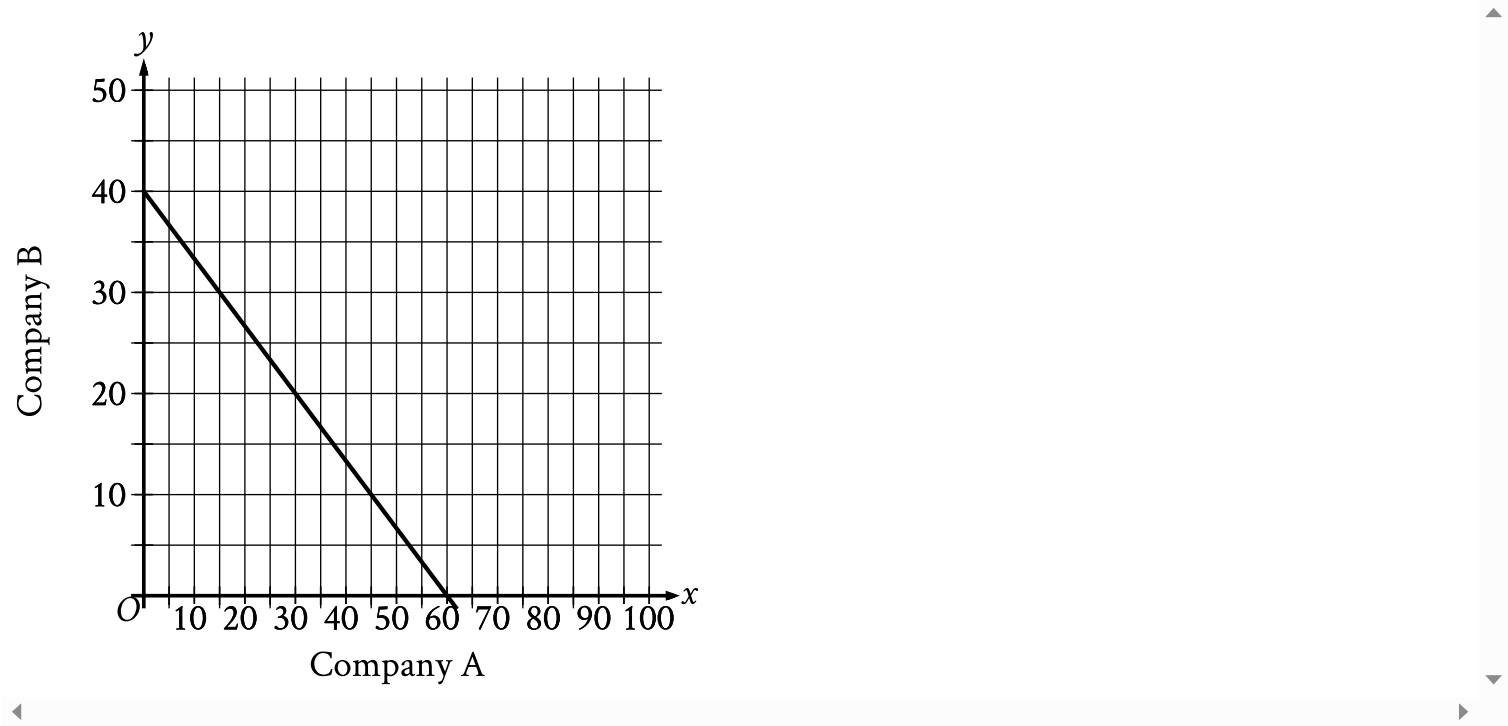
The correct answer is **6**. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The first equation in the given system is $\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$. Multiplying each side of this equation by **12** yields $18y - 3x = 8 - 18y$. Adding **18y** to each side of this equation yields $36y - 3x = 8$, or $-3x + 36y = 8$. The second equation in the given system is $\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$. Multiplying each side of this equation by **2** yields $x + 3 = 2py + 9$. Subtracting $2py$ from each side of this equation yields $x + 3 - 2py = 9$. Subtracting **3** from each side of this equation yields $x - 2py = 6$. Therefore, the two equations in the given system, written in standard form, are $-3x + 36y = 8$ and $x - 2py = 6$. As previously stated, if this system has no solution, the lines represented by the equations in the xy -plane are parallel and distinct, meaning the proportion $\frac{1}{-3} = \frac{-2p}{36}$, or $-\frac{1}{3} = -\frac{p}{18}$, is true and the proportion $\frac{6}{8} = \frac{1}{-3}$ is not true. The proportion $\frac{6}{8} = \frac{1}{-3}$ is not true. Multiplying each side of the true proportion, $-\frac{1}{3} = -\frac{p}{18}$, by **-18** yields **6** = p . Therefore, if the system has no solution, then the value of p is **6**.

Question Difficulty: Hard

Question ID 75dc50c6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 75dc50c6



The graph shows the relationship between the number of shares of stock from Company A, x , and the number of shares of stock from Company B, y , that Simone can purchase. Which equation could represent this relationship?

- A. $y = 8x + 12$
- B. $8x + 12y = 480$
- C. $y = 12x + 8$
- D. $12x + 8y = 480$

ID: 75dc50c6 Answer

Correct Answer: B

Rationale

Choice B is correct. The graph shown is a line passing through the points $(0, 40)$ and $(60, 0)$. Since the relationship between x and y is linear, if two points on the graph make a linear equation true, then the equation represents the relationship. Substituting 0 for x and 40 for y in the equation in choice B, $8x + 12y = 480$, yields $8(0) + 12(40) = 480$, or $480 = 480$, which is true. Substituting 60 for x and 0 for y in the equation $8x + 12y = 480$ yields $8(60) + 12(0) = 480$, or $480 = 480$, which is true. Therefore, the equation $8x + 12y = 480$ represents the relationship between x and y .

Choice A is incorrect. The point $(0, 40)$ is not on the graph of this equation, since $40 = 8(0) + 12$, or $40 = 12$, is not true.

Choice C is incorrect. The point $(0, 40)$ is not on the graph of this equation, since $40 = 12(0) + 8$, or $40 = 8$, is not true.

Choice D is incorrect. The point $(0, 40)$ is not on the graph of this equation, since $12(0) + 8(40) = 480$, or $320 = 480$, is not true.

Question ID 2b225f92

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 2b225f92

A bus traveled on the highway and on local roads to complete a trip of **160 miles**. The trip took **4 hours**. The bus traveled at an average speed of **55 miles per hour (mph)** on the highway and an average speed of **25 mph** on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

- A. $55x + 25y = 4$
 $x + y = 160$
- B. $55x + 25y = 160$
 $x + y = 4$
- C. $25x + 55y = 4$
 $x + y = 160$
- D. $25x + 55y = 160$
 $x + y = 4$

ID: 2b225f92 Answer

Correct Answer: B

Rationale

Choice B is correct. If the bus traveled at an average speed of **55 miles per hour (mph)** on the highway for x hours, then the bus traveled **$55x$** miles on the highway. If the bus traveled at an average speed of **25 mph** on local roads for y hours, then the bus traveled **$25y$** miles on local roads. It's given that the trip was **160** miles. This can be represented by the equation **$55x + 25y = 160$** . It's also given that the trip took **4** hours. This can be represented by the equation **$x + y = 4$** . Therefore, the system consisting of the equations **$55x + 25y = 160$** and **$x + y = 4$** represents this situation.

Choice A is incorrect. This system of equations represents a situation where the trip was **4** miles and took **160** hours.

Choice C is incorrect. This system of equations represents a situation where the trip was **4** miles and took **160** hours, and the bus traveled at an average speed of **25 mph** on the highway and **55 mph** on local roads.

Choice D is incorrect. This system of equations represents a situation where the bus traveled at an average speed of **25 mph** on the highway and **55 mph** on local roads.

Question Difficulty: Hard

Question ID 19014c22

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 19014c22

Line k is defined by $y = 3x + 15$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

- A. $-\frac{1}{3}$
- B. $-\frac{1}{12}$
- C. $-\frac{1}{18}$
- D. $-\frac{1}{45}$

ID: 19014c22 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that line j is perpendicular to line k in the xy -plane. It follows that the slope of line j is the opposite reciprocal of the slope of line k . The equation for line k is written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept of the line. It follows that the slope of line k is 3 . The opposite reciprocal of a number is -1 divided by the number. Thus, the opposite reciprocal of 3 is $-\frac{1}{3}$. Therefore, the slope of line j is $-\frac{1}{3}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 36b3a5b7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 36b3a5b7

The pressure exerted on a scuba diver at sea level is **14.70 pounds per square inch (psi)**. For each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by **0.44 psi**. What is the total pressure, in **psi**, exerted on the scuba diver at **105** feet below sea level?

- A. **60.90**
- B. **31.50**
- C. **14.70**
- D. **0.44**

ID: 36b3a5b7 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the pressure exerted on a scuba diver at sea level is **14.70 pounds per square inch (psi)**. It's also given that for each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by **0.44 psi**. The total pressure, in **psi**, exerted on the scuba diver at x feet below sea level can be represented by the expression **$0.44x + 14.70$** . Substituting **105** for x in this expression yields **$0.44(105) + 14.70$** , or **60.90**. Therefore, the total pressure exerted on the scuba diver at **105** feet below sea level is **60.90 psi**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pressure, in **psi**, exerted on the scuba diver at sea level, not at **105** feet below sea level.

Choice D is incorrect. This is the rate by which the pressure, in **psi**, exerted on the scuba diver increases for each foot the scuba diver descends below sea level.

Question Difficulty: Medium

Question ID 45f00d75

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 45f00d75

$$\begin{aligned}y - 9x &= 13 \\ 5x &= 2y\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(\frac{5}{2}, 1)$
- B. $(1, \frac{2}{5})$
- C. $(-2, -5)$
- D. $(-5, -2)$

ID: 45f00d75 Answer

Correct Answer: C

Rationale

Choice C is correct. Adding $9x$ to both sides of the first equation in the given system yields $y = 9x + 13$. Substituting the expression $9x + 13$ for y in the second equation in the given system yields $5x = 2(9x + 13)$. Distributing the 2 on the right-hand side of this equation yields $5x = 18x + 26$. Subtracting $18x$ from both sides of this equation yields $-13x = 26$. Dividing both sides of this equation by -13 yields $x = -2$. Substituting -2 for x in the equation $y = 9x + 13$ yields $y = 9(-2) + 13$, or $y = -5$. Therefore, the solution (x, y) to the given system of equations is $(-2, -5)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution (y, x) , not (x, y) , to the given system of equations.

Question Difficulty: Medium

Question ID b2d554a4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: b2d554a4

$$\begin{aligned} 48x - 64y &= 48y + 24 \\ ry &= \frac{1}{8} - 12x \end{aligned}$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

ID: b2d554a4 Answer

Correct Answer: -28

Rationale

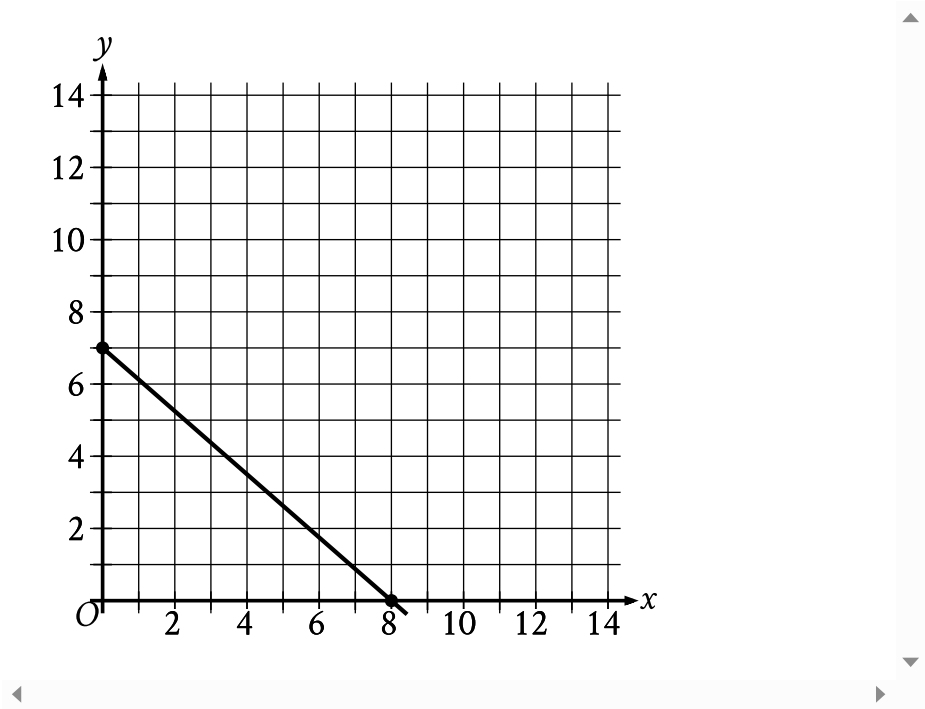
The correct answer is -28 . A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in the form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients for x and y in the other equation. The first equation in the given system, $48x - 64y = 48y + 24$, can be written in the form $Ax + By = C$ by subtracting $48y$ from both sides of the equation to yield $48x - 112y = 24$. The second equation in the given system, $ry = \frac{1}{8} - 12x$, can be written in the form $Ax + By = C$ by adding $12x$ to both sides of the equation to yield $12x + ry = \frac{1}{8}$. The coefficient of x in the second equation is $\frac{1}{4}$ times the coefficient of x in the first equation. That is, $48\left(\frac{1}{4}\right) = 12$. For the lines to be parallel, the coefficient of y in the second equation must also be $\frac{1}{4}$ times the coefficient of y in the first equation. Therefore, $-112\left(\frac{1}{4}\right) = r$, or $-28 = r$. Thus, if the given system has no solution, the value of r is -28 .

Question Difficulty: Hard

Question ID 0989eea4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 0989eea4



The point with coordinates $(d, 4)$ lies on the line shown. What is the value of d ?

- A. $\frac{7}{2}$
- B. $\frac{26}{7}$
- C. $\frac{24}{7}$
- D. $\frac{27}{8}$

ID: 0989eea4 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given from the graph that the points $(0, 7)$ and $(8, 0)$ lie on the line. For two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(0, 7)$ for (x_1, y_1) and $(8, 0)$ for (x_2, y_2) in this formula, the slope of the line can be calculated as $m = \frac{0 - 7}{8 - 0}$, or $m = -\frac{7}{8}$. It's also given that the point $(d, 4)$ lies on the line. Substituting $(d, 4)$ for (x_1, y_1) , $(8, 0)$ for (x_2, y_2) , and $-\frac{7}{8}$ for m in the slope formula yields $-\frac{7}{8} = \frac{0 - 4}{8 - d}$, or $-\frac{7}{8} = \frac{-4}{8 - d}$. Multiplying both sides of this equation by $8 - d$ yields $-\frac{7}{8}(8 - d) = -4$. Expanding the left-hand side of this equation yields $-7 + \frac{7}{8}d = -4$. Adding 7 to both sides of this equation yields $\frac{7}{8}d = 3$. Multiplying both sides of this equation by $\frac{8}{7}$ yields $d = \frac{24}{7}$. Thus, the value of d is $\frac{24}{7}$.

- Choice A is incorrect. This is the value of y when $x = 4$.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice D is incorrect and may result from conceptual or calculation errors.

Question ID 3108220a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 3108220a

The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

- A. 50
- B. 57
- C. 80
- D. 110

ID: 3108220a Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the function f is defined by $f(x) = 25x + 30$. Substituting 2 for x in this equation yields $f(2) = 25(2) + 30$, which is equivalent to $f(2) = 50 + 30$, or $f(2) = 80$. Therefore, the value of $f(x)$ is 80 when $x = 2$.

Choice A is incorrect. This is the value of $25(2)$, not $25(2) + 30$.

Choice B is incorrect. This is the value of $25 + 2 + 30$, not $25(2) + 30$.

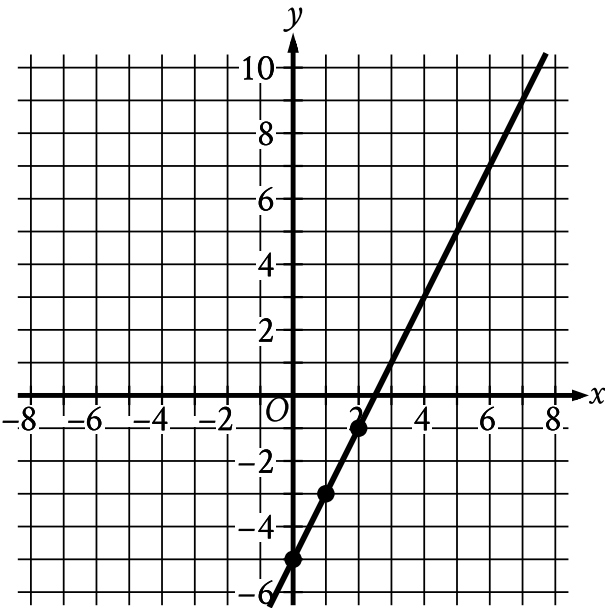
Choice D is incorrect. This is the value of $(25 + 30)(2)$, not $25(2) + 30$.

Question Difficulty: Easy

Question ID 55614835

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 55614835



The graph shows the linear relationship between x and y . Which table gives three values of x and their corresponding values of y for this relationship?

A.

x	y
0	0
1	-7
2	-9

B.

x	y
0	0
1	-3
2	-1

C.

x	y
0	-5
1	-7
2	-9

D.

x	y
0	-5
1	-3
2	-1

ID: 55614835 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the graph shows the linear relationship between x and y . The given graph passes through the points $(0, -5)$, $(1, -3)$, and $(2, -1)$. It follows that when $x = 0$, the corresponding value of y is -5 , when $x = 1$, the corresponding value of y is -3 , and when $x = 2$, the corresponding value of y is -1 . Of the given choices, only the table in choice D gives these three values of x and their corresponding values of y for the relationship shown in the graph.

Choice A is incorrect. This table represents a relationship between x and y such that the graph passes through the points $(0, 0)$, $(1, -7)$, and $(2, -9)$.

Choice B is incorrect. This table represents a relationship between x and y such that the graph passes through the points $(0, 0)$, $(1, -3)$, and $(2, -1)$.

Choice C is incorrect. This table represents a linear relationship between x and y such that the graph passes through the points $(0, -5)$, $(1, -7)$, and $(2, -9)$.

Question Difficulty: Easy

Question ID d6ded01f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: d6ded01f

$$\frac{12x+28}{4} - \frac{s}{13} = r(x - 8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s ?

ID: d6ded01f Answer

Correct Answer: 403

Rationale

The correct answer is **403**. For a linear equation in one variable to have infinitely many solutions, the coefficients of the variable must be equal on both sides of the equation and the constant terms must also be equal on both sides of the equation. The given equation can be rewritten as $\frac{4(3x+7)}{4} - \frac{s}{13} = r(x - 8)$, or $3x + 7 - \frac{s}{13} = r(x - 8)$. Applying the distributive property to the right-hand side of this equation yields $3x + 7 - \frac{s}{13} = rx - 8r$. For this equation to have infinitely many solutions, the coefficients of x must be equal, so it follows that $3 = r$. Additionally, the constant terms must be equal, which means $7 - \frac{s}{13} = -8r$. Substituting 3 for r in this equation yields $7 - \frac{s}{13} = -8(3)$, or $7 - \frac{s}{13} = -24$. Adding $\frac{s}{13}$ to both sides of this equation yields $7 = -24 + \frac{s}{13}$. Adding 24 to both sides of this equation yields $31 = \frac{s}{13}$. Multiplying both sides of this equation by 13 yields $403 = s$. Therefore, if the equation has infinitely many solutions, the value of s is **403**.

Question Difficulty: Hard

Question ID f55b70f4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: f55b70f4

A wire with a length of **106** inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is **6** more than **4** times the value of y . What is the value of x ?

- A. **25**
- B. **28**
- C. **56**
- D. **86**

ID: f55b70f4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a wire with a length of **106** inches is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. This can be represented by the equation $x + y = 106$. It's also given that the value of x is **6** more than **4** times the value of y . This can be represented by the equation $x = 4y + 6$. Substituting $4y + 6$ for x in the equation $x + y = 106$ yields $4y + 6 + y = 106$, or $5y + 6 = 106$. Subtracting **6** from each side of this equation yields $5y = 100$. Dividing each side of this equation by **5** yields $y = 20$. Substituting **20** for y in the equation $x = 4y + 6$ yields $x = 4(20) + 6$, or $x = 86$.

Choice A is incorrect. This value represents less than half of the total length of **106** inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is **6** more than **4** times the value of y .

Choice B is incorrect. This value represents less than half of the total length of **106** inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is **6** more than **4** times the value of y .

Choice C is incorrect. This represents a part that is **6** more than the length of the other part, rather than **6** more than **4** times the length of the other part.

Question Difficulty: Hard

Question ID 1269f3df

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 1269f3df

A company that creates and sells tape dispensers calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. The equation $15,000 = 2.00x - 4,500$ represents this situation for a month where x tape dispensers are created and sold. Which statement is the best interpretation of $2.00x$ in this context?

- A. The monthly sales revenue, in dollars, from selling x tape dispensers
- B. The monthly sales revenue, in dollars, from each tape dispenser sold
- C. The monthly cost, in dollars, of creating each tape dispenser
- D. The monthly cost, in dollars, of creating x tape dispensers

ID: 1269f3df Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the equation $15,000 = 2.00x - 4,500$ represents this situation for a month where x tape dispensers are created and sold. It’s also given that the company calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. It follows that $2.00x$ represents the monthly sales revenue, in dollars. Therefore, the best interpretation of $2.00x$ in this context is the monthly sales revenue from selling x tape dispensers.

Choice B is incorrect. This is the best interpretation of 2.00 , not $2.00x$.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of $4,500$, not $2.00x$.

Question Difficulty: Medium

Question ID 8b7fd4ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 8b7fd4ac

$$f(x) = 8x + 4$$

The function f gives the estimated height, in feet, of a willow tree x years after its height was first measured. Which statement is the best interpretation of 4 in this context?

- A. The tree will be measured each year for 4 years.
- B. The tree is estimated to grow to a maximum height of 4 feet.
- C. The estimated height of the tree increased by 4 feet each year.
- D. The estimated height of the tree was 4 feet when it was first measured.

ID: 8b7fd4ac Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the function $f(x) = 8x + 4$ gives the estimated height, in feet, of a willow tree x years after its height was first measured. For a function defined by an equation of the form $f(x) = mx + b$, where m and b are constants, b represents the value of $f(x)$ when $x = 0$. It follows that in the given function, 4 represents the value of $f(x)$ when $x = 0$. Therefore, the best interpretation of 4 in this context is that the estimated height of the tree was 4 feet when it was first measured.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 46be5680

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 46be5680

Store A sells raspberries for \$5.50 per pint and blackberries for \$3.00 per pint. Store B sells raspberries for \$6.50 per pint and blackberries for \$8.00 per pint. A certain purchase of raspberries and blackberries would cost \$37.00 at Store A or \$66.00 at Store B. How many pints of blackberries are in this purchase?

- A. 4
- B. 5
- C. 8
- D. 12

ID: 46be5680 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that Store A sells raspberries for \$5.50 per pint and blackberries for \$3.00 per pint, and a certain purchase of raspberries and blackberries at Store A would cost \$37.00. It’s also given that Store B sells raspberries for \$6.50 per pint and blackberries for \$8.00 per pint, and this purchase of raspberries and blackberries at Store B would cost \$66.00. Let r represent the number of pints of raspberries and b represent the number of pints of blackberries in this purchase. The equation $5.50r + 3.00b = 37.00$ represents this purchase of raspberries and blackberries from Store A and the equation $6.50r + 8.00b = 66.00$ represents this purchase of raspberries and blackberries from Store B. Solving the system of equations by elimination gives the value of r and the value of b that make the system of equations true. Multiplying both sides of the equation for Store A by 6.5 yields $(5.50r)(6.5) + (3.00b)(6.5) = (37.00)(6.5)$, or $35.75r + 19.5b = 240.5$. Multiplying both sides of the equation for Store B by 5.5 yields $(6.50r)(5.5) + (8.00b)(5.5) = (66.00)(5.5)$, or $35.75r + 44b = 363$. Subtracting both sides of the equation for Store A, $35.75r + 19.5b = 240.5$, from the corresponding sides of the equation for Store B, $35.75r + 44b = 363$, yields $(35.75r - 35.75r) + (44b - 19.5b) = (363 - 240.5)$, or $24.5b = 122.5$. Dividing both sides of this equation by 24.5 yields $b = 5$. Thus, 5 pints of blackberries are in this purchase.

Choice A is incorrect. This is the number of pints of raspberries, not blackberries, in the purchase.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID faa89fd2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: faa89fd2

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than **100** centimeters. Each of the following inequalities represents the possible heights h , in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights h , in centimeters, for a tall plant species?

- A. $106 < h < 158$
- B. $80 < h < 100$
- C. $42 < h < 87$
- D. $17 < h < 85$

ID: faa89fd2 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a particular botanist classifies a species of plant as tall if its typical height when fully grown is more than **100** centimeters. The inequality $106 < h < 158$ represents possible heights h , in centimeters, for a plant species when fully grown where h is between **106** and **158** centimeters. Since all values of h in this inequality are greater than **100** centimeters, this inequality represents the possible heights for a tall plant species.

Choice B is incorrect. This inequality represents possible heights h , in centimeters, for a plant species when fully grown where h is between **80** and **100** centimeters, not more than **100** centimeters.

Choice C is incorrect. This inequality represents possible heights h , in centimeters, for a plant species when fully grown where h is between **42** and **87** centimeters, not more than **100** centimeters.

Choice D is incorrect. This inequality represents possible heights h , in centimeters, for a plant species when fully grown where h is between **17** and **85** centimeters, not more than **100** centimeters.

Question Difficulty: Medium

Question ID 4ca7a295

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 4ca7a295

$$\begin{aligned} -12x + 14y &= 36 \\ -6x + 7y &= -18 \end{aligned}$$

How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 4ca7a295 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has zero solutions if the lines representing the equations in the xy -plane are distinct and parallel. Two lines are distinct and parallel if they have the same slope but different y -intercepts. Each equation in the given system can be written in slope-intercept form $y = mx + b$, where m is the slope of the line representing the equation in the xy -plane and $(0, b)$ is the y -intercept. Adding $12x$ to both sides of the first equation in the given system of equations, $-12x + 14y = 36$, yields $14y = 12x + 36$. Dividing both sides of this equation by 14 yields $y = \frac{6}{7}x + \frac{18}{7}$. It follows that the first equation in the given system of equations has a slope of $\frac{6}{7}$ and a y -intercept of $(0, \frac{18}{7})$. Adding $6x$ to both sides of the second equation in the given system of equations, $-6x + 7y = -18$, yields $7y = 6x - 18$. Dividing both sides of this equation by 7 yields $y = \frac{6}{7}x - \frac{18}{7}$. It follows that the second equation in the given system of equations has a slope of $\frac{6}{7}$ and a y -intercept of $(0, -\frac{18}{7})$. Since the slopes of these lines are the same and the y -intercepts are different, it follows that the given system of equations has zero solutions.

Alternate approach: To solve the system by elimination, multiplying the second equation in the given system of equations, $-6x + 7y = -18$, by -2 yields $12x - 14y = 36$. Adding this equation to the first equation in the given system of equations, $-12x + 14y = 36$, yields $(-12x + 12x) + (-14y + 14y) = 36 + 36$, or $0 = 72$. Since this equation isn't true, the given system of equations has zero solutions.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ac181113

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: ac181113

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the y -intercept of the graph of $y = f(x)$ in the xy -plane in this context?

- A. Caleb used approximately 5 fluid ounces of juice for each popsicle.
- B. Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.
- C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.
- D. Caleb used approximately 30 fluid ounces of juice for each popsicle.

ID: ac181113 Answer

Correct Answer: C

Rationale

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m represents the slope and b represents the y -intercept, $(0, b)$, of the line of $y = f(x)$ in the xy -plane. The function $f(x) = -5x + 30$ is linear. Therefore, the graph of the given function $y = f(x)$ in the xy -plane has a y -intercept of $(0, 30)$. It's given that $f(x)$ gives the approximate volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. It follows that the y -intercept of $(0, 30)$ means that Caleb had approximately 30 fluid ounces of juice remaining after making 0 popsicles. In other words, Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

Choice A is incorrect. This is an interpretation of the slope, rather than the y -intercept, of the graph of $y = f(x)$ in the xy -plane.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 2d54169d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 2d54169d

A piece of wire with a length of **32** inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is **4** more than **3** times the value of y . What is the value of x ?

ID: 2d54169d Answer

Correct Answer: 25

Rationale

The correct answer is **25**. It's given that a piece of wire has a length of **32** inches and is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. It follows that the equation $x + y = 32$ represents this situation. It's also given that the value of x is **4** more than **3** times the value of y , or $x = 3y + 4$. Substituting $3y + 4$ for x in the equation $x + y = 32$ yields $3y + 4 + y = 32$. Combining like terms on the left-hand side of this equation yields $4y + 4 = 32$. Subtracting **4** from both sides of this equation yields $4y = 28$. Dividing both sides of this equation by **4** yields $y = 7$. Substituting **7** for y in the equation $x = 3y + 4$ yields $x = 3(7) + 4$, or $x = 25$. Therefore, the value of x is **25**.

Question Difficulty: Hard

Question ID bd243919

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: bd243919

The equation $40x + 20y = 160$ represents the number of sweaters, x , and number of shirts, y , that Yesenia purchased for \$160. If Yesenia purchased 2 sweaters, how many shirts did she purchase?

- A. 3
- B. 4
- C. 8
- D. 40

ID: bd243919 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $40x + 20y = 160$ represents the number of sweaters, x , and the number of shirts, y , that Yesenia purchased for \$160. If Yesenia purchased 2 sweaters, the number of shirts she purchased can be calculated by substituting 2 for x in the given equation, which yields $40(2) + 20y = 160$, or $80 + 20y = 160$. Subtracting 80 from both sides of this equation yields $20y = 80$. Dividing both sides of this equation by 20 yields $y = 4$. Therefore, if Yesenia purchased 2 sweaters, she purchased 4 shirts.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of shirts Yesenia purchased if she purchased 0 sweaters.

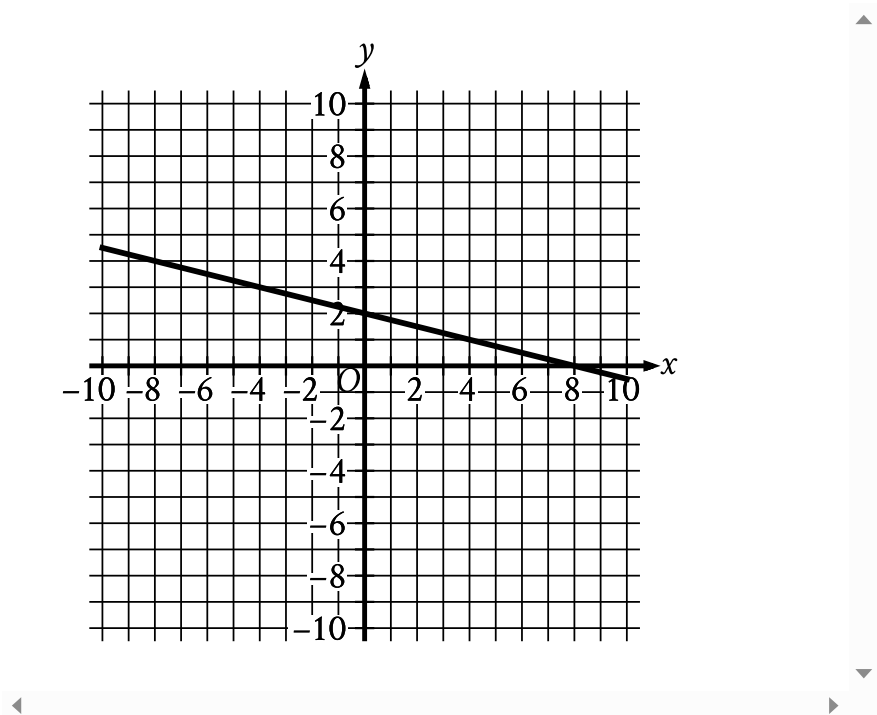
Choice D is incorrect. This is the price, in dollars, for each sweater, not the number of shirts Yesenia purchased.

Question Difficulty: Medium

Question ID 4522c150

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 4522c150



The graph of $y = f(x) + 14$ is shown. Which equation defines function f ?

- A. $f(x) = -\frac{1}{4}x - 12$
- B. $f(x) = -\frac{1}{4}x + 16$
- C. $f(x) = -\frac{1}{4}x + 2$
- D. $f(x) = -\frac{1}{4}x - 14$

ID: 4522c150 Answer

Correct Answer: A

Rationale

Choice A is correct. An equation for the graph shown can be written in slope-intercept form $y = mx + b$, where m is the slope of the graph and its y -intercept is $(0, b)$. Since the y -intercept of the graph shown is $(0, 2)$, the value of b is 2 . Since the graph also passes through the point $(4, 1)$, the slope can be calculated as $\frac{1-2}{4-0}$, or $-\frac{1}{4}$. Therefore, the value of m is $-\frac{1}{4}$. Substituting $-\frac{1}{4}$ for m and 2 for b in the equation $y = mx + b$ yields $y = -\frac{1}{4}x + 2$. It's given that an equation for the graph shown is $y = f(x) + 14$. Substituting $f(x) + 14$ for y in the equation $y = -\frac{1}{4}x + 2$ yields $f(x) + 14 = -\frac{1}{4}x + 2$. Subtracting 14 from both sides of this equation yields $f(x) = -\frac{1}{4}x - 12$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID df3b20f0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: df3b20f0

$$\begin{aligned}y &= 4x + 1 \\ 4y &= 15x - 8\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

ID: df3b20f0 Answer

Correct Answer: 35

Rationale

The correct answer is **35**. The first equation in the given system of equations defines y as $4x + 1$. Substituting $4x + 1$ for y in the second equation in the given system of equations yields $4(4x + 1) = 15x - 8$. Applying the distributive property on the left-hand side of this equation yields $16x + 4 = 15x - 8$. Subtracting $15x$ from each side of this equation yields $x + 4 = -8$. Subtracting 4 from each side of this equation yields $x = -12$. Substituting -12 for x in the first equation of the given system of equations yields $y = 4(-12) + 1$, or $y = -47$. Substituting -12 for x and -47 for y into the expression $x - y$ yields $-12 - (-47)$, or **35**.

Question Difficulty: Hard

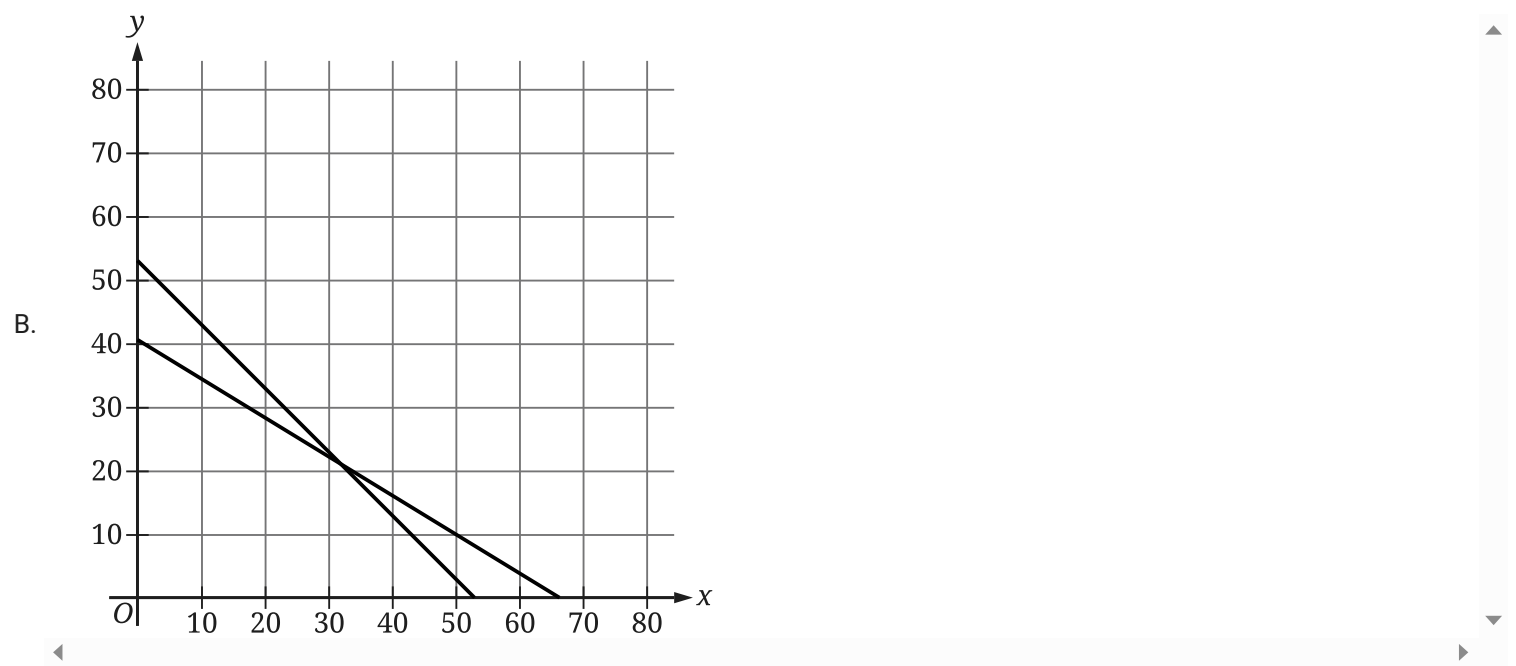
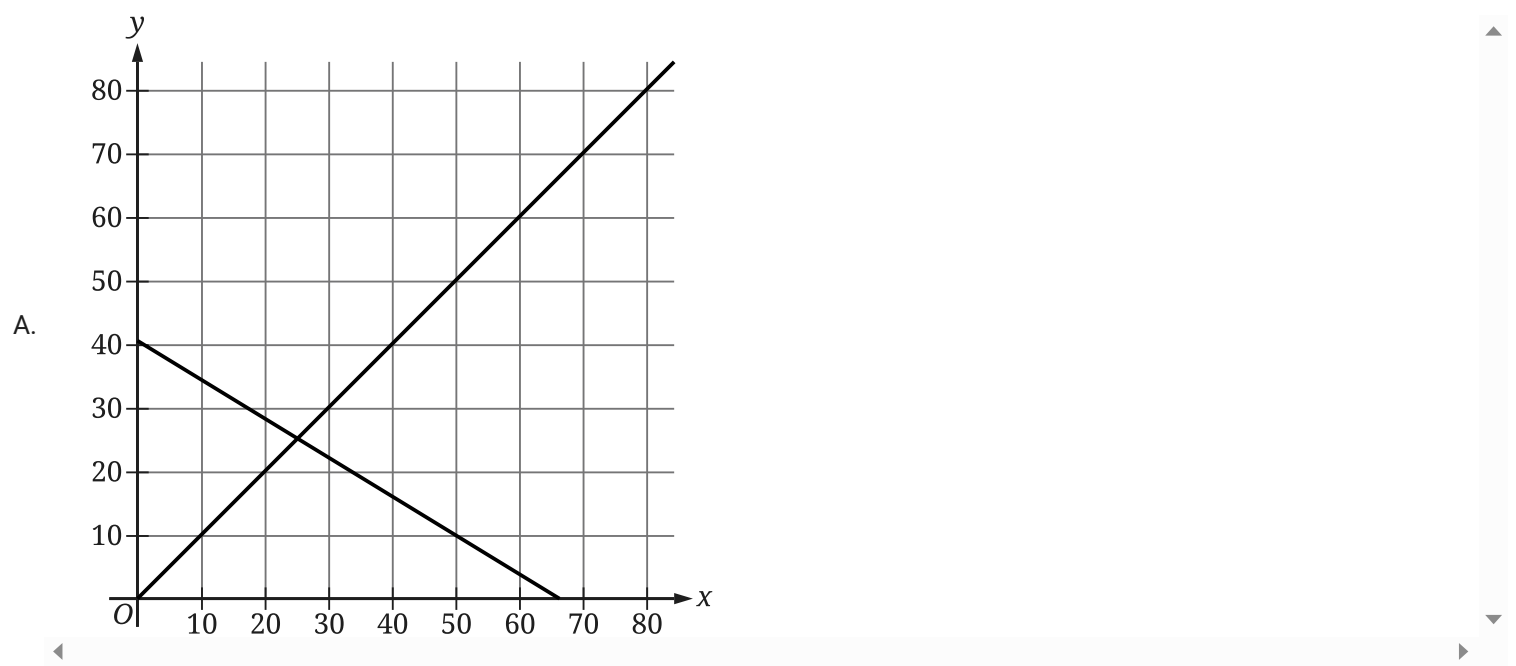
Question ID f7500040

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

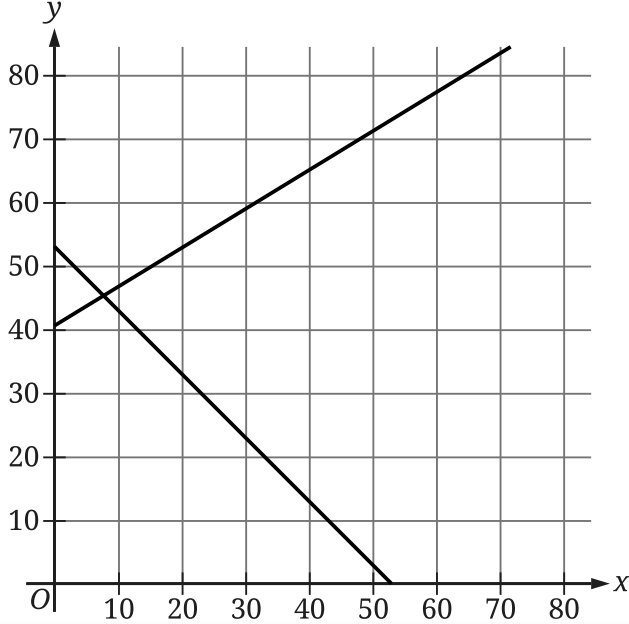
ID: f7500040

$$x + y = 53$$
$$11x + 18y = 730$$

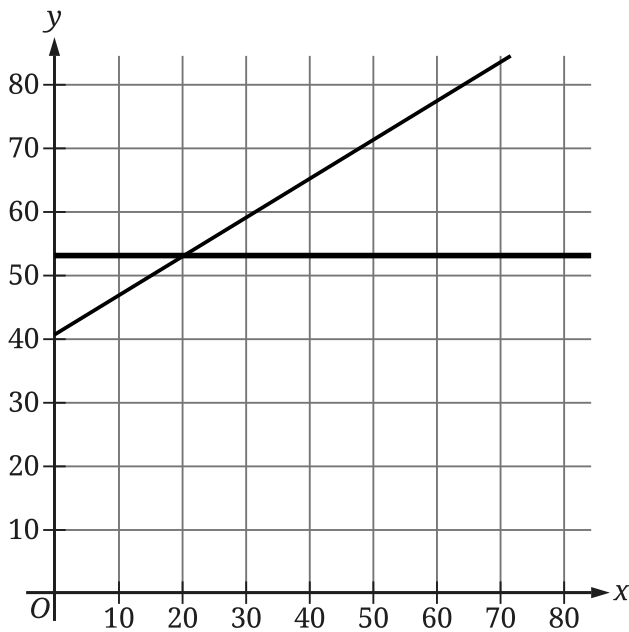
The given equations represent the possible numbers of beach chairs, x , and umbrellas, y , rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?



C.



D.



ID: f7500040 Answer

Correct Answer: B

Rationale

Choice B is correct. The intercepts of the graph of each equation can be found by substituting 0 for one variable at a time. For the equation $x + y = 53$, substituting 0 for y yields $x = 53$, so the x-intercept of the graph of the equation is $(53, 0)$. Substituting 0 for x in this equation yields $y = 53$, so the y-intercept of the graph of the equation is $(0, 53)$. Thus, the graph of the equation $x + y = 53$ has intercepts at $(53, 0)$ and $(0, 53)$. For the equation $11x + 18y = 730$, substituting 0 for y yields $x = \frac{730}{11}$, so the x-intercept of the graph of the equation is approximately $(66, 0)$. Substituting 0 for x in this equation yields $y = \frac{730}{18}$, so the y-intercept of the graph of the equation is approximately $(0, 41)$. Thus, the graph of the equation $11x + 18y = 730$ has intercepts at approximately $(66, 0)$ and $(0, 41)$. Choice B is a graph that shows one line with intercepts at $(0, 53)$ and $(53, 0)$ and another line with intercepts at approximately $(0, 41)$ and $(66, 0)$. Thus, choice B represents this situation.

Choice A is incorrect. This graph represents the equations $y = x$ and $11x + 18y = 730$.

Choice C is incorrect. This graph represents the equations $x + y = 53$ and $-11x + 18y = 730$.

Choice D is incorrect. This graph represents the equations $y = 53$ and $-11x + 18y = 730$.

Question Difficulty: Medium

Question ID bc1e9b9b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: bc1e9b9b

x	$f(x)$
0	29
1	32
2	35

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

- A. $f(x) = 3x + 29$
- B. $f(x) = 29x + 32$
- C. $f(x) = 35x + 29$
- D. $f(x) = 32x + 35$

ID: bc1e9b9b Answer

Correct Answer: A

Rationale

Choice A is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m and b are constants. It's given in the table that when $x = 0$, $f(x) = 29$. Substituting 0 for x and 29 for $f(x)$ in the equation $f(x) = mx + b$ yields $29 = m(0) + b$, or $29 = b$. Substituting 29 for b in the equation $f(x) = mx + b$ yields $f(x) = mx + 29$. It's also given in the table that when $x = 1$, $f(x) = 32$. Substituting 1 for x and 32 for $f(x)$ in the equation $f(x) = mx + 29$ yields $32 = m(1) + 29$, or $32 = m + 29$. Subtracting 29 from both sides of this equation yields $3 = m$. Substituting 3 for m in the equation $f(x) = mx + 29$ yields $f(x) = 3x + 29$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID bef391e6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: bef391e6

$x + y = 20$

One of the two equations in a system of linear equations is given. The system has exactly one solution. Which equation could be the second equation in the system?

- A. $3x + 3y = 60$
- B. $-x - y = 40$
- C. $3x + 3y = 14$
- D. $x - y = 20$

ID: bef391e6 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has exactly one solution if the graphs of the lines represented by the equations in the xy -plane have different slopes. In the xy -plane, the slope of a line defined by an equation in the form $Ax + By = C$, where A , B , and C are constants, is $-\frac{A}{B}$. It's given that one equation in the system is $x + y = 20$. It follows that for this equation, $A = 1$ and $B = 1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{1}$, or -1 . For the equation in choice D, $x - y = 20$, $A = 1$ and $B = -1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{(-1)}$, or 1 . Since the slopes are different, it follows that the system of $x + y = 20$ and $x - y = 20$ has exactly one solution. Thus, the second equation in the system could be $x - y = 20$.

Choice A is incorrect. The equation $3x + 3y = 60$ can be rewritten as $x + y = 20$. Since $x + y = 20$ is the same as the given equation, the system of $3x + 3y = 60$ and $x + y = 20$ has infinitely many solutions rather than exactly one solution.

Choice B is incorrect. The equation $-x - y = 40$ can be rewritten as $x + y = -40$. The graph of this line has a slope of -1 . Since the graphs of the lines represented by the equations $x + y = -40$ and $x + y = 20$ have the same slope in the xy -plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Choice C is incorrect. The equation $3x + 3y = 14$ can be rewritten as $x + y = \frac{14}{3}$. The graph of this line has a slope of -1 . Since the graphs of the lines represented by the equations $x + y = \frac{14}{3}$ and $x + y = 20$ have the same slope in the xy -plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Question Difficulty: Hard

Question ID bd329fe0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: bd329fe0

$$\begin{aligned}x + y &= 18 \\ 5y &= x\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(15, 3)$
- B. $(16, 2)$
- C. $(17, 1)$
- D. $(18, 0)$

ID: bd329fe0 Answer

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system defines the value of x as $5y$. Substituting $5y$ for x into the first equation yields $5y + y = 18$ or $6y = 18$. Dividing each side of this equation by 6 yields $y = 3$. Substituting 3 for y in the second equation yields $5(3) = x$ or $x = 15$. Therefore, the solution (x, y) to the given system of equations is $(15, 3)$.

Choice B is incorrect. Substituting 16 for x and 2 for y in the second equation yields $5(2) = 16$, which is not true. Therefore, $(16, 2)$ is not a solution to the given system of equations.

Choice C is incorrect. Substituting 17 for x and 1 for y in the second equation yields $5(1) = 17$, which is not true. Therefore, $(17, 1)$ is not a solution to the given system of equations.

Choice D is incorrect. Substituting 18 for x and 0 for y in the second equation yields $5(0) = 18$, which is not true. Therefore, $(18, 0)$ is not a solution to the given system of equations.

Question Difficulty: Easy

Question ID 1ef88808

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 1ef88808

$$\begin{aligned}x + 3 &= -2y + 5 \\ x - 3 &= 2y + 7\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $2x$?

- A. -2
- B. 6
- C. 12
- D. 24

ID: 1ef88808 Answer

Correct Answer: C

Rationale

Choice C is correct. Adding the second equation in the given system to the first equation in the given system yields $(x + 3) + (x - 3) = (-2y + 5) + (2y + 7)$. Adding like terms in this equation yields $2x = 12$. Thus, the value of $2x$ is 12 .

Choice A is incorrect. This is the value of y , not $2x$.

Choice B is incorrect. This is the value of x , not $2x$.

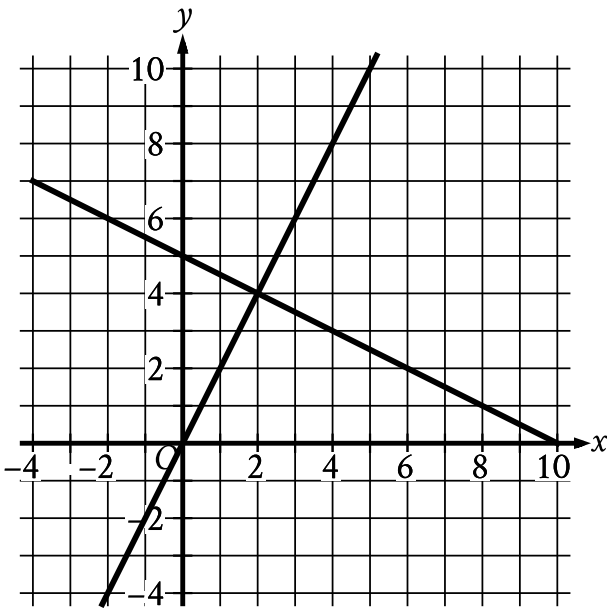
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID abe39d7f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: abe39d7f



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(0, 5)$
- B. $(2, 4)$
- C. $(5, 10)$
- D. $(10, 0)$

ID: abe39d7f Answer

Correct Answer: B

Rationale

Choice B is correct. A solution to a system of equations must be the solution to each equation in the system. It follows that if (x, y) is a solution to the system, the point (x, y) lies on the graph in the xy -plane of each equation in the system. The point that lies on each graph of the system of linear equations shown is their intersection point $(2, 4)$. Therefore, the solution to the system is $(2, 4)$.

Choice A is incorrect. The point $(0, 5)$ lies on one, but not both, of the graphs of the linear equations shown.

Choice C is incorrect. The point $(5, 10)$ lies on one, but not both, of the graphs of the linear equations shown.

Choice D is incorrect. The point $(10, 0)$ lies on one, but not both, of the graphs of the linear equations shown.

Question Difficulty: Easy

Question ID a1849c73

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a1849c73

$$\begin{aligned} 7x + 6y &= 5 \\ 28x + 24y &= 20 \end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $(r, -\frac{6r}{7} + \frac{5}{7})$
- B. $(r, \frac{7r}{6} + \frac{5}{6})$
- C. $(\frac{r}{4} + 5, -\frac{r}{4} + 20)$
- D. $(-\frac{6r}{7} + \frac{5}{7}, r)$

ID: a1849c73 Answer

Correct Answer: D

Rationale

Choice D is correct. Dividing each side of the second equation in the given system by 4 yields $7x + 6y = 5$. It follows that the two equations in the given system are equivalent and any point that lies on the graph of one equation will also lie on the graph of the other equation. Substituting r for y in the equation $7x + 6y = 5$ yields $7x + 6r = 5$. Subtracting $6r$ from each side of this equation yields $7x = -6r + 5$. Dividing each side of this equation by 7 yields $x = -\frac{6r}{7} + \frac{5}{7}$. Therefore, the point $(-\frac{6r}{7} + \frac{5}{7}, r)$ lies on the graph of each equation in the xy -plane for each real number r .

Choice A is incorrect. Substituting r for x in the equation $7x + 6y = 5$ yields $7r + 6y = 5$. Subtracting $7r$ from each side of this equation yields $6y = -7r + 5$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{6} + \frac{5}{6}$. Therefore, the point $(r, -\frac{7r}{6} + \frac{5}{6})$, not the point $(r, -\frac{6r}{7} + \frac{5}{7})$, lies on the graph of each equation.

Choice B is incorrect. Substituting r for x in the equation $7x + 6y = 5$ yields $7r + 6y = 5$. Subtracting $7r$ from each side of this equation yields $6y = -7r + 5$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{6} + \frac{5}{6}$. Therefore, the point $(r, -\frac{7r}{6} + \frac{5}{6})$, not the point $(r, \frac{7r}{6} + \frac{5}{6})$, lies on the graph of each equation.

Choice C is incorrect. Substituting $\frac{r}{4} + 5$ for x in the equation $7x + 6y = 5$ yields $7(\frac{r}{4} + 5) + 6y = 5$, or $(\frac{7r}{4} + 35) + 6y = 5$. Subtracting $(\frac{7r}{4} + 35)$ from each side of this equation yields $6y = -\frac{7r}{4} - 35 + 5$, or $6y = -\frac{7r}{4} - 30$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{24} - 5$. Therefore, the point $(\frac{r}{4} + 5, -\frac{7r}{24} - 5)$, not the point $(\frac{r}{4} + 5, -\frac{r}{4} + 20)$, lies on the graph of each equation.

Question Difficulty: Hard

Question ID bc62c9b6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: bc62c9b6

For a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water, for a total of **240** liters of water. Which equation represents this situation?

- A. $x + 3y = 240$
- B. $x + y = 240$
- C. $3x + 3y = 240$
- D. $3x + y = 240$

ID: bc62c9b6 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that for a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water. Since the group bought x one-liter bottles of water, the total number of liters bought from x one-liter bottles of water is represented as **1x**, or x . Since the group bought y three-liter bottles of water, the total number of liters bought from y three-liter bottles of water is represented as **3y**. It's given that the group bought a total of **240** liters; thus, the equation $x + 3y = 240$ represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the group bought x three-liter bottles of water and y one-liter bottles of water, for a total of **240** liters of water.

Question Difficulty: Easy

Question ID 4de6314a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 4de6314a

A total of **165** people contributed to a charity event as either a donor or a volunteer. **130** people contributed as a donor. How many people contributed as a volunteer?

- A. **35**
- B. **130**
- C. **165**
- D. **330**

ID: 4de6314a Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a total of **165** people contributed to a charity event as either a donor or a volunteer. It's also given that **130** people contributed as a donor. It follows that **165 — 130**, or **35**, people contributed as a volunteer.

Choice B is incorrect. This is the number of people who contributed as a donor, not a volunteer.

Choice C is incorrect. This is the total number of people who contributed as either a donor or a volunteer, not the number of people who contributed as a volunteer.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID d8bd7b78

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d8bd7b78

In the xy -plane, line k passes through the points $(8, 6)$ and $(9, 12)$ and is defined by the equation $y = mx + b$. What is the value of b ?

ID: d8bd7b78 Answer

Correct Answer: -42

Rationale

The correct answer is -42 . For a linear equation in the form $y = mx + b$, m represents the slope and b represents the y -coordinate of the y -intercept of the graph in the xy -plane. The slope of the graph of a line containing any two points, (x_1, y_1) and (x_2, y_2) , can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that in the xy -plane, line k passes through the points $(8, 6)$ and $(9, 12)$. Substituting $(8, 6)$ and $(9, 12)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{12 - 6}{9 - 8}$, or $m = 6$. Since line k passes through the point $(9, 12)$ and the slope of the line is 6 , substituting 9 for x , 12 for y , and 6 for m in the equation $y = mx + b$ yields $12 = (6)(9) + b$, or $12 = 54 + b$. Subtracting 54 from both sides of this equation yields $-42 = b$. Therefore, the value of b is -42 .

Question Difficulty: Hard

Question ID 2e47c476

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 2e47c476

A geologist estimates that the volume of a slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet. The geologist also estimates that the slab of granite weighs **165** pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the slab of granite?

- A. $165 - 15.7 < x < 165 - 12.7$
- B. $165 + 12.7 < x < 165 + 15.7$
- C. $165(12.7) < x < 165(15.7)$
- D. $\frac{165}{15.7} < x < \frac{165}{12.7}$

ID: 2e47c476 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the estimated volume of the slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet. It's also given that the estimated weight per cubic foot of volume of that slab is **165** pounds. The estimated total weight of the slab of granite, in pounds, can be calculated by multiplying the estimated volume by the estimated weight per cubic foot. Therefore, if the estimated volume of the slab of granite is **12.7** cubic feet, its estimated total weight is **165(12.7)** pounds, and if the estimated volume of the slab of granite is **15.7** cubic feet, its estimated total weight is **165(15.7)** pounds. Since the estimated volume of the slab of granite is greater than **12.7** cubic feet but less than **15.7** cubic feet, the estimated total weight x , in pounds, must be greater than **165(12.7)** pounds and less than **165(15.7)** pounds. This situation can be represented by the inequality $165(12.7) < x < 165(15.7)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 304668b4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 304668b4

What is the slope of the graph of $y = \frac{5x}{13} - 23$ in the xy -plane?

ID: 304668b4 Answer

Correct Answer: .3846, 5/13

Rationale

The correct answer is $\frac{5}{13}$. The graph of a line in the xy -plane can be represented by the equation $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept. The given equation can be written as $y = \left(\frac{5}{13}\right)x - 23$. Therefore, the slope of the graph of this equation in the xy -plane is $\frac{5}{13}$. Note that 5/13, .3846, 0.385, and 0.384 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 35f40980

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 35f40980

A small business owner budgets **\$2,200** to purchase candles. The owner must purchase a minimum of **200** candles to maintain the discounted pricing. If the owner pays **\$4.90** per candle to purchase small candles and **\$11.60** per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

ID: 35f40980 Answer

Correct Answer: 182

Rationale

The correct answer is **182**. Let s represent the number of small candles the owner can purchase, and let ℓ represent the number of large candles the owner can purchase. It's given that the owner pays **\$4.90** per candle to purchase small candles and **\$11.60** per candle to purchase large candles. Therefore, the owner pays $4.90s$ dollars for s small candles and 11.60ℓ dollars for ℓ large candles, which means the owner pays a total of $4.90s + 11.60\ell$ dollars to purchase candles. It's given that the owner budgets **\$2,200** to purchase candles. Therefore, $4.90s + 11.60\ell \leq 2,200$. It's also given that the owner must purchase a minimum of **200** candles. Therefore, $s + \ell \geq 200$. The inequalities $4.90s + 11.60\ell \leq 2,200$ and $s + \ell \geq 200$ can be combined into one compound inequality by rewriting the second inequality so that its left-hand side is equivalent to the left-hand side of the first inequality. Subtracting ℓ from both sides of the inequality $s + \ell \geq 200$ yields $s \geq 200 - \ell$. Multiplying both sides of this inequality by **4.90** yields $4.90s \geq 4.90(200 - \ell)$, or $4.90s \geq 980 - 4.90\ell$. Adding **11.60** ℓ to both sides of this inequality yields $4.90s + 11.60\ell \geq 980 - 4.90\ell + 11.60\ell$, or $4.90s + 11.60\ell \geq 980 + 6.70\ell$. This inequality can be combined with the inequality $4.90s + 11.60\ell \leq 2,200$, which yields the compound inequality $980 + 6.70\ell \leq 4.90s + 11.60\ell \leq 2,200$. It follows that $980 + 6.70\ell \leq 2,200$. Subtracting **980** from both sides of this inequality yields $6.70\ell \leq 2,200$. Dividing both sides of this inequality by **6.70** yields approximately $\ell \leq 182.09$. Since the number of large candles the owner purchases must be a whole number, the maximum number of large candles the owner can purchase is the largest whole number less than **182.09**, which is **182**.

Question Difficulty: Hard

Question ID 7808c7b0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 7808c7b0

The point $(8, 2)$ in the xy -plane is a solution to which of the following systems of inequalities?

A. $x > 0$
 $y > 0$

B. $x > 0$
 $y < 0$

C. $x < 0$
 $y > 0$

D. $x < 0$
 $y < 0$

ID: 7808c7b0 Answer

Correct Answer: A

Rationale

Choice A is correct. The given point, $(8, 2)$, is located in the first quadrant in the xy -plane. The system of inequalities in choice A represents all the points in the first quadrant in the xy -plane. Therefore, $(8, 2)$ is a solution to the system of inequalities in choice A.

Alternate approach: Substituting 8 for x in the first inequality in choice A, $x > 0$, yields $8 > 0$, which is true. Substituting 2 for y in the second inequality in choice A, $y > 0$, yields $2 > 0$, which is true. Since the coordinates of the point $(8, 2)$ make the inequalities $x > 0$ and $y > 0$ true, the point $(8, 2)$ is a solution to the system of inequalities consisting of $x > 0$ and $y > 0$.

Choice B is incorrect. This system of inequalities represents all the points in the fourth quadrant, not the first quadrant, in the xy -plane.

Choice C is incorrect. This system of inequalities represents all the points in the second quadrant, not the first quadrant, in the xy -plane.

Choice D is incorrect. This system of inequalities represents all the points in the third quadrant, not the first quadrant, in the xy -plane.

Question Difficulty: Medium

Question ID b882793f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: b882793f

$y = 2x + 10$
 $y = 2x - 1$

At how many points do the graphs of the given equations intersect in the xy -plane?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

ID: b882793f Answer

Correct Answer: A

Rationale

Choice A is correct. A system of two linear equations in two variables, x and y , has zero points of intersection if the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, are distinct if the y -coordinates of their y -intercepts, b , are different and are parallel if their slopes, m , are the same. For the two equations in the given system, $y = 2x + 10$ and $y = 2x - 1$, the values of b are 10 and -1 , respectively, and the values of m are both 2 . Since the values of b are different, the graphs of these lines have different y -coordinates of the y -intercept and are distinct. Since the values of m are the same, the graphs of these lines have the same slope and are parallel. Therefore, the graphs of the given equations are lines that intersect at zero points in the xy -plane.

Choice B is incorrect. The graphs of a system of two linear equations have exactly one point of intersection if the lines represented by the equations have different slopes. Since the given equations represent lines with the same slope, there is not exactly one intersection point.

Choice C is incorrect. The graphs of a system of two linear equations can never have exactly two intersection points.

Choice D is incorrect. The graphs of a system of two linear equations have infinitely many intersection points when the lines represented by the equations have the same slope and the same y -coordinate of the y -intercept. Since the given equations represent lines with different y -coordinates of their y -intercepts, there are not infinitely many intersection points.

Question Difficulty: Medium

Question ID 1a85e94a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 1a85e94a

$$8x + 7y = 9$$
$$24x + 21y = 27$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $\left(r, -\frac{8r}{7} + \frac{9}{7}\right)$
- B. $\left(-\frac{8r}{7} + \frac{9}{7}, r\right)$
- C. $\left(-\frac{8r}{7} + 9, \frac{8r}{7} + 27\right)$
- D. $\left(\frac{r}{3} + 9, -\frac{r}{3} + 27\right)$

ID: 1a85e94a Answer

Correct Answer: A

Rationale

Choice A is correct. Dividing both sides of the second equation in the given system by 3 yields $8x + 7y = 9$, which is the first equation in the given system. Therefore, the first and second equations represent the same line in the xy -plane. If the x - and y -coordinates of a point satisfy an equation, the point lies on the graph of the equation in the xy -plane. Choice A is a point with x -coordinate r and y -coordinate $-\frac{8r}{7} + \frac{9}{7}$. Substituting r for x and $-\frac{8r}{7} + \frac{9}{7}$ for y in the equation $8x + 7y = 9$ yields $8r + 7\left(-\frac{8r}{7} + \frac{9}{7}\right) = 9$. Applying the distributive property to the left-hand side of this equation yields $8r - 8r + 9 = 9$. Combining like terms on the left-hand side of this equation yields $9 = 9$, so the coordinates of the point $\left(r, -\frac{8}{7}r + \frac{9}{7}\right)$ satisfy both equations in the given system. Therefore, for each real number r , the point $\left(r, -\frac{8}{7}r + \frac{9}{7}\right)$ lies on the graph of each equation in the xy -plane for the given system.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3127ac95

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 3127ac95

A city employee will plant two types of bushes, azaleas and boxwoods, in a park. There will be no more than **164** total bushes planted, and the number of azaleas planted will be at most three times the number of boxwoods planted. Which of the following systems of inequalities best represents this situation, where ***a*** is the number of azaleas that will be planted, and ***b*** is the number of boxwoods that will be planted?

- A. $a + b \geq 164$
 $3a \geq b$
- B. $a + b \geq 164$
 $a \leq 3b$
- C. $a + b \leq 164$
 $3a \geq b$
- D. $a + b \leq 164$
 $a \leq 3b$

ID: 3127ac95 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that a city employee will plant azaleas and boxwoods in a park, where ***a*** is the number of azaleas that will be planted and ***b*** is the number of boxwoods that will be planted. It’s also given that there will be no more than **164** total bushes planted, which can be represented by the inequality $a + b \leq 164$. It’s also given that the number of azaleas planted will be at most three times the number of boxwoods planted, which can be represented by the inequality $a \leq 3b$. Therefore, the system of inequalities containing $a + b \leq 164$ and $a \leq 3b$ best represents this situation.

Choice A is incorrect. The inequality $a + b \geq 164$ represents a situation where at least **164** total bushes will be planted, not that there will be no more than **164** total bushes planted. Also, the inequality $3a \geq b$ represents a situation where the number of boxwoods that will be planted is at most three times the number of azaleas that will be planted, not that the number of azaleas planted will be at most three times the number of boxwoods planted.

Choice B is incorrect. The inequality $a + b \geq 164$ represents a situation where at least **164** total bushes will be planted, not that there will be no more than **164** total bushes planted.

Choice C is incorrect. The inequality $3a \geq b$ represents a situation where the number of boxwoods that will be planted is at most three times the number of azaleas that will be planted, not that the number of azaleas planted will be at most three times the number of boxwoods planted.

Question Difficulty: Hard

Question ID f69d8c8f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: f69d8c8f

x	$f(x)$
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. If $h(x) = f(x) - 13$, which equation defines h ?

- A. $h(x) = 5x - 4$
- B. $h(x) = 5x + 7$
- C. $h(x) = 5x + 9$
- D. $h(x) = 5x + 20$

ID: f69d8c8f Answer

Correct Answer: B

Rationale

Choice B is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m and b are constants. It's given in the table that when $x = -4$, $f(x) = 0$. Substituting -4 for x and 0 for $f(x)$ in the equation $f(x) = mx + b$ yields $0 = m(-4) + b$, or $0 = -4m + b$. Adding $4m$ to both sides of this equation yields $4m = b$. Substituting $4m$ for b in the equation $f(x) = mx + b$ yields $f(x) = mx + 4m$. It's also given in the table that when $x = -\frac{19}{5}$, $f(x) = 1$. Substituting $-\frac{19}{5}$ for x and 1 for $f(x)$ in the equation $f(x) = mx + 4m$ yields $1 = m(-\frac{19}{5}) + 4m$, or $1 = \frac{1}{5}m$. Multiplying both sides of this equation by 5 yields $m = 5$. Substituting 5 for m in the equation $f(x) = mx + 4m$ yields $f(x) = 5x + 4(5)$, or $f(x) = 5x + 20$. If $h(x) = f(x) - 13$, substituting $5x + 20$ for $f(x)$ in this equation yields $h(x) = (5x + 20) - 13$, or $h(x) = 5x + 7$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is an equation that defines the linear function f , not h .

Question Difficulty: Hard

Question ID 389861fc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 389861fc

The function h is defined by $h(x) = x + 200$. What is the value of $h(50)$?

- A. 200
- B. 250
- C. 10,000
- D. 50,200

ID: 389861fc Answer

Correct Answer: B

Rationale

Choice B is correct. Substituting 50 for x in the given function yields $h(50) = 50 + 200$, or $h(50) = 250$. Therefore, the value of $h(50)$ is 250.

Choice A is incorrect. This is the value of $h(0)$.

Choice C is incorrect. This is the value of $h(9,800)$.

Choice D is incorrect. This is the value of $h(50,000)$.

Question Difficulty: Easy

Question ID f193e75a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: f193e75a

If $5x = 20$, what is the value of $15x$?

- A. 7
- B. 12
- C. 23
- D. 60

ID: f193e75a Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $5x = 20$. Multiplying both sides of this equation by 3 yields $15x = 60$. Therefore, the value of $15x$ is 60.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 7ea92487

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 7ea92487

Josie purchased a pass that cost **\$19.50** to tour a nature preserve. She also purchased **10** identical stickers to give to her nieces and nephews. The pass and the stickers cost **\$62.00** total. What was the cost of one sticker?

- A. **\$1.95**
- B. **\$4.25**
- C. **\$6.20**
- D. **\$8.15**

ID: 7ea92487 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that Josie purchased a pass to tour a nature preserve and **10** identical stickers for **\$62.00** total. It's also given that the pass cost **\$19.50**. Let x represent the cost, in dollars, of one sticker. This relationship can be represented by the equation $19.50 + 10x = 62$. Subtracting **19.50** from both sides of this equation yields $10x = 42.50$. Dividing both sides of this equation by **10** yields $x = 4.25$. Therefore, the cost of one sticker was **\$4.25**.

Choice A is incorrect. This is the cost of one sticker if Josie purchased **10** identical stickers for **\$19.50**, not a pass and **10** identical stickers for **\$62.00**.

Choice C is incorrect. This is the cost of one sticker if Josie purchased **10** identical stickers for **\$62.00**, not a pass and **10** identical stickers for **\$62.00**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 73da0278

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 73da0278

The combined original price for a mirror and a vase is \$60. After a 25% discount to the mirror and a 45% discount to the vase are applied, the combined sale price for the two items is \$39. Which system of equations gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase?

- A.

$m + v = 60$
 $0.55m + 0.75v = 39$
- B.

$m + v = 60$
 $0.45m + 0.25v = 39$
- C.

$m + v = 60$
 $0.75m + 0.55v = 39$
- D.

$m + v = 60$
 $0.25m + 0.45v = 39$

ID: 73da0278 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that m represents the original price, in dollars, of the mirror, and v represents the original price, in dollars, of the vase. It's also given that the combined original price for the mirror and the vase is \$60. This can be represented by the equation $m + v = 60$. After a 25% discount to the mirror is applied, the sale price of the mirror is 75% of its original price. This can be represented by the expression $0.75m$. After a 45% discount to the vase is applied, the sale price of the vase is 55% of its original price. This can be represented by the expression $0.55v$. It's given that the combined sale price for the two items is \$39. This can be represented by the equation $0.75m + 0.55v = 39$. Therefore, the system of equations consisting of the equations $m + v = 60$ and $0.75m + 0.55v = 39$ gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase.

Choice A is incorrect. The second equation in this system of equations represents a 45% discount to the mirror and a 25% discount to the vase.

Choice B is incorrect. The second equation in this system of equations represents a 55% discount to the mirror and a 75% discount to the vase.

Choice D is incorrect. The second equation in this system of equations represents a 75% discount to the mirror and a 55% discount to the vase.

Question Difficulty: Hard

Question ID d737500a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: d737500a

If $x = 7$, what is the value of $x + 20$?

- A. 13
- B. 20
- C. 27
- D. 34

ID: d737500a Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that $x = 7$. Substituting 7 for x into the given expression $x + 20$ yields $7 + 20$, which is equivalent to 27.

Choice A is incorrect. This is the value of $x + 6$.

Choice B is incorrect. This is the value of $x + 13$.

Choice D is incorrect. This is the value of $x + 27$.

Question Difficulty: Easy

Question ID 58d03572

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 58d03572

$$\begin{aligned}\frac{7}{8}y - \frac{5}{8}x &= \frac{4}{7} - \frac{7}{8}y \\ \frac{5}{4}x + \frac{7}{4} &= py + \frac{15}{4}\end{aligned}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

ID: 58d03572 Answer

Correct Answer: 3.5, 7/2

Rationale

The correct answer is $\frac{7}{2}$. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. Two lines represented by equations in standard form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system, $\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$, can be written in standard form by adding $\frac{7}{8}y$ to both sides of the equation, which yields $\frac{14}{8}y - \frac{5}{8}x = \frac{4}{7}$, or $-\frac{5}{8}x + \frac{14}{8}y = \frac{4}{7}$. Multiplying each term in this equation by -8 yields $5x - 14y = -\frac{32}{7}$. The second equation in the given system, $\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$, can be written in standard form by subtracting $\frac{7}{4}$ and py from both sides of the equation, which yields $\frac{5}{4}x - py = \frac{8}{4}$. Multiplying each term in this equation by 4 yields $5x - 4py = 8$. The coefficient of x in the first equation, $5x - 14y = -\frac{32}{7}$, is equal to the coefficient of x in the second equation, $5x - 4py = 8$. For the lines to be parallel, and for the coefficients for x and y in one equation to be proportional to the corresponding coefficients in the other equation, the coefficient of y in the second equation must also be equal to the coefficient of y in the first equation. Therefore, $-14 = -4p$. Dividing both sides of this equation by -4 yields $\frac{-14}{-4} = p$, or $p = \frac{7}{2}$. Therefore, if the given system of equations has no solution, the value of p is $\frac{7}{2}$. Note that 7/2 and 3.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID d0d3c29c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: d0d3c29c

$$\begin{aligned}y &= -3x \\ 4x + y &= 15\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. 1
- B. 5
- C. 15
- D. 45

ID: d0d3c29c Answer

Correct Answer: C

Rationale

Choice C is correct. The given system of linear equations can be solved by the substitution method. Substituting $-3x$ for y from the first equation in the given system into the second equation yields $4x + (-3x) = 15$, or $x = 15$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the absolute value of y , not the value of x .

Question Difficulty: Medium

Question ID 4990b468

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 4990b468

The owner of a bakery is giving away free muffins every hour. The equation $f(x) = 108 - 9x$ gives the number of free muffins, $f(x)$, that remain to be given away after x hours. Which of the following is the best interpretation of $f(12) = 0$ in this context?

- A. The owner of the bakery has 12 free muffins to give away.
- B. The owner of the bakery will give away 12 free muffins in 9 hours.
- C. It will take 12 hours to give away all the free muffins.
- D. It will take 108 hours to give away 12 free muffins.

ID: 4990b468 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $f(x) = 108 - 9x$ represents the number of free muffins that remain to be given away after x hours. For the given equation $f(12) = 0$, when the value of x is 12, the value of $f(x)$ is 0. It follows that after 12 hours there are 0 free muffins remaining to be given away. Therefore, based on this context, it will take 12 hours to give away all the free muffins.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This represents giving away 12 muffins over 108 hours rather than giving away all 108 muffins in 12 hours.

Question Difficulty: Medium

Question ID 0bb9d90d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 0bb9d90d

x	y
−6	65
−3	56
3	38
6	29

The table shows four values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $9x + 3y = 141$
- B. $9x + 3y = 3$
- C. $3x + 9y = 141$
- D. $3x + 9y = 3$

ID: 0bb9d90d Answer

Correct Answer: A

Rationale

Choice A is correct. An equation representing the linear relationship between x and y can be written in slope-intercept form $y = mx + b$, where m is the slope of the graph of the equation in the xy -plane and $(0, b)$ is the y -intercept. The slope, m , can be calculated using two ordered pairs, (x_1, y_1) and (x_2, y_2) , and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the ordered pairs $(-6, 65)$ and $(6, 29)$ from the table for (x_1, y_1) and (x_2, y_2) , respectively, in this formula yields $m = \frac{29 - 65}{6 - (-6)}$, which is equivalent to $m = \frac{-36}{12}$, or $m = -3$. Substituting -3 for m in the formula $y = mx + b$ yields $y = -3x + b$. Substituting the point $(-6, 65)$ into this equation yields $65 = -3(-6) + b$, or $65 = 18 + b$. Subtracting 18 from both sides of this equation yields $47 = b$. Substituting 47 for b in the equation $y = -3x + b$ yields $y = -3x + 47$. Adding $3x$ to both sides of this equation yields $3x + y = 47$. Multiplying both sides of this equation by 3 yields $9x + 3y = 141$.

Choice B is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $9(-6) + 3(65) = 3$, or $141 = 3$, which is false.

Choice C is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $3(-6) + 9(65) = 141$, or $567 = 141$, which is false.

Choice D is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $3(-6) + 9(65) = 3$, or $567 = 3$, which is false.

Question Difficulty: Medium

Question ID a0903fed

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a0903fed

$$\begin{aligned}y &= 12x - 20 \\ y &= 28\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(4, 28)$
- B. $(20, 28)$
- C. $(28, 4)$
- D. $(28, 20)$

ID: a0903fed Answer

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system is $y = 28$. Substituting **28** for y in the first equation in the given system yields $28 = 12x - 20$. Adding **20** to both sides of this equation yields $48 = 12x$. Dividing both sides of this equation by **12** yields $4 = x$. Therefore, the solution (x, y) to the given system of equations is **$(4, 28)$** .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the solution (y, x) , not (x, y) , to the given system of equations.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID f762f4c4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: f762f4c4

$$f(x) = x + b$$

For the linear function f , b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b ?

- A. -30
- B. $-\frac{1}{30}$
- C. $\frac{1}{30}$
- D. 30

ID: f762f4c4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that when $x = 0$, $f(x) = 30$. Substituting 0 for x and 30 for $f(x)$ in the given function yields $30 = 0 + b$, or $30 = b$. Therefore, the value of b is 30 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b6c5bfb3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b6c5bfb3

In the xy -plane, the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$. Which equation defines f , where $y = f(x)$?

- A. $f(x) = 2x + 42$
- B. $f(x) = 32x + 36$
- C. $f(x) = 4x + 2$
- D. $f(x) = 8x + 2$

ID: b6c5bfb3 Answer

Correct Answer: C

Rationale

Choice C is correct. In the xy -plane, the graph of a linear function can be written in the form $f(x) = mx + b$, where m represents the slope and $(0, b)$ represents the y -intercept of the graph of $y = f(x)$. It's given that the graph of the linear function f , where $y = f(x)$, in the xy -plane contains the point $(0, 2)$. Thus, $b = 2$. The slope of the graph of a line containing any two points (x_1, y_1) and (x_2, y_2) can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$, it follows that the slope of the graph of the line containing these points is $m = \frac{34 - 2}{8 - 0}$, or $m = 4$. Substituting 4 for m and 2 for b in $f(x) = mx + b$ yields $f(x) = 4x + 2$.

Choice A is incorrect. This function represents a graph with a slope of 2 and a y -intercept of $(0, 42)$.

Choice B is incorrect. This function represents a graph with a slope of 32 and a y -intercept of $(0, 36)$.

Choice D is incorrect. This function represents a graph with a slope of 8 and a y -intercept of $(0, 2)$.

Question Difficulty: Medium

Question ID e97f2db3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: e97f2db3

$$\begin{aligned}\frac{2}{5}x + \frac{7}{5}y &= \frac{2}{7} \\ gx + ky &= \frac{5}{2}\end{aligned}$$

In the given system of equations, g and k are constants. The system has infinitely many solutions. What is the value of $\frac{g}{k}$?

ID: e97f2db3 Answer

Correct Answer: .2857, 2/7

Rationale

The correct answer is $\frac{2}{7}$. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions if and only if the two linear equations are equivalent. Multiplying each side of the first equation in the system by $\frac{35}{4}$ yields $\frac{35}{4}\left(\frac{2}{5}x + \frac{7}{5}y\right) = \frac{35}{4}\left(\frac{2}{7}\right)$, or $\frac{7}{2}x + \frac{49}{4}y = \frac{5}{2}$. Since this equation is equivalent to the second equation and has the same right side as the second equation, the coefficients of x and y , respectively, should also be the same. It follows that $g = \frac{7}{2}$ and $k = \frac{49}{4}$. Therefore, the value of $\frac{g}{k}$ is $\frac{\frac{7}{2}}{\frac{49}{4}}$, or $\frac{2}{7}$. Note that 2/7, .2857, 0.285, and 0.286 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 76f44a77

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 76f44a77

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

- A. 8
- B. 9
- C. 64
- D. 80

ID: 76f44a77 Answer

Correct Answer: B

Rationale

Choice B is correct. Substituting 72 for $f(x)$ in the given function yields $72 = 8x$. Dividing each side of this equation by 8 yields $9 = x$. Therefore, $f(x) = 72$ when the value of x is 9 .

Choice A is incorrect. This is the value of x for which $f(x) = 64$, not $f(x) = 72$.

Choice C is incorrect. This is the value of x for which $f(x) = 512$, not $f(x) = 72$.

Choice D is incorrect. This is the value of x for which $f(x) = 640$, not $f(x) = 72$.

Question Difficulty: Easy

Question ID 0fef875d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 0fef875d

For a party, **50** dinner rolls are needed. Dinner rolls are sold in packages of **12**. What is the minimum number of packages that should be bought for the party?

ID: 0fef875d Answer

Correct Answer: 5

Rationale

The correct answer is **5**. Let **p** represent the number of packages of dinner rolls that should be bought for the party. It's given that dinner rolls are sold in packages of **12**. Therefore, **$12p$** represents the number of dinner rolls that should be bought for the party. It's also given that **50** dinner rolls are needed; therefore, **$12p \geq 50$** . Dividing both sides of this inequality by **12** yields **$p \geq \frac{50}{12}$** , or approximately **$p \geq 4.17$** . Since the number of packages of dinner rolls must be a whole number, the minimum number of packages that should be bought for the party is **5**.

Question Difficulty: Medium

Question ID f9b3676d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: f9b3676d

$$\begin{aligned} 3y &= 4x + 17 \\ -3y &= 9x - 23 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $39x$?

- A. -18
- B. -6
- C. 6
- D. 18

ID: f9b3676d Answer

Correct Answer: D

Rationale

Choice D is correct. Adding the second equation to the first equation in the given system of equations yields $3y - 3y = 4x + 9x + 17 - 23$, or $0 = 13x - 6$. Adding 6 to each side of this equation yields $6 = 13x$. Multiplying each side of this equation by 3 yields $18 = 39x$. Therefore, the value of $39x$ is 18 .

Choice A is incorrect. This is the value of $-39x$, not $39x$.

Choice B is incorrect. This is the value of $-13x$, not $39x$.

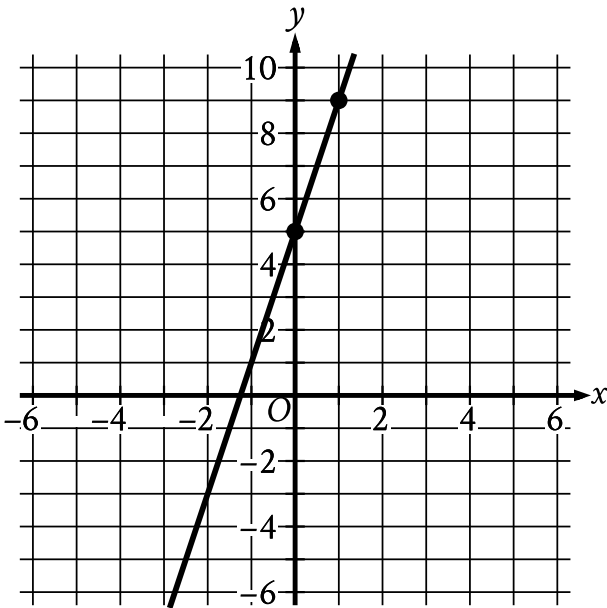
Choice C is incorrect. This is the value of $13x$, not $39x$.

Question Difficulty: Hard

Question ID 143330a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 143330a1



Line j is shown in the xy -plane. Line k (not shown) is parallel to line j . What is the slope of line k ?

ID: 143330a1 Answer

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that line k is parallel to line j . It follows that the slope of line k is equal to the slope of line j . Given two points on a line in the xy -plane, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. In the xy -plane shown, the points $(0, 5)$ and $(1, 9)$ are on line j . It follows that the slope of line j is $\frac{9 - 5}{1 - 0}$, or **4**. Since the slope of line j is equal to the slope of line k , the slope of line k is also **4**.

Question Difficulty: Medium

Question ID f75e6a40

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: f75e6a40

$$\begin{aligned}x &= 10 \\ y &= x + 21\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 2.1
- B. 10
- C. 21
- D. 31

ID: f75e6a40 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given by the first equation in the given system of equations that $x = 10$. Substituting **10** for x in the second equation in the given system yields $y = 10 + 21$, or $y = 31$. Therefore, the value of y is **31**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not the value of y .

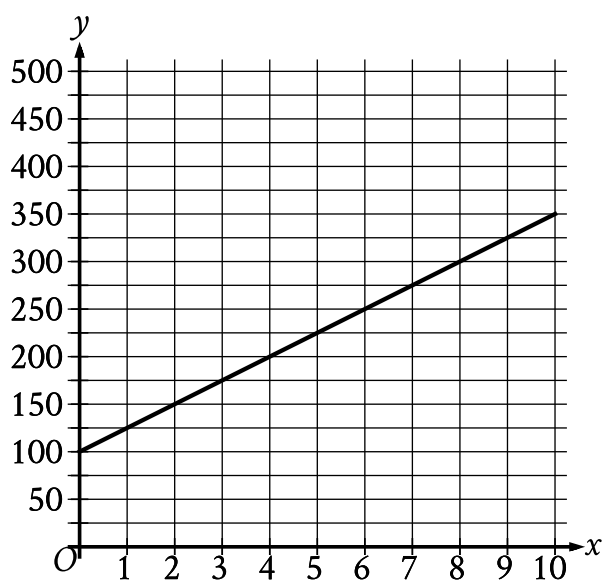
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID d9958f0c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d9958f0c



The graph of the function f , where $y = f(x)$, gives the total cost y , in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

- A. Each game costs \$25.
- B. The video game system costs \$100.
- C. The video game system costs \$25.
- D. Each game costs \$100.

ID: d9958f0c Answer

Correct Answer: A

Rationale

Choice A is correct. The given graph is a line, and the slope of a line is defined as the change in the value of y for each increase in the value of x by 1. It's given that y represents the total cost, in dollars, and that x represents the number of games. Therefore, the change in the value of y for each increase in the value of x by 1 represents the change in total cost, in dollars, for each increase in the number of games by 1. In other words, the slope represents the cost, in dollars, per game. The graph shows that when the value of x increases from 0 to 1, the value of y increases from 100 to 125. It follows that the slope is 25, or the cost per game is \$25. Thus, the best interpretation of the slope of the graph is that each game costs \$25.

Choice B is incorrect. This is an interpretation of the y -intercept of the graph rather than the slope of the graph.

Choice C is incorrect. The slope of the graph is the cost per game, not the cost of the video game system.

Choice D is incorrect. Each game costs \$25, not \$100.

Question ID fac6ead2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: fac6ead2

The cost of renting a large canopy tent for up to **10** days is **\$430** for the first day and **\$215** for each additional day. Which of the following equations gives the cost y , in dollars, of renting the tent for x days, where x is a positive integer and $x \leq 10$?

- A. $y = 215x + 215$
- B. $y = 430x - 215$
- C. $y = 430x + 215$
- D. $y = 215x + 430$

ID: fac6ead2 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the cost of renting a large canopy tent is **\$430** for the first day and **\$215** for each additional day for up to **10** days. For x days of renting the tent, the cost includes **\$430** for the first day and **\$215** for each of the $(x - 1)$ additional days. It follows that the cost y , in dollars, of renting the tent can be expressed as $y = 430 + 215(x - 1)$, which is equivalent to $y = 430 + 215x - 215$, or $y = 215x + 215$. Therefore, the equation $y = 215x + 215$ gives the cost of renting the tent for x days, where x is a positive integer and $x \leq 10$.

Choice B is incorrect. This equation represents a situation where the cost of renting the tent for the first day is **\$215**, not **\$430**, and the cost for each additional day is **\$430**, not **\$215**.

Choice C is incorrect. This equation represents a situation where the cost of renting the tent for the first day is **\$645**, not **\$430**, and the cost for each additional day is **\$430**, not **\$215**.

Choice D is incorrect. This equation represents a situation where the cost of renting the tent for the first day is **\$645**, not **\$430**.

Question Difficulty: Hard

Question ID 2827595d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 2827595d

$8x + 11y = 170$

The equation gives the possible combinations of the number of **2009** premium grade Log Cabin Pennies, x , and the number of **1996** select grade Lincoln Pennies, y , in a collection that is worth a total of **\$170**. If there are **6 1996** select grade Lincoln Pennies in the collection, how many **2009** premium grade Log Cabin Pennies are in the collection?

ID: 2827595d Answer

Correct Answer: 13

Rationale

The correct answer is **13**. It's given that the equation $8x + 11y = 170$ gives the possible combinations of the number of **2009** premium grade Log Cabin Pennies, x , and the number of **1996** select grade Lincoln Pennies, y , in a collection that is worth a total of **\$170**. It's also given that there are **6 1996** select grade Lincoln Pennies in the collection, and thus $y = 6$. Substituting **6** for y in the given equation yields $8x + 11(6) = 170$, or $8x + 66 = 170$. Subtracting **66** from both sides of this equation yields $8x = 104$. Dividing both sides of this equation by **8** yields $x = 13$. Therefore, there are **13 2009** premium grade Log Cabin Pennies in the collection.

Question Difficulty: Medium

Question ID 740a23e7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 740a23e7

$k + 12 = 336$

What is the solution to the given equation?

- A. 28
- B. 324
- C. 348
- D. 4,032

ID: 740a23e7 Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 12 from both sides of the given equation yields $k = 324$. Therefore, the solution to the given equation is 324.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 1ac409af

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 1ac409af

A neighborhood consists of a **2**-hectare park and a **35**-hectare residential area. The total number of trees in the neighborhood is **3,934**. The equation **$2x + 35y = 3,934$** represents this situation. Which of the following is the best interpretation of x in this context?

- A. The average number of trees per hectare in the park
- B. The average number of trees per hectare in the residential area
- C. The total number of trees in the park
- D. The total number of trees in the residential area

ID: 1ac409af Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a neighborhood consists of a **2**-hectare park and a **35**-hectare residential area and that the total number of trees in the neighborhood is **3,934**. It's also given that the equation **$2x + 35y = 3,934$** represents this situation. Since the total number of trees for a given area can be determined by taking the number of hectares times the average number of trees per hectare, this must mean that the terms **$2x$** and **$35y$** correspond to the number of trees in the park and in the residential area, respectively. Since **$2x$** corresponds to the number of trees in the park, and **2** is the size of the park, in hectares, **x** must represent the average number of trees per hectare in the park.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 4c580285

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 4c580285

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

ID: 4c580285 Answer

Correct Answer: 30

Rationale

The correct answer is **30**. The value of $f(x)$ when $x = 4$ can be found by substituting **4** for x in the given equation $f(x) = 7x + 2$. This yields $f(4) = 7(4) + 2$, or $f(4) = 30$. Therefore, when $x = 4$, the value of $f(x)$ is **30**.

Question Difficulty: Easy

Question ID 42258aab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 42258aab

The y-intercept of the graph of $12x + 2y = 18$ in the xy-plane is $(0, y)$. What is the value of y ?

ID: 42258aab Answer

Correct Answer: 9

Rationale

The correct answer is **9**. It's given that the y-intercept of the graph of $12x + 2y = 18$ in the xy-plane is $(0, y)$. Substituting **0** for x in the equation $12x + 2y = 18$ yields $12(0) + 2y = 18$, or $2y = 18$. Dividing both sides of this equation by **2** yields $y = 9$. Therefore, the value of y is **9**.

Question Difficulty: Medium

Question ID 4325992e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 4325992e

$0.10x + 0.20y = 0.17(x + y)$

The equation gives a volume x , in gallons, of a **10%** solution that could be mixed with a volume y , in gallons, of a **20%** solution to produce a **17%** solution. According to this equation, what volume, in gallons, of the **20%** solution could be mixed with **90.0** gallons of the **10%** solution to produce a **17%** solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

ID: 4325992e Answer

Correct Answer: 210

Rationale

The correct answer is **210**. It's given that the equation $0.10x + 0.20y = 0.17(x + y)$ represents mixing a **10%** solution and a **20%** solution to obtain a **17%** solution. It's also given that x represents the volume, in gallons, of the **10%** solution. If **90.0** gallons of the **10%** solution is used, it follows that $x = 90.0$. Substituting **90.0** for x in the given equation yields $0.10(90.0) + 0.20y = 0.17(90.0 + y)$, or $9 + 0.20y = 15.3 + 0.17y$. Subtracting $0.17y$ from both sides of this equation yields $9 + 0.03y = 15.3$. Subtracting **9** from both sides of this equation yields $0.03y = 6.3$. Dividing both sides of this equation by **0.03** yields $y = 210$. Therefore, the volume of the **20%** solution that could be mixed with **90.0** gallons of the **10%** solution to produce a **17%** solution is **210** gallons.

Question Difficulty: Hard

Question ID a4695b90

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a4695b90

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is (a, b) . What is the value of b ?

ID: a4695b90 Answer

Correct Answer: 25

Rationale

The correct answer is **25**. Subtracting the second equation from the first equation in the given system of equations yields $(2a - 2a) + (8b - 4b) = 198 - 98$, which is equivalent to $0 + 4b = 100$, or $4b = 100$. Dividing each side of this equation by **4** yields $b = 25$.

Question Difficulty: Medium

Question ID 104c58b2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 104c58b2

$x(r - 9) + 4 = 19x + 27$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

- A. 4
- B. 9
- C. 23
- D. 28

ID: 104c58b2 Answer

Correct Answer: D

Rationale

Choice D is correct. In the given equation, $x(r - 9) + 4 = 19x + 27$, r is a positive integer. Distributing on the left-hand side of this equation yields $rx - 9x + 4 = 19x + 27$. Subtracting 4 from both sides of this equation yields $rx - 9x = 19x + 23$. Subtracting $19x$ from both sides of this equation yields $rx - 28x = 23$, which can be written as $(r - 28)x = 23$. For this equation to have exactly one solution, the coefficient of x , $(r - 28)$, cannot be equal to 0. Therefore, 28 CANNOT be the value of r .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 95e42caf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 95e42caf

If $f(x) = x + 7$ and $g(x) = 7x$, what is the value of $4f(2) - g(2)$?

- A. -5
- B. 1
- C. **22**
- D. 28

ID: 95e42caf Answer

Correct Answer: C

Rationale

Choice C is correct. The value of $f(2)$ can be found by substituting 2 for x in the given equation $f(x) = x + 7$, which yields $f(2) = 2 + 7$, or $f(2) = 9$. The value of $g(2)$ can be found by substituting 2 for x in the given equation $g(x) = 7x$, which yields $g(2) = 7(2)$, or $g(2) = 14$. The value of the expression $4f(2) - g(2)$ can be found by substituting the corresponding values into the expression, which gives $4(9) - 14$. This expression is equivalent to $36 - 14$, or 22 .

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $4f(2) - g(2)$.

Choice B is incorrect and may result from calculating $4f(2)$ as $4(2) + 7$, rather than $4(2 + 7)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8229671d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 8229671d

$13x = 112 - x$

What value of x is the solution to the given equation?

ID: 8229671d Answer

Correct Answer: 8

Rationale

The correct answer is **8**. Adding x to both sides of the given equation yields **$14x = 112$** . Dividing both sides of this equation by **14** yields **$x = 8$** .

Question Difficulty: Medium

Question ID a5aa159f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a5aa159f

$$\begin{aligned} 6x + 7y &= 28 \\ 2x + 2y &= 10 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. -2
- B. 7
- C. 14
- D. 18

ID: a5aa159f Answer

Correct Answer: A

Rationale

Choice A is correct. The given system of linear equations can be solved by the elimination method. Multiplying each side of the second equation in the given system by 3 yields $(2x + 2y)(3) = (10)(3)$, or $6x + 6y = 30$. Subtracting this equation from the first equation in the given system yields $(6x + 7y) - (6x + 6y) = (28) - (30)$, which is equivalent to $(6x - 6x) + (7y - 6y) = 28 - 30$, or $y = -2$.

Choice B is incorrect. This is the value of x , not the value of y .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 144d0888

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 144d0888

The y-intercept of the graph of $y = -6x - 32$ in the xy -plane is $(0, y)$. What is the value of y ?

ID: 144d0888 Answer

Correct Answer: -32

Rationale

The correct answer is -32 . It's given that the y-intercept of the graph of $y = -6x - 32$ is $(0, y)$. Substituting 0 for x in this equation yields $y = -6(0) - 32$, or $y = -32$. Therefore, the value of y that corresponds to the y-intercept of the graph of $y = -6x - 32$ in the xy -plane is -32 .

Question Difficulty: Medium

Question ID 3f2e31bd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 3f2e31bd

Line r in the xy -plane has a slope of 4 and passes through the point $(0, 6)$. Which equation defines line r ?

- A. $y = -6x + 4$
- B. $y = 6x + 4$
- C. $y = 4x - 6$
- D. $y = 4x + 6$

ID: 3f2e31bd Answer

Correct Answer: D

Rationale

Choice D is correct. A line in the xy -plane with a slope of m and a y -intercept of $(0, b)$ can be defined by an equation in the form $y = mx + b$. It's given that line r has a slope of 4 and passes through the point $(0, 6)$. It follows that $m = 4$ and $b = 6$. Substituting 4 for m and 6 for b in the equation $y = mx + b$ yields $y = 4x + 6$. Therefore, the equation $y = 4x + 6$ defines line r .

Choice A is incorrect. This equation defines a line that has a slope of -6 , not 4 , and passes through the point $(0, 4)$, not $(0, 6)$.

Choice B is incorrect. This equation defines a line that has a slope of 6 , not 4 , and passes through the point $(0, 4)$, not $(0, 6)$.

Choice C is incorrect. This equation defines a line that passes through the point $(0, -6)$, not $(0, 6)$.

Question Difficulty: Medium

Question ID 39423fbf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 39423fbf

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 39 and passes through the point $(0, 0)$. Which equation defines f ?

- A. $f(x) = -39x$
- B. $f(x) = \frac{1}{39}x$
- C. $f(x) = x - 39$
- D. $f(x) = 39x$

ID: 39423fbf Answer

Correct Answer: D

Rationale

Choice D is correct. An equation defining a linear function can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. It's given that the graph of $y = f(x)$ has a slope of 39 , so $m = 39$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 0)$, so $b = 0$. Substituting 39 for m and 0 for b in $f(x) = mx + b$ yields $f(x) = 39x + 0$, or $f(x) = 39x$. Thus, the equation that defines f is $f(x) = 39x$.

Choice A is incorrect. This equation defines a function whose graph has a slope of -39 , not 39 .

Choice B is incorrect. This equation defines a function whose graph has a slope of $\frac{1}{39}$, not 39 .

Choice C is incorrect. This equation defines a function whose graph has a slope of 1 , not 39 , and passes through the point $(0, -39)$, not $(0, 0)$.

Question Difficulty: Medium

Question ID 0461a3a4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 0461a3a4

A batch of banana milkshakes consists of **4** cups of ice cream and **2** bananas and has **1,114 milligrams (mg)** of calcium. There is **276 mg** of calcium in **1** cup of the ice cream used to make this batch of milkshakes. How much calcium, **in mg**, is in **1** banana?

- A. **5**
- B. **10**
- C. **419**
- D. **1,104**

ID: 0461a3a4 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that a batch of banana milkshakes consists of **4** cups of ice cream and **2** bananas and has **1,114** mg of calcium. It’s also given that there is **276** mg of calcium in **1** cup of the ice cream used to make this batch of milkshakes. It follows that the total amount of calcium in **4** cups of ice cream is **4(276)**, or **1,104** mg. Let ***x*** represent the amount of calcium, in mg, in **1** banana. It follows that the total amount of calcium in **2** bananas is **2*x*** mg. Since the batch of banana milkshakes has **1,114** mg of calcium, the equation **1,104 + 2*x* = 1,114** represents this situation. Subtracting **1,104** from both sides of this equation yields **2*x* = 10**. Dividing both sides of this equation by **2** yields ***x* = 5**. Therefore, the amount of calcium in **1** banana is **5** mg.

Choice B is incorrect. This is the amount of calcium, in mg, in **2** bananas, not in **1** banana.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the amount of calcium, in mg, in **4** cups of ice cream, not in **1** banana.

Question Difficulty: Medium

Question ID a28ee3f1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: a28ee3f1

In **2010**, a swim club had a total of **35** swimmers, each classified as either advanced or intermediate. From **2010** to **2020**, the number of advanced swimmers in the club increased by approximately **53%**, and the number of intermediate swimmers in the club increased by approximately **44%**. The total number of swimmers in the club increased by approximately **49%**. Which equation best represents this situation, where ***a*** represents the number of advanced swimmers in the club in **2010** and ***b*** represents the number of intermediate swimmers in the club in **2010**?

- A. $1.53a + 1.49b = 35(1.44)$
- B. $1.49a + 0.53b = 35(1.44)$
- C. $1.53a + 1.44b = 35(1.49)$
- D. $1.44a + 1.53b = 35(1.49)$

ID: a28ee3f1 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that in **2010**, a swim club had a total of **35** swimmers, each classified as either advanced or intermediate, and that ***a*** represents the number of advanced swimmers in **2010** and ***b*** represents the number of intermediate swimmers in **2010**. It's also given that from **2010** to **2020**, the number of advanced swimmers in the club increased by approximately **53%** and the number of intermediate swimmers in the club increased by approximately **44%**. Thus, in **2020**, the approximate number of advanced swimmers in the club can be represented as **$1.53a$** and the approximate number of intermediate swimmers in the club can be represented as **$1.44b$** . It's given that the total number of swimmers in the club increased by approximately **49%** from **2010** to **2020**. Since the club had **35** swimmers in **2010**, it follows that the total number of swimmers in **2020** can be represented as **$35(1.49)$** . Since the sum of the number of advanced swimmers in **2020** and the number of intermediate swimmers in **2020** equals the total number of swimmers in **2020**, the equation **$1.53a + 1.44b = 35(1.49)$** best represents this situation.

Choice A is incorrect. This equation represents a situation where the number of intermediate swimmers in the club in **2020** increased by approximately **49%**, not **44%**, and the total number of swimmers in the club in **2020** increased by approximately **44%**, not **49%**.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the number of advanced swimmers in the club in **2020** increased by approximately **44%**, not **53%**, and the number of intermediate swimmers in the club in **2020** increased by approximately **53%**, not **44%**.

Question Difficulty: Medium

Question ID 843533bc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 843533bc

$y = x + 4$

Which table gives three values of x and their corresponding values of y for the given equation?

- A.

x	y
0	4
1	5
2	6
- B.

x	y
0	6
1	5
2	4
- C.

x	y
0	2
1	1
2	0
- D.

x	y
0	0
1	1
2	2

ID: 843533bc Answer

Correct Answer: A

Rationale

Choice A is correct. Substituting **0** for x into the given equation yields $y = 0 + 4$, or $y = 4$. Therefore, when $x = 0$, the corresponding value of y for the given equation is **4**. Substituting **1** for x into the given equation yields $y = 1 + 4$, or $y = 5$. Therefore, when $x = 1$, the corresponding value of y for the given equation is **5**. Substituting **2** for x into the given equation yields $y = 2 + 4$, or $y = 6$. Therefore, when $x = 2$, the corresponding value of y for the given equation is **6**. Of the choices given, only the table in choice A gives these three values of x and their corresponding values of y for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 6$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 2$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = x$.

Question Difficulty: Easy

Question ID 83c21f13

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 83c21f13

Each side of a **30**-sided polygon has one of three lengths. The number of sides with length **8 centimeters (cm)** is **5** times the number of sides ***n*** with length **3 cm**. There are **6** sides with length **4 cm**. Which equation must be true for the value of ***n***?

- A. $5n + 6 = 30$
- B. $6n + 6 = 30$
- C. $8n + 3n + 4n = 30$
- D. $8(5n) + 3n + 4(6) = 30$

ID: 83c21f13 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that each side of a **30**-sided polygon has one of three lengths. It's also given that the number of sides with length **8 centimeters (cm)** is **5** times the number of sides ***n*** with length **3 cm**. Therefore, there are **$5 \times n$** , or **$5n$** , sides with length **8 cm**. It's also given that there are **6** sides with length **4 cm**. Therefore, the number of **3 cm**, **4 cm**, and **8 cm** sides are ***n***, **6**, and **$5n$** , respectively. Since there are a total of **30** sides, the equation **$n + 6 + 5n = 30$** represents this situation. Combining like terms on the left-hand side of this equation yields **$6n + 6 = 30$** . Therefore, the equation that must be true for the value of ***n*** is **$6n + 6 = 30$** .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID dc85c551

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: dc85c551

$d = 16t$

The given equation represents the distance d , in inches, where t represents the number of seconds since an object started moving. Which of the following is the best interpretation of **16** in this context?

- A. The object moved a total of **16** inches.
- B. The object moved a total of **16t** inches.
- C. The object is moving at a rate of **16** inches per second.
- D. The object is moving at a rate of $\frac{1}{16}$ inches per second.

ID: dc85c551 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that in the equation $d = 16t$, d represents the distance, in inches, and t represents the number of seconds since an object started moving. In this equation, t is being multiplied by **16**. This means that the object’s distance increases by **16** inches each second. Therefore, the best interpretation of **16** in this context is that the object is moving at a rate of **16** inches per second.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is the best interpretation of **16t**, rather than **16**, in this context.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 15bcfa0b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 15bcfa0b

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

ID: 15bcfa0b Answer

Correct Answer: 2

Rationale

The correct answer is **2**. Substituting **8** for $f(x)$ in the given equation yields $8 = 4x$. Dividing the left- and right-hand sides of this equation by **4** yields $x = 2$. Therefore, the value of x is **2** when $f(x) = 8$.

Question Difficulty: Easy

Question ID 8df58923

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 8df58923

Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the xy -plane. What is the slope of line j ?

- A. $\frac{7}{17}$
- B. $\frac{17}{7}$
- C. 4
- D. 17

ID: 8df58923 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that line k is defined by $y = \frac{17x}{7} + 4$. For an equation of a line written in the form $y = mx + b$, m is the slope of the line and b is the y -coordinate of the y -intercept of the line. It follows that the slope of line k is $\frac{17}{7}$. It's also given that line j is parallel to line k in the xy -plane. Since parallel lines have equal slopes, line j also has a slope of $\frac{17}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the y -coordinate of the y -intercept of line k , not the slope of line j .

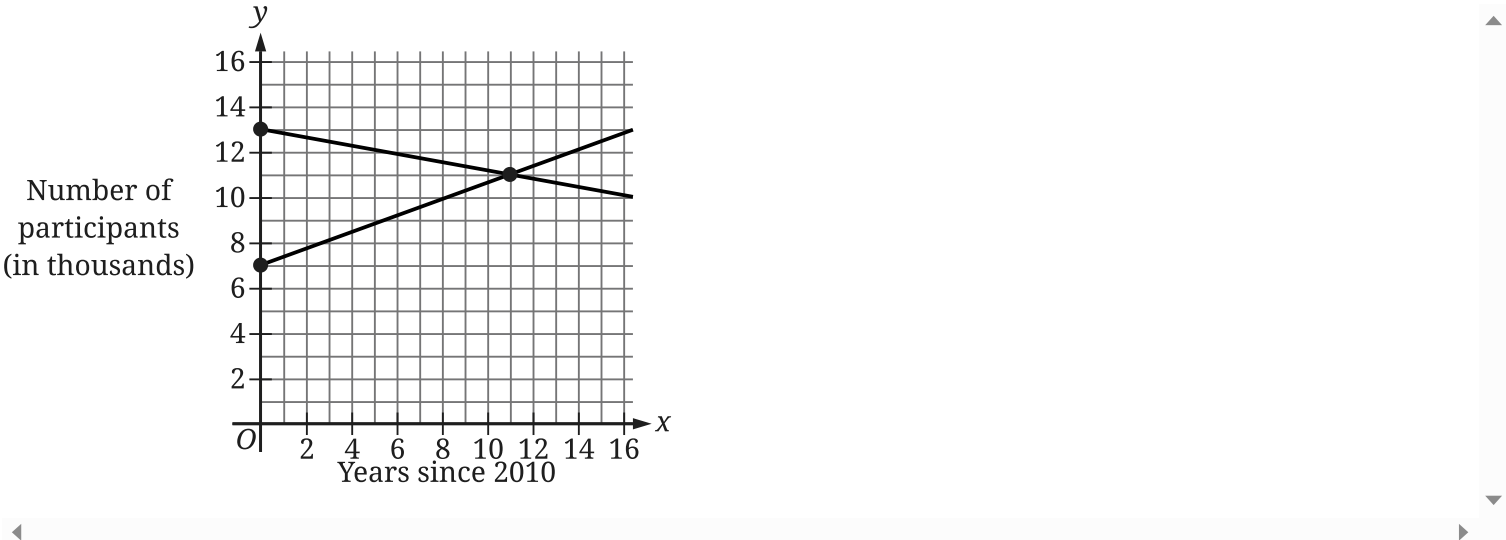
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b8898a93

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: b8898a93



The functions f and g model the number of participants, in thousands, in two different programs x years since 2010. The graphs of $y = f(x)$ and $y = g(x)$ are shown. Which of the following could represent functions f and g ?

- A. $f(x) = -\frac{4}{11}x + 13$
 $g(x) = \frac{2}{11}x + 7$
- B. $f(x) = -\frac{4}{11}x + 7$
 $g(x) = \frac{2}{11}x + 13$
- C. $f(x) = -\frac{2}{11}x + 13$
 $g(x) = \frac{4}{11}x + 7$
- D. $f(x) = -\frac{2}{11}x + 7$
 $g(x) = \frac{4}{11}x + 13$

ID: b8898a93 Answer

Correct Answer: C

Rationale

Choice C is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has slope m , where $m = \frac{y_2 - y_1}{x_2 - x_1}$. The equation of the line can be written in the form $y = mx + b$, where b is the y -intercept represented by $(0, b)$. One of the lines shown in the graph passes through the points $(0, 7)$ and $(11, 11)$. Substituting 0 for x_1 , 7 for y_1 , 11 for x_2 , and 11 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{11 - 7}{11 - 0}$, or $m = \frac{4}{11}$. Substituting $\frac{4}{11}$ for m and 7 for b in the equation $y = mx + b$ yields $y = \frac{4}{11}x + 7$. Therefore, an equation of this line is $y = \frac{4}{11}x + 7$. Similarly, the other line shown in the graph passes through the points $(0, 13)$ and $(11, 11)$. Substituting 0 for x_1 , 13 for y_1 , 11 for x_2 , and 11 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{11 - 13}{11 - 0}$, or $m = \frac{-2}{11}$. Substituting $\frac{-2}{11}$ for m and 13 for b in the equation $y = mx + b$ yields $y = \frac{-2}{11}x + 13$. Therefore, an equation of this line is $y = \frac{-2}{11}x + 13$. So, the functions f and g could be represented by $f(x) = \frac{-2}{11}x + 13$ and $g(x) = \frac{4}{11}x + 7$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID bfab48a4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: bfab48a4

In a chess tournament, each participant earns **1** point for each game the participant plays that ends in a draw and **3** points for each game the participant wins. A certain participant in this tournament has earned **41** points. Which equation represents this situation, where ***d*** represents the number of games this participant has played that ended in a draw and ***w*** represents the number of games this participant has won?

- A. $d + 3w = 41$
- B. $3d + w = 41$
- C. $d + \frac{w}{3} = 41$
- D. $\frac{d}{3} + w = 41$

ID: bfab48a4 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that each participant earns **1** point for each game the participant plays that ends in a draw and **3** points for each game the participant wins. Since ***d*** represents the number of games that a certain participant has played that ended in a draw and ***w*** represents the number of games this participant has won, this participant has earned a total of **$1d + 3w$** , or **$d + 3w$** , points. Since this participant earned **41** points, the equation **$d + 3w = 41$** represents this situation.

Choice B is incorrect. This equation represents the situation where the participant earned **3** points for each game played that ended in a draw and **1** point for each game won and earned **41** points.

Choice C is incorrect. This equation represents the situation where the participant earned **1** point for each game played that ended in a draw and $\frac{1}{3}$ point for each game won and earned **41** points.

Choice D is incorrect. This equation represents the situation where the participant earned $\frac{1}{3}$ point for each game played that ended in a draw and **1** point for each game won and earned **41** points.

Question Difficulty: Easy

Question ID 028a4b2c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 028a4b2c

$$\begin{aligned}x &= 5 \\ y &= x - 8\end{aligned}$$

Which of the following points (x, y) is the solution to the given system of equations in the xy -plane?

- A. $(0, 0)$
- B. $(5, -3)$
- C. $(5, -8)$
- D. $(5, 8)$

ID: 028a4b2c Answer

Correct Answer: B

Rationale

Choice B is correct. A solution to a system of equations in the xy -plane is a point (x, y) that lies on the graph of each equation in the system. The first equation given is $x = 5$. Substituting 5 for x in the second given equation yields $y = 5 - 8$, or $y = -3$. It follows that in the xy -plane, the point $(5, -3)$ lies on the graph of each equation in the system. Therefore, the solution to the given system of equations in the xy -plane is $(5, -3)$.

Choice A is incorrect. The point $(0, 0)$ doesn't lie on the graph of either equation in the given system.

Choice C is incorrect. The point $(5, -8)$ doesn't lie on the graph of the second equation in the given system.

Choice D is incorrect. The point $(5, 8)$ doesn't lie on the graph of the second equation in the given system.

Question Difficulty: Easy

Question ID 29057d17

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 29057d17

An event planner is planning a party. It costs the event planner a onetime fee of **\$35** to rent the venue and **\$10.25** per attendee. The event planner has a budget of **\$300**. What is the greatest number of attendees possible without exceeding the budget?

ID: 29057d17 Answer

Correct Answer: 25

Rationale

The correct answer is **25**. The total cost of the party is found by adding the onetime fee of the venue to the cost per attendee times the number of attendees. Let x be the number of attendees. The expression $35 + 10.25x$ thus represents the total cost of the party. It's given that the budget is **\$300**, so this situation can be represented by the inequality $35 + 10.25x \leq 300$. Subtracting **35** from both sides of this inequality gives $10.25x \leq 265$. Dividing both sides of this inequality by **10.25** results in approximately $x \leq 25.854$. Since the question is stated in terms of attendees, rounding **25.854** down to the greatest whole number gives the greatest number of attendees possible, which is **25**.

Question Difficulty: Medium

Question ID eb330004

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: eb330004

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is **\$25.40**, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

- A. $f(x) = 25.40(x - 1) + 12.70$
- B. $f(x) = 25.40(x - 2) + 12.70$
- C. $f(x) = 25.40(x - 1) + 12.70(x - 2)$
- D. $f(x) = 25.40(x - 2) + 12.70(x - 1)$

ID: eb330004 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the regular price of a session at a yoga studio is **\$25.40**. It's also given that the yoga studio is offering a deal where the first session is free, or **\$0**; that the second session is half off the regular price, or **$\$25.40 - (\frac{1}{2})(\$25.40)$** , which is equal to **\$12.70**; and that the remaining sessions are at the regular price, **\$25.40**. Since x represents the total number of sessions and two sessions have already been accounted for, the number of remaining sessions can be represented by $x - 2$. The total cost, in dollars, of purchasing x sessions using this deal can be represented by the function $f(x) = 0 + 12.70 + 25.40(x - 2)$, or $f(x) = 25.40(x - 2) + 12.70$.

Choice A is incorrect. The function represents the total cost of purchasing x sessions when the first session is half off the regular price and the remaining sessions are at the regular price.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5eeef20e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 5eeef20e

A veterinarian recommends that each day a certain rabbit should eat **25** calories per pound of the rabbit’s weight, plus an additional **11** calories. Which equation represents this situation, where ***c*** is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit’s weight is ***x*** pounds?

- A. **$c = 25x$**
- B. **$c = 36x$**
- C. **$c = 11x + 25$**
- D. **$c = 25x + 11$**

ID: 5eeef20e Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that a veterinarian recommends that each day the rabbit should eat **25** calories per pound of the rabbit’s weight, plus an additional **11** calories. If the rabbit’s weight is ***x*** pounds, then multiplying **25** calories per pound by the rabbit’s weight, ***x*** pounds, yields **25*x*** calories. Adding the additional **11** calories that the rabbit should eat each day yields **25*x* + 11** calories. It’s given that ***c*** is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit’s weight is ***x*** pounds. Therefore, this situation can be represented by the equation **$c = 25x + 11$** .

Choice A is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **25** calories per pound of the rabbit’s weight.

Choice B is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **25 + 11**, or **36**, calories per pound of the rabbit’s weight.

Choice C is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat **11** calories per pound of the rabbit’s weight, plus an additional **25** calories.

Question Difficulty: Easy

Question ID d507e7ba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: d507e7ba

A team hosting an event to raise money for new uniforms plans to sell at least **140** tickets before this event and at least **220** tickets during this event to raise a total of at least **\$5,820** from all tickets sold. The price of a ticket during this event is **\$3** less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

- A. $140(x + 3) + 220x \leq 5,820$
- B. $140(x + 3) + 220x \geq 5,820$
- C. $140(x - 3) + 220x \leq 5,820$
- D. $140(x - 3) + 220x \geq 5,820$

ID: d507e7ba Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that a team plans to sell at least **140** tickets before an event and at least **220** tickets during the event to raise a total of at least **\$5,820** from all tickets sold. It's also given that the price of a ticket during the event is **\$3** less than the price of a ticket before the event and that x represents the price, in dollars, of a ticket sold during the event. It follows that $x + 3$ represents the price, in dollars, of a ticket sold before the event. The expression $140(x + 3)$ represents the planned revenue, in dollars, from the tickets sold before the event, and the expression $220x$ represents the planned revenue, in dollars, from the tickets sold during the event. Thus, the expression $140(x + 3) + 220x$ represents the planned revenue, in dollars, from all tickets sold. Since the team plans to raise a total of at least **\$5,820** from all tickets sold, the total revenue must be at least **\$5,820**. Therefore, the inequality $140(x + 3) + 220x \geq 5,820$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which the team raises a total of at most **\$5,820** from all tickets sold.

Choice C is incorrect. This inequality represents a situation in which the price of a ticket before the event is **\$3** less, rather than **\$3** more, than the price of a ticket during the event and the team raises a total of at most **\$5,820** from all tickets sold.

Choice D is incorrect. This inequality represents a situation in which the price of a ticket before the event is **\$3** less, rather than **\$3** more, than the price of a ticket during the event.

Question Difficulty: Hard

Question ID a96e353c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: a96e353c

Terry has a goal to collect at least **16** items per week for a donation drive. This week, Terry has collected **4** items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

- A. $4 - x \leq 16$
- B. $4 + x \leq 16$
- C. $4 - x \geq 16$
- D. $4 + x \geq 16$

ID: a96e353c Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that this week, Terry has collected **4** items. It follows that the expression $4 + x$ represents the total number of items Terry will have collected after collecting x additional items. It's also given that Terry's goal is to collect at least **16** items per week. Therefore, the inequality $4 + x \geq 16$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This inequality represents a situation where the goal is to collect at most, not at least, **16** items per week.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 805354f5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 805354f5

Line k is defined by $y = \frac{1}{4}x + 1$. Line j is parallel to line k in the xy -plane. What is the slope of j ?

ID: 805354f5 Answer

Correct Answer: .25, 1/4

Rationale

The correct answer is $\frac{1}{4}$. It's given that line k is defined by $y = \frac{1}{4}x + 1$. It's also given that line j is parallel to line k in the xy -plane. A line in the xy -plane represented by an equation in slope-intercept form $y = mx + b$ has a slope of m and a y -intercept of $(0, b)$. Therefore, the slope of line k is $\frac{1}{4}$. Since parallel lines have equal slopes, the slope of line j is $\frac{1}{4}$. Note that 1/4 and .25 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 8edd9405

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 8edd9405

The functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x-coordinate of the x-intercept of the graph of $y = h(x)$ in the xy-plane?

ID: 8edd9405 Answer

Correct Answer: -12

Rationale

The correct answer is -12 . It's given that the functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, then substituting $\frac{1}{4}x - 9$ for $f(x)$ and $\frac{3}{4}x + 21$ for $g(x)$ in this function yields $h(x) = \frac{1}{4}x - 9 + \frac{3}{4}x + 21$. This can be rewritten as $h(x) = \frac{4}{4}x + 12$, or $h(x) = x + 12$. The x-intercept of a graph in the xy-plane is the point on the graph where $y = 0$. The equation representing the graph of $y = h(x)$ is $y = x + 12$. Substituting 0 for y in this equation yields $0 = x + 12$. Subtracting 12 from both sides of this equation yields $-12 = x$, or $x = -12$. Therefore, the x-coordinate of the x-intercept of the graph of $y = h(x)$ in the xy-plane is -12 .

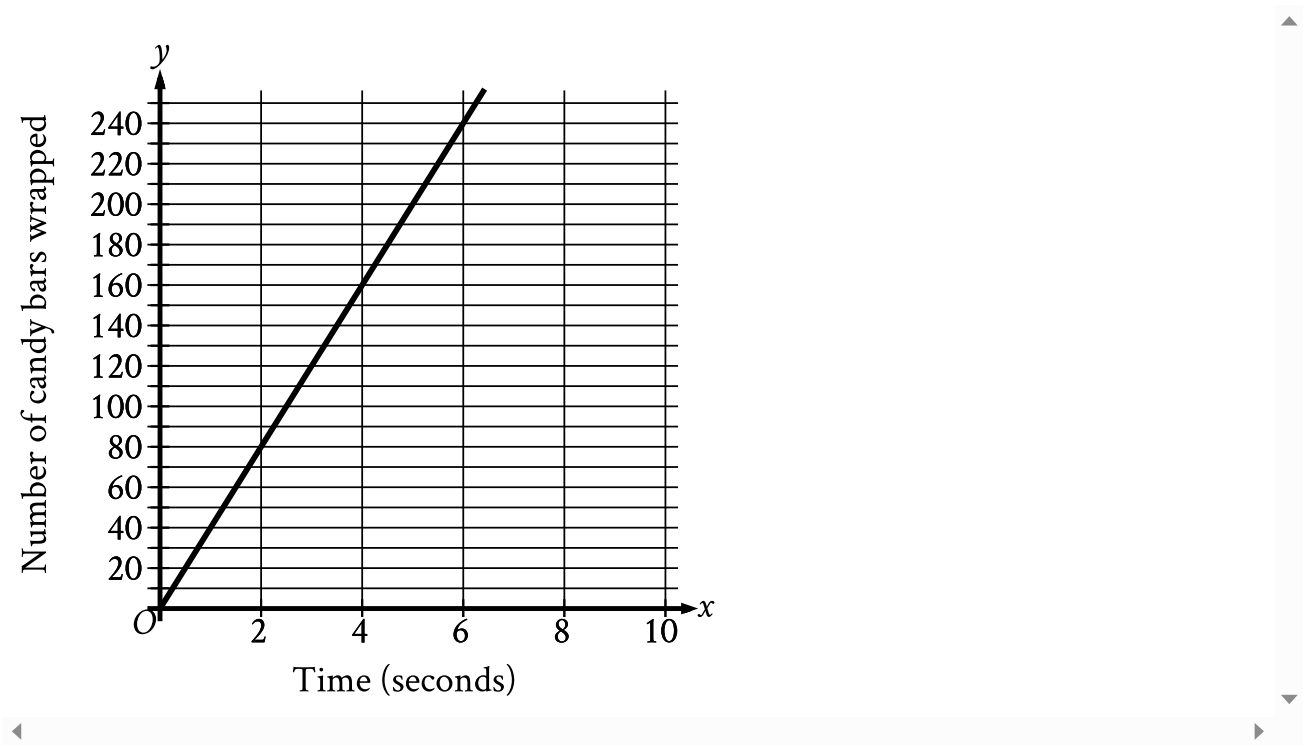
Question Difficulty: Hard

Question ID b15c634e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b15c634e

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

- A. 2
- B. 40
- C. 78
- D. 80

ID: b15c634e Answer

Correct Answer: B

Rationale

Choice B is correct. For the graph shown, the x -axis represents time, in seconds, and the y -axis represents the number of candy bars wrapped. The slope of a line in the xy -plane is the change in y for each 1-unit increase in x . It follows that the slope of the graph shown represents the estimated number of candy bars the machine wraps with a label per second. The slope, m , of a line in the xy -plane can be found using any two points, (x_1, y_1) and (x_2, y_2) , on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shown passes through the points $(0, 0)$ and $(2, 80)$. Substituting these points for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{80 - 0}{2 - 0}$, which is equivalent to $m = \frac{80}{2}$, or $m = 40$. Therefore, the estimated number of candy bars the machine wraps with a label per second is 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID bfaa932c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: bfaa932c

$2x + 9y = 7$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

- I. $3x + 13.5y = 10.5$
- II. $3x - 13.5y = 10.5$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: bfaa932c Answer

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables has at least one solution if the equations represent the same line or lines that intersect at exactly one point. Statement I gives the equation $3x + 13.5y = 10.5$. Multiplying both sides of the given equation, $2x + 9y = 7$, by $\frac{3}{2}$ yields $3x + 13.5y = 10.5$, so the equation in statement I is equivalent to the given equation. Therefore, the system formed by these two equations represents the same line and has infinitely many solutions. Statement II gives the equation $3x - 13.5y = 10.5$. Adding the right- and left-hand sides of $3x + 13.5y = 10.5$ and $3x - 13.5y = 10.5$ yields $3x + 13.5y + 3x - 13.5y = 10.5 + 10.5$, or $6x = 21$. Dividing both sides of this equation by 6 yields $x = \frac{21}{6}$. Therefore, the lines represented by these equations intersect at exactly one point, where $x = \frac{21}{6}$, and the system formed by the given equation and the equation in statement II has one solution. It follows that both the equation in statement I and the equation in statement II could be the other equation in a system of equations with at least one solution.

Choice A is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice B is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

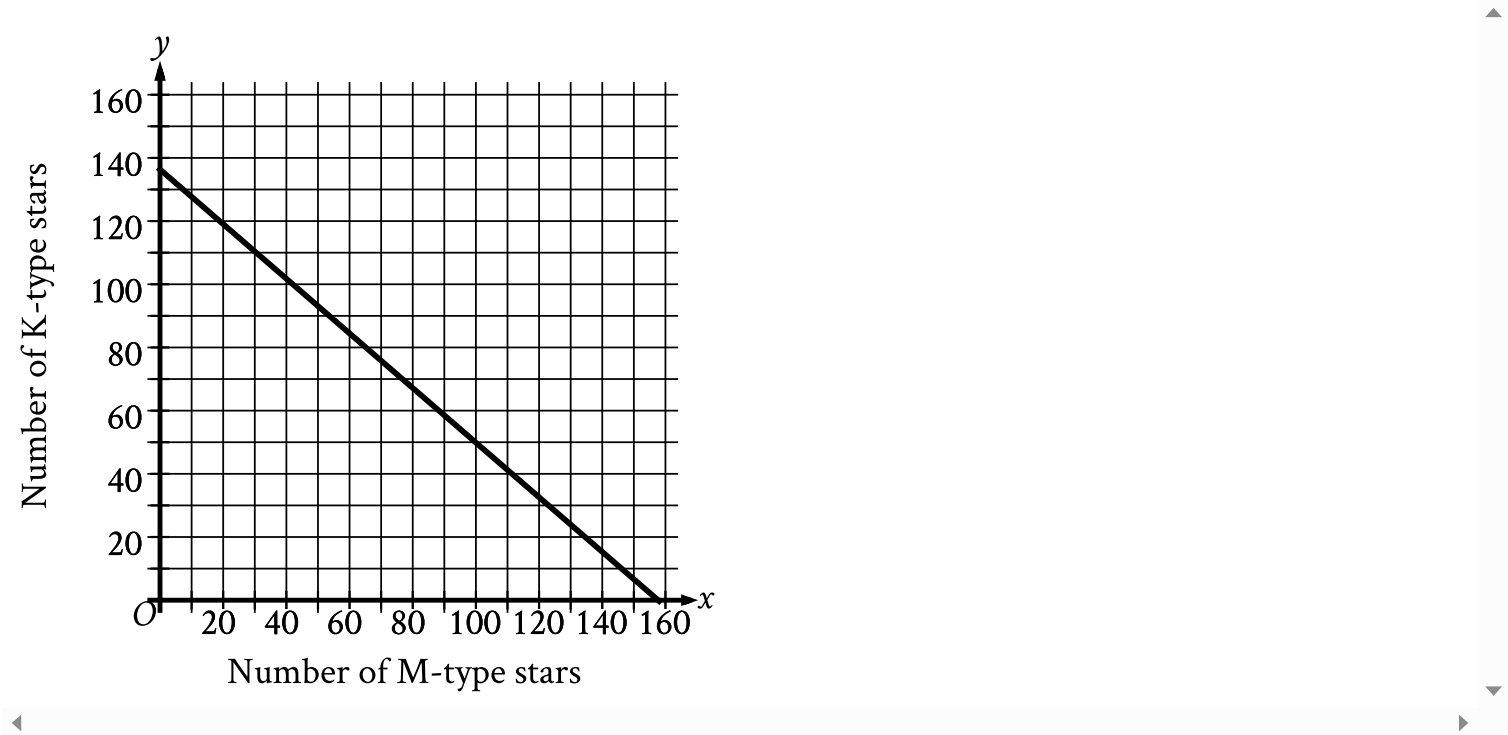
Choice D is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Question Difficulty: Hard

Question ID 9f591345

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 9f591345



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is **127,882 quettagrams**. The graph shown models the possible combinations of the number of M-type stars, x , and K-type stars, y , that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

- A. 811
- B. 938
- C. 51,904
- D. 75,978

ID: 9f591345 Answer

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the x-intercept is approximately **(158, 0)**. Therefore, if there are **0** K-type stars in the cluster, then there are approximately **158** M-type stars in the cluster. It's given that the total mass of the stars in this cluster is **127,882** quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127,882}{158}$ quettagrams, or approximately **809.38** quettagrams. Thus, of the given choices, **811** is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID be7c80b1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: be7c80b1

$2\ell + 2w \leq 27$

A rectangle has length ℓ and width w . The inequality gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to **27**. Which statement is the best interpretation of $(\ell, w) = (8, 3)$ in this context?

- A. If the rectangle has length **3** and width **8**, its perimeter is less than or equal to **27**.
- B. If the rectangle has length **8** and width **3**, its perimeter is less than or equal to **27**.
- C. If the rectangle has length **3** and width **8**, its perimeter is greater than or equal to **27**.
- D. If the rectangle has length **8** and width **3**, its perimeter is greater than or equal to **27**.

ID: be7c80b1 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that a rectangle has length ℓ and width w , and the inequality $2\ell + 2w \leq 27$ gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to **27**. To determine the best interpretation of $(\ell, w) = (8, 3)$ in this context, the values can be substituted in the given inequality. Substituting **8** for ℓ and **3** for w in this inequality yields $2(8) + 2(3) \leq 27$, which is equivalent to $16 + 6 \leq 27$, or $22 \leq 27$. Since this inequality is true, if the rectangle has length **8** and width **3**, its perimeter is less than or equal to **27**.

Choice A is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length **8** and width **3**, not length **3** and width **8**.

Choice C is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length **8** and width **3**, not length **3** and width **8**.

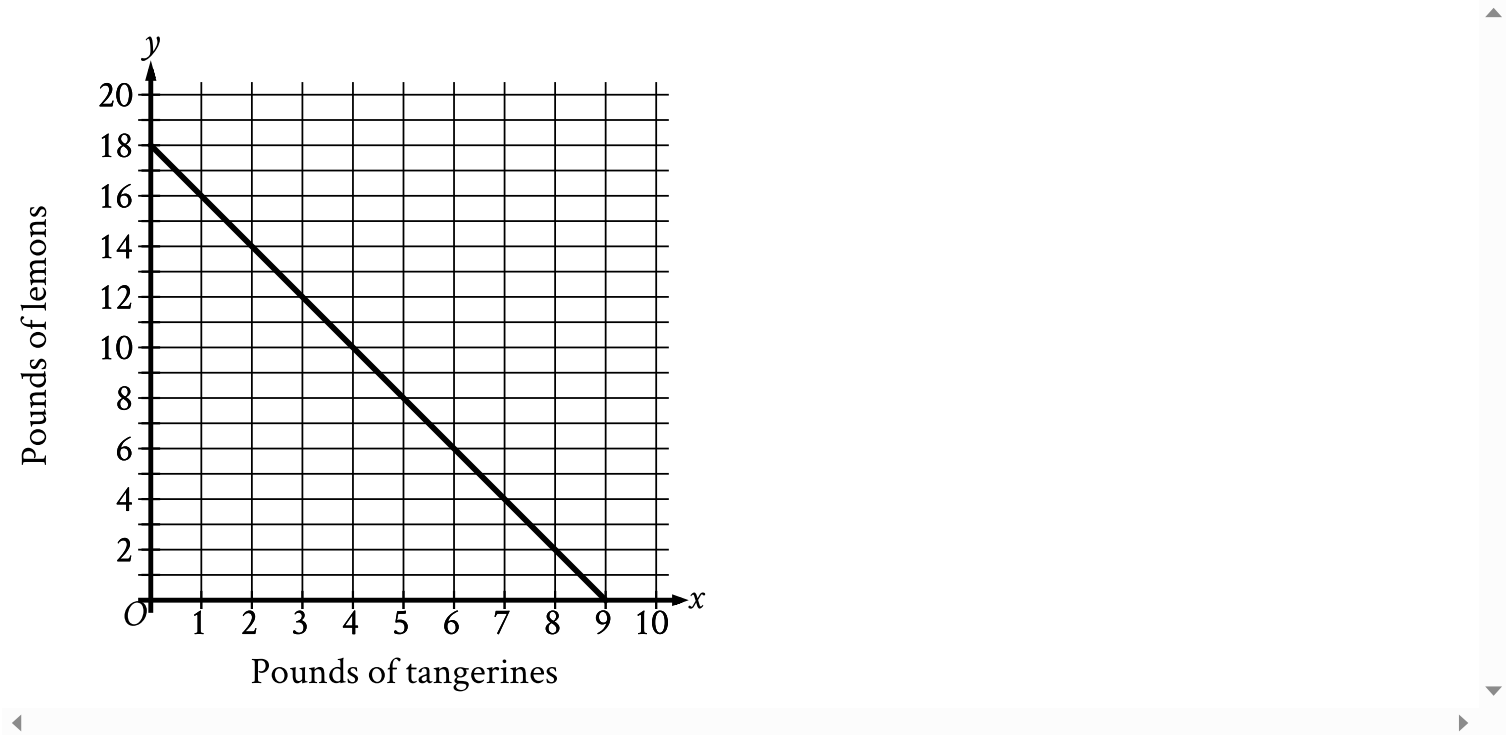
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID f355fc06

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: f355fc06



The graph shows the possible combinations of the number of pounds of tangerines and lemons that could be purchased for **\$18** at a certain store. If Melvin purchased lemons and **4** pounds of tangerines for a total of **\$18**, how many pounds of lemons did he purchase?

- A. 7
- B. 10
- C. 14
- D. 16

ID: f355fc06 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph shows the possible combinations of the number of pounds of tangerines, x , and the number of pounds of lemons, y , that could be purchased for **\$18** at a certain store. If Melvin purchased lemons and **4** pounds of tangerines for a total of **\$18**, the number of pounds of lemons he purchased is represented by the y -coordinate of the point on the graph where $x = 4$. For the graph shown, when $x = 4$, $y = 10$. Therefore, if Melvin purchased lemons and **4** pounds of tangerines for a total of **\$18**, then he purchased **10** pounds of lemons.

Choice A is incorrect. This is the number of pounds of tangerines Melvin purchased if he purchased tangerines and **4** pounds of lemons for a total of **\$18**.

Choice C is incorrect. This is the number of pounds of lemons Melvin purchased if he purchased lemons and **2** pounds of tangerines for a total of **\$18**.

Choice D is incorrect. This is the number of pounds of lemons Melvin purchased if he purchased lemons and **1** pound of tangerines for a total of **\$18**.

Question Difficulty: Easy

Question ID 81fa9041

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 81fa9041

$-3x + 21px = 84$

In the given equation, p is a constant. The equation has no solution. What is the value of p ?

- A. 0
- B. $\frac{1}{7}$
- C. $\frac{4}{3}$
- D. 4

ID: 81fa9041 Answer

Correct Answer: B

Rationale

Choice B is correct. A linear equation in one variable has no solution if and only if the equation is false; that is, when there is no value of x that produces a true statement. It's given that in the equation $-3x + 21px = 84$, p is a constant and the equation has no solution for x . Therefore, the value of the constant p is one that results in a false equation. Factoring out the common factor of $-3x$ on the left-hand side of the given equation yields $-3x(1 - 7p) = 84$. Dividing both sides of this equation by -3 yields $x(1 - 7p) = -28$. Dividing both sides of this equation by $(1 - 7p)$ yields $x = \frac{-28}{1-7p}$. This equation is false if and only if $1 - 7p = 0$. Adding $7p$ to both sides of $1 - 7p = 0$ yields $1 = 7p$. Dividing both sides of this equation by 7 yields $\frac{1}{7} = p$. It follows that the equation $x = \frac{-28}{1-7p}$ is false if and only if $p = \frac{1}{7}$. Therefore, the given equation has no solution if and only if the value of p is $\frac{1}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f77140e4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: f77140e4

$y > 7x - 4$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	13
5	27
8	48

B.

x	y
3	17
5	31
8	52

C.

x	y
3	21
5	27
8	52

D.

x	y
3	21
5	35
8	56

ID: f77140e4 Answer

Correct Answer: D

Rationale

Choice D is correct. A solution (x, y) to the given inequality is a value of x and the corresponding value of y such that the value of y is greater than the value of $7x - 4$. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting 3 for x in the given inequality yields $y > 7(3) - 4$, or $y > 17$. Substituting 5 for x in the given inequality yields $y > 7(5) - 4$, or $y > 31$. Substituting 8 for x in the given inequality yields $y > 7(8) - 4$, or $y > 52$. Therefore, when $x = 3$, $x = 5$, and $x = 8$, the

corresponding values of y must be greater than **17**, greater than **31**, and greater than **52**, respectively. In the table in choice D, when $x = 3$, the corresponding value of y is **21**, which is greater than **17**; when $x = 5$, the corresponding value of y is **35**, which is greater than **31**; when $x = 8$, the corresponding value of y is **56**, which is greater than **52**. Of the given choices, only choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b771895f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b771895f

For the linear function g , the graph of $y = g(x)$ in the xy -plane has a slope of 2 and passes through the point $(1, 14)$. Which equation defines g ?

- A. $g(x) = 2x$
- B. $g(x) = 2x + 2$
- C. $g(x) = 2x + 12$
- D. $g(x) = 2x + 14$

ID: b771895f Answer

Correct Answer: C

Rationale

Choice C is correct. An equation defining a linear function can be written in the form $g(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = g(x)$ in the xy -plane. It's given that the graph of $y = g(x)$ has a slope of 2 . Therefore, $m = 2$. It's also given that the graph of $y = g(x)$ passes through the point $(1, 14)$. It follows that when $x = 1$, $g(x) = 14$. Substituting 1 for x , 14 for $g(x)$, and 2 for m in the equation $g(x) = mx + b$ yields $14 = 2(1) + b$, or $14 = 2 + b$. Subtracting 2 from each side of this equation yields $12 = b$. Therefore, $b = 12$. Substituting 2 for m and 12 for b in the equation $g(x) = mx + b$ yields $g(x) = 2x + 12$. Therefore, the equation that defines g is $g(x) = 2x + 12$.

Choice A is incorrect. For this function, the graph of $y = g(x)$ in the xy -plane passes through the point $(1, 2)$, not $(1, 14)$.

Choice B is incorrect. For this function, the graph of $y = g(x)$ in the xy -plane passes through the point $(1, 4)$, not $(1, 14)$.

Choice D is incorrect. For this function, the graph of $y = g(x)$ in the xy -plane passes through the point $(1, 16)$, not $(1, 14)$.

Question Difficulty: Medium

Question ID ce480966

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: ce480966

$$h(x) = x + b$$

For the linear function h , b is a constant and $h(0) = 45$. What is the value of b ?

ID: ce480966 Answer

Correct Answer: 45

Rationale

The correct answer is **45**. It's given that $h(0) = 45$. Therefore, for the given function h , when $x = 0$, $h(x) = 45$. Substituting **0** for x and **45** for $h(x)$ in the given function, $h(x) = x + b$, yields $45 = 0 + b$, or $45 = b$. Therefore, the value of b is **45**.

Question Difficulty: Medium

Question ID 03238eb2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 03238eb2

$$\begin{aligned} 2(8x) + 4(7y) &= 12 \\ -2(8x) + 4(7y) &= 12 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $8x + 7y$?

ID: 03238eb2 Answer

Correct Answer: 3

Rationale

The correct answer is **3**. Adding the second equation to the first equation in the given system of equations yields $(2(8x) - 2(8x)) + (4(7y) + 4(7y)) = 12 + 12$, or $8(7y) = 24$. Dividing both sides of this equation by **8** yields $7y = 3$. Substituting **3** for $7y$ in the first equation, $2(8x) + 4(7y) = 12$, yields $2(8x) + 4(3) = 12$, or $2(8x) + 12 = 12$. Subtracting **12** from both sides of this equation yields $2(8x) = 0$. Dividing both sides of this equation by **2** yields $8x = 0$. Substituting **0** for $8x$ and **3** for $7y$ in the expression $8x + 7y$ yields $0 + 3$, or **3**. Therefore, the value of $8x + 7y$ is **3**.

Question Difficulty: Hard

Question ID eae227e1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: eae227e1

In the xy -plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 2$. If line ℓ also passes through the point $(3, d)$, what is the value of d ?

ID: eae227e1 Answer

Correct Answer: 24

Rationale

The correct answer is **24**. A line in the xy -plane can be defined by the equation $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept of the line. It's given that line ℓ passes through the point $(0, 0)$. Therefore, the y -coordinate of the y -intercept of line ℓ is **0**. It's given that line ℓ is parallel to the line represented by the equation $y = 8x + 2$. Since parallel lines have the same slope, it follows that the slope of line ℓ is **8**. Therefore, line ℓ can be defined by an equation in the form $y = mx + b$, where $m = 8$ and $b = 0$. Substituting **8** for m and **0** for b in $y = mx + b$ yields the equation $y = 8x + 0$, or $y = 8x$. If line ℓ passes through the point $(3, d)$, then when $x = 3, y = d$ for the equation $y = 8x$. Substituting **3** for x and d for y in the equation $y = 8x$ yields $d = 8(3)$, or $d = 24$.

Question Difficulty: Hard

Question ID c4b690d2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c4b690d2

$$3x + 5(x + 4) = 76$$

What value of x is the solution to the given equation?

- A. 7
- B. 8
- C. 56
- D. 72

ID: c4b690d2 Answer

Correct Answer: A

Rationale

Choice A is correct. Applying the distributive property on the left-hand side of the given equation yields $3x + 5x + 20 = 76$, or $8x + 20 = 76$. Subtracting 20 from each side of this equation yields $8x = 56$. Dividing each side of this equation by 8 yields $x = 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution to the equation $x + 4 = 76$, not $3x + 5(x + 4) = 76$.

Question Difficulty: Easy

Question ID 652cb78d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 652cb78d

$$\begin{aligned}2x + 3y &= 7 \\ 10x + 15y &= 35\end{aligned}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

- A. $(\frac{r}{5} + 7, -\frac{r}{5} + 35)$
- B. $(-\frac{3r}{2} + \frac{7}{2}, r)$
- C. $(r, \frac{2r}{3} + \frac{7}{3})$
- D. $(r, -\frac{3r}{2} + \frac{7}{2})$

ID: 652cb78d Answer

Correct Answer: B

Rationale

Choice B is correct. The two given equations are equivalent because the second equation can be obtained from the first equation by multiplying each side of the equation by 5. Thus, the graphs of the equations are coincident, so if a point lies on the graph of one of the equations, it also lies on the graph of the other equation. A point (x, y) lies on the graph of an equation in the xy -plane if and only if this point represents a solution to the equation. It is sufficient, therefore, to find the point that represents a solution to the first given equation. Substituting the x - and y -coordinates of choice B, $-\frac{3r}{2} + \frac{7}{2}$ and r , for x and y , respectively, in the first equation yields $2(-\frac{3r}{2} + \frac{7}{2}) + 3r = 7$, which is equivalent to $-3r + 7 + 3r = 7$, or $7 = 7$. Therefore, the point $(-\frac{3r}{2} + \frac{7}{2}, r)$ represents a solution to the first equation and thus lies on the graph of each equation in the xy -plane for the given system.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e84ae4e3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: e84ae4e3

If $5 - 7(2 - 4x) = 16 - 8(2 - 4x)$, what is the value of $2 - 4x$?

ID: e84ae4e3 Answer

Correct Answer: 11

Rationale

The correct answer is **11**. Subtracting **5** from each side of the given equation yields $-7(2 - 4x) = 11 - 8(2 - 4x)$. Adding $8(2 - 4x)$ to each side of this equation yields $2 - 4x = 11$. Therefore, the value of $2 - 4x$ is **11**.

Question Difficulty: Hard

Question ID 3bf5b5b3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 3bf5b5b3

In the linear function f , $f(0) = 8$ and $f(1) = 12$. Which equation defines f ?

- A. $f(x) = 12x + 8$
- B. $f(x) = 4x$
- C. $f(x) = 4x + 12$
- D. $f(x) = 4x + 8$

ID: 3bf5b5b3 Answer

Correct Answer: D

Rationale

Choice D is correct. Since f is a linear function, it can be defined by an equation of the form $f(x) = ax + b$, where a and b are constants. It's given that $f(0) = 8$. Substituting 0 for x and 8 for $f(x)$ in the equation $f(x) = ax + b$ yields $8 = a(0) + b$, or $8 = b$. Substituting 8 for b in the equation $f(x) = ax + b$ yields $f(x) = ax + 8$. It's given that $f(1) = 12$. Substituting 1 for x and 12 for $f(x)$ in the equation $f(x) = ax + 8$ yields $12 = a(1) + 8$, or $12 = a + 8$. Subtracting 8 from both sides of this equation yields $a = 4$. Substituting 4 for a in the equation $f(x) = ax + 8$ yields $f(x) = 4x + 8$. Therefore, an equation that defines f is $f(x) = 4x + 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b5963a10

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b5963a10

The function f is defined by $f(x) = \frac{7}{10}x + 55$. What is the value of $f(20)$?

ID: b5963a10 Answer

Correct Answer: 69

Rationale

The correct answer is **69**. The value of $f(20)$ can be found by evaluating the function $f(x) = \frac{7}{10}x + 55$ for $x = 20$. Substituting **20** for x in this function yields $f(20) = \frac{7}{10}(20) + 55$, or $f(20) = 69$. Therefore, the value of $f(20)$ is **69**.

Question Difficulty: Medium

Question ID 13f677a4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 13f677a4

A train traveled **173** miles in the first **3** hours of a trip and traveled at an average speed of **53** miles per hour for the remainder of the trip. Which equation gives the total number of miles, ***y***, the train traveled if the trip was ***x*** hours long and ***x* > 3**?

- A. ***y* = 53*x* + 173**
- B. ***y* = 53*x* + 14**
- C. ***y* = 14*x* + 226**
- D. ***y* = 14*x* + 53**

ID: 13f677a4 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that a train traveled **173** miles in the first **3** hours of a trip. It's also given that the train traveled at an average speed of **53** miles per hour for the remainder of the trip. Since the trip was ***x*** hours long, it follows that the remainder of the trip was ***x* − 3** hours. Therefore, **53(*x* − 3)** represents the distance, in miles, the train traveled for the remainder of the trip. Adding **173** to **53(*x* − 3)** gives the total number of miles, ***y***, the train traveled if the trip was ***x*** hours long and ***x* > 3**. It follows that ***y* = 173 + 53(*x* − 3)**, which is equivalent to ***y* = 173 + 53*x* − 159**, or ***y* = 53*x* + 14**.

Choice A is incorrect. This equation gives the total number of miles, ***y***, the train traveled if the trip was ***x* + 3**, not ***x***, hours long.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c98d15e1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c98d15e1

$x + 40 = 95$

What value of x is the solution to the given equation?

ID: c98d15e1 Answer

Correct Answer: 55

Rationale

The correct answer is **55**. Subtracting **40** from both sides of the given equation yields $x = 55$. Therefore, the value of x is **55**.

Question Difficulty: Easy

Question ID 144e15c9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 144e15c9

The function f is defined by $f(x) = \frac{x+15}{5}$, and $f(a) = 10$, where a is a constant. What is the value of a ?

- A. 5
- B. 10
- C. 35
- D. 65

ID: 144e15c9 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x) = \frac{x+15}{5}$ and $f(a) = 10$, where a is a constant. Therefore, for the given function f , when $x = a$, $f(x) = 10$. Substituting a for x and 10 for $f(x)$ in the given function f yields $10 = \frac{a+15}{5}$. Multiplying both sides of this equation by 5 yields $50 = a + 15$. Subtracting 15 from both sides of this equation yields $35 = a$. Therefore, the value of a is 35 .

Choice A is incorrect. This is the value of a if $f(a) = 4$.

Choice B is incorrect. This is the value of a if $f(a) = 5$.

Choice D is incorrect. This is the value of a if $f(a) = 16$.

Question Difficulty: Hard

Question ID 3fc0113d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 3fc0113d

In the xy -plane, line k has a slope of 5 and a y -intercept of $(0, -35)$. What is the x -coordinate of the x -intercept of line k ?

ID: 3fc0113d Answer

Correct Answer: 7

Rationale

The correct answer is 7 . An equation of a line in the xy -plane can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y -intercept of the line. It's given that line k has a slope of 5 and a y -intercept of $(0, -35)$. Therefore, $m = 5$ and $b = -35$. Substituting 5 for m and -35 for b in the equation $y = mx + b$ yields $y = 5x - 35$. The x -intercept of a line in the xy -plane is the point where the line intersects the x -axis, which is a point with a y -coordinate of 0 . Substituting 0 for y in the equation $y = 5x - 35$ yields $0 = 5x - 35$. Adding 35 to both sides of this equation yields $35 = 5x$. Dividing both sides of this equation by 5 yields $7 = x$. Therefore, the x -coordinate of the x -intercept of line k is 7 .

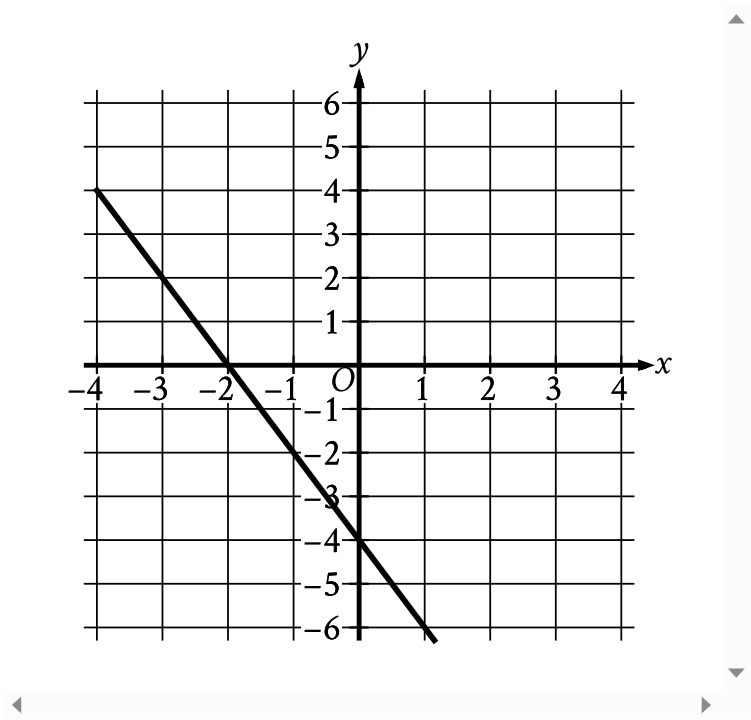
Question Difficulty: Medium

Question ID b9e821cb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b9e821cb

The graph of $y = f(x) - 11$ is shown.



Which equation defines the linear function f ?

- A. $f(x) = -13x - 11$
- B. $f(x) = -2x + 7$
- C. $f(x) = -13x + 7$
- D. $f(x) = -2x - 11$

ID: b9e821cb Answer

Correct Answer: B

Rationale

Choice B is correct. The graph of a line in the xy -plane can be represented by the equation $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y -intercept. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $(-1, -2)$ and $(0, -4)$. Substituting $(-1, -2)$ and $(0, -4)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{-4 - (-2)}{0 - (-1)}$, which is equivalent to $m = \frac{-2}{1}$, or $m = -2$. Since the line shown passes through the point $(0, -4)$, it follows that $b = -4$. Substituting -2 for m and -4 for b in the equation $y = mx + b$ yields $y = -2x - 4$. It's given that the graph shown is the graph of $y = f(x) - 11$. Substituting $-2x - 4$ for y in the equation $y = f(x) - 11$ yields $-2x - 4 = f(x) - 11$. Adding 11 to both sides of this equation yields $-2x + 7 = f(x)$. Therefore, the equation $f(x) = -2x + 7$ defines the linear function f .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f82800cb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: f82800cb

For the linear function h , $h(0) = 0$ and $h(11) = 10$. Which equation defines h ?

- A. $h(x) = \frac{10}{11}x$
- B. $h(x) = \frac{10}{11}x + 10$
- C. $h(x) = \frac{11}{10}x$
- D. $h(x) = \frac{11}{10}x + 11$

ID: f82800cb Answer

Correct Answer: A

Rationale

Choice A is correct. Since h is a linear function, it can be defined by an equation of the form $h(x) = mx + b$, where m and b are constants. It's given that $h(0) = 0$. Substituting 0 for x and 0 for $h(x)$ in the equation $h(x) = mx + b$ yields $0 = 0m + b$, or $0 = b$. Substituting 0 for b in the equation $h(x) = mx + b$ yields $h(x) = mx$. It's also given that $h(11) = 10$. Substituting 11 for x and 10 for $h(x)$ in the equation $h(x) = mx$ yields $10 = 11m$. Dividing both sides of this equation by 11 yields $m = \frac{10}{11}$. Substituting $\frac{10}{11}$ for m in the equation $h(x) = mx$ yields $h(x) = \frac{10}{11}x$. Therefore, an equation that defines h is $h(x) = \frac{10}{11}x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

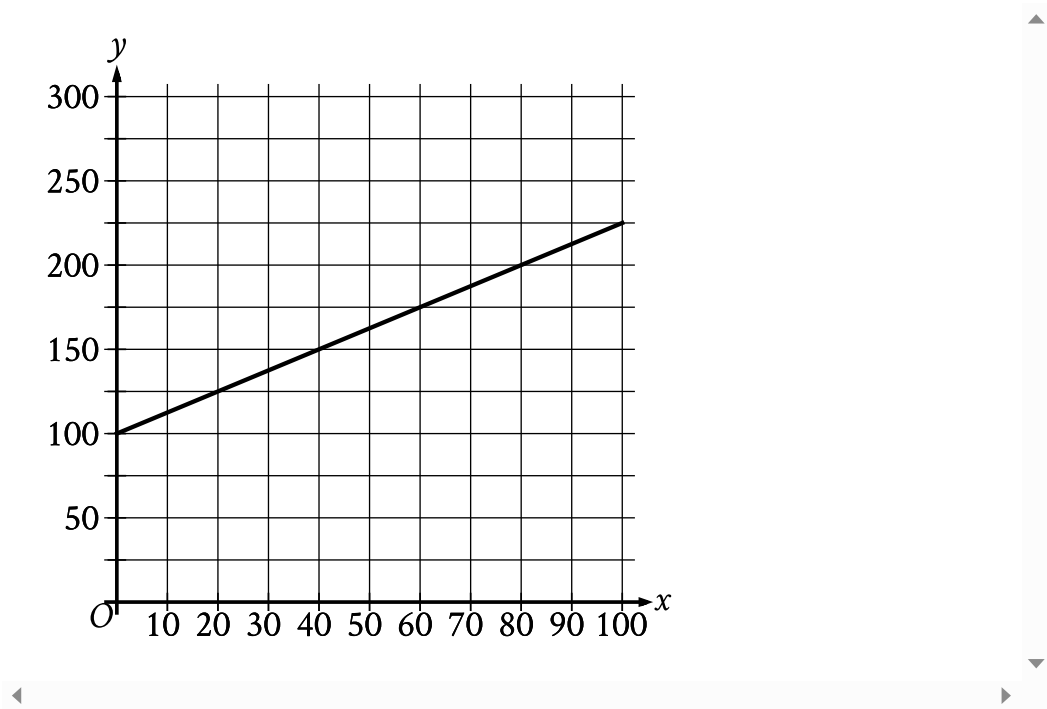
Question Difficulty: Medium

Question ID a4e1df2e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: a4e1df2e

The cost y , in dollars, for a manufacturer to make x rings is represented by the line shown.



What is the cost, in dollars, for the manufacturer to make **60** rings?

- A. 100
- B. 125
- C. 175
- D. 225

ID: a4e1df2e Answer

Correct Answer: C

Rationale

Choice C is correct. The line shown represents the cost y , in dollars, for a manufacturer to make x rings. For the line shown, the x -axis represents the number of rings made by the manufacturer and the y -axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make **60** rings is represented by the y -coordinate of the point on the line that has an x -coordinate of **60**. The point on the line with an x -coordinate of **60** has a y -coordinate of **175**. Therefore, the cost, in dollars, for the manufacturer to make **60** rings is **175**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make **20** rings.

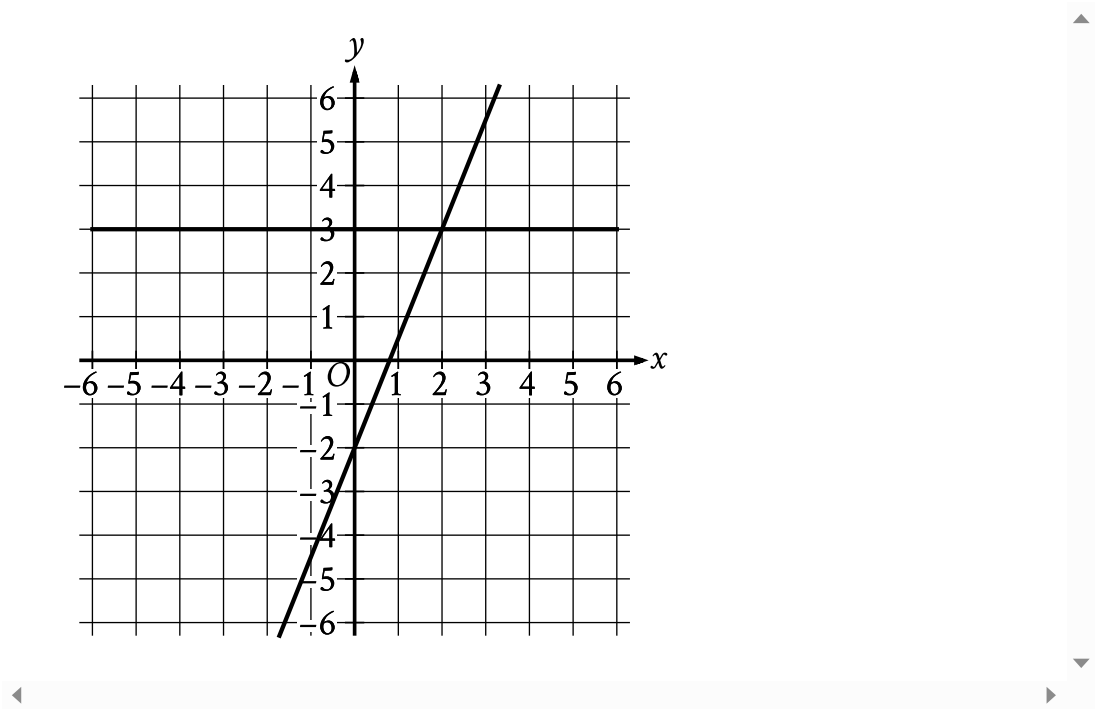
Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make **100** rings.

Question Difficulty: Easy

Question ID 1e5390fd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 1e5390fd



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(0, 3)$
- B. $(1, 3)$
- C. $(2, 3)$
- D. $(3, 3)$

ID: 1e5390fd Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 3$, or at the point $(2, 3)$. Therefore, the solution (x, y) to the system is $(2, 3)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a5eac396

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: a5eac396

$$2(kx - n) = -\frac{28}{15}x - \frac{36}{19}$$

In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k ?

ID: a5eac396 Answer

Correct Answer: -.9333, -14/15

Rationale

The correct answer is $-\frac{14}{15}$. A linear equation in the form $ax + b = cx + d$ has no solution only when the coefficients of x on each side of the equation are equal and the constant terms are not equal. Dividing both sides of the given equation by 2 yields $kx - n = -\frac{28}{30}x - \frac{36}{38}$, or $kx - n = -\frac{14}{15}x - \frac{18}{19}$. Since it's given that the equation has no solution, the coefficient of x on both sides of this equation must be equal, and the constant terms on both sides of this equation must not be equal. Since $\frac{18}{19} < 1$, and it's given that $n > 1$, the second condition is true. Thus, k must be equal to $-\frac{14}{15}$. Note that -14/15, -.9333, and -0.933 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 61853d97

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 61853d97

$y < x$
 $x < 22$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A.

x	y
19	18
20	19
21	20

B.

x	y
19	20
20	21
21	22

C.

x	y
23	22
24	23
25	24

D.

x	y
23	24
24	25
25	26

ID: 61853d97 Answer

Correct Answer: A

Rationale

Choice A is correct. The inequality $y < x$ indicates that for any solution to the given system of inequalities, the value of x must be greater than the corresponding value of y . The inequality $x < 22$ indicates that for any solution to the given system of inequalities, the value of x must be less than **22**. Of the given choices, only choice A contains values of x that are each greater

than the corresponding value of y and less than **22**. Therefore, for choice A, all the values of x and their corresponding values of y are solutions to the given system of inequalities.

Choice B is incorrect. The values in this table aren't solutions to the inequality $y < x$.

Choice C is incorrect. The values in this table aren't solutions to the inequality $x < \mathbf{22}$.

Choice D is incorrect. The values in this table aren't solutions to the inequality $y < x$ or the inequality $x < \mathbf{22}$.

Question Difficulty: Medium

Question ID dd42a36b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: dd42a36b

The perimeter of an isosceles triangle is **83** inches. Each of the two congruent sides of the triangle has a length of **24** inches. What is the length, in inches, of the third side?

ID: dd42a36b Answer

Correct Answer: 35

Rationale

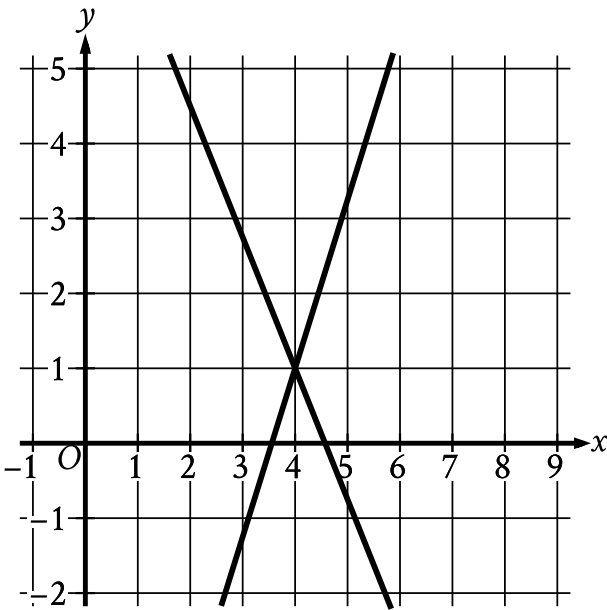
The correct answer is **35**. It’s given that the perimeter of an isosceles triangle is **83** inches and that each of the two congruent sides has a length of **24** inches. The perimeter of a triangle is the sum of the lengths of its three sides. The equation **$24 + 24 + x = 83$** can be used to represent this situation, where **x** is the length, in inches, of the third side. Combining like terms on the left-hand side of this equation yields **$48 + x = 83$** . Subtracting **48** from both sides of this equation yields **$x = 35$** . Therefore, the length, in inches, of the third side is **35**.

Question Difficulty: Easy

Question ID 25d0931c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 25d0931c



The graph of a system of linear equations is shown. The solution to the system is (x, y) . What is the value of x ?

ID: 25d0931c Answer

Correct Answer: 4

Rationale

The correct answer is **4**. A solution to a system of equations must satisfy each equation in the system. It follows that if (x, y) is a solution to the system, the point (x, y) lies on the graph in the xy -plane of each equation in the system. According to the graph, the point (x, y) that lies on the graph of each equation in the system is $(4, 1)$. Therefore, the solution to the system is $(4, 1)$. It follows that the value of x is **4**.

Question Difficulty: Easy

Question ID f7711f35

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: f7711f35

In the xy -plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 18x + 2$. If line s also passes through the point $(4, d)$, what is the value of d ?

- A. 2
- B. 18
- C. 72
- D. 74

ID: f7711f35 Answer

Correct Answer: C

Rationale

Choice C is correct. A line in the xy -plane can be represented by an equation of the form $y = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the line. It's given that line s passes through the point $(0, 0)$. Therefore, the y -coordinate of the y -intercept of line s is 0 . It's also given that line s is parallel to the line represented by the equation $y = 18x + 2$. Since parallel lines have the same slope, it follows that the slope of line s is 18 . Therefore, line s can be represented by the equation $y = mx + b$, where $m = 18$ and $b = 0$. Substituting 18 for m and 0 for b in $y = mx + b$ yields the equation $y = 18x + 0$, or $y = 18x$. If line s passes through the point $(4, d)$, then when $x = 4$, $y = d$ for the equation $y = 18x$. Substituting 4 for x and d for y in this equation yields $d = 18(4)$, or $d = 72$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the line represented by the equation $y = 18x + 2$.

Choice B is incorrect. This is the slope of the line represented by the equation $y = 18x + 2$.

Choice D is incorrect. The line represented by the equation $y = 18x + 2$, not line s , passes through the point $(4, 74)$.

Question Difficulty: Hard

Question ID 4b153444

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 4b153444

$$\begin{aligned}5x &= 15 \\ -4x + y &= -2\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

- A. -17
- B. -13
- C. 13
- D. 17

ID: 4b153444 Answer

Correct Answer: C

Rationale

Choice C is correct. Adding the second equation of the given system to the first equation yields $5x + (-4x + y) = 15 + (-2)$, which is equivalent to $x + y = 13$. So the value of $x + y$ is 13 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $-(x + y)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID e28c3fbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: e28c3fbe

The length of a rectangle is **50** inches and the width is x inches. The perimeter is at most **210** inches. Which inequality represents this situation?

- A. $2x + 100 \leq 210$
- B. $2x + 100 \geq 210$
- C. $2x + 50 \leq 210$
- D. $2x + 50 \geq 210$

ID: e28c3fbe Answer

Correct Answer: A

Rationale

Choice A is correct. The perimeter of a rectangle is equal to the sum of **2** times its length and **2** times its width. It's given that the rectangle's length is **50** inches and the width is x inches. Therefore, the perimeter, in inches, is $2(50) + 2x$, or $100 + 2x$, which is equivalent to $2x + 100$. It's given that the perimeter is at most **210** inches; therefore, $2x + 100 \leq 210$ represents this situation.

Choice B is incorrect. This inequality represents a situation where the perimeter is at least, rather than at most, **210** inches.

Choice C is incorrect. This inequality represents a situation where **2** times the length, rather than the length, is **50** inches.

Choice D is incorrect. This inequality represents a situation where **2** times the length, rather than the length, is **50** inches, and the perimeter is at least, rather than at most, **210** inches.

Question Difficulty: Medium

Question ID 75a8e183

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 75a8e183

$$y = 6x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which equation could be the second equation in this system?

- A. $y = 2(6x) + 3$
- B. $y = 2(6x + 3)$
- C. $2(y) = 2(6x) + 3$
- D. $2(y) = 2(6x + 3)$

ID: 75a8e183 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. When one equation is a multiple of another equation, the two equations are equivalent. Multiplying each side of the given equation by 2 yields $2(y) = 2(6x + 3)$. Thus, $2(y) = 2(6x + 3)$ is equivalent to the given equation and could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution rather than infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than infinitely many solutions.

Choice C is incorrect. The system consisting of this equation and the given equation has no solutions rather than infinitely many solutions.

Question Difficulty: Medium

Question ID c71c5c71

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c71c5c71

Nasir bought **9** storage bins that were each the same price. He used a coupon for **\$63** off the entire purchase. The cost for the entire purchase after using the coupon was **\$27**. What was the original price, in dollars, for **1** storage bin?

ID: c71c5c71 Answer

Correct Answer: 10

Rationale

The correct answer is **10**. It's given that the cost for the entire purchase was **\$27** after a coupon was used for **\$63** off the entire purchase. Adding the amount of the coupon to the purchase price yields $27 + 63 = 90$. Thus, the cost for the entire purchase before using the coupon was **\$90**. It's given that Nasir bought **9** storage bins. The original price for **1** storage bin can be found by dividing the total cost by **9**. Therefore, the original price, in dollars, for **1** storage bin is $\frac{90}{9}$, or **10**.

Question Difficulty: Medium

Question ID 086840dd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 086840dd

The equation $h = \frac{9(v-273.15)}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of **467.33** degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

ID: 086840dd Answer

Correct Answer: 515

Rationale

The correct answer is **515**. It's given that the equation $h = \frac{9(v-273.15)}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. Substituting **467.33** for h in the given equation yields $467.33 = \frac{9(v-273.15)}{5} + 32$. Subtracting **32** from both sides of this equation yields $435.33 = \frac{9(v-273.15)}{5}$. Multiplying both sides of this equation by **5** yields $2,176.65 = 9(v - 273.15)$. Dividing both sides of this equation by **9** yields $241.85 = v - 273.15$. Adding **273.15** to both sides of this equation yields $515 = v$. Therefore, if a substance has a temperature of **467.33** degrees Fahrenheit, the corresponding temperature, in kelvins, of this substance is **515**.

Question Difficulty: Hard

Question ID 51105ab7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 51105ab7

$$\begin{aligned}y &= -\frac{1}{5}x \\ y &= \frac{1}{7}x\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -5
- B. 0
- C. 2
- D. 7

ID: 51105ab7 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given by the first equation in the system that $y = -\frac{1}{5}x$. Substituting $-\frac{1}{5}x$ for y in the second equation in the system, $y = \frac{1}{7}x$, yields $-\frac{1}{5}x = \frac{1}{7}x$. Adding $-\frac{1}{5}x$ to both sides of this equation yields $0 = \frac{1}{7}x + \frac{1}{5}x$, which is equivalent to $0 = \frac{5}{35}x + \frac{7}{35}x$, or $0 = \frac{12}{35}x$. Multiplying both sides of this equation by $\frac{35}{12}$ yields $0 = x$. Therefore, the value of x is 0 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID d5a53f84

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: d5a53f84

A food truck buys forks for **\$0.04** each and plates for **\$0.48** each. The total cost of x forks and y plates is **\$661.76**. Which equation represents this situation?

- A. $0.48x - 0.04y = 661.76$
- B. $0.04x - 0.48y = 661.76$
- C. $0.48x + 0.04y = 661.76$
- D. $0.04x + 0.48y = 661.76$

ID: d5a53f84 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the food truck buys forks for **\$0.04** each. Therefore, the cost, in dollars, of x forks can be represented by the expression **$0.04x$** . It's also given that the food truck buys plates for **\$0.48** each. Therefore, the cost, in dollars, of y plates can be represented by the expression **$0.48y$** . Since the total cost of x forks and y plates is **\$661.76**, the equation **$0.04x + 0.48y = 661.76$** represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation in which the food truck buys forks for **\$0.48** each and plates for **\$0.04** each.

Question Difficulty: Easy

Question ID 36ba6526

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 36ba6526

$$\begin{aligned}\frac{7}{2}x + 6y &= 25 \\ \frac{5}{2}x + 6y &= 23\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $\frac{17}{2}x + 18y$?

- A. 2
- B. 3
- C. 48
- D. 71

ID: 36ba6526 Answer

Correct Answer: D

Rationale

Choice D is correct. Multiplying the second equation in the given system by 2 yields $\frac{10}{2}x + 12y = 46$. Adding this equation to the first equation in the system yields $(\frac{7}{2}x + 6y) + (\frac{10}{2}x + 12y) = 25 + 46$, which is equivalent to $(\frac{7}{2}x + \frac{10}{2}x) + (6y + 12y) = 25 + 46$, or $\frac{17}{2}x + 18y = 71$. Therefore, the value of $\frac{17}{2}x + 18y$ is 71.

Choice A is incorrect. This is the value of x , not the value of $\frac{17}{2}x + 18y$.

Choice B is incorrect. This is the value of y , not the value of $\frac{17}{2}x + 18y$.

Choice C is incorrect. This the value of $(\frac{7}{2}x + 6y) + (\frac{5}{2}x + 6y)$, or $6x + 12y$, not the value of $\frac{17}{2}x + 18y$.

Question Difficulty: Hard

Question ID 7ae90ec8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 7ae90ec8

A line in the xy -plane has a slope of 9 and passes through the point $(0, -5)$. The equation $y = px + r$ defines the line, where p and r are constants. What is the value of p ?

ID: 7ae90ec8 Answer

Correct Answer: 9

Rationale

The correct answer is 9 . It's given that the equation $y = px + r$ defines the line. In this equation, p represents the slope of the line and r represents the y -coordinate of the y -intercept of the line. It's given that the line has a slope of 9 . Therefore, the value of p is 9 .

Question Difficulty: Medium

Question ID 374cb937

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 374cb937

Last week, an interior designer earned a total of **\$1,258** from consulting for x hours and drawing up plans for y hours. The equation $68x + 85y = 1,258$ represents this situation. Which of the following is the best interpretation of **68** in this context?

- A. The interior designer earned **\$68** per hour consulting last week.
- B. The interior designer worked **68** hours drawing up plans last week.
- C. The interior designer earned **\$68** per hour drawing up plans last week.
- D. The interior designer worked **68** hours consulting last week.

ID: 374cb937 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $68x + 85y = 1,258$ represents the situation where an interior designer earned a total of **\$1,258** last week from consulting for x hours and drawing up plans for y hours. Thus, $68x$ represents the amount earned, in dollars, from consulting for x hours, and $85y$ represents the amount earned, in dollars, from drawing up plans for y hours. Since $68x$ represents the amount earned, in dollars, from consulting for x hours, it follows that the interior designer earned **\$68** per hour consulting last week.

Choice B is incorrect. The interior designer worked y hours, not **68** hours, drawing up plans last week.

Choice C is incorrect. The interior designer earned **\$85** per hour, not **\$68** per hour, drawing up plans last week.

Choice D is incorrect. The interior designer worked x hours, not **68** hours, consulting last week.

Question Difficulty: Medium

Question ID c120f4f4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c120f4f4

What value of x is the solution to the equation $16x + 24 = 24x$?

- A. -4
- B. $\frac{3}{10}$
- C. $\frac{1}{3}$
- D. 3

ID: c120f4f4 Answer

Correct Answer: D

Rationale

Choice D is correct. Subtracting $16x$ from both sides of the given equation yields $24 = 8x$. Dividing both sides of this equation by 8 yields $3 = x$. Therefore, the solution to the given equation is 3 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID e7af391e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: e7af391e

$$3(kx + 13) = \frac{48}{17}x + 36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k ?

ID: e7af391e Answer

Correct Answer: .9411, .9412, 16/17

Rationale

The correct answer is $\frac{16}{17}$. It's given that the equation $3(kx + 13) = \frac{48}{17}x + 36$ has no solution. A linear equation in the form $ax + b = cx + d$, where a, b, c , and d are constants, has no solution only when the coefficients of x on each side of the equation are equal and the constant terms aren't equal. Dividing both sides of the given equation by 3 yields $kx + 13 = \frac{48}{51}x + \frac{36}{3}$, or $kx + 13 = \frac{16}{17}x + 12$. Since the coefficients of x on each side of the equation must be equal, it follows that the value of k is $\frac{16}{17}$. Note that 16/17, .9411, .9412, and 0.941 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 77ccd38a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 77ccd38a

$$\begin{aligned}x + y &= 125 \\x + y + y &= 155\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 77ccd38a Answer

Correct Answer: 30

Rationale

The correct answer is **30**. It's given that $x + y = 125$. Substituting **125** for $x + y$ in the second equation of the given system yields $125 + y = 155$. Subtracting **125** from both sides of this equation yields $y = 30$. Therefore, the value of y is **30**.

Question Difficulty: Medium

Question ID 4d802587

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 4d802587

A linear model estimates the population of a city from **1991** to **2015**. The model estimates the population was **57** thousand in **1991**, **224** thousand in **2011**, and x thousand in **2015**. To the nearest whole number, what is the value of x ?

ID: 4d802587 Answer

Correct Answer: 257

Rationale

The correct answer is **257**. It's given that a linear model estimates the population of a city from **1991** to **2015**. Since the population can be estimated using a linear model, it follows that there is a constant rate of change for the model. It's also given that the model estimates the population was **57** thousand in **1991**, **224** thousand in **2011**, and x thousand in **2015**. The change in the population between **2011** and **1991** is $224 - 57$, or **167**, thousand. The change in the number of years between **2011** and **1991** is $2011 - 1991$, or **20**, years. Dividing **167** by **20** gives $167/20$, or **8.35**, thousand per year. Thus, the change in population per year from **1991** to **2015** estimated by the model is **8.35** thousand. The change in the number of years between **2015** and **2011** is $2015 - 2011$, or **4**, years. Multiplying the change in population per year by the change in number of years yields the increase in population from **2011** to **2015** estimated by the model: $(8.35)(4)$, or **33.4**, thousand. Adding the change in population from **2011** to **2015** estimated by the model to the estimated population in **2011** yields the estimated population in **2015**. Thus, the estimated population in **2015** is $33.4 + 224$, or **257.4**, thousand. Therefore to the nearest whole number, the value of x is **257**.

Question Difficulty: Hard

Question ID b9133e6d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b9133e6d

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{4}$ and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = \frac{1}{4}x + 5$
- B. $f(x) = \frac{1}{4}x + \frac{1}{5}$
- C. $f(x) = \frac{1}{4}x - \frac{5}{4}$
- D. $f(x) = \frac{1}{4}x - 5$

ID: b9133e6d Answer

Correct Answer: A

Rationale

Choice A is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{4}$. Therefore, $m = \frac{1}{4}$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting $\frac{1}{4}$ for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = \frac{1}{4}x + 5$.

Choice B is incorrect. This equation defines a function whose graph has a y -intercept of $(0, \frac{1}{5})$, not $(0, 5)$.

Choice C is incorrect. This equation defines a function whose graph has a y -intercept of $(0, -\frac{5}{4})$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a y -intercept of $(0, -5)$, not $(0, 5)$.

Question Difficulty: Easy

Question ID 531af159

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 531af159

If $\frac{6}{7}p + 18 = 54$, what is the value of $7p$?

ID: 531af159 Answer

Correct Answer: 294

Rationale

The correct answer is **294**. Subtracting **18** from each side of the given equation yields $\frac{6}{7}p = 36$. Multiplying each side of this equation by $\frac{7}{6}$ yields $p = 42$. Multiplying each side of this equation by **7** yields $7p = 294$. Therefore, the value of $7p$ is **294**.

Question Difficulty: Hard

Question ID e5b7b067

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: e5b7b067

$$\begin{aligned} 4x &= 20 \\ -3x + y &= -7 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

- A. -27
- B. -13
- C. 13
- D. 27

ID: e5b7b067 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $4x = 20$ and $-3x + y = -7$ is a system of equations with a solution (x, y) . Adding the second equation in the given system to the first equation yields $4x + (-3x + y) = 20 + (-7)$, which is equivalent to $x + y = 13$. Thus, the value of $x + y$ is 13 .

Choice A is incorrect. This represents the value of $-2(x + y) - 1$.

Choice B is incorrect. This represents the value of $-(x + y)$.

Choice D is incorrect. This represents the value of $2(x + y) + 1$.

Question Difficulty: Medium

Question ID a7481d1d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a7481d1d

$$\begin{aligned} 4x - 3y &= 5 \\ x &= 8 \end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(8, 9)$
- B. $(8, -24)$
- C. $(8, -9)$
- D. $(8, 24)$

ID: a7481d1d Answer

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system is $x = 8$. Substituting 8 for x in the first equation in the given system yields $4(8) - 3y = 5$, or $32 - 3y = 5$. Subtracting 32 from both sides of this equation yields $-3y = -27$. Dividing both sides of this equation by -3 yields $y = 9$. Therefore, the solution (x, y) to the given system of equations is $(8, 9)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

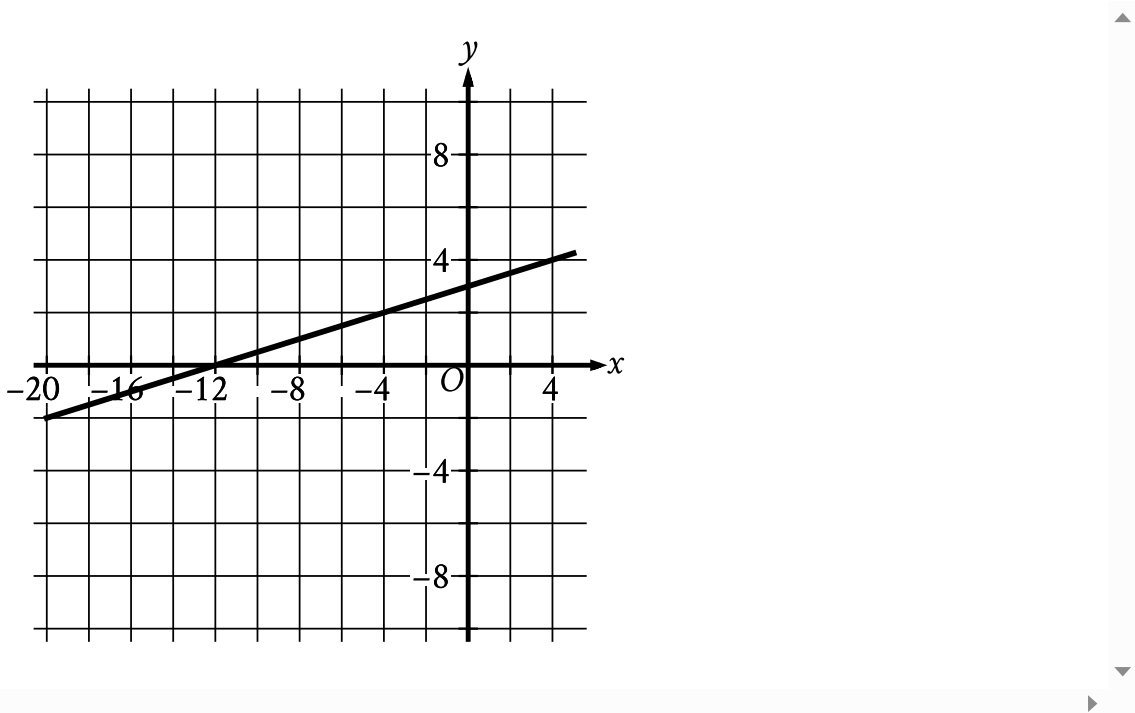
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 7d9d3fb4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 7d9d3fb4



The graph of the linear function f is shown, where $y = f(x)$. What is the x-intercept of the graph of f ?

- A. $(-12, 0)$
- B. $(0, 0)$
- C. $(\frac{1}{4}, 0)$
- D. $(12, 0)$

ID: 7d9d3fb4 Answer

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of a graph is the point where the graph intersects the x-axis. The graph of function f , where $y = f(x)$, intersects the x-axis at $(-12, 0)$. Therefore, the x-intercept of the graph of f is $(-12, 0)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 6a564367

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 6a564367

$x + y = 350$

The given equation relates the total number of maple trees, x , and the total number of birch trees, y , planted in a **14**-acre forest preserve. If **245** maple trees were planted in the forest preserve, how many birch trees were planted in the forest preserve?

- A. **14**
- B. **25**
- C. **105**
- D. **245**

ID: 6a564367 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the equation $x + y = 350$ relates the total number of maple trees, x , and the total number of birch trees, y , planted in a **14**-acre forest preserve. It’s also given that **245** maple trees were planted in the forest preserve. Substituting **245** for x in the given equation yields $245 + y = 350$. Subtracting **245** from both sides of this equation yields $y = 105$. Therefore, **105** birch trees were planted in the forest preserve.

Choice A is incorrect. This is the number of acres in the forest preserve, not the number of birch trees planted in the forest preserve.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the number of maple trees planted in the forest preserve, not the number of birch trees planted in the forest preserve.

Question Difficulty: Easy

Question ID c206d65e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c206d65e

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid **\$2** for the box of cereal and **\$1.90** per pound for the strawberries. If Lorenzo paid a total of **\$9.60** for the box of cereal and the strawberries, which of the following equations can be used to find ***p***, the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

- A. $1.90p + 2 = 9.60$
- B. $1.90p - 2 = 9.60$
- C. $1.90 + 2p = 9.60$
- D. $1.90 - 2p = 9.60$

ID: c206d65e Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that ***p*** represents the number of pounds of strawberries Lorenzo purchased and Lorenzo paid **\$1.90** per pound for the strawberries. It follows that the total amount, in dollars, Lorenzo paid for strawberries can be represented by **$1.90p$** . It's given that Lorenzo paid **\$2** for the box of cereal. If Lorenzo paid a total of **\$9.60** for the box of cereal and strawberries, it follows that the equation **$1.90p + 2 = 9.60$** can be used to find ***p***.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 4c37c185

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 4c37c185

A tow truck travels at a speed of at least **25** miles per hour but no more than **45** miles per hour for a section of a road. Which inequality represents this situation, where x is the speed of the tow truck, in miles per hour, on this section of the road?

- A. $x \geq 25$
- B. $x \geq 45$
- C. $25 \leq x \leq 45$
- D. $x \leq 70$

ID: 4c37c185 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that a tow truck travels at a speed of at least **25** miles per hour but no more than **45** miles per hour. It's also given that x represents the speed of the tow truck, in miles per hour. At least **25** miles per hour means $x \geq 25$, and no more than **45** miles per hour means $x \leq 45$. Therefore, the inequality that represents this situation is $25 \leq x \leq 45$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID db7305ee

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: db7305ee

If $7x = 28$, what is the value of $8x$?

- A. 21
- B. 32
- C. 168
- D. 224

ID: db7305ee Answer

Correct Answer: B

Rationale

Choice B is correct. Dividing both sides of the given equation $7x = 28$ by 7 yields $x = 4$. Substituting 4 for x in the expression $8x$ yields $8(4)$, which is equivalent to 32.

Choice A is incorrect. This is the value of $\frac{21}{4}x$.

Choice C is incorrect. This is the value of $42x$.

Choice D is incorrect. This is the value of $56x$.

Question Difficulty: Easy

Question ID 8bf18c83

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 8bf18c83

$$j(x) = mx + 144$$

For the linear function j , m is a constant and $j(12) = 18$. What is the value of $j(10)$?

ID: 8bf18c83 Answer

Correct Answer: 39

Rationale

The correct answer is **39**. It's given that for the linear function j , m is a constant and $j(12) = 18$. Substituting **12** for x and **18** for $j(x)$ in the given equation yields $18 = m(12) + 144$. Subtracting **144** from both sides of this equation yields $-126 = m(12)$. Dividing both sides of this equation by **12** yields $-10.5 = m$. Substituting -10.5 for m in the given equation, $j(x) = mx + 144$, yields $j(x) = -10.5x + 144$. Substituting **10** for x in this equation yields $j(10) = (-10.5)(10) + 144$, or $j(10) = 39$. Therefore, the value of $j(10)$ is **39**.

Question Difficulty: Hard

Question ID f993c1b5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: f993c1b5

Julissa needs at least **100** hours of flight time to get her private pilot certification. If Julissa already has **86** hours of flight time, what is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification?

- A. **14**
- B. **76**
- C. **86**
- D. **186**

ID: f993c1b5 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that Julissa already has **86** hours of flight time. Let x represent the number of additional hours of flight time Julissa needs to get her private pilot certification. After completing x hours of flight time, Julissa will have completed a total of $86 + x$ hours of flight time. It's given that Julissa needs at least **100** hours of flight time to get her private pilot certification. Therefore, $86 + x \geq 100$. Subtracting **86** from both sides of this inequality yields $x \geq 14$. Thus, **14** is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of hours of flight time Julissa already has, rather than the minimum number of additional hours of flight time Julissa needs.

Choice D is incorrect. This is the number of hours of flight time Julissa will have if she completes **100** more hours of flight time, rather than the minimum number of additional hours of flight time Julissa needs.

Question Difficulty: Easy

Question ID 310c2cc7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 310c2cc7

For a particular car, the linear function f gives the predicted power, in brake horsepower (**bhp**), for engine speeds between **1,000 revolutions per minute (rpm)** and **6,000 rpm**. According to this function, the car's predicted power is **433 bhp** at an engine speed of **3,331 rpm** and **600 bhp** at an engine speed of **4,500 rpm**. The equation $f(x) = \frac{1}{7}(x - a) + 433$ defines f , where x is the engine speed, **in rpm**, and a is a constant. What is the value of a ?

ID: 310c2cc7 Answer

Correct Answer: 3331

Rationale

The correct answer is **3,331**. For engine speeds between **1,000 revolutions per minute (rpm)** and **6,000 rpm**, it's given that the equation $f(x) = \frac{1}{7}(x - a) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (**bhp**), at an engine speed of x **rpm**, where a is a constant. It's also given that the car's predicted power is **433 bhp** at an engine speed of **3,331 rpm**. Substituting **3,331** for x and **433** for $f(x)$ in the equation yields $433 = \frac{1}{7}(3,331 - a) + 433$. Subtracting **433** from both sides of this equation yields $0 = \frac{1}{7}(3,331 - a)$. Multiplying both sides of this equation by **7** yields $0 = 3,331 - a$. Adding a to both sides of this equation yields $a = 3,331$. Thus, the value of a is **3,331**.

Question Difficulty: Hard

Question ID 49a3d731

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 49a3d731

$3x = 36y - 45$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

- A. $x = 4y$
- B. $\frac{1}{3}x = 4y$
- C. $x = 12y - 15$
- D. $\frac{1}{3}x = 12y - 15$

ID: 49a3d731 Answer

Correct Answer: B

Rationale

Choice B is correct. A system of two linear equations in two variables, x and y , has no solution when the lines in the xy -plane representing the equations are parallel and distinct. Two lines are parallel and distinct if their slopes are the same and their y -intercepts are different. The slope of the graph of the given equation, $3x = 36y - 45$, in the xy -plane can be found by rewriting the equation in the form $y = mx + b$, where m is the slope of the graph and $(0, b)$ is the y -intercept. Adding 45 to each side of the given equation yields $3x + 45 = 36y$. Dividing each side of this equation by 36 yields $\frac{1}{12}x + \frac{5}{4} = y$, or $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the slope of the graph of the given equation is $\frac{1}{12}$ and the y -intercept is $(0, \frac{5}{4})$. Therefore, the graph of the second equation in the system must also have a slope of $\frac{1}{12}$, but must not have a y -intercept of $(0, \frac{5}{4})$. Multiplying each side of the equation given in choice B by $\frac{1}{4}$ yields $\frac{1}{12}x = y$, or $y = \frac{1}{12}x$. It follows that the graph representing the equation in choice B has a slope of $\frac{1}{12}$ and a y -intercept of $(0, 0)$. Since the slopes of the graphs of the two equations are equal and the y -intercepts of the graphs of the two equations are different, the equation in choice B could be the second equation in the system.

Choice A is incorrect. This equation can be rewritten as $y = \frac{1}{4}x$. It follows that the graph of this equation has a slope of $\frac{1}{4}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Choice C is incorrect. This equation can be rewritten as $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{12}$ and a y -intercept of $(0, \frac{5}{4})$, so the system consisting of this equation and the given equation has infinitely many solutions, rather than no solution.

Choice D is incorrect. This equation can be rewritten as $y = \frac{1}{36}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{36}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Question Difficulty: Hard

Question ID 9e86a911

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 9e86a911

In the xy -plane, line p has a slope of $-\frac{5}{3}$ and an x -intercept of $(-6, 0)$. What is the y -coordinate of the y -intercept of line p ?

ID: 9e86a911 Answer

Correct Answer: -10

Rationale

The correct answer is -10 . A line in the xy -plane can be represented by the equation $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept. It's given that line p has a slope of $-\frac{5}{3}$. Therefore, $m = -\frac{5}{3}$. It's also given that line p has an x -intercept of $(-6, 0)$. Therefore, when $x = -6, y = 0$. Substituting $-\frac{5}{3}$ for $m, -6$ for x , and 0 for y in the equation $y = mx + b$ yields $0 = (-\frac{5}{3})(-6) + b$, which is equivalent to $0 = 10 + b$. Subtracting 10 from both sides of this equation yields $-10 = b$. Therefore, the y -coordinate of the y -intercept of line p is -10 .

Question Difficulty: Hard

Question ID 2c16fe41

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 2c16fe41

If $3x - 8 = 7$, what is the value of $3x + 8$?

- A. -1
- B. 5
- C. 13
- D. 23

ID: 2c16fe41 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $3x - 8 = 7$. Adding 8 to both sides of this equation yields $3x = 15$. Adding 8 to both sides of this equation yields $3x + 8 = 23$. Therefore, the value of $3x + 8$ is 23 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $3x + 8$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 114dc543

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 114dc543

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

ID: 114dc543 Answer

Correct Answer: .5, 1/2

Rationale

The correct answer is $\frac{1}{2}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the xy -plane defined by this equation. It's given that line p is defined by $4y + 8x = 6$. Subtracting $8x$ from both sides of this equation yields $4y = -8x + 6$. Dividing both sides of this equation by 4 yields $y = -\frac{8}{4}x + \frac{6}{4}$, or $y = -2x + \frac{3}{2}$. Thus, the slope of line p is -2 . If line r is perpendicular to line p , then the slope of line r is the negative reciprocal of the slope of line p . The negative reciprocal of -2 is $-\frac{1}{(-2)} = \frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 551de84f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 551de84f

Which of the following systems of linear equations has no solution?

- A. $y = 6x + 3$
 $y = 6x + 9$
- B. $y = 10$
 $y = 10x + 10$
- C. $y = 14x + 14$
 $y = 10x + 14$
- D. $x = 3$
 $y = 10$

ID: 551de84f Answer

Correct Answer: A

Rationale

Choice A is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, are parallel if their slopes, m , are the same and are distinct if their y -coordinates of the y -intercepts, b , are different. In the equations $y = 6x + 3$ and $y = 6x + 9$, the values of m are each 6 , and the values of b are 3 and 9 , respectively. Since the slopes of these lines are the same and the y -coordinates of the y -intercepts are different, it follows that the system of linear equations in choice A has no solution.

Choice B is incorrect. The two lines represented by these equations are a horizontal line and a line with a slope of 10 that have the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 10)$, rather than no solution.

Choice C is incorrect. The two lines represented by these equations have different slopes and the same y -coordinate of the y -intercept. Therefore, this system has a solution, $(0, 14)$, rather than no solution.

Choice D is incorrect. The two lines represented by these equations are a vertical line and a horizontal line. Therefore, this system has a solution, $(3, 10)$, rather than no solution.

Question Difficulty: Hard

Question ID eed1dbb6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: eed1dbb6

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area x , in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define f ?

- A. $f(x) = 0.6x$
- B. $f(x) = 1.8x$
- C. $f(x) = 2,268x$
- D. $f(x) = 4,536x$

ID: eed1dbb6 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the table represents the relationship between the area x , in square feet, of the roof of a shopping center and the amount of water $f(x)$, in gallons, drained from the roof into the gutters. Every choice represents this relationship with an equation defining f in the form $f(x) = mx$, where m is a constant rate of change. The value of m can be determined by dividing both sides of the equation by x . Each of three pairs of x and $f(x)$ in the table yield $m = 1.8$, since $\frac{4,536}{2,520} = 1.8$, $\frac{6,804}{3,780} = 1.8$, and $\frac{9,072}{5,040} = 1.8$. Therefore, the equation $f(x) = 1.8x$ could define f .

Choice A is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $0.6(2,520)$, or 1,512, gallons, not the 4,536 gallons shown in the table.

Choice C is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $2,268(2,520)$, or 5,715,360, gallons, not the 4,536 gallons shown in the table.

Choice D is incorrect. For the roof with an area of 2,520 square feet, this equation would yield $4,536(2,520)$, or 11,430,720, gallons, not the 4,536 gallons shown in the table.

Question Difficulty: Medium

Question ID 5f69b45d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 5f69b45d

What value of p satisfies the equation $5p + 180 = 250$?

- A. 14
- B. 65
- C. 86
- D. 250

ID: 5f69b45d Answer

Correct Answer: A

Rationale

Choice A is correct. Subtracting 180 from both sides of the given equation yields $5p = 70$. Dividing both sides of this equation by 5 yields $p = 14$. Therefore, the value of p that satisfies the equation $5p + 180 = 250$ is 14.

Choice B is incorrect. This value of p satisfies the equation $5p + 180 = 505$.

Choice C is incorrect. This value of p satisfies the equation $5p + 180 = 610$.

Choice D is incorrect. This value of p satisfies the equation $5p + 180 = 1,430$.

Question Difficulty: Easy

Question ID e21c2169

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: e21c2169

$$\begin{aligned}y - 6 &= \frac{4}{5}(x + 5) \\ y - v &= \frac{4}{5}(x + 10)\end{aligned}$$

In the given system of equations, v is a constant. The system has infinitely many solutions. What is the value of v ?

- A. 0
- B. 2
- C. 4
- D. 12

ID: e21c2169 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Adding **6** to both sides of $y - 6 = \frac{4}{5}(x + 5)$ yields $y = \frac{4}{5}(x + 5) + 6$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{20}{5} + 6$, or $y = \frac{4}{5}x + 10$. Adding v to both sides of $y - v = \frac{4}{5}(x + 10)$ yields $y = \frac{4}{5}(x + 10) + v$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{40}{5} + v$, or $y = \frac{4}{5}x + 8 + v$. For the equations $y = \frac{4}{5}x + 10$ and $y = \frac{4}{5}x + 8 + v$ to be equivalent, it follows that $10 = 8 + v$. Subtracting **8** from both sides of this equation yields $2 = v$. Therefore, the value of v is **2**.

Choice A is incorrect. If $v = 0$, the system of equations would have no solution, not infinitely many solutions.

Choice C is incorrect. If $v = 4$, the system of equations would have no solution, not infinitely many solutions.

Choice D is incorrect. If $v = 12$, the system of equations would have no solution, not infinitely many solutions.

Question Difficulty: Hard

Question ID 60f35c6d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 60f35c6d

$8x - 7x + 130 = 260$

What value of x is the solution to the given equation?

ID: 60f35c6d Answer

Correct Answer: 130

Rationale

The correct answer is **130**. It's given that $8x - 7x + 130 = 260$. Combining like terms on the left-hand side of this equation yields $x + 130 = 260$. Subtracting **130** from each side of this equation yields $x = 130$. Therefore, the value of x that's the solution to the given equation is **130**.

Question Difficulty: Easy

Question ID 0e2e504f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 0e2e504f

The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

- A. $(-2, 0)$
- B. $(0, -2)$
- C. $(0, \frac{1}{10})$
- D. $(\frac{1}{10}, 0)$

ID: 0e2e504f Answer

Correct Answer: B

Rationale

Choice B is correct. The y-intercept of the graph of a function in the xy-plane is the point on the graph where $x = 0$. It's given that $f(x) = \frac{1}{10}x - 2$. Substituting 0 for x in this equation yields $f(0) = \frac{1}{10}(0) - 2$, or $f(0) = -2$. Since it's given that $y = f(x)$, it follows that $y = -2$ when $x = 0$. Therefore, the y-intercept of the graph of $y = f(x)$ in the xy-plane is $(0, -2)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 556e4caa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 556e4caa

3 more than 8 times a number x is equal to 83. Which equation represents this situation?

- A. $(3)(8)x = 83$
- B. $8x = 83 + 3$
- C. $3x + 8 = 83$
- D. $8x + 3 = 83$

ID: 556e4caa Answer

Correct Answer: D

Rationale

Choice D is correct. The given phrase “8 times a number x ” can be represented by the expression $8x$. The given phrase “3 more than” indicates an increase of 3 to a quantity. Therefore “3 more than 8 times a number x ” can be represented by the expression $8x + 3$. Since it’s given that 3 more than 8 times a number x is equal to 83, it follows that $8x + 3$ is equal to 83, or $8x + 3 = 83$. Therefore, the equation that represents this situation is $8x + 3 = 83$.

Choice A is incorrect. This equation represents 3 times the quantity 8 times a number x is equal to 83.

Choice B is incorrect. This equation represents 8 times a number x is equal to 3 more than 83.

Choice C is incorrect. This equation represents 8 more than 3 times a number x is equal to 83.

Question Difficulty: Easy

Question ID 5ca4fe7f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 5ca4fe7f

$y > 13x - 18$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52
8	91

ID: 5ca4fe7f Answer

Correct Answer: D

Rationale

Choice D is correct. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting **3** for x in the given inequality yields $y > 13(3) - 18$, or $y > 21$. Therefore, when $x = 3$, the corresponding value of y is greater than **21**. Substituting **5** for x in the given inequality yields $y > 13(5) - 18$, or $y > 47$. Therefore, when $x = 5$, the corresponding value of y is greater than **47**. Substituting **8** for x in the given inequality yields $y > 13(8) - 18$, or $y > 86$. Therefore, when $x = 8$, the corresponding value of y is greater than **86**. For the table in choice D, when $x = 3$, the corresponding value of y is **26**, which is greater than **21**; when $x = 5$, the corresponding value of y is **52**, which is greater than **47**; when $x = 8$, the corresponding value of y is **91**, which is greater than **86**. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. In the table for choice A, when $x = 3$, the corresponding value of y is **21**, which is not greater than **21**; when $x = 5$, the corresponding value of y is **47**, which is not greater than **47**; when $x = 8$, the corresponding value of y is **86**, which is not greater than **86**.

Choice B is incorrect. In the table for choice B, when $x = 5$, the corresponding value of y is **42**, which is not greater than **47**; when $x = 8$, the corresponding value of y is **86**, which is not greater than **86**.

Choice C is incorrect. In the table for choice C, when $x = 3$, the corresponding value of y is **16**, which is not greater than **21**; when $x = 5$, the corresponding value of y is **42**, which is not greater than **47**; when $x = 8$, the corresponding value of y is **81**, which is not greater than **86**.

Question Difficulty: Hard

Question ID 88917ace

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 88917ace

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, **in centimeters (cm)**, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of **39** in this context?

- A. The estimated height, **in cm**, of the fluid at the start of the experiment
- B. The estimated height, **in cm**, of the fluid at the end of the experiment
- C. The estimated change in the height, **in cm**, of the fluid each day
- D. The estimated number of days for all the fluid to evaporate

ID: 88917ace Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters (cm), of the fluid in the beaker x days after the start of the experiment. For a function defined by an equation of the form $f(x) = b + mx$, where m and b are constants, b represents the value of $f(x)$ when $x = 0$. It follows that in the given function, **39** represents the value of $f(x)$ when $x = 0$. Since $x = 0$ represents the start of the experiment, then the best interpretation of **39** in this context is that the estimated height, in cm, of the fluid was **39** at the start of the experiment.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. The estimated change in the height, in cm, of the fluid each day is **0.18**, not **39**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 77830a4f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 77830a4f

$$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

- A. 0.05
- B. 12.1
- C. 20
- D. 242.0

ID: 77830a4f Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function g models the number of gallons that remain from a full gas tank in a car after driving m miles. In the given function $g(m) = -0.05m + 12.1$, the coefficient of m is -0.05 . This means that for every increase in the value of m by 1, the value of $g(m)$ decreases by 0.05. It follows that for each mile driven, there is a decrease of 0.05 gallons of gasoline. Therefore, 0.05 gallons of gasoline are used to drive each mile.

Choice B is incorrect and represents the number of gallons of gasoline in a full gas tank.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 937f7437

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 937f7437

$$\begin{aligned}s + 7r &= 27 \\ r &= 3\end{aligned}$$

What is the solution (r, s) to the given system of equations?

- A. $(6, 3)$
- B. $(3, 6)$
- C. $(3, 27)$
- D. $(27, 3)$

ID: 937f7437 Answer

Correct Answer: B

Rationale

Choice B is correct. The second equation in the given system is $r = 3$. Substituting 3 for r in the first equation in the given system yields $s + 7(3) = 27$, or $s + 21 = 27$. Subtracting 21 from both sides of this equation yields $s = 6$. Therefore, the solution (r, s) to the given system of equations is $(3, 6)$.

Choice A is incorrect. This is the solution (s, r) , not (r, s) , to the given system of equations.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 79b568d3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 79b568d3

$$\begin{aligned} 3x &= 12 \\ -3x + y &= -6 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. -3
- B. 6
- C. 18
- D. 30

ID: 79b568d3 Answer

Correct Answer: B

Rationale

Choice B is correct. Adding the second equation in the given system to the first equation in the given system yields $3x + (-3x + y) = 12 + (-6)$, which is equivalent to $y = 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c20160de

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: c20160de

Monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit (°F)**. If a monarch butterfly's body temperature is **51.3°F**, what is the minimum increase needed in its body temperature, in °F, so that it can fly?

- A. **1.3**
- B. **3.7**
- C. **5.0**
- D. **6.3**

ID: c20160de Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit (°F)**. Let x represent the minimum increase needed in the monarch butterfly's body temperature to fly. If the monarch butterfly's body temperature is **51.3°F**, the inequality $51.3 + x \geq 55.0$ represents this situation. Subtracting **51.3** from both sides of this inequality yields $x \geq 3.7$. Therefore, if the monarch butterfly's body temperature is **51.3°F**, the minimum increase needed in its body temperature, in °F, so that it can fly is **3.7**.

Choice A is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **53.7°F**, not **51.3°F**.

Choice C is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **50.0°F**, not **51.3°F**.

Choice D is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is **48.7°F**, not **51.3°F**.

Question Difficulty: Easy

Question ID c5c0dd21

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c5c0dd21

$16x + 30 = 190$

Which equation has the same solution as the given equation?

- A. $16x = 30$
- B. $16x = 130$
- C. $16x = 160$
- D. $16x = 190$

ID: c5c0dd21 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $16x + 30 = 190$. Subtracting 30 from each side of this equation yields $16x = 160$. Therefore, the equation $16x = 160$ is equivalent to the given equation and has the same solution.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 392863a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 392863a5

For a restaurant owner to purchase a pizza oven at a total price of **13,000** dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of **200** in this context?

- A. The amount, in dollars, of the down payment
- B. The amount, in dollars, of each fixed monthly payment
- C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments
- D. The total number of fixed monthly payments

ID: 392863a5 Answer

Correct Answer: B

Rationale

Choice B is correct. For a restaurant owner to purchase a pizza oven at a total price of **13,000** dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. It's given that the equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. In this equation, **2,800** is a constant term, which represents the onetime down payment, in dollars, and **200** is the coefficient of t . Multiplying **200** by t gives the total amount, in dollars, paid by the fixed monthly payments. Therefore, the best interpretation of **200** in this context is the amount, in dollars, of each fixed monthly payment.

Choice A is incorrect. This is the best interpretation of **2,800**, not **200**, in this context.

Choice C is incorrect. This is the best interpretation of **13,000**, not **200**, in this context.

Choice D is incorrect. This is the best interpretation of the variable t , not **200**, in this context.

Question Difficulty: Medium

Question ID 532da541

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 532da541

The sum of a number x and 7 is twice as large as a number y . The number y is 3 less than the number x . Which system of equations describes this situation?

- A. $x + 7 = 2y$
 $y = x - 3$
- B. $x + 7 = 2y$
 $y = 3 - x$
- C. $2(x + 7) = y$
 $y = x - 3$
- D. $2(x + 7) = y$
 $y = 3 - x$

ID: 532da541 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the sum of a number x and 7 is twice as large as a number y . This can be described by the equation $x + 7 = 2y$. It's also given that the number y is 3 less than the number x . This can be described by the equation $y = x - 3$. Therefore, the system consisting of the equations $x + 7 = 2y$ and $y = x - 3$ describes this situation.

Choice B is incorrect. The equation $y = 3 - x$ describes a situation where the number y is x less than 3 .

Choice C is incorrect. The equation $2(x + 7) = y$ describes a situation where the number y is twice the sum of a number x and 7 .

Choice D is incorrect. The equation $2(x + 7) = y$ describes a situation where the number y is twice the sum of a number x and 7 , and the equation $y = 3 - x$ describes a situation where a number y is x less than 3 .

Question Difficulty: Hard

Question ID 7af4dc7f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 7af4dc7f

$(p + 3) + 8 = 10$

What value of p is the solution to the given equation?

- A. -1
- B. 5
- C. 15
- D. 21

ID: 7af4dc7f Answer

Correct Answer: A

Rationale

Choice A is correct. Subtracting 8 from both sides of the given equation yields $p + 3 = 2$. Subtracting 3 from both sides of this equation yields $p = -1$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3c80bb57

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 3c80bb57

The total cost $f(x)$, in dollars, to lease a car for **36** months from a particular car dealership is given by $f(x) = 36x + 1,000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is **\$400**?

- A. **\$13,400**
- B. **\$13,000**
- C. **\$15,400**
- D. **\$37,400**

ID: 3c80bb57 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x)$ is the total cost, in dollars, to lease a car from this dealership with a monthly payment of x dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is **\$400** is represented by the value of $f(x)$ when $x = 400$. Substituting **400** for x in the equation $f(x) = 36x + 1,000$ yields $f(400) = 36(400) + 1,000$, or $f(400) = 15,400$. Thus, when the monthly payment is **\$400**, the total cost to lease a car is **\$15,400**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 729bbc01

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 729bbc01

$$\begin{aligned}y &> 14 \\ 4x + y &< 18\end{aligned}$$

The point $(x, 53)$ is a solution to the system of inequalities in the xy -plane. Which of the following could be the value of x ?

- A. -9
- B. -5
- C. 5
- D. 9

ID: 729bbc01 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the point $(x, 53)$ is a solution to the given system of inequalities in the xy -plane. This means that the coordinates of the point, when substituted for the variables x and y , make both of the inequalities in the system true. Substituting 53 for y in the inequality $y > 14$ yields $53 > 14$, which is true. Substituting 53 for y in the inequality $4x + y < 18$ yields $4x + 53 < 18$. Subtracting 53 from both sides of this inequality yields $4x < -35$. Dividing both sides of this inequality by 4 yields $x < -8.75$. Therefore, x must be a value less than -8.75 . Of the given choices, only -9 is less than -8.75 .

Choice B is incorrect. Substituting -5 for x and 53 for y in the inequality $4x + y < 18$ yields $4(-5) + 53 < 18$, or $33 < 18$, which is not true.

Choice C is incorrect. Substituting 5 for x and 53 for y in the inequality $4x + y < 18$ yields $4(5) + 53 < 18$, or $73 < 18$, which is not true.

Choice D is incorrect. Substituting 9 for x and 53 for y in the inequality $4x + y < 18$ yields $4(9) + 53 < 18$, or $89 < 18$, which is not true.

Question Difficulty: Hard

Question ID 55d10558

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 55d10558

If $8x = 6$, what is the value of $72x$?

- A. 3
- B. 15
- C. 54
- D. 57

ID: 55d10558 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $8x = 6$. Multiplying each side of this equation by 9 yields $72x = 54$. Therefore, the value of $72x$ is 54.

Choice A is incorrect. This is the value of $4x$, not $72x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c7eb4ad9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: c7eb4ad9

If $4x + 2 = 12$, what is the value of $16x + 8$?

- A. 40
- B. 48
- C. 56
- D. 60

ID: c7eb4ad9 Answer

Correct Answer: B

Rationale

Choice B is correct. Multiplying both sides of the given equation by 4 yields $(4)(4x + 2) = (4)(12)$, or $16x + 8 = 48$. Therefore, the value of $16x + 8$ is 48.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 7c71d844

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 7c71d844

$$w(t) = 300 - 4t$$

The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

- A. 300
- B. 296
- C. 75
- D. 4

ID: 7c71d844 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. The given function $w(t) = 300 - 4t$ can be rewritten as $w(t) = -4t + 300$. Thus, for each increase of t by 1, the value of $w(t)$ decreases by $4(1)$, or 4. Therefore, the predicted volume, in milliliters, draining from the container each second is 4 milliliters.

Choice A is incorrect. This is the amount of liquid, in milliliters, in the container before the liquid begins draining.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 49785a71

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 49785a71

$4x + 12 = \frac{a(x+3)}{2}$

In the given equation, *a* is a constant. If the equation has infinitely many solutions, what is the value of *a*?

- A. 0
- B. 3
- C. 8
- D. 12

ID: 49785a71 Answer

Correct Answer: C

Rationale

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields $8x + 24 = a(x + 3)$. Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as $8(x + 3) = a(x + 3)$. The two sides of this equation are equivalent when $a = 8$. Therefore, if the given equation has infinitely many solutions, the value of *a* is 8.

Alternate approach: If the given equation, $4x + 12 = \frac{a(x+3)}{2}$, has infinitely many solutions, then both sides of this equation are equal for any value of *x*. If $x = 0$, then substituting 0 for *x* in the given equation yields $4(0) + 12 = \frac{a(0+3)}{2}$, or $12 = \frac{3}{2}a$. Dividing both sides of this equation by $\frac{3}{2}$ yields $8 = a$.

Choice A is incorrect. If the value of *a* is 0, the given equation is equivalent to $4x + 12 = 0$, which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of *a* is 3, the given equation is equivalent to $4x + 12 = \frac{3(x+3)}{2}$, or $4x + 12 = \frac{3}{2}x + \frac{9}{2}$, which has one solution, not infinitely many solutions.

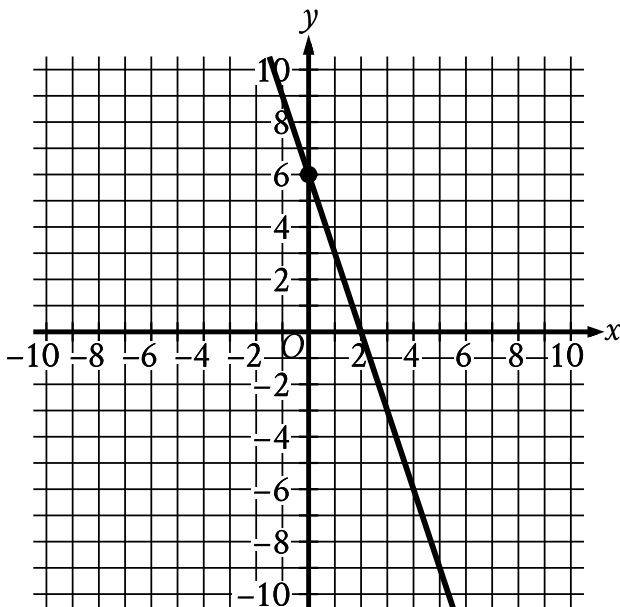
Choice D is incorrect. If the value of *a* is 12, the given equation is equivalent to $4x + 12 = \frac{12(x+3)}{2}$, or $4x + 12 = 6x + 18$, which has one solution, not infinitely many solutions.

Question Difficulty: Hard

Question ID 173484b2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 173484b2



The graph shows a linear relationship between x and y . Which equation represents this relationship, where R is a positive constant?

- A. $Rx + 18y = 36$
- B. $Rx - 18y = -36$
- C. $18x + Ry = 36$
- D. $18x - Ry = -36$

ID: 173484b2 Answer

Correct Answer: C

Rationale

Choice C is correct. The equation representing the linear relationship shown can be written in slope-intercept form $y = mx + b$, where m is the slope and $(0, b)$ is the y-intercept of the line. The line shown passes through the points $(0, 6)$ and $(2, 0)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(0, 6)$ and $(2, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in this equation yields $m = \frac{0 - 6}{2 - 0}$, which is equivalent to $m = -\frac{6}{2}$, or $m = -3$. Since $(0, 6)$ is the y-intercept, it follows that $b = 6$. Substituting -3 for m and 6 for b in the equation $y = mx + b$ yields $y = -3x + 6$. Adding $3x$ to both sides of this equation yields $3x + y = 6$. Multiplying this equation by 6 yields $18x + 6y = 36$. It follows that the equation $18x + Ry = 36$, where R is a positive constant, represents this relationship.

Choice A is incorrect. The graph of this relationship passes through the point $(0, 2)$, not $(0, 6)$.

Choice B is incorrect. The graph of this relationship passes through the point $(0, 2)$, not $(0, 6)$.

Choice D is incorrect. The graph of this relationship passes through the point $(-2, 0)$, not $(2, 0)$.

Question Difficulty: Hard

Question ID 61ab5e49

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 61ab5e49

The function f is defined by $f(x) = 80 - 6x$. What is the value of $f(7)$?

- A. 13
- B. 38
- C. 74
- D. 81

ID: 61ab5e49 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that function f is defined by $f(x) = 80 - 6x$. The value of $f(7)$ can be found by substituting 7 for x in the given function, which yields $f(7) = 80 - 6(7)$, or $f(7) = 80 - 42$, which is equivalent to $f(7) = 38$. Therefore, the value of $f(7)$ is 38.

Choice A is incorrect. This is the value of $80 - 67$, not $80 - 6(7)$.

Choice C is incorrect. This is the value of $80 - 6(1)$, not $80 - 6(7)$.

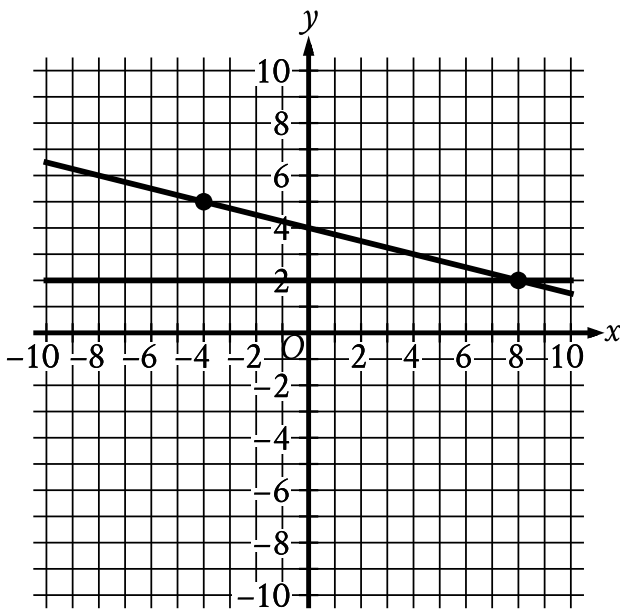
Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 6(7)$.

Question Difficulty: Easy

Question ID ffe7118f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: ffe7118f



If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions (x, y) will the resulting system of three equations have?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

ID: ffe7118f Answer

Correct Answer: A

Rationale

Choice A is correct. A solution to a system of equations must satisfy each equation in the system. It follows that if an ordered pair (x, y) is a solution to the system, the point (x, y) lies on the graph in the xy -plane of each equation in the system. The only point that lies on each graph of the system of two linear equations shown is their intersection point $(8, 2)$. It follows that if a new graph of three linear equations is created using the system of equations shown and the graph of $x + 4y = -16$, this system has either zero solutions or one solution, the point $(8, 2)$. Substituting 8 for x and 2 for y in the equation $x + 4y = -16$ yields $8 + 4(2) = -16$, or $16 = -16$. Since this equation is not true, the point $(8, 2)$ does not lie on the graph of $x + 4y = -16$. Therefore, $(8, 2)$ is not a solution to the system of three equations. It follows that there are zero solutions to this system.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4453d265

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 4453d265

The number y is **84** less than the number x . Which equation represents the relationship between x and y ?

- A. $y = x + 84$
- B. $y = \frac{1}{84}x$
- C. $y = 84x$
- D. $y = x - 84$

ID: 4453d265 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the number y is **84** less than the number x . A number that's **84** less than the number x is equivalent to **84** subtracted from the number x , or $x - 84$. Therefore, the equation $y = x - 84$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 685ba18f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 685ba18f

A manufacturing plant makes **10**-inch, **9**-inch, and **7**-inch frying pans. During a certain day, the number of **10**-inch frying pans that the manufacturing plant makes is **4** times the number ***n*** of **9**-inch frying pans it makes, and the number of **7**-inch frying pans it makes is **10**. During this day, the manufacturing plant makes **100** frying pans total. Which equation represents this situation?

- A. $10(4n) + 9n + 7(10) = 100$
- B. $10n + 9n + 7n = 100$
- C. $4n + 10 = 100$
- D. $5n + 10 = 100$

ID: 685ba18f Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day, the number of **9**-inch frying pans the manufacturing plant makes is ***n*** and the number of **7**-inch frying pans it makes is **10**. It's also given that during this day the number of **10**-inch frying pans that the manufacturing plant makes is **4** times the number of **9**-inch frying pans, or ***4n***. Therefore, the total number of **7**-inch, **9**-inch, and **10**-inch frying pans the manufacturing plant makes is ***n* + 10 + 4*n***, or ***5n* + 10**. It's given that during this day the manufacturing plant makes **100** frying pans total. Thus, the equation ***5n* + 10 = 100** represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

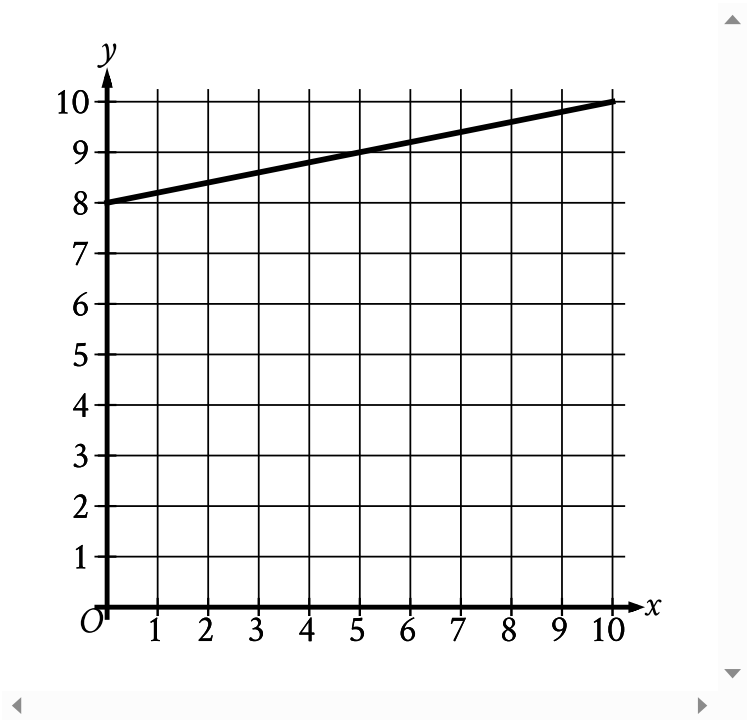
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 176d7796

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 176d7796



What is the y-intercept of the line graphed?

- A. $(0, -8)$
- B. $(0, -\frac{1}{8})$
- C. $(0, 0)$
- D. $(0, 8)$

ID: 176d7796 Answer

Correct Answer: D

Rationale

Choice D is correct. The y-intercept of a line graphed in the xy-plane is the point where the line intersects the y-axis. The line graphed intersects the y-axis at the point $(0, 8)$. Therefore, the y-intercept of the line graphed is $(0, 8)$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 79d9c537

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 79d9c537

A teacher is creating an assignment worth **70** points. The assignment will consist of questions worth **1** point and questions worth **3** points. Which equation represents this situation, where ***x*** represents the number of **1**-point questions and ***y*** represents the number of **3**-point questions?

- A. **$4xy = 70$**
- B. **$4(x + y) = 70$**
- C. **$3x + y = 70$**
- D. **$x + 3y = 70$**

ID: 79d9c537 Answer

Correct Answer: D

Rationale

Choice D is correct. Since ***x*** represents the number of **1**-point questions and ***y*** represents the number of **3**-point questions, the assignment is worth a total of **$1 \cdot x + 3 \cdot y$** , or **$x + 3y$** , points. Since the assignment is worth **70** points, the equation **$x + 3y = 70$** represents this situation.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 07cab4f0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 07cab4f0

A store sells two different-sized containers of a certain Greek yogurt. The store's sales of this Greek yogurt totaled **1,277.94** dollars last month. The equation $5.48x + 7.30y = 1,277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. According to the equation, which of the following represents the price, in dollars, of each smaller container?

- A. **5.48**
- B. $7.30y$
- C. **7.30**
- D. $5.48x$

ID: 07cab4f0 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the store's sales of a certain Greek yogurt totaled **1,277.94** dollars last month. It's also given that the equation $5.48x + 7.30y = 1,277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. Since x represents the number of smaller containers of yogurt sold, the expression $5.48x$ represents the total sales, in dollars, from smaller containers of yogurt. This means that x smaller containers of yogurt were sold at a price of **5.48** dollars each. Therefore, according to the equation, **5.48** represents the price, in dollars, of each smaller container.

Choice B is incorrect. This expression represents the total sales, in dollars, from selling y larger containers of yogurt.

Choice C is incorrect. This value represents the price, in dollars, of each larger container of yogurt.

Choice D is incorrect. This expression represents the total sales, in dollars, from selling x smaller containers of yogurt.

Question Difficulty: Medium

Question ID a854ff94

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: a854ff94

$2.5b + 5r = 80$

The given equation describes the relationship between the number of birds, b , and the number of reptiles, r , that can be cared for at a pet care business on a given day. If the business cares for **16** reptiles on a given day, how many birds can it care for on this day?

- A. **0**
- B. **5**
- C. **40**
- D. **80**

ID: a854ff94 Answer

Correct Answer: A

Rationale

Choice A is correct. The number of birds can be found by calculating the value of b when $r = 16$ in the given equation. Substituting **16** for r in the given equation yields $2.5b + 5(16) = 80$, or $2.5b + 80 = 80$. Subtracting **80** from both sides of this equation yields $2.5b = 0$. Dividing both sides of this equation by **2.5** yields $b = 0$. Therefore, if the business cares for **16** reptiles on a given day, it can care for **0** birds on this day.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

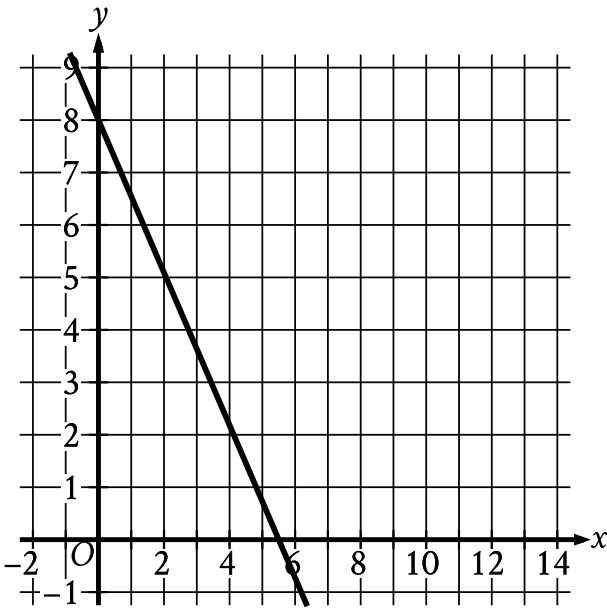
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 99235d83

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 99235d83



The graph of the linear function f is shown, where $y = f(x)$. What is the y-intercept of the graph of f ?

- A. $(0, 0)$
- B. $(0, -\frac{16}{11})$
- C. $(0, -8)$
- D. $(0, 8)$

ID: 99235d83 Answer

Correct Answer: D

Rationale

Choice D is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The graph of function f shown intersects the y-axis at the point $(0, 8)$. Therefore, the y-intercept of the graph of f is $(0, 8)$.

Choice A is incorrect. This is the point where the x-axis, not the graph of f , intersects the y-axis.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 52484f52

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 52484f52

The function g is defined by $g(x) = 4x - 6$. What is the value of $g(-7)$?

- A. -34
- B. -22
- C. $-\frac{13}{4}$
- D. $-\frac{1}{4}$

ID: 52484f52 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function g is defined by $g(x) = 4x - 6$. Substituting -7 for x into the given equation yields $g(-7) = 4(-7) - 6$, or $g(-7) = -34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $g(x) = -7$, not the value of $g(-7)$.

Question Difficulty: Medium

Question ID 7c69b87f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 7c69b87f

What is the slope of the graph of $y = \frac{1}{4}(27x + 15) + 7x$ in the xy -plane?

ID: 7c69b87f Answer

Correct Answer: 13.75, 55/4

Rationale

The correct answer is $\frac{55}{4}$. In the xy -plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a y -intercept of $(0, b)$. Applying the distributive property to the right-hand side of the given equation yields $y = \frac{27}{4}x + \frac{15}{4} + 7x$. Combining like terms yields $y = \frac{55}{4}x + \frac{15}{4}$. This equation is in the form $y = mx + b$, where $m = \frac{55}{4}$ and $b = \frac{15}{4}$. It follows that the slope of the graph of $y = \frac{1}{4}(27x + 15) + 7x$ in the xy -plane is $\frac{55}{4}$. Note that 55/4 and 13.75 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID c1cabad5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: c1cabad5

The cost of renting a carpet cleaner is **\$52** for the first day and **\$26** for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

- A. $C(d) = 26d + 26$
- B. $C(d) = 26d + 52$
- C. $C(d) = 52d - 26$
- D. $C(d) = 52d + 78$

ID: c1cabad5 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the cost of renting a carpet cleaner is **\$52** for the first day and **\$26** for each additional day. Therefore, the cost $C(d)$, in dollars, of renting the carpet cleaner for d days is the sum of the cost for the first day, **\$52**, and the cost for the additional $d - 1$ days, **\$26**($d - 1$). It follows that $C(d) = 52 + 26(d - 1)$, which is equivalent to $C(d) = 52 + 26d - 26$, or $C(d) = 26d + 26$.

Choice B is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is **\$78**, not **\$52**, for the first day and **\$26** for each additional day.

Choice C is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is **\$26**, not **\$52**, for the first day and **\$52**, not **\$26**, for each additional day.

Choice D is incorrect. This function gives the cost of renting a carpet cleaner for d days if the cost is **\$130**, not **\$52**, for the first day and **\$52**, not **\$26**, for each additional day.

Question Difficulty: Hard

Question ID 674f3a63

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 674f3a63

The function f is defined by $f(x) = \frac{1}{2}(x + 6)$. What is the value of $f(4)$?

- A. 20
- B. 12
- C. 10
- D. 5

ID: 674f3a63 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the function f is defined by $f(x) = \frac{1}{2}(x + 6)$. Substituting 4 for x in the given function yields $f(4) = \frac{1}{2}(4 + 6)$, or $f(4) = 5$. Therefore, the value of $f(4)$ is 5 .

Choice A is incorrect. This is the value of $2(4 + 6)$, not $\frac{1}{2}(4 + 6)$.

Choice B is incorrect. This is the value of $2 + (4 + 6)$, not $\frac{1}{2}(4 + 6)$.

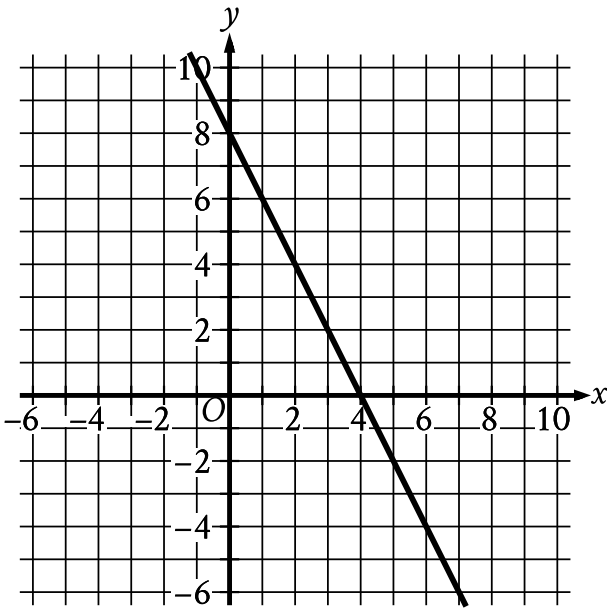
Choice C is incorrect. This is the value of $4 + 6$, not $\frac{1}{2}(4 + 6)$.

Question Difficulty: Medium

Question ID a9bdafcd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: a9bdafcd



The graph of the linear function f is shown, where $y = f(x)$. Which equation defines function f ?

- A. $f(x) = -2x - 8$
- B. $f(x) = -2x + 8$
- C. $f(x) = 2x - 8$
- D. $f(x) = 2x + 8$

ID: a9bdafcd Answer

Correct Answer: B

Rationale

Choice B is correct. A linear function can be defined by the equation $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y-intercept of the graph of $y = f(x)$ in the xy -plane. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $(4, 0)$ and $(0, 8)$. Substituting $(4, 0)$ for (x_1, y_1) and $(0, 8)$ for (x_2, y_2) in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{8 - 0}{0 - 4}$, or $m = -2$. Since the line shown passes through the point $(0, 8)$, it follows that $b = 8$. Substituting -2 for m and 8 for b in the equation $f(x) = mx + b$ yields $f(x) = -2x + 8$. Therefore, the equation $f(x) = -2x + 8$ defines function f .

Choice A is incorrect. This equation defines a linear function whose graph has a slope of -2 and a y-intercept of $(0, -8)$, not $(0, 8)$.

Choice C is incorrect. This equation defines a linear function whose graph has a slope of 2 , not -2 , and a y-intercept of $(0, -8)$, not $(0, 8)$.

Choice D is incorrect. This equation defines a linear function whose graph has a slope of **2**, not **−2**, and a y -intercept of **(0, 8)**.

Question Difficulty: Medium

Question ID f3a8f54e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: f3a8f54e

$$3x + 6 = 4y$$

$$3x + 4 = 2y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: f3a8f54e Answer

Correct Answer: 1

Rationale

The correct answer is **1**. Subtracting the second equation from the first equation in the given system of equations yields $(3x - 3x) + (6 - 4) = 4y - 2y$, which is equivalent to $0 + 2 = 2y$, or $2 = 2y$. Dividing each side of this equation by **2** yields $1 = y$.

Question Difficulty: Medium

Question ID 3757bc97

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 3757bc97

Henry receives a **\$60.00** gift card to pay for movies online. He uses his gift card to buy **3** movies for **\$7.50** each. If he spends the rest of his gift card balance on renting movies for **\$1.50** each, how many movies can Henry rent?

- A. 10
- B. 25
- C. 35
- D. 40

ID: 3757bc97 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that Henry uses his **\$60.00** gift card to buy **3** movies for **\$7.50** each. Therefore, Henry spends **3(\$7.50)**, or **\$22.50**, of his **\$60.00** gift card to buy **3** movies. After buying **3** movies with his **\$60.00** gift card, Henry has a gift card balance of **\$60.00 — \$22.50**, or **\$37.50**. It's also given that Henry spends the rest of his gift card balance on renting movies for **\$1.50** each. Therefore, Henry can rent $\frac{\$37.50}{\$1.50}$, or **25**, movies.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c0f646aa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: c0f646aa

$y = 7x + 18$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

- A. $-21x + y = 54$
- B. $-21x + y = 18$
- C. $-7x + y = 21$
- D. $-7x + y = 18$

ID: c0f646aa Answer

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the equations in the xy -plane are two distinct and parallel lines, which don't intersect. Two lines in the xy -plane are distinct and parallel if their slopes are the same and the y -coordinates of their y -intercepts are different. A linear equation can be written in slope-intercept form, $y = mx + b$, where m is the slope of the line in the xy -plane and b is the y -coordinate of the y -intercept. Choice C can be written in slope-intercept form by adding $7x$ to both sides of the equation, which gives $y = 7x + 21$. In the equations $y = 7x + 18$ and $y = 7x + 21$, the values of m are each 7 , and the values of b are 18 and 21 , respectively. Since the slopes of these lines are the same and the y -coordinates of the y -intercepts are different, it follows that choice C could be the second equation in the system.

Choice A is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 54$. In the equations $y = 7x + 18$ and $y = 21x + 54$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice B is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 18$. In the equations $y = 7x + 18$ and $y = 21x + 18$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice D is incorrect. Adding $7x$ to both sides of this equation gives $y = 7x + 18$. Since this equation is the same as the given equation, these two lines are not distinct. Therefore, the corresponding system of equations has infinitely many solutions.

Question Difficulty: Hard

Question ID 309f1dc6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 309f1dc6

$$\begin{aligned} y &\leq x + 7 \\ y &\geq -2x - 1 \end{aligned}$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- A. $(-14, 0)$
- B. $(0, -14)$
- C. $(0, 14)$
- D. $(14, 0)$

ID: 309f1dc6 Answer

Correct Answer: D

Rationale

Choice D is correct. A point (x, y) is a solution to a system of inequalities in the xy -plane if substituting the x -coordinate and the y -coordinate of the point for x and y , respectively, in each inequality makes both of the inequalities true. Substituting the x -coordinate and the y -coordinate of choice D, **14** and **0**, for x and y , respectively, in the first inequality in the given system, $y \leq x + 7$, yields $0 \leq 14 + 7$, or $0 \leq 21$, which is true. Substituting **14** for x and **0** for y in the second inequality in the given system, $y \geq -2x - 1$, yields $0 \geq -2(14) - 1$, or $0 \geq -29$, which is true. Therefore, the point **$(14, 0)$** is a solution to the given system of inequalities in the xy -plane.

Choice A is incorrect. Substituting **-14** for x and **0** for y in the inequality $y \leq x + 7$ yields $0 \leq -14 + 7$, or $0 \leq -7$, which is not true.

Choice B is incorrect. Substituting **0** for x and **-14** for y in the inequality $y \geq -2x - 1$ yields $-14 \geq -2(0) - 1$, or $-14 \geq -1$, which is not true.

Choice C is incorrect. Substituting **0** for x and **14** for y in the inequality $y \leq x + 7$ yields $14 \leq 0 + 7$, or $14 \leq 7$, which is not true.

Question Difficulty: Hard

Question ID 21526fbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 21526fbe

If $4x = 3$, what is the value of $24x$?

- A. $\frac{9}{2}$
- B. 6
- C. 18
- D. 72

ID: 21526fbe Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $4x = 3$. Multiplying each side of this equation by 6 yields $24x = 18$. Therefore, the value of $24x$ is 18.

Choice A is incorrect. This is the value of $6x$, not $24x$.

Choice B is incorrect. This is the value of $8x$, not $24x$.

Choice D is incorrect. This is the value of $96x$, not $24x$.

Question Difficulty: Medium

Question ID 09f4035b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 09f4035b

Davio bought some potatoes and celery. The potatoes cost **\$0.69** per pound, and the celery cost **\$0.99** per pound. If Davio spent **\$5.34** in total and bought twice as many pounds of celery as pounds of potatoes, how many pounds of celery did Davio buy?

- A. **2**
- B. **2.5**
- C. **2.67**
- D. **4**

ID: 09f4035b Answer

Correct Answer: D

Rationale

Choice D is correct. Let p represent the number of pounds of potatoes and let c represent the number of pounds of celery that Davio bought. It's given that potatoes cost **\$0.69** per pound and celery costs **\$0.99** per pound. If Davio spent **\$5.34** in total, then the equation $0.69p + 0.99c = 5.34$ represents this situation. It's also given that Davio bought twice as many pounds of celery as pounds of potatoes; therefore, $c = 2p$. Substituting $2p$ for c in the equation $0.69p + 0.99c = 5.34$ yields $0.69p + 0.99(2p) = 5.34$, which is equivalent to $0.69p + 1.98p = 5.34$, or $2.67p = 5.34$. Dividing both sides of this equation by **2.67** yields $p = 2$. Substituting **2** for p in the equation $c = 2p$ yields $c = 2(2)$, or $c = 4$. Therefore, Davio bought **4** pounds of celery.

Choice A is incorrect. This is the number of pounds of potatoes, not the number of pounds of celery, Davio bought.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 87bd4e09

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 87bd4e09

The function f is defined by the equation $f(x) = 100x + 2$. What is the value of $f(x)$ when $x = 9$?

- A. 111
- B. 118
- C. 900
- D. 902

ID: 87bd4e09 Answer

Correct Answer: D

Rationale

Choice D is correct. Substituting 9 for x in the given equation yields $f(9) = 100(9) + 2$, or $f(9) = 902$. Therefore, the value of $f(x)$ when $x = 9$ is 902.

Choice A is incorrect. This is the value of $f(x)$ when $x = 1.09$.

Choice B is incorrect. This is the value of $f(x)$ when $x = 1.16$.

Choice C is incorrect. This is the value of $f(x)$ when $x = 8.98$.

Question Difficulty: Easy

Question ID 6bd32629

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 6bd32629

$$f(x) = 7x + 1$$

The function gives the total number of people on a company retreat with x managers. What is the total number of people on a company retreat with 7 managers?

ID: 6bd32629 Answer

Correct Answer: 50

Rationale

The correct answer is 50. It's given that the function f gives the total number of people on a company retreat with x managers. It's also given that 7 managers are on the company retreat. Substituting 7 for x in the given function yields $f(7) = 7(7) + 1$, or $f(7) = 50$. Therefore, there are a total of 50 people on a company retreat with 7 managers.

Question Difficulty: Easy

Question ID d37842cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: d37842cc

$$\begin{aligned}y &= 4 \\x &= y + 6\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. 10
- B. 6
- C. 4
- D. 2

ID: d37842cc Answer

Correct Answer: A

Rationale

Choice A is correct. According to the first equation in the given system, $y = 4$. Substituting 4 for y in the second equation in the given system yields $x = 4 + 6$, or $x = 10$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of y , not x .

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 6a8610a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 6a8610a5

If $y = 5x + 10$, what is the value of y when $x = 8$?

ID: 6a8610a5 Answer

Correct Answer: 50

Rationale

The correct answer is **50**. Substituting **8** for x in the given equation yields $y = 5(8) + 10$, or $y = 50$. Therefore, the value of y is **50** when $x = 8$.

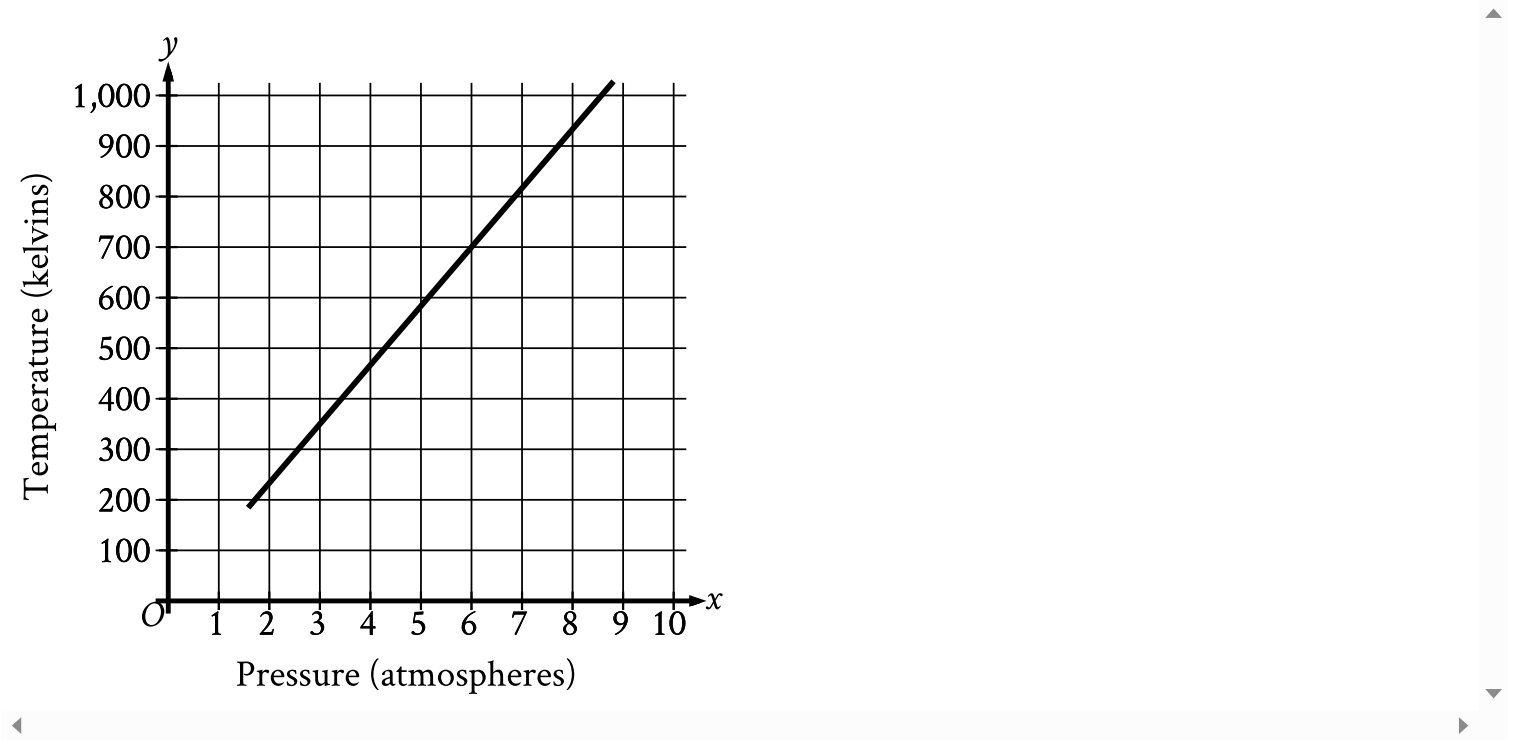
Question Difficulty: Easy

Question ID 8b896724

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 8b896724

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y , in kelvins, of the oxygen gas when its pressure is x atmospheres.



What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres?

- A. **6**
- B. **60**
- C. **700**
- D. **760**

ID: 8b896724 Answer

Correct Answer: C

Rationale

Choice C is correct. For the graph shown, the x -axis represents pressure, in atmospheres, and the y -axis represents temperature, in kelvins. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres is represented by the y -coordinate of the point on the graph that has an x -coordinate of **6**. The point on the graph with an x -coordinate of **6** has a y -coordinate of approximately **700**. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres is **700**.

Choice A is incorrect. This is the pressure, in atmospheres, not the estimated temperature, in kelvins, of the oxygen gas when its pressure is **6** atmospheres.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 59e1fbbe

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 59e1fbbe

A number x is at most 2 less than 3 times the value of y . If the value of y is -4 , what is the greatest possible value of x ?

ID: 59e1fbbe Answer

Correct Answer: -14

Rationale

The correct answer is -14 . It's given that a number x is at most 2 less than 3 times the value of y . Therefore, x is less than or equal to 2 less than 3 times the value of y . The expression $3y$ represents 3 times the value of y . The expression $3y - 2$ represents 2 less than 3 times the value of y . Therefore, x is less than or equal to $3y - 2$. This can be shown by the inequality $x \leq 3y - 2$. Substituting -4 for y in this inequality yields $x \leq 3(-4) - 2$ or, $x \leq -14$. Therefore, if the value of y is -4 , the greatest possible value of x is -14 .

Question Difficulty: Hard

Question ID 2c8566ba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 2c8566ba

x	y
-12	-45
6	45

The table shows two values of x and their corresponding values of y . The graph of the linear equation representing this relationship passes through the point $(\frac{1}{4}, a)$. What is the value of a ?

ID: 2c8566ba Answer

Correct Answer: 16.25, 65/4

Rationale

The correct answer is $\frac{65}{4}$. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m and b are constants. It's given in the table that when $x = -12$, $y = -45$. Substituting -12 for x and -45 for y in the equation $y = mx + b$ yields $-45 = -12m + b$, which can be rewritten as $-45 + 12m = b$. It's also given in the table that when $x = 6$, $y = 45$. Substituting 6 for x and 45 for y in the equation $y = mx + b$ yields $45 = 6m + b$, which can be rewritten as $45 - 6m = b$. Substituting $-45 + 12m$ for b in this equation yields $45 - 6m = -45 + 12m$. Adding $6m$ to both sides of this equation yields $45 = -45 + 18m$. Adding 45 to both sides of this equation yields $90 = 18m$. Dividing both sides of this equation by 18 yields $5 = m$, or $m = 5$. Substituting 5 for m , -12 for x , and -45 for y in the equation $y = mx + b$ yields $-45 = 5(-12) + b$, or $-45 = -60 + b$. Adding 60 to both sides of this equation yields $15 = b$. Therefore, $m = 5$ and $b = 15$. Substituting 5 for m and 15 for b in the equation $y = mx + b$ yields $y = 5x + 15$. Thus, the equation $y = 5x + 15$ represents the linear relationship between x and y . It's also given that the graph of the linear equation representing this relationship passes through the point $(\frac{1}{4}, a)$. Substituting $\frac{1}{4}$ for x and a for y in the equation $y = 5x + 15$ yields $a = 5(\frac{1}{4}) + 15$, which is equivalent to $a = \frac{5}{4} + 15$, or $a = \frac{65}{4}$. Note that 65/4 and 16.25 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 82297d55

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 82297d55

The function f is defined by $f(x) = 7x - 84$. What is the x-intercept of the graph of $y = f(x)$ in the xy-plane?

- A. $(-12, 0)$
- B. $(-7, 0)$
- C. $(7, 0)$
- D. $(12, 0)$

ID: 82297d55 Answer

Correct Answer: D

Rationale

Choice D is correct. The given function f is a linear function. Therefore, the graph of $y = f(x)$ in the xy-plane has one x-intercept at the point $(k, 0)$, where k is a constant. Substituting 0 for $f(x)$ and k for x in the given function yields $0 = 7k - 84$. Adding 84 to both sides of this equation yields $84 = 7k$. Dividing both sides of this equation by 7 yields $12 = k$. Therefore, the x-intercept of the graph of $y = f(x)$ in the xy-plane is $(12, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID adf5e14b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: adf5e14b

The function f is defined by $f(x) = -9x + 9$. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

ID: adf5e14b Answer

Correct Answer: 9

Rationale

The correct answer is **9**. The y -intercept of the graph of $y = f(x)$ in the xy -plane is the point where the graph of $y = f(x)$ crosses the y -axis, which occurs at $x = 0$. It's given that the function f is defined by $f(x) = -9x + 9$. Substituting **0** for x and y for $f(x)$ in this equation yields $y = -9(0) + 9$, or $y = 9$. It follows that $y = 9$ when $x = 0$ and that the y -intercept of the graph of $y = f(x)$ in the xy -plane is **(0, 9)**. Therefore, the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane is **9**.

Question Difficulty: Medium

Question ID 07c14769

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 07c14769

Which system of linear equations has no solution?

- A.

$-2x + 3y = -9$

$2x - 3y = 9$
- B.

$2x - 3y = 9$

$3x + 4y = 10$
- C.

$2x - 3y = 9$

$-6x + 9y = -27$
- D.

$-2x + 3y = 9$

$4x - 6y = 18$

ID: 07c14769 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of linear equations can be solved by the elimination method. Multiplying the equation $-2x + 3y = 9$ by 2 yields $-4x + 6y = 18$. Adding this equation to the equation $4x - 6y = 18$ yields $0 = 36$, which has no solution. It follows that the system of linear equations consisting of $-2x + 3y = 9$ and $4x - 6y = 18$ has no solution.

Choice A is incorrect. This system of linear equations has infinitely many solutions, rather than no solution.

Choice B is incorrect. This system of linear equations has one solution, rather than no solution.

Choice C is incorrect. This system of linear equations has infinitely many solutions, rather than no solution.

Question Difficulty: Hard

Question ID 149c8821

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 149c8821

$8x = 88$

What value of x is the solution to the given equation?

- A. 11
- B. 80
- C. 96
- D. 704

ID: 149c8821 Answer

Correct Answer: A

Rationale

Choice A is correct. Dividing both sides of the given equation by 8 yields $x = 11$. Therefore, 11 is the solution to the given equation.

Choice B is incorrect. This is the solution to the equation $x + 8 = 88$.

Choice C is incorrect. This is the solution to the equation $x - 8 = 88$.

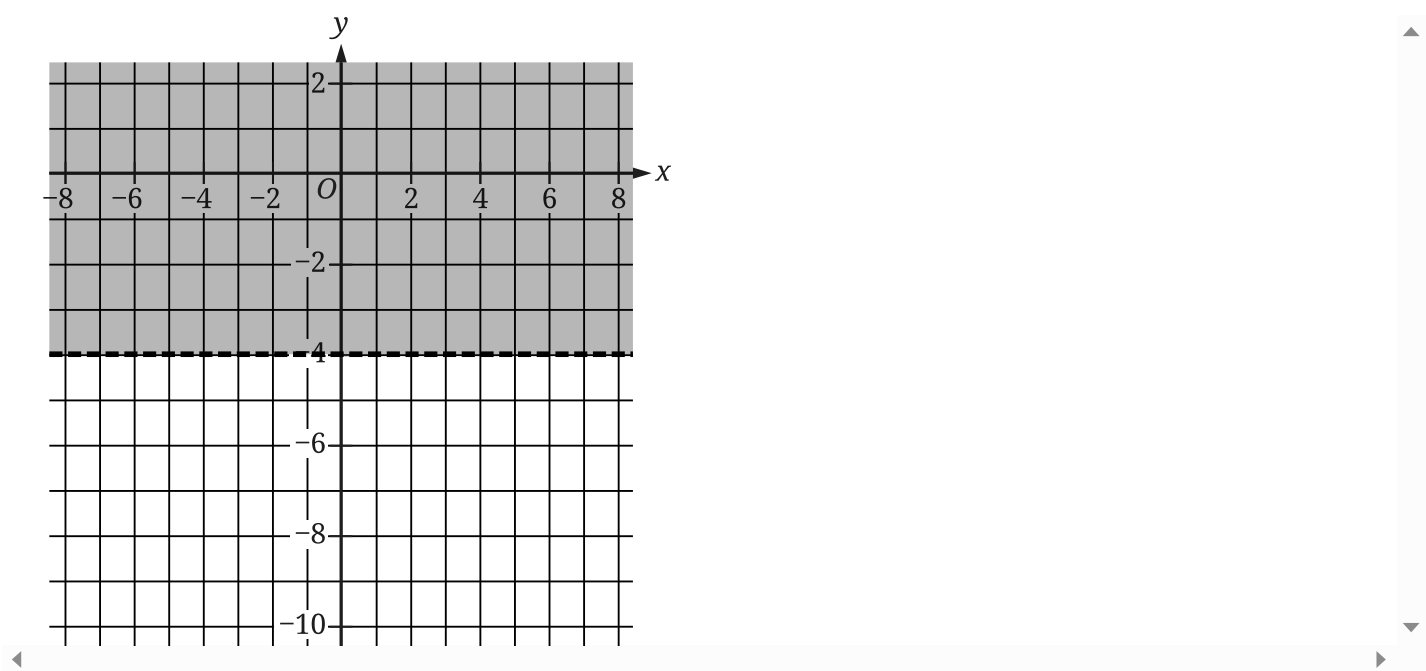
Choice D is incorrect. This is the solution to the equation $\frac{x}{8} = 88$.

Question Difficulty: Easy

Question ID cc9da54c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: cc9da54c



The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r ?

- A. 15
- B. 4
- C. -4
- D. -15

ID: cc9da54c Answer

Correct Answer: D

Rationale

Choice D is correct. The line representing the boundary of the shaded region shown is a horizontal line; therefore, the equation of the line can be written as $y = -4$. Since the shaded region represents all of the points above this boundary line, it follows that the shaded region represents the solutions to the inequality $y > -4$. It's given that the shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. The inequality $y > -4$ can be rewritten in the form $ry < 60$ by multiplying both sides of the inequality by -15 , which yields $(-15)y < 60$. Therefore, the value of r is -15 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

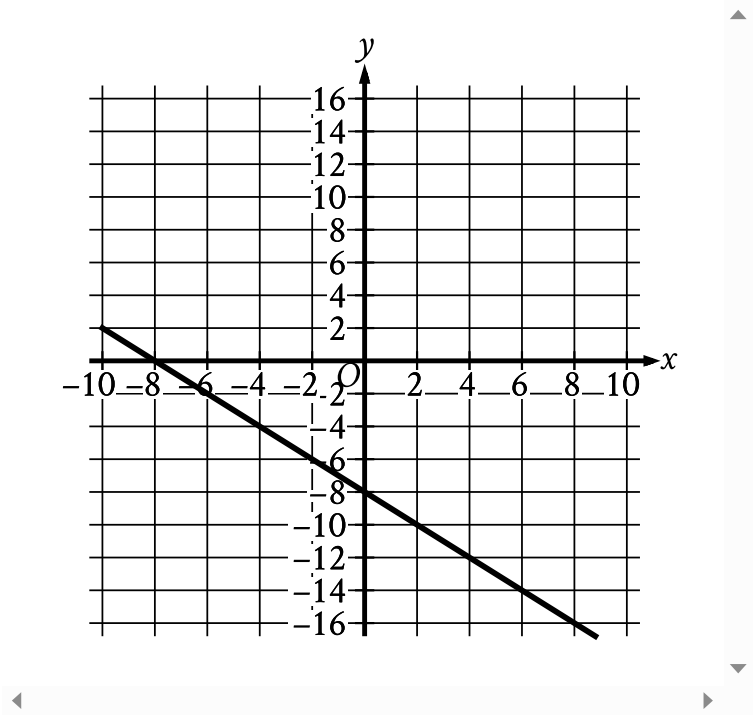
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ea3f862c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: ea3f862c



What is an equation of the graph shown?

- A. $y = -2x - 8$
- B. $y = x - 8$
- C. $y = -x - 8$
- D. $y = 2x - 8$

ID: ea3f862c Answer

Correct Answer: C

Rationale

Choice C is correct. An equation of a line can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y -intercept of the line. The line shown passes through the point $(0, -8)$, so $b = -8$. The line shown also passes through the point $(-8, 0)$. The slope, m , of a line passing through two points (x_1, y_1) and (x_2, y_2) can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. For the points $(0, -8)$ and $(-8, 0)$, this gives $m = \frac{(-8) - 0}{0 - (-8)}$, or $m = -1$. Substituting -8 for b and -1 for m in $y = mx + b$ yields $y = (-1)x + (-8)$, or $y = -x - 8$. Therefore, an equation of the graph shown is $y = -x - 8$.

Choice A is incorrect. This is an equation of a line with a slope of -2 , not -1 .

Choice B is incorrect. This is an equation of a line with a slope of 1 , not -1 .

Choice D is incorrect. This is an equation of a line with a slope of 2 , not -1 .

Question Difficulty: Medium

Question ID 273c9f49

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 273c9f49

The function h is defined by $h(x) = 3x - 7$. What is the value of $h(-2)$?

- A. -13
- B. -10
- C. 10
- D. 13

ID: 273c9f49 Answer

Correct Answer: A

Rationale

Choice A is correct. The value of $h(-2)$ can be found by substituting -2 for x in the equation defining h . Substituting -2 for x in $h(x) = 3x - 7$ yields $h(-2) = 3(-2) - 7$, or $h(-2) = -13$. Therefore, the value of $h(-2)$ is -13 .

Choice B is incorrect. This is the value of $h(-1)$, not $h(-2)$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 5ddc0ad1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 5ddc0ad1

Brian saves $\frac{2}{5}$ of the **\$215** he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in **9** weeks?

ID: 5ddc0ad1 Answer

Correct Answer: 774

Rationale

The correct answer is **774**. It's given that Brian saves $\frac{2}{5}$ of the **\$215** he earns each week from his job. Therefore, Brian saves $(\frac{2}{5})(\$215)$, or **\$86**, per week. If Brian continues to save at this rate of **\$86** per week for **9** weeks, then he will save a total of $(9)(86)$, or **774**, dollars.

Question Difficulty: Medium

Question ID 675325b9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 675325b9

Hana deposited a fixed amount into her bank account each month. The function $f(t) = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of **25** in this context?

- A. With each monthly deposit, the amount in Hana's bank account increased by **\$25**.
- B. Before Hana made any monthly deposits, the amount in her bank account was **\$25**.
- C. After **1** monthly deposit, the amount in Hana's bank account was **\$25**.
- D. Hana made a total of **25** monthly deposits.

ID: 675325b9 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that t represents the number of monthly deposits. In the given function $f(t) = 100 + 25t$, the coefficient of t is **25**. This means that for every increase in the value of t by **1**, the value of $f(t)$ increases by **25**. It follows that with each monthly deposit, the amount in Hana's bank account increased by **\$25**.

Choice B is incorrect. Before Hana made any monthly deposits, the amount in her bank account was **\$100**.

Choice C is incorrect. After **1** monthly deposit, the amount in Hana's bank account was **\$125**.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID cf9a5572

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: cf9a5572

$$\begin{aligned} 4x - 6y &= 10y + 2 \\ ty &= \frac{1}{2} + 2x \end{aligned}$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t ?

ID: cf9a5572 Answer

Correct Answer: 8

Rationale

The correct answer is 8. The given system of equations can be solved using the elimination method. Multiplying both sides of the second equation in the given system by -2 yields $-2ty = -1 - 4x$, or $-1 - 4x = -2ty$. Adding this equation to the first equation in the given system, $4x - 6y = 10y + 2$, yields $(4x - 6y) + (-1 - 4x) = (10y + 2) + (-2ty)$, or $-1 - 6y = 10y - 2ty + 2$. Subtracting $10y$ from both sides of this equation yields $(-1 - 6y) - (10y) = (10y - 2ty + 2) - (10y)$, or $-1 - 16y = -2ty + 2$. If the given system has no solution, then the equation $-1 - 16y = -2ty + 2$ has no solution. If this equation has no solution, the coefficients of y on each side of the equation, -16 and $-2t$, must be equal, which yields the equation $-16 = -2t$. Dividing both sides of this equation by -2 yields $8 = t$. Thus, if the system has no solution, the value of t is 8.

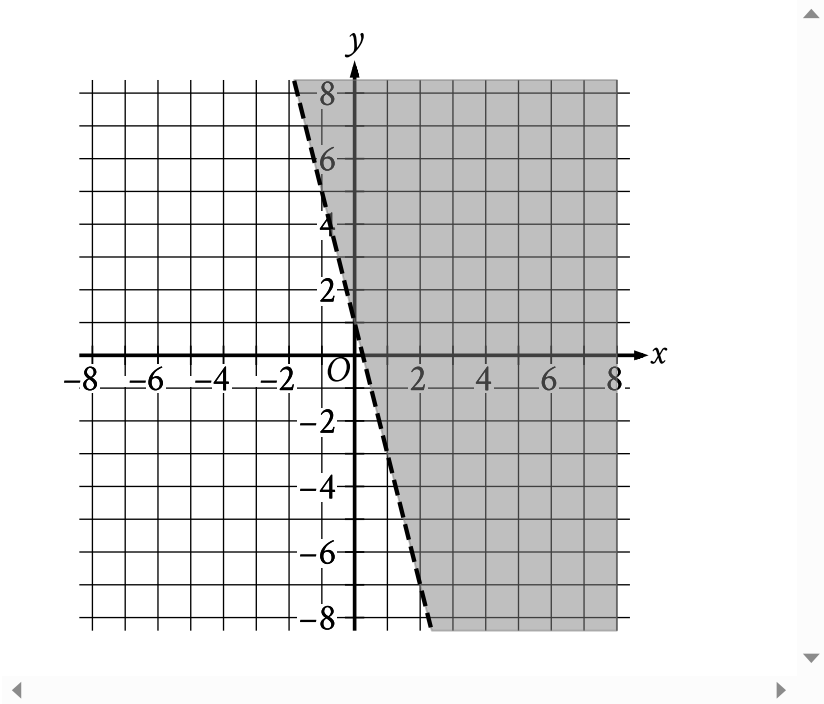
Alternate approach: A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are parallel and distinct. Lines represented by equations in the form $Ax + By = C$, where A , B , and C are constant terms, are parallel if the ratio of the x -coefficients is equal to the ratio of the y -coefficients, and distinct if the ratio of the x -coefficients are not equal to the ratio of the constant terms. Subtracting $10y$ from both sides of the first equation in the given system yields $(4x - 6y) - (10y) = (10y + 2) - (10y)$, or $4x - 16y = 2$. Subtracting $2x$ from both sides of the second equation in the given system yields $(ty) - (2x) = (\frac{1}{2} + 2x) - (2x)$, or $-2x + ty = \frac{1}{2}$. The ratio of the x -coefficients for these equations is $-\frac{2}{4}$, or $-\frac{1}{2}$. The ratio of the y -coefficients for these equations is $-\frac{t}{16}$. The ratio of the constant terms for these equations is $\frac{1/2}{2}$, or $\frac{1}{4}$. Since the ratio of the x -coefficients, $-\frac{1}{2}$, is not equal to the ratio of the constants, $\frac{1}{4}$, the lines represented by the equations are distinct. Setting the ratio of the x -coefficients equal to the ratio of the y -coefficients yields $-\frac{1}{2} = -\frac{t}{16}$. Multiplying both sides of this equation by -16 yields $(-\frac{1}{2})(-16) = (-\frac{t}{16})(-16)$, or $t = 8$. Therefore, when $t = 8$, the lines represented by these equations are parallel. Thus, if the system has no solution, the value of t is 8.

Question Difficulty: Hard

Question ID 7ce498a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 7ce498a1



The shaded region shown represents the solutions to which inequality?

- A. $y < 1 + 4x$
- B. $y < 1 - 4x$
- C. $y > 1 + 4x$
- D. $y > 1 - 4x$

ID: 7ce498a1 Answer

Correct Answer: D

Rationale

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in slope-intercept form $y = b + mx$, where m is the slope and $(0, b)$ is the y-intercept of the line. For the graph shown, the boundary line passes through the points $(0, 1)$ and $(1, -3)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, 1)$ and $(1, -3)$ for (x_1, y_1) and (x_2, y_2) in this equation yields $m = \frac{-3 - 1}{1 - 0}$, which is equivalent to $m = \frac{-4}{1}$, or $m = -4$. Since the point $(0, 1)$ represents the y-intercept, it follows that $b = 1$. Substituting -4 for m and 1 for b in the equation $y = b + mx$ yields $y = 1 - 4x$ as the equation of the boundary line. Since the shaded region represents all the points above this boundary line, it follows that the shaded region shown represents the solutions to the inequality $y > 1 - 4x$.

Choice A is incorrect. This inequality represents a region below, not above, a boundary line with a slope of 4 , not -4 .

Choice B is incorrect. This inequality represents a region below, not above, the boundary line shown.

Choice C is incorrect. This inequality represents a region whose boundary line has a slope of **4**, not **−4**.

Question Difficulty: Hard

Question ID 800e59a0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 800e59a0

$$\begin{aligned}x &= 4 \\ y &= 5 - x\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 1
- B. 4
- C. 5
- D. 9

ID: 800e59a0 Answer

Correct Answer: A

Rationale

Choice A is correct. The first equation in the given system of equations is $x = 4$. Substituting **4** for x in the second equation in the given system of equations yields $y = 5 - 4$, or $y = 1$.

Choice B is incorrect. This is the value of x in the solution to the given system of equations, not the value of y .

Choice C is incorrect and may result from conceptual or calculation errors.

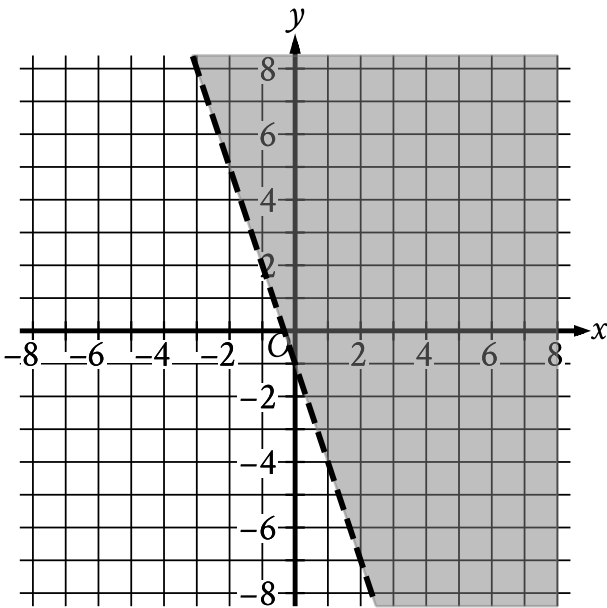
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 09834b8b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 09834b8b



The shaded region shown represents the solutions to which inequality?

- A. $y < -1 + 3x$
- B. $y < -1 - 3x$
- C. $y > -1 + 3x$
- D. $y > -1 - 3x$

ID: 09834b8b Answer

Correct Answer: D

Rationale

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in slope-intercept form $y = b + mx$, where m is the slope and $(0, b)$ is the y-intercept of the line. For the graph shown, the boundary line passes through the points $(0, -1)$ and $(1, -4)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, -1)$ and $(1, -4)$ for (x_1, y_1) and (x_2, y_2) in this equation yields $m = \frac{-4 - (-1)}{1 - 0}$, which is equivalent to $m = \frac{-3}{1}$, or $m = -3$. Since the point $(0, -1)$ represents the y-intercept, it follows that $b = -1$. Substituting -3 for m and -1 for b in the equation $y = b + mx$ yields $y = -1 - 3x$ as the equation of the boundary line. Since the shaded region represents all the points above this boundary line, it follows that the shaded region shown represents the solutions to the inequality $y > -1 - 3x$.

Choice A is incorrect. This inequality represents a region below, not above, a boundary line with a slope of 3 , not -3 .

Choice B is incorrect. This inequality represents a region below, not above, the boundary line shown.

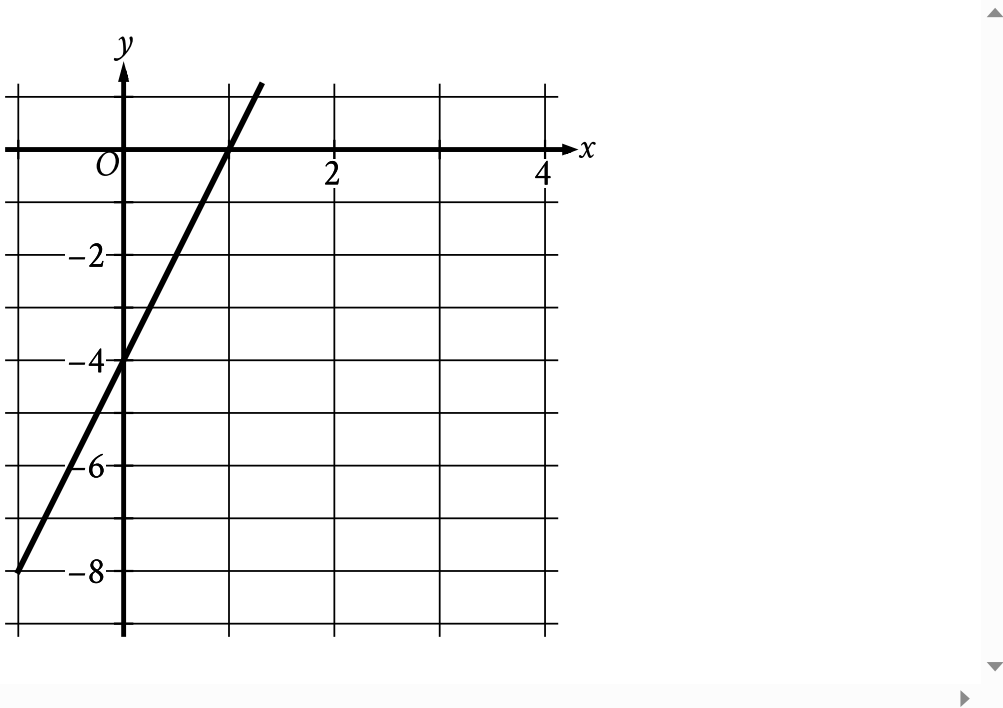
Choice C is incorrect. This inequality represents a region whose boundary line has a slope of **3**, not **−3**.

Question Difficulty: Hard

Question ID f717089b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: f717089b



The graph of the function f is shown, where $y = f(x)$. What is the y -intercept of the graph?

- A. $(0, -1)$
- B. $(0, -4)$
- C. $(0, 1)$
- D. $(0, 4)$

ID: f717089b Answer

Correct Answer: B

Rationale

Choice B is correct. The y -intercept of a graph is the point where the graph intersects the y -axis. The graph of function f shown intersects the y -axis at the point $(0, -4)$. Therefore, the y -intercept of the graph is $(0, -4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

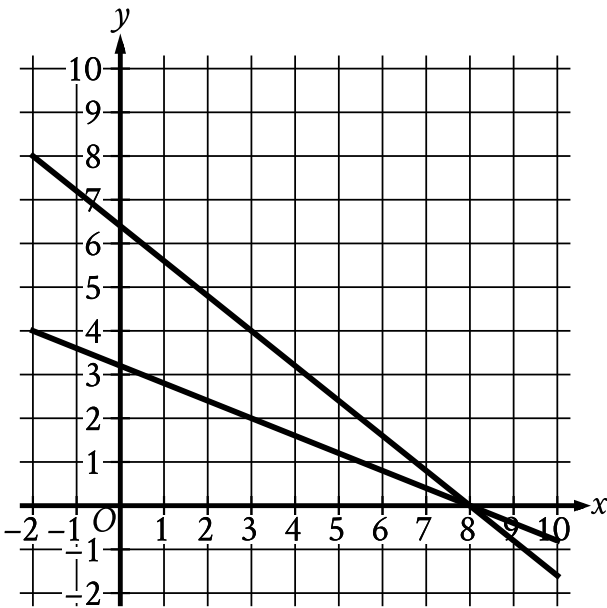
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 5d5d0dc7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 5d5d0dc7



What system of linear equations is represented by the lines shown?

- A. $8x + 4y = 32$
 $-10x - 4y = -64$
- B. $8x - 4y = 32$
 $-10x + 4y = -64$
- C. $4x - 10y = 32$
 $-8x + 10y = -64$
- D. $4x + 10y = 32$
 $-8x - 10y = -64$

ID: 5d5d0dc7 Answer

Correct Answer: D

Rationale

Choice D is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has slope m , where $m = \frac{y_2 - y_1}{x_2 - x_1}$, and can be defined by an equation of the form $y - y_1 = m(x - x_1)$. One of the lines shown in the graph passes through the points $(8, 0)$ and $(3, 4)$. Substituting 8 for x_1 , 0 for y_1 , 3 for x_2 , and 4 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{4 - 0}{3 - 8}$, or $m = -\frac{4}{5}$. Substituting $-\frac{4}{5}$ for m , 8 for x_1 and 0 for y_1 in the equation $y - y_1 = m(x - x_1)$ yields $y - 0 = -\frac{4}{5}(x - 8)$, which is equivalent to $y = -\frac{4}{5}x + \frac{32}{5}$. Adding $\frac{4}{5}x$ to both sides of this equation yields $\frac{4}{5}x + y = \frac{32}{5}$. Multiplying both sides of this equation by -10 yields $-8x - 10y = -64$. Therefore, an equation of this line is $-8x - 10y = -64$. Similarly, the other line shown in the graph passes through the points $(8, 0)$ and $(3, 2)$. Substituting 8 for x_1 , 0 for y_1 , 3 for x_2 , and 2 for y_2 in the

equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{2 - 0}{3 - 8}$, or $m = -\frac{2}{5}$. Substituting $-\frac{2}{5}$ for m , 8 for x_1 , and 0 for y_1 in the equation $y - y_1 = m(x - x_1)$ yields $y - 0 = -\frac{2}{5}(x - 8)$, which is equivalent to $y = -\frac{2}{5}x + \frac{16}{5}$. Adding $\frac{2}{5}x$ to both sides of this equation yields $\frac{2}{5}x + y = \frac{16}{5}$. Multiplying both sides of this equation by 10 yields $4x + 10y = 32$. Therefore, an equation of this line is $4x + 10y = 32$. So, the system of linear equations represented by the lines shown is $4x + 10y = 32$ and $-8x - 10y = -64$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 30135558

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 30135558

$y < 6x + 2$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36
7	48

ID: 30135558 Answer

Correct Answer: C

Rationale

Choice C is correct. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting **3** for x in the given inequality yields $y < 6(3) + 2$, or $y < 20$. Therefore, when $x = 3$, the corresponding value of y is less than **20**. Substituting **5** for x in the given inequality yields $y < 6(5) + 2$, or $y < 32$. Therefore, when $x = 5$, the corresponding value of y is less than **32**. Substituting **7** for x in the given inequality yields $y < 6(7) + 2$, or $y < 44$. Therefore, when $x = 7$, the corresponding value of y

is less than **44**. For the table in choice C, when $x = 3$, the corresponding value of y is **16**, which is less than **20**; when $x = 5$, the corresponding value of y is **28**, which is less than **32**; when $x = 7$, the corresponding value of y is **40**, which is less than **44**. Therefore, the table in choice C gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. In the table for choice A, when $x = 3$, the corresponding value of y is **20**, which is not less than **20**; when $x = 5$, the corresponding value of y is **32**, which is not less than **32**; when $x = 7$, the corresponding value of y is **44**, which is not less than **44**.

Choice B is incorrect. In the table for choice B, when $x = 5$, the corresponding value of y is **36**, which is not less than **32**.

Choice D is incorrect. In the table for choice D, when $x = 3$, the corresponding value of y is **24**, which is not less than **20**; when $x = 5$, the corresponding value of y is **36**, which is not less than **32**; when $x = 7$, the corresponding value of y is **48**, which is not less than **44**.

Question Difficulty: Hard

Question ID 65b23ff2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 65b23ff2

$g(x) = 3(14x - 15)$

What is the y-coordinate of the y-intercept of the graph of $y = g(x) - 2$ in the xy-plane?

- A. ~~−47~~
- B. ~~−45~~
- C. ~~−17~~
- D. ~~−15~~

ID: 65b23ff2 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = 3(14x - 15)$. Substituting $3(14x - 15)$ for $g(x)$ in the equation $y = g(x) - 2$ yields $y = 3(14x - 15) - 2$. The y-intercept of the graph of a function in the xy-plane is the point where the function crosses the y-axis, which occurs when $x = 0$. Substituting 0 for x in the equation $y = 3(14x - 15) - 2$ yields $y = 3(14(0) - 15) - 2$. It follows that $y = -45 - 2$, or $y = -47$. Therefore, the y-intercept of the graph of $y = g(x) - 2$ in the xy-plane is $(0, -47)$, and the y-coordinate of the y-intercept of the graph is -47 .

Choice B is incorrect. This is the y-coordinate of the y-intercept of the graph of $y = g(x)$ in the xy-plane, not $y = g(x) - 2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6bee1d56

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 6bee1d56

$-15x + 25y = 65$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. Which of the following could be the second equation in the system?

- A. $12x + 20y = 52$
- B. $12x + 20y = -52$
- C. $-12x + 20y = 52$
- D. $-12x + 20y = -52$

ID: 6bee1d56 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Dividing both sides of the given equation by 5 yields $-3x + 5y = 13$. Dividing both sides of choice C by 4 also yields $-3x + 5y = 13$, so choice C is equivalent to the given equation. Thus, choice C could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution, not infinitely many solutions.

Choice D is incorrect. The system consisting of this equation and the given equation has no solution, not infinitely many solutions.

Question Difficulty: Hard

Question ID 17e93ce3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 17e93ce3

For a **3**-week period in a town in Illinois, the lowest recorded temperature was **31** degrees Fahrenheit (**°F**) and the highest recorded temperature was **67°F**. Which inequality is true for any recorded temperature t , in **°F**, in this town for this **3**-week period?

- A. $t \geq 98$
- B. $t \geq 67$
- C. $31 \leq t \leq 67$
- D. $t \leq 31$

ID: 17e93ce3 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that for a **3**-week period in a town in Illinois, the lowest recorded temperature was **31°F** and the highest recorded temperature was **67°F**. It follows that the inequality $31 \leq t \leq 67$ is true for any recorded temperature t , in **°F**, in this town for this **3**-week period.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID f66cba39

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: f66cba39

Sean rents a tent at a cost of **\$11** per day plus a onetime insurance fee of **\$10**. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

- A. $c = 11(d + 10)$
- B. $c = 10(d + 11)$
- C. $c = 11d + 10$
- D. $c = 10d + 11$

ID: f66cba39 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the cost of renting a tent is **\$11** per day for d days. Multiplying the rental cost by the number of days yields **\$11 d** , which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of **\$10** to the rental cost of **\$11 d** gives the total cost c , in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was **\$($d + 11$)** for **10** days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was **\$10** and the onetime fee was **\$11**.

Question Difficulty: Easy

Question ID 7eaf38ce

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 7eaf38ce

$y = \frac{2}{7}x + 3$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

ID: 7eaf38ce Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the system has infinitely many solutions. The graphs of two lines in the xy -plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, have infinitely many solutions if their slopes, m , are the same and if their y -coordinates of the y -intercepts, b , are also the same. The first equation in the given system is $y = \frac{2}{7}x + 3$. For this equation, the slope is $\frac{2}{7}$ and the y -coordinate of the y -intercept is 3 . If the second equation is in the form $y = mx + b$, then for the two equations to be equivalent, the values of m and b in the second equation must equal the corresponding values in the first equation. Therefore, the second equation must have a slope, m , of $\frac{2}{7}$, and a y -coordinate of the y -intercept, b , of 3 . Thus, the value of b is 3 .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 1c016d37

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 1c016d37

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of **\$641**, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil, ***x***, and the number of tons of crushed stone, ***y***, Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

ID: 1c016d37 Answer

Correct Answer: 0.25, 1/4

Rationale

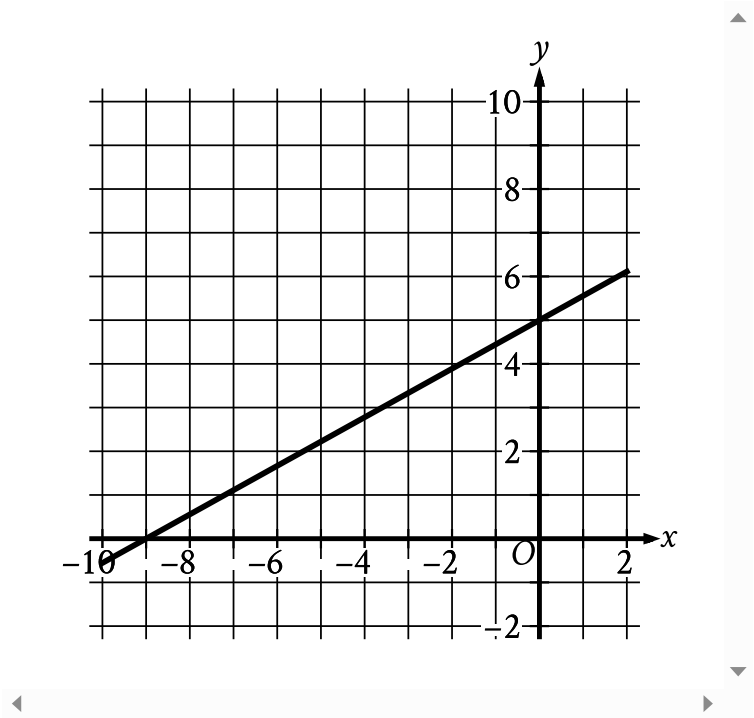
The correct answer is **.25**. It's given that the topsoil and crushed stone Isabel ordered for her garden cost a total of **\$641**. It's also given that the equation **$24.5x + 24.75y = 641$** represents the relationship between the number of cubic yards of topsoil, ***x***, and the number of tons of crushed stone, ***y***, that Isabel ordered. Since ***x*** represents the number of cubic yards of topsoil ordered, **$24.5x$** represents the total cost, in dollars, of the topsoil, and the cost per cubic yard of topsoil is **\$24.50**. Similarly, since ***y*** represents the number of tons of crushed stone ordered, **$24.75y$** represents the total cost, in dollars, of crushed stone ordered, and the cost per ton of crushed stone is **\$24.75**. Therefore, a ton of crushed stone cost Isabel **$24.75 - 24.50$** , or **0.25**, more dollars than a cubic yard of topsoil. Note that .25 and 1/4 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID e5dafa84

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: e5dafa84



What is the y-intercept of the line graphed?

- A. $(-5, 0)$
- B. $(0, 0)$
- C. $(0, 5)$
- D. $(0, 9)$

ID: e5dafa84 Answer

Correct Answer: C

Rationale

Choice C is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The line graphed intersects the y-axis at the point $(0, 5)$. Therefore, the y-intercept of the line graphed is $(0, 5)$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 2e4cdcad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 2e4cdcad

Ty set a goal to walk at least **24** kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of **4** kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

- A. **4**
- B. **6**
- C. **20**
- D. **24**

ID: 2e4cdcad Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that Ty plans to walk at an average speed of **4** kilometers per hour. The number of kilometers Ty will walk is determined by the expression **4s**, where **s** is the number of hours Ty walks. The given goal of at least **24** kilometers means that the inequality **4s ≥ 24** represents the situation. Dividing both sides of this inequality by **4** gives **s ≥ 6**, which corresponds to a minimum of **6** hours Ty must walk.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID be1cb6f1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: be1cb6f1

x	y
-6	$n + 184$
-3	$n + 92$
0	n

The table shows three values of x and their corresponding values of y , where n is a constant, for the linear relationship between x and y . What is the slope of the line that represents this relationship in the xy -plane?

- A. $-\frac{92}{3}$
- B. $-\frac{3}{92}$
- C. $\frac{n+92}{-3}$
- D. $\frac{2n-92}{3}$

ID: be1cb6f1 Answer

Correct Answer: A

Rationale

Choice A is correct. The slope, m , of a line in the xy -plane can be found using two points on the line, (x_1, y_1) and (x_2, y_2) , and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Based on the given table, the line representing the relationship between x and y in the xy -plane passes through the points $(-6, n + 184)$, $(-3, n + 92)$, and $(0, n)$, where n is a constant. Substituting two of these points, $(-3, n + 92)$ and $(0, n)$, for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{n - (n + 92)}{0 - (-3)}$, which is equivalent to $m = \frac{n - n - 92}{0 + 3}$, or $m = -\frac{92}{3}$. Therefore, the slope of the line that represents this relationship in the xy -plane is $-\frac{92}{3}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID cba73ea2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: cba73ea2

$$\begin{aligned}y &= 3x + 9 \\ 3y &= 8x - 6\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

- A. -123
- B. -33
- C. 3
- D. 57

ID: cba73ea2 Answer

Correct Answer: D

Rationale

Choice D is correct. The first equation in the given system of equations defines y as $3x + 9$. Substituting $3x + 9$ for y in the second equation in the given system of equations yields $3(3x + 9) = 8x - 6$. Applying the distributive property on the left-hand side of this equation yields $9x + 27 = 8x - 6$. Subtracting $8x$ from both sides of this equation yields $x + 27 = -6$. Subtracting 27 from both sides of this equation yields $x = -33$. Substituting -33 for x in the first equation of the given system of equations yields $y = 3(-33) + 9$, or $y = -90$. Substituting -33 for x and -90 for y into the expression $x - y$ yields $-33 - (-90)$, or 57 .

Choice A is incorrect. This is the value of $x + y$, not $x - y$.

Choice B is incorrect. This is the value of x , not $x - y$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e348737d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: e348737d

For groups of **25** or more people, a museum charges **\$21** per person for the first **25** people and **\$14** for each additional person. Which function f gives the total charge, in dollars, for a tour group with n people, where $n \geq 25$?

- A. $f(n) = 14n + 175$
- B. $f(n) = 14n + 525$
- C. $f(n) = 35n - 350$
- D. $f(n) = 14n + 21$

ID: e348737d Answer

Correct Answer: A

Rationale

Choice A is correct. A tour group with n people, where $n \geq 25$, can be split into two subgroups: the first **25** people and the additional $n - 25$ people. Since the museum charges **\$21** per person for the first **25** people and **\$14** for each additional person, the charge for the first **25** people is **\$21(25)** and the charge for the additional $n - 25$ people is **\$14($n - 25$)**. Therefore, the total charge, in dollars, is given by the function $f(n) = 21(25) + 14(n - 25)$, or $f(n) = 14n + 175$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e4c9754f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: e4c9754f

What is the slope of the graph of $10x - 5y = -12$ in the xy -plane?

- A. -2
- B. $-\frac{5}{6}$
- C. $\frac{5}{6}$
- D. 2

ID: e4c9754f Answer

Correct Answer: D

Rationale

Choice D is correct. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy -plane and $(0, b)$ is the y -intercept. Subtracting $10x$ from each side of the given equation, $10x - 5y = -12$, yields $-5y = -10x - 12$. Dividing each side of this equation by -5 yields $y = 2x + \frac{12}{5}$. This equation is in the form $y = mx + b$, where $m = 2$. Therefore, the slope of the graph of the given equation in the xy -plane is 2 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8b6409ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 8b6409ac

A mixture consisting of only vitamin D and calcium has a total mass of **150** grams. The mass of vitamin D in the mixture is **50** grams. What is the mass, in grams, of calcium in the mixture?

- A. **200**
- B. **150**
- C. **100**
- D. **50**

ID: 8b6409ac Answer

Correct Answer: C

Rationale

Choice C is correct. Let *d* represent the mass, in grams, of vitamin D in the mixture, and let *c* represent the mass, in grams, of calcium in the mixture. It's given that the mixture consists of only vitamin D and calcium and that the total mass of the mixture is **150** grams. Therefore, the equation $d + c = 150$ represents this situation. It's also given that the mass of vitamin D in the mixture is **50** grams. Substituting **50** for *d* in the equation $d + c = 150$ yields $50 + c = 150$. Subtracting **50** from both sides of this equation yields $c = 100$. Therefore, the mass of calcium in the mixture is **100** grams.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the total mass, in grams, of the mixture, not the mass, in grams, of calcium in the mixture.

Choice D is incorrect. This is the mass, in grams, of vitamin D in the mixture, not the mass, in grams, of calcium in the mixture.

Question Difficulty: Easy

Question ID 6b2a1159

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 6b2a1159

A chemist combines water and acetic acid to make a mixture with a volume of **56 milliliters (mL)**. The volume of acetic acid in the mixture is **10 mL**. What is the volume of water, in **mL**, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

ID: 6b2a1159 Answer

Correct Answer: 46

Rationale

The correct answer is **46**. It's given that a chemist combines water and acetic acid to make a mixture with a volume of **56** milliliters (mL) and that the volume of acetic acid in the mixture is **10** mL. Let x represent the volume of water, in mL, in the mixture. The equation $x + 10 = 56$ represents this situation. Subtracting **10** from both sides of this equation yields $x = 46$. Therefore, the volume of water, in mL, in the mixture is **46**.

Question Difficulty: Medium

Question ID 3ce8c43b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 3ce8c43b

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by **9.10** kelvins, by how much did the temperature increase, in degrees Fahrenheit?

- A. **16.38**
- B. **48.38**
- C. **475.29**
- D. **507.29**

ID: 3ce8c43b Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function $F(x) = \frac{9}{5}(x - 273.15) + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by **9.10** kelvins means that the value of x increased by **9.10** kelvins. It follows that an increase in x by **9.10** increases $F(x)$ by $\frac{9}{5}(9.10)$, or **16.38**. Therefore, if a temperature increased by **9.10** kelvins, the temperature increased by **16.38** degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 904f3cac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 904f3cac

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy -plane and passes through the point $(0, 5)$. Which equation defines line j ?

- A. $y = 6x + 5$
- B. $y = -5x + 5$
- C. $y = -6x + 5$
- D. $y = 5x + 5$

ID: 904f3cac Answer

Correct Answer: A

Rationale

Choice A is correct. An equation that defines a line in the xy -plane can be written as $y = mx + b$, where m is the slope of the line and $(0, b)$ is its y -intercept. It's given that line k is defined by the equation $y = 6x + 4$; therefore, the slope of line k is 6 . Since line j is parallel to line k in the xy -plane and parallel lines have equal slopes, it follows that the slope of line j is 6 . It's also given that line j passes through the point $(0, 5)$, which is its y -intercept. Substituting 6 for m and 5 for b in the equation $y = mx + b$ yields $y = 6x + 5$. Therefore, the equation $y = 6x + 5$ defines line j .

Choice B is incorrect. This equation defines a line that has a slope of -5 , not 6 .

Choice C is incorrect. This equation defines a line that has a slope of -6 , not 6 .

Choice D is incorrect. This equation defines a line that has a slope of 5 , not 6 .

Question Difficulty: Medium

Question ID db849464

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: db849464

If $2(5x) = 6$, what is the value of $5x$?

- A. $\frac{4}{5}$
- B. $\frac{5}{3}$
- C. **3**
- D. **4**

ID: db849464 Answer

Correct Answer: C

Rationale

Choice C is correct. Dividing both sides of the given equation by **2** yields $5x = 3$. Therefore, the value of $5x$ is **3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b2d83661

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b2d83661

A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

- A. $f(x) = 60x + 100$
- B. $f(x) = 60x + 220$
- C. $f(x) = 80x$
- D. $f(x) = 80x + 220$

ID: b2d83661 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. Let n represent the hourly fee for each additional hour after the first two hours. Since it's given that x is the number of hours of repair, it follows that the charge generated by the hourly fee after the first two hours can be represented by the expression $n(x - 2)$. Therefore, the total cost, in dollars, for x hours of repair is $f(x) = 220 + n(x - 2)$. It's given that the total cost for 5 hours of repair is \$400. Substituting 5 for x and 400 for $f(x)$ into the equation $f(x) = 220 + n(x - 2)$ yields $400 = 220 + n(5 - 2)$, or $400 = 220 + 3n$. Subtracting 220 from both sides of this equation yields $180 = 3n$. Dividing both sides of this equation by 3 yields $n = 60$. Substituting 60 for n in the equation $f(x) = 220 + n(x - 2)$ yields $f(x) = 220 + 60(x - 2)$, which is equivalent to $f(x) = 220 + 60x - 120$, or $f(x) = 60x + 100$. Therefore, the total cost, in dollars, for x hours of repair is $f(x) = 60x + 100$.

Choice B is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges \$340, rather than \$220, for the first two hours of repair.

Choice C is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges \$160, rather than \$220, for the first two hours of repair, and an hourly fee of \$80, rather than \$60, after the first two hours.

Choice D is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges \$380, rather than \$220, for the first two hours of repair, and an hourly fee of \$80, rather than \$60, after the first two hours.

Question Difficulty: Hard

Question ID aae8b8c7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: aae8b8c7

A shipment consists of **5**-pound boxes and **10**-pound boxes with a total weight of **220** pounds. There are **13** **10**-pound boxes in the shipment. How many **5**-pound boxes are in the shipment?

- A. **5**
- B. **10**
- C. **13**
- D. **18**

ID: aae8b8c7 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the shipment consists of **5**-pound boxes and **10**-pound boxes with a total weight of **220** pounds. Let *x* represent the number of **5**-pound boxes and *y* represent the number of **10**-pound boxes in the shipment. Therefore, the equation **5*x* + 10*y* = 220** represents this situation. It's given that there are **13** **10**-pound boxes in the shipment. Substituting **13** for *y* in the equation **5*x* + 10*y* = 220** yields **5*x* + 10(13) = 220**, or **5*x* + 130 = 220**. Subtracting **130** from both sides of this equation yields **5*x* = 90**. Dividing both sides of this equation by **5** yields **18**. Thus, there are **18** **5**-pound boxes in the shipment.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of **10**-pound boxes in the shipment.

Question Difficulty: Easy

Question ID 12a4a51b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 12a4a51b

A proposal for a new library was included on an election ballot. A radio show stated that **3** times as many people voted in favor of the proposal as people who voted against it. A social media post reported that **15,000** more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

- A. **7,500**
- B. **15,000**
- C. **22,500**
- D. **45,000**

ID: 12a4a51b Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a radio show stated that **3** times as many people voted in favor of the proposal as people who voted against it. Let **x** represent the number of people who voted against the proposal. It follows that **$3x$** is the number of people who voted in favor of the proposal and **$3x - x$** , or **$2x$** , is how many more people voted in favor of the proposal than voted against it. It's also given that a social media post reported that **15,000** more people voted in favor of the proposal than voted against it. Thus, **$2x = 15,000$** . Since **$2x = 15,000$** , the value of **x** must be half of **15,000**, or **7,500**. Therefore, **7,500** people voted against the proposal.

Choice B is incorrect. This is how many more people voted in favor of the proposal than voted against it, not the number of people who voted against the proposal.

Choice C is incorrect. This is the number of people who voted in favor of the proposal, not the number of people who voted against the proposal.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 32112ba4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 32112ba4

If $2x = 12$, what is the value of $9x$?

ID: 32112ba4 Answer

Correct Answer: 54

Rationale

The correct answer is **54**. Dividing both sides of the given equation by **2** yields $x = 6$. Multiplying both sides of this equation by **9** yields $9x = 54$. Thus, the value of $9x$ is **54**.

Question Difficulty: Easy

Question ID 63a9331e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 63a9331e

For the linear function h , the graph of $y = h(x)$ in the xy -plane passes through the points $(7, 21)$ and $(9, 25)$. Which equation defines h ?

- A. $h(x) = \frac{1}{2}x - \frac{7}{2}$
- B. $h(x) = 2x + 7$
- C. $h(x) = 7x + 21$
- D. $h(x) = 9x + 25$

ID: 63a9331e Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph of the linear function h , where $y = h(x)$, passes through the points $(7, 21)$ and $(9, 25)$ in the xy -plane. An equation defining h can be written in the form $y = mx + b$, where $y = h(x)$, m represents the slope of the graph in the xy -plane, and b represents the y -coordinate of the y -intercept of the graph. The slope can be found using any two points, (x_1, y_1) and (x_2, y_2) , and the formula $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$. Substituting $(7, 21)$ and $(9, 25)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{25 - 21}{9 - 7}$, which is equivalent to $m = \frac{4}{2}$, or $m = 2$. Substituting 2 for m and $(7, 21)$ for (x, y) in the equation $y = mx + b$ yields $21 = (2)(7) + b$, or $21 = 14 + b$. Subtracting 14 from each side of this equation yields $7 = b$. Substituting 2 for m and 7 for b in the equation $y = mx + b$ yields $y = 2x + 7$. Since $y = h(x)$, it follows that the equation that defines h is $h(x) = 2x + 7$.

Choice A is incorrect. For this function, the graph of $y = h(x)$ in the xy -plane would pass through $(7, 0)$, not $(7, 21)$, and $(9, 1)$, not $(9, 25)$.

Choice C is incorrect. For this function, the graph of $y = h(x)$ in the xy -plane would pass through $(7, 70)$, not $(7, 21)$, and $(9, 84)$, not $(9, 25)$.

Choice D is incorrect. For this function, the graph of $y = h(x)$ in the xy -plane would pass through $(7, 88)$, not $(7, 21)$, and $(9, 106)$, not $(9, 25)$.

Question Difficulty: Medium

Question ID 8ee3ff82

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 8ee3ff82

The cost to rent a commercial fishing boat from a certain company is **\$950** for the first **2** hours and an additional **\$50** per hour for each hour after the first **2** hours. If the total cost to rent the commercial fishing boat from the company for t hours, where $t > 2$, is **\$1,100**, which equation represents this situation?

- A. $950(t - 2) + 50t = 1,100$
- B. $950(2t) + 50t = 1,100$
- C. $950 + 50(t - 2) = 1,100$
- D. $950 + 50(2t) = 1,100$

ID: 8ee3ff82 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the cost to rent a commercial fishing boat is **\$950** for the first **2** hours and an additional **\$50** per hour for each hour after the first **2** hours. It's also given that t represents the total number of hours and $t > 2$. Therefore, the number of additional hours after the first **2** hours can be represented with the expression $t - 2$. The cost for these additional hours is **\$50** per hour, so the cost for the additional hours can be represented by the expression $50(t - 2)$. The total cost can be calculated by adding the cost for the first **2** hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 2)$. It's also given that the total cost to rent the commercial fishing boat from the company for t hours is **\$1,100**. Thus, the equation that represents this situation is $950 + 50(t - 2) = 1,100$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 7a495244

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 7a495244

Line r is defined by the equation $4x - 9y = 3$. Line s is parallel to line r in the xy -plane. What is the slope of line s ?

- A. $\frac{9}{4}$
- B. $\frac{4}{9}$
- C. -4
- D. -9

ID: 7a495244 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that line s is parallel to line r in the xy -plane. This means that the slope of line r is equal to the slope of line s . Line r is defined by the equation $4x - 9y = 3$. This equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line and b represents the y -coordinate of the y -intercept of the line. Subtracting $4x$ from both sides of the equation $4x - 9y = 3$ yields $-9y = -4x + 3$. Dividing both sides of this equation by -9 yields $y = \frac{4}{9}x - \frac{1}{3}$. Therefore, the slope of line r is $\frac{4}{9}$. Since parallel lines have equal slopes, the slope of line s is also $\frac{4}{9}$.

Choice A is incorrect. This is the reciprocal of the slope of line s , not the slope of line s .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID f08b80b6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: f08b80b6

$$5y = 10x + 11$$

$$-5y = 5x - 21$$

The solution to the given system of equations is (x, y) . What is the value of $30x$?

ID: f08b80b6 Answer

Correct Answer: 20

Rationale

The correct answer is **20**. Adding the first equation to the second equation in the given system yields $5y - 5y = 10x + 5x + 11 - 21$, or $0 = 15x - 10$. Adding **10** to both sides of this equation yields $10 = 15x$. Multiplying both sides of this equation by **2** yields $20 = 30x$. Therefore, the value of $30x$ is **20**.

Question Difficulty: Hard

Question ID 57622fc3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 57622fc3

The equation $x + y = 1,440$ represents the number of minutes of daylight (between sunrise and sunset), x , and the number of minutes of non-daylight, y , on a particular day in Oak Park, Illinois. If this day has **670** minutes of daylight, how many minutes of non-daylight does it have?

- A. **670**
- B. **770**
- C. **1,373**
- D. **1,440**

ID: 57622fc3 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the equation $x + y = 1,440$ represents the number of minutes of daylight, x , and the number of minutes of non-daylight, y , on a particular day in Oak Park, Illinois. It’s also given that this day has **670** minutes of daylight. Substituting **670** for x in the equation $x + y = 1,440$ yields $670 + y = 1,440$. Subtracting **670** from both sides of this equation yields $y = 770$. Therefore, this day has **770** minutes of non-daylight.

Choice A is incorrect. This is the number of minutes of daylight, not non-daylight, on this day.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the total number of minutes of daylight and non-daylight.

Question Difficulty: Easy

Question ID cba7456d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: cba7456d

If $\frac{x-5}{7} = \frac{x-5}{9}$, the value of $x - 5$ is between which of the following pairs of values?

- A. -9 and -7
- B. -3 and 3
- C. 4.5 and 5.5
- D. 6.75 and 9.25

ID: cba7456d Answer

Correct Answer: B

Rationale

Choice B is correct. Multiplying both sides of the given equation by $(7)(9)$, or **63**, yields $(63)\left(\frac{x-5}{7}\right) = (63)\left(\frac{x-5}{9}\right)$, or $9(x - 5) = 7(x - 5)$. Subtracting $7(x - 5)$ from both sides of this equation yields $2(x - 5) = 0$. Dividing both sides of this equation by **2** yields $x - 5 = 0$. Therefore, if $\frac{x-5}{7} = \frac{x-5}{9}$, then the value of $x - 5$ is **0**. It follows that of the given choices, the value of $x - 5$ is between -3 and **3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5f546ecc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 5f546ecc

If $5(x + 4) = 4(x + 4) + 29$, what is the value of $x + 4$?

- A. -4
- B. 25
- C. 29
- D. 33

ID: 5f546ecc Answer

Correct Answer: C

Rationale

Choice C is correct. Subtracting $4(x + 4)$ from both sides of the given equation yields $1(x + 4) = 29$, or $x + 4 = 29$. Therefore, the value of $x + 4$ is 29 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $x + 4$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3f613d99

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 3f613d99

$$\begin{aligned}x + 2y &= 6 \\ x - 2y &= 4\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. 2.5
- B. 5
- C. 6
- D. 10

ID: 3f613d99 Answer

Correct Answer: B

Rationale

Choice B is correct. Adding the first equation to the second equation in the given system yields $(x + 2y) + (x - 2y) = 6 + 4$, or $(x + x) + (2y - 2y) = 10$. Combining like terms in this equation yields $2x = 10$. Dividing both sides of this equation by 2 yields $x = 5$. Thus, the value of x is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $2x$, not x .

Question Difficulty: Hard

Question ID 82feeb71

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 82feeb71

For the function f , $f(cx) = x - 8$ for all values of x , where c is a positive constant. If $f(2) = 35$, what is the value of c ?

ID: 82feeb71 Answer

Correct Answer: .0465, 2/43

Rationale

The correct answer is $\frac{2}{43}$. It's given that $f(cx) = x - 8$ for all values of x , where c is a positive constant, and $f(2) = 35$. Therefore, for the given function f , $cx = 2$. Dividing both sides of this equation by c yields $x = \frac{2}{c}$. Substituting $\frac{2}{c}$ for x in the equation $f(cx) = x - 8$ yields $f(\frac{2c}{c}) = \frac{2}{c} - 8$, or $f(2) = \frac{2}{c} - 8$. Since it's given that $f(2) = 35$, substituting 35 for $f(2)$ yields $35 = \frac{2}{c} - 8$. Adding 8 to both sides of this equation yields $43 = \frac{2}{c}$. Multiplying both sides of this equation by c yields $43c = 2$. Dividing both sides of this equation by 43 yields $c = \frac{2}{43}$. Note that 2/43, .0465, 0.046, and 0.047 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID d4915b30

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: d4915b30

In the xy -plane, line t passes through the points $(0, 9)$ and $(1, 17)$. Which equation defines line t ?

- A. $y = \frac{1}{8}x + 9$
- B. $y = x + \frac{1}{8}$
- C. $y = x + 8$
- D. $y = 8x + 9$

ID: d4915b30 Answer

Correct Answer: D

Rationale

Choice D is correct. An equation defining a line in the xy -plane can be written in the form $y = mx + b$, where m represents the slope and $(0, b)$ represents the y -intercept of the line. It's given that line t passes through the point $(0, 9)$; therefore, $b = 9$. The slope, m , of a line can be found using any two points on the line, (x_1, y_1) and (x_2, y_2) , and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(0, 9)$ and $(1, 17)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{17 - 9}{1 - 0}$, or $m = 8$. Substituting 8 for m and 9 for b in the equation $y = mx + b$ yields $y = 8x + 9$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 5897227e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 5897227e

$$f(x) = 2x + 244$$

The given function f represents the perimeter, in **centimeters (cm)**, of a rectangle with a length of x **cm** and a fixed width. What is the width, in **cm**, of the rectangle?

- A. 2
- B. 122
- C. 244
- D. 488

ID: 5897227e Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = 2x + 244$ represents the perimeter, in **centimeters (cm)**, of a rectangle with a length of x **cm** and a fixed width. If w represents a fixed width, in **cm**, then the perimeter, in **cm**, of a rectangle with a length of x **cm** and a fixed width of w **cm** can be given by the function $f(x) = 2x + 2w$. Therefore, $2x + 2w = 2x + 244$. Subtracting $2x$ from both sides of this equation yields $2w = 244$. Dividing both sides of this equation by 2 yields $w = 122$. Therefore, the width, in **cm**, of the rectangle is **122**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 1f65f0a8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 1f65f0a8

The function f is defined by $f(x) = \frac{9}{7}x + \frac{8}{7}$. For what value of x does $f(x) = 5$?

ID: 1f65f0a8 Answer

Correct Answer: 3

Rationale

The correct answer is **3**. Substituting **5** for $f(x)$ in the given function yields $5 = \frac{9}{7}x + \frac{8}{7}$. Multiplying each side of this equation by **7** yields $7(5) = 7(\frac{9}{7}x + \frac{8}{7})$, or $35 = 9x + 8$. Subtracting **8** from each side of this equation yields $27 = 9x$. Dividing each side of this equation by **9** yields $3 = x$. Therefore, $f(x) = 5$ when the value of x is **3**.

Question Difficulty: Medium

Question ID 6e7e28ad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 6e7e28ad

The function f is defined by $f(x) = 6.2x$. What is the value of $f(x)$ when $x = 10$?

- A. 72
- B. 62
- C. 17.2
- D. 16.2

ID: 6e7e28ad Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 6.2x$. The value of $f(x)$ when $x = 10$ can be found by substituting 10 for x in $f(x) = 6.2x$, which yields $f(10) = 6.2(10)$, or $f(10) = 62$. Therefore, when $x = 10$, the value of $f(x)$ is 62.

Choice A is incorrect. This is the value of $f(10) + 10$, not the value of $f(x)$ when $x = 10$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $6.2 + 10$, not the value of $f(x)$ when $x = 10$.

Question Difficulty: Easy

Question ID b6e45cd0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b6e45cd0

Sam has a flag with an area of **110 square inches (in²)**. Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by **64 in²**. Which equation gives the flag's total area ***y***, in square inches, after Sam sews on ***x*** pieces of nylon?

- A. ***y* = 64*x***
- B. ***y* = 64 + 110*x***
- C. ***y* = 110 + 64*x***
- D. ***y* = 174*x***

ID: b6e45cd0 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that Sam has a flag with an area of **110 square inches (in²)** and that each piece of colored nylon Sam sews on increases the flag's area by **64 in²**. If Sam sews on ***x*** pieces of nylon and each piece increases the area by **64 in²**, it follows that the expression **64*x*** represents the total increase in area, in square inches, from sewing on ***x*** pieces of nylon. Adding this increase in area, **64*x* in²**, to the original area of the flag, **110 in²**, gives the total area of the flag. Therefore, the equation that gives the flag's total area ***y***, in square inches, after Sam sews on ***x*** pieces of nylon is ***y* = 110 + 64*x***.

Choice A is incorrect. This equation represents the total area, in square inches, of ***x*** pieces of colored nylon.

Choice B is incorrect and may result from conceptual or calculation errors.

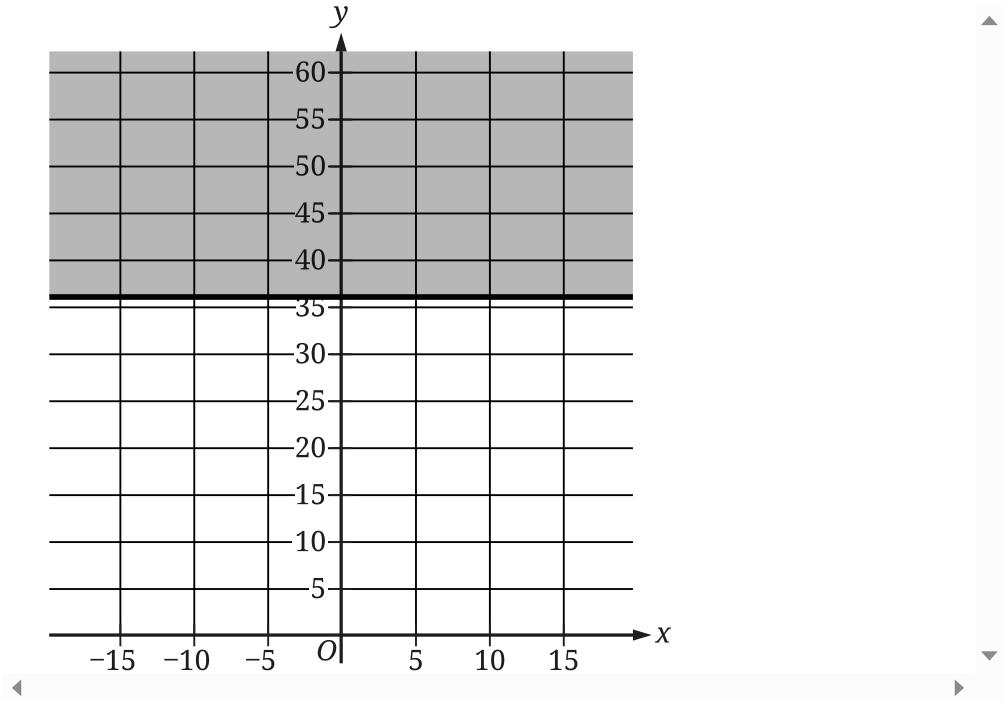
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 393baf87

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 393baf87



The shaded region shown in the graph represents all the solutions to which inequality?

- A. $x \leq 36$
- B. $x \geq 36$
- C. $y \leq 36$
- D. $y \geq 36$

ID: 393baf87 Answer

Correct Answer: D

Rationale

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in the form $y = b$, where $(0, b)$ is the y -intercept of the line in the xy -plane. For the graph shown in the xy -plane, the boundary line passes through the points $(-15, 36)$ and $(0, 36)$. Thus, the equation of the boundary line is $y = 36$. Since the shaded region represents all the points on and above this boundary line, it follows that the graph represents all the solutions to the inequality $y \geq 36$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID e0bf6e4e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: e0bf6e4e

What is the slope of the graph of $y = \frac{1}{3}(29x + 10) + 5x$ in the xy -plane?

ID: e0bf6e4e Answer

Correct Answer: 14.66, 14.67, 44/3

Rationale

The correct answer is $\frac{44}{3}$. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy -plane and $(0, b)$ is the y -intercept. Distributing the $\frac{1}{3}$ in the equation $y = \frac{1}{3}(29x + 10) + 5x$ yields $y = \frac{29}{3}x + \frac{10}{3} + 5x$. Combining like terms on the right-hand side of this equation yields $y = \frac{44}{3}x + \frac{10}{3}$. This equation is in the form $y = mx + b$, where $m = \frac{44}{3}$ and $b = \frac{10}{3}$. Therefore, the slope of the graph of the given equation in the xy -plane is $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 224d2503

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 224d2503

The table gives the number of hours, h , of labor and a plumber’s total charge $f(h)$, in dollars, for two different jobs.

h	$f(h)$
1	155
3	285

There is a linear relationship between h and $f(h)$. Which equation represents this relationship?

- A. $f(h) = 25h + 130$
- B. $f(h) = 130h + 25$
- C. $f(h) = 65h + 90$
- D. $f(h) = 90h + 65$

ID: 224d2503 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that there is a linear relationship between a plumber's hours of labor, h , and the plumber's total charge $f(h)$, in dollars. It follows that the relationship can be represented by an equation of the form $f(h) = mh + b$, where m is the rate of change of the function f and b is a constant. The rate of change of f can be calculated by dividing the difference in two values of $f(h)$ by the difference in the corresponding values of h . Based on the values given in the table, the rate of change of f is $\frac{285-155}{3-1}$, or 65. Substituting 65 for m in the equation $f(h) = mh + b$ yields $f(h) = 65h + b$. The value of b can be found by substituting any value of h and its corresponding value of $f(h)$ for h and $f(h)$, respectively, in this equation. Substituting 1 for h and 155 for $f(h)$ yields $155 = 65(1) + b$, or $155 = 65 + b$. Subtracting 65 from both sides of this equation yields $90 = b$. Substituting 90 for b in the equation $f(h) = 65h + b$ yields $f(h) = 65h + 90$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 4a58cb67

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 4a58cb67

$9r = 7(r + 7)$

What value of r is the solution to the given equation?

ID: 4a58cb67 Answer

Correct Answer: 24.5, 49/2

Rationale

The correct answer is $\frac{49}{2}$. It's given that $9r = 7(r + 7)$. Applying the distributive property on the right-hand side of this equation yields $9r = 7r + 49$. Subtracting $7r$ from both sides of this equation yields $2r = 49$. Dividing both sides of this equation by 2 yields $r = \frac{49}{2}$. Therefore, the solution to the given equation is $\frac{49}{2}$. Note that 49/2 and 24.5 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 02e51c3d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 02e51c3d

A manager is responsible for ordering supplies for a shaved ice shop. The shop's inventory starts with **4,500** paper cups, and the manager estimates that **70** of these paper cups are used each day. Based on this estimate, in how many days will the supply of paper cups reach **1,700**?

- A. 20
- B. 40
- C. 60
- D. 80

ID: 02e51c3d Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the shop's inventory starts with **4,500** paper cups and that the manager estimates that **70** of these paper cups are used each day. Let x represent the number of days in which the estimated supply of paper cups will reach **1,700**. The equation $4,500 - 70x = 1,700$ represents this situation. Subtracting **4,500** from both sides of this equation yields $-70x = -2,800$. Dividing both sides of this equation by -70 yields $x = 40$. Therefore, based on this estimate, the supply of paper cups will reach **1,700** in **40** days.

Choice A is incorrect. After **20** days, the estimated supply of paper cups would be $4,500 - 70(20)$, or **3,100** cups, not **1,700** cups.

Choice C is incorrect. After **60** days, the estimated supply of paper cups would be $4,500 - 70(60)$, or **300** cups, not **1,700** cups.

Choice D is incorrect. After **80** days, the estimated supply of paper cups would be $4,500 - 70(80)$, or $-1,100$ cups, which isn't possible.

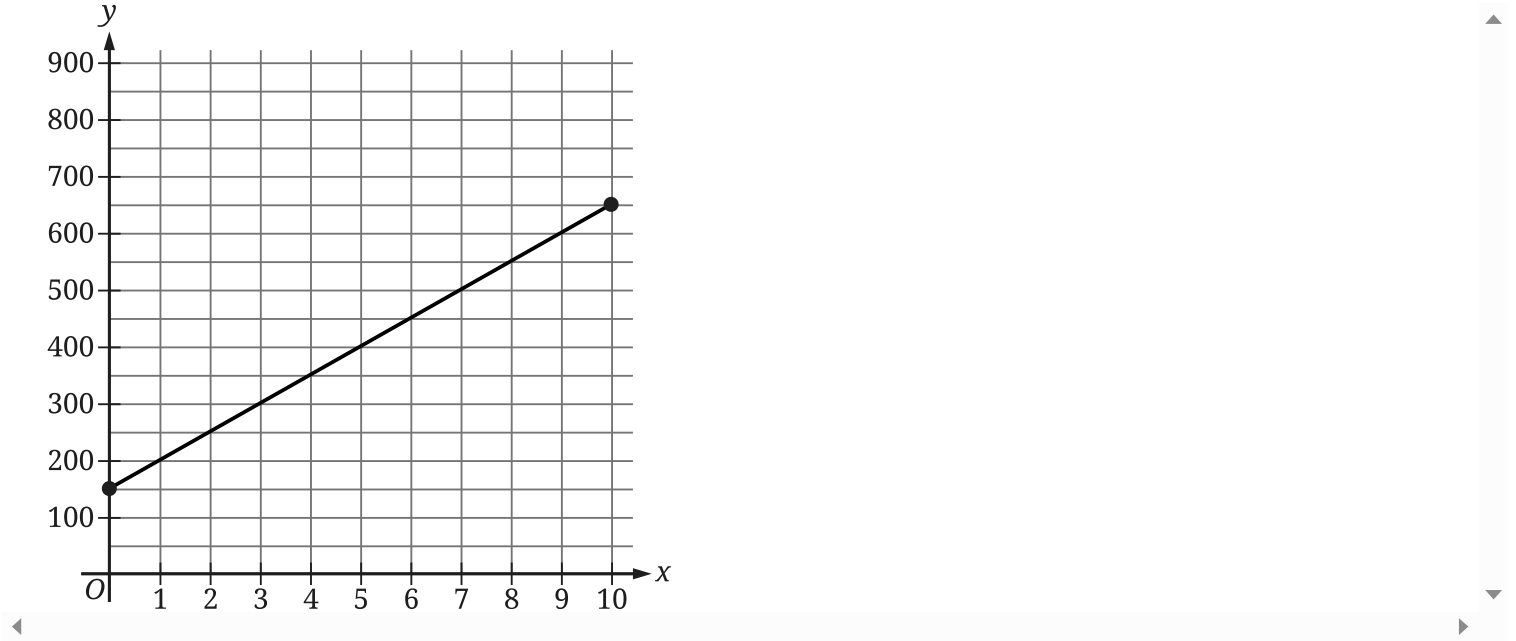
Question Difficulty: Medium

Question ID c25f8f4d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: c25f8f4d

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y , in meters, Mary had walked on the trail x minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the **10** minutes after leaving the trail sign?

ID: c25f8f4d Answer

Correct Answer: 500

Rationale

The correct answer is **500**. It's given that the graph shows the total distance y , in meters, Mary had walked on the trail x minutes after leaving the trail sign. The graph represents a linear relationship between time and distance, so the distance walked in a given time interval corresponds to the change in the y -values over that interval. At $x = 0$, the graph shows a y -value of **150**, and at $x = 10$, the graph shows a y -value of **650**. The distance Mary walked in the **10** minutes after leaving the trail sign is the difference between these y -values, which can be written as **650** $-$ **150**, or **500**. Therefore, the distance, in meters, Mary walked in the **10** minutes after leaving the trail sign is **500**.

Question Difficulty: Hard

Question ID 32be8a82

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 32be8a82

In the xy -plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

- A. The x -axis
- B. The region where $x > 0$ and $y > 0$
- C. The region where $x < 0$ and $y < 0$
- D. The region where $x < 0$ and $y > 0$

ID: 32be8a82 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the solution set consists of points in the xy -plane where $5x - 7y > 35$, which can also be written as $5x + (-7y) > 35$. If (x, y) is a point in the region where $x < 0$ and $y > 0$, then x is a negative number and y is a positive number. If x is a negative number, then the value of $5x$ is a negative number. If y is a positive number, then the value of $-7y$ is a negative number. It follows that the expression $5x + (-7y)$ is the sum of two negative numbers and is therefore negative. If the value of $5x + (-7y)$ is negative, then the value of $5x + (-7y)$ cannot be greater than 35 . Thus, the region where $x < 0$ and $y > 0$ does not contain any points that are part of the solution set to $5x - 7y > 35$.

Choice A is incorrect. The x -axis contains the point $(10, 0)$, which is part of the solution set to $5x - 7y > 35$, since $5(10) - 7(0) > 35$, or $50 > 35$, is true.

Choice B is incorrect. The region where $x > 0$ and $y > 0$ contains the point $(10, 1)$, which is part of the solution set to $5x - 7y > 35$, since $5(10) - 7(1) > 35$, or $43 > 35$, is true.

Choice C is incorrect. The region where $x < 0$ and $y < 0$ contains the point $(-1, -10)$, which is part of the solution set to $5x - 7y > 35$, since $5(-1) - 7(-10) > 35$, or $65 > 35$, is true.

Question Difficulty: Hard

Question ID 31499031

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 31499031

$$5G + 45R = 380$$

At a school fair, students can win colored tokens that are worth a different number of points depending on the color. One student won G green tokens and R red tokens worth a total of 380 points. The given equation represents this situation. How many more points is a red token worth than a green token?

ID: 31499031 Answer

Correct Answer: 40

Rationale

The correct answer is 40 . It's given that $5G + 45R = 380$, where G is the number of green tokens and R is the number of red tokens won by one student and these tokens are worth a total of 380 points. Since the equation represents the situation where the student won points with green tokens and red tokens for a total of 380 points, each term on the left-hand side of the equation represents the number of points won for one of the colors. Since the coefficient of G in the given equation is 5 , a green token must be worth 5 points. Similarly, since the coefficient of R in the given equation is 45 , a red token must be worth 45 points. Therefore, a red token is worth $45 - 5$ points, or 40 points, more than a green token.

Question Difficulty: Hard

Question ID 139fec63

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 139fec63

The cost to rent a bus from Company X is \$950 for the first 3 hours and an additional \$50 per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is \$1,150, which equation represents this situation?

- A. $950(t - 3) + 50t = 1,150$
- B. $950(3t) + 50t = 1,150$
- C. $950 + 50(t - 3) = 1,150$
- D. $950 + 50(3t) = 1,150$

ID: 139fec63 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the cost to rent a bus is \$950 for the first 3 hours and an additional \$50 per hour for each hour after the first 3 hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first 3 hours can be represented by the expression $t - 3$. The cost for these additional hours is \$50 per hour, so the cost for the additional hours can be represented by the expression $50(t - 3)$. The total cost can be calculated by adding the cost for the first 3 hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 3)$. It's also given that the total cost to rent the bus from the company for t hours is \$1,150. Thus, the equation that represents this situation is $950 + 50(t - 3) = 1,150$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID f235b54c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: f235b54c

x	y
$-2s$	24
$-s$	21
s	15

The table shows three values of x and their corresponding values of y , where s is a constant. There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $sx + 3y = 18s$
- B. $3x + sy = 18s$
- C. $3x + sy = 18$
- D. $sx + 3y = 18$

ID: f235b54c Answer

Correct Answer: B

Rationale

Choice B is correct. The linear relationship between x and y can be represented by an equation of the form $y - y_1 = m(x - x_1)$, where m is the slope of the graph of the equation in the xy -plane and (x_1, y_1) is a point on the graph. The slope of a line can be found using two points on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Each value of x and its corresponding value of y in the table can be represented by a point (x, y) . Substituting the points $(-s, 21)$ and $(s, 15)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{15 - 21}{s - (-s)}$, which gives $m = \frac{-6}{2s}$, or $m = -\frac{3}{s}$. Substituting $-\frac{3}{s}$ for m and the point $(s, 15)$ for (x_1, y_1) in the equation $y - y_1 = m(x - x_1)$ yields $y - 15 = -\frac{3}{s}(x - s)$. Distributing $-\frac{3}{s}$ on the right-hand side of this equation yields $y - 15 = -\frac{3x}{s} + 3$. Adding 15 to each side of this equation yields $y = -\frac{3x}{s} + 18$. Multiplying each side of this equation by s yields $sy = -3x + 18s$. Adding $3x$ to each side of this equation yields $3x + sy = 18s$. Therefore, the equation $3x + sy = 18s$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 8e338d65

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 8e338d65

$5(t + 3) - 7(t + 3) = 38$

What value of t is the solution to the given equation?

ID: 8e338d65 Answer

Correct Answer: -22

Rationale

The correct answer is -22 . The given equation can be rewritten as $-2(t + 3) = 38$. Dividing both sides of this equation by -2 yields $t + 3 = -19$. Subtracting 3 from both sides of this equation yields $t = -22$. Therefore, -22 is the value of t that is the solution to the given equation.

Question Difficulty: Hard

Question ID bf9315c0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: bf9315c0

Gabriella deposits **\$35** in a savings account at the end of each week. At the beginning of the **1st** week of a year there was **\$600** in that savings account. How much money, in dollars, will be in the account at the end of the **4th** week of that year?

- A. **460**
- B. **635**
- C. **639**
- D. **740**

ID: bf9315c0 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that at the beginning of the **1st** week of the year there was **\$600** in a savings account and Gabriella deposits **\$35** in that savings account at the end of each week. Therefore, the amount of money, in dollars, in the savings account at the end of the **4th** week of that year is **$600 + 4(35)$** , or **740**.

Choice A is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the **4th** week if Gabriella withdraws, rather than deposits, **\$35** at the end of each week.

Choice B is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the **1st** week, not the **4th** week.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8bc32738

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 8bc32738

$$y = \frac{1}{3}x - 14$$
$$y = -x + 18$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: 8bc32738 Answer

Correct Answer: 24

Rationale

The correct answer is **24**. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is $y = \frac{1}{3}x - 14$. Substituting $\frac{1}{3}x - 14$ for y in the second equation in the given system yields $\frac{1}{3}x - 14 = -x + 18$. Adding **14** to both sides of this equation yields $\frac{1}{3}x = -x + 32$. Adding x to both sides of this equation yields $\frac{4}{3}x = 32$. Multiplying both sides of this equation by $\frac{3}{4}$ yields $x = 24$.

Question Difficulty: Hard

Question ID 017427ff

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 017427ff

On the first day of a semester, a film club has **90** members. Each day after the first day of the semester, **10** new members join the film club. If no members leave the film club, how many total members will the film club have **4** days after the first day of the semester?

- A. **400**
- B. **130**
- C. **94**
- D. **90**

ID: 017427ff Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the film club has **90** members on the first day of a semester, and **10** new members join the film club each day after the first day of the semester. This means that after **4** days, 4×10 , or **40**, new members will have joined the club. Adding **40** members to the original **90** club members yields **130** members. Thus, the film club will have **130** total members **4** days after the first day of the semester.

Choice A is incorrect. This is the number of members that will have joined the film club **4** days after the first day of the semester if **100** new members, not **10**, join the film club each day.

Choice C is incorrect. This is the number of members the film club will have **4** days after the first day of the semester if **1** new member, not **10**, joins the film club each day.

Choice D is incorrect. This is the number of members the film club has on the first day of the semester.

Question Difficulty: Easy

Question ID fc285932

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: fc285932

The total cost, in dollars, to rent a surfboard consists of a **\$25** service fee and a **\$10** per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of **\$75** to rent the surfboard. Which inequality represents this situation?

- A. $10t \leq 75$
- B. $10 + 25t \leq 75$
- C. $25t \leq 75$
- D. $25 + 10t \leq 75$

ID: fc285932 Answer

Correct Answer: D

Rationale

Choice D is correct. The cost of the rental fee depends on the number of hours the surfboard is rented. Multiplying t hours by **10** dollars per hour yields a rental fee of $10t$ dollars. The total cost of the rental consists of the rental fee plus the **25** dollar service fee, which yields a total cost of $25 + 10t$ dollars. Since the person intends to spend a maximum of **75** dollars to rent the surfboard, the total cost must be at most **75** dollars. Therefore, the inequality $25 + 10t \leq 75$ represents this situation.

Choice A is incorrect. This represents a situation where the rental fee, not the total cost, is at most **75** dollars.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 0ef62ebb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 0ef62ebb

A cleaning service that cleans both offices and homes can clean at most **14** places per day. Which inequality represents this situation, where ***f*** is the number of offices and ***h*** is the number of homes?

- A. $f + h \leq 14$
- B. $f + h \geq 14$
- C. $f - h \leq 14$
- D. $f - h \geq 14$

ID: 0ef62ebb Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the cleaning service cleans both offices and homes, where ***f*** is the number of offices and ***h*** is the number of homes the cleaning service can clean per day. Therefore, the expression **$f + h$** represents the number of places the cleaning service can clean per day. It's also given that the cleaning service can clean at most **14** places per day. Since **$f + h$** represents the number of places the cleaning service can clean per day and the service can clean at most **14** places per day, it follows that the inequality **$f + h \leq 14$** represents this situation.

Choice B is incorrect. This inequality represents a cleaning service that cleans at least **14** places per day.

Choice C is incorrect. This inequality represents a cleaning service that cleans at most **14** more offices than homes per day.

Choice D is incorrect. This inequality represents a cleaning service that cleans at least **14** more offices than homes per day.

Question Difficulty: Medium

Question ID 9559faae

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 9559faae

$$\begin{aligned}y &= -\frac{1}{9}x \\ y &= \frac{1}{2}x\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -9
- B. -7
- C. 0
- D. 2

ID: 9559faae Answer

Correct Answer: C

Rationale

Choice C is correct. It's given by the first equation in the system that $y = -\frac{1}{9}x$. Substituting $-\frac{1}{9}x$ for y in the second equation in the system yields $-\frac{1}{9}x = \frac{1}{2}x$. Multiplying the left-hand side of this equation by $\frac{2}{2}$ and the right-hand side by $\frac{9}{9}$ yields $-\frac{2}{18}x = \frac{9}{18}x$. Adding $\frac{2}{18}x$ to both sides of this equation yields $0 = \frac{11}{18}x$. Multiplying both sides of this equation by $\frac{18}{11}$ yields $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3c94b861

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 3c94b861

A candle is made of **17** ounces of wax. When the candle is burning, the amount of wax in the candle decreases by **1** ounce every **4** hours. If **6** ounces of wax remain in this candle, for how many hours has it been burning?

- A. **3**
- B. **6**
- C. **24**
- D. **44**

ID: 3c94b861 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the candle starts with **17** ounces of wax and has **6** ounces of wax remaining after a period of time has passed. The amount of wax the candle has lost during the time period can be found by subtracting the remaining amount of wax from the amount of wax the candle was made of, which yields **17 – 6** ounces, or **11** ounces. This means the candle loses **11** ounces of wax during that period of time. It’s given that the amount of wax decreases by **1** ounce every **4** hours. If ***h*** represents the number of hours the candle has been burning, it follows that $\frac{1}{4} = \frac{11}{h}$. Multiplying both sides of this equation by **4*h*** yields ***h* = 44**. Therefore, the candle has been burning for **44** hours.

Choice A is incorrect and may result from using the equation $\frac{1}{4} = \frac{h}{11}$ rather than $\frac{1}{4} = \frac{11}{h}$ to represent the situation, and then rounding to the nearest whole number.

Choice B is incorrect. This is the amount of wax, in ounces, remaining in the candle, not the number of hours it has been burning.

Choice C is incorrect and may result from using the equation $\frac{1}{4} = \frac{6}{h}$ rather than $\frac{1}{4} = \frac{11}{h}$ to represent the situation.

Question Difficulty: Medium

Question ID b3ff677c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: b3ff677c

A principal used a total of **25** flags that were either blue or yellow for field day. The principal used **20** blue flags. How many yellow flags were used?

- A. **5**
- B. **20**
- C. **25**
- D. **30**

ID: b3ff677c Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a principal used a total of **25** blue flags and yellow flags. It's also given that of the **25** flags used, **20** flags were blue. Subtracting the number of blue flags used from the total number of flags used results in the number of yellow flags used. It follows that the number of yellow flags used is **25 — 20**, or **5**.

Choice B is incorrect. This is the number of blue flags used.

Choice C is incorrect. This is the total number of flags used.

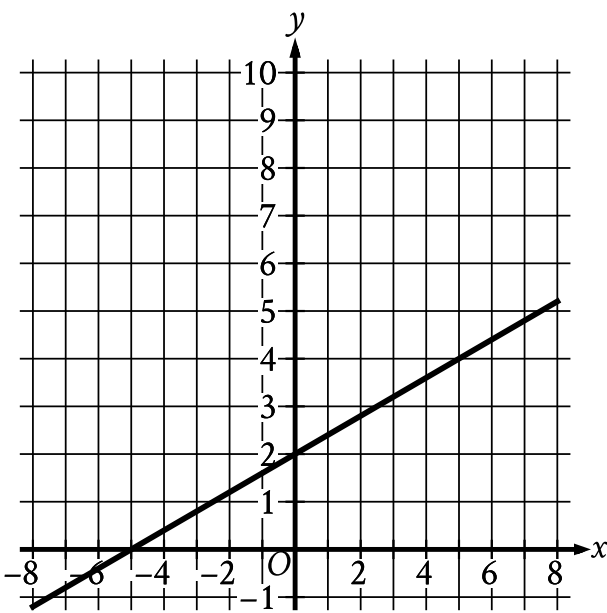
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 1e0a38e7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 1e0a38e7



The graph of the linear function f is shown. What is the y-intercept of the graph of $y = f(x)$?

- A. $(-5, 0)$
- B. $(2, 0)$
- C. $(0, 2)$
- D. $(0, -5)$

ID: 1e0a38e7 Answer

Correct Answer: C

Rationale

Choice C is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The graph of $y = f(x)$ shown intersects the y-axis at the point $(0, 2)$. Therefore, the y-intercept of the graph of $y = f(x)$ is $(0, 2)$.

Choice A is incorrect. This is the x-intercept, not the y-intercept, of the graph of $y = f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 68b9bd32

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 68b9bd32

Line k is defined by $y = -\frac{17}{3}x + 5$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

ID: 68b9bd32 Answer

Correct Answer: .1764, .1765, 3/17

Rationale

The correct answer is $\frac{3}{17}$. It's given that line j is perpendicular to line k in the xy -plane. This means that the slope of line j is the negative reciprocal of the slope of line k . The equation of line k , $y = -\frac{17}{3}x + 5$, is written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept of the line. It follows that the slope of line k is $-\frac{17}{3}$. The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of $-\frac{17}{3}$ is $\frac{-1}{-\frac{17}{3}}$, or $\frac{3}{17}$. Thus, the slope of line j is $\frac{3}{17}$. Note that 3/17, .1764, .1765, and 0.176 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID a18fe14c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: a18fe14c

$y < 6x + 2$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	20
5	32
7	44

B.

x	y
3	16
5	36
7	40

C.

x	y
3	16
5	28
7	40

D.

x	y
3	24
5	36
7	48

ID: a18fe14c Answer

Correct Answer: C

Rationale

Choice C is correct. All the tables in the choices have the same three values of x , so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting **3** for x in the given inequality yields $y < 6(3) + 2$, or $y < 20$. Therefore, when $x = 3$, the corresponding value of y is less than **20**. Substituting **5** for x in the given inequality yields $y < 6(5) + 2$, or $y < 32$. Therefore, when $x = 5$, the corresponding value of y is less than **32**. Substituting **7** for x in the given inequality yields $y < 6(7) + 2$, or $y < 44$. Therefore, when $x = 7$, the corresponding value of y

is less than **44**. For the table in choice C, when $x = 3$, the corresponding value of y is **16**, which is less than **20**; when $x = 5$, the corresponding value of y is **28**, which is less than **32**; when $x = 7$, the corresponding value of y is **40**, which is less than **44**. Therefore, the table in choice C gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. In the table for choice A, when $x = 3$, the corresponding value of y is **20**, which is not less than **20**; when $x = 5$, the corresponding value of y is **32**, which is not less than **32**; when $x = 7$, the corresponding value of y is **44**, which is not less than **44**.

Choice B is incorrect. In the table for choice B, when $x = 5$, the corresponding value of y is **36**, which is not less than **32**.

Choice D is incorrect. In the table for choice D, when $x = 3$, the corresponding value of y is **24**, which is not less than **20**; when $x = 5$, the corresponding value of y is **36**, which is not less than **32**; when $x = 7$, the corresponding value of y is **48**, which is not less than **44**.

Question Difficulty: Hard

Question ID 593de6f7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 593de6f7

How many solutions does the equation $10(15x - 9) = -15(6 - 10x)$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 593de6f7 Answer

Correct Answer: C

Rationale

Choice C is correct. Applying the distributive property to each side of the given equation yields $150x - 90 = -90 + 150x$. Applying the commutative property of addition to the right-hand side of this equation yields $150x - 90 = 150x - 90$. Since the two sides of the equation are equivalent, this equation is true for any value of x . Therefore, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 221b6f45

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 221b6f45

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t , in minutes, it takes Carly to travel a certain distance d , in kilometers. Which equation could represent this linear relationship?

- A. $t = 4d$
- B. $t = \frac{1}{25}d$
- C. $t = 25d$
- D. $t = \frac{1}{4}d$

ID: 221b6f45 Answer

Correct Answer: C

Rationale

Choice C is correct. The average time t , in minutes, it takes Carly to travel a certain distance d , in kilometers, is given in the table. This linear relationship can be represented by an equation in the form $t = ad + b$, where a and b are constants. The table shows that it takes Carly an average time of 8 minutes to travel 0.32 kilometers. Substituting 8 for t and 0.32 for d in the equation $t = ad + b$ yields $8 = 0.32a + b$. Subtracting $0.32a$ from both sides of this equation yields $8 - 0.32a = b$. The table also shows that it takes Carly an average time of 14 minutes to travel 0.56 kilometers. Substituting 14 for t and 0.56 for d in the equation $t = ad + b$ yields $14 = 0.56a + b$. Subtracting $0.56a$ from both sides of this equation yields $14 - 0.56a = b$. Substituting $8 - 0.32a$ for b in this equation yields $14 - 0.56a = 8 - 0.32a$. Subtracting 8 from both sides of this equation yields $6 - 0.56a = -0.32a$. Adding $0.56a$ to both sides of this equation yields $6 = 0.24a$. Dividing both sides of this equation by 0.24 yields $25 = a$. Substituting 25 for a in the equation $8 = 0.32a + b$ yields $8 = 0.32(25) + b$, or $8 = 8 + b$. Subtracting 8 from both sides of this equation yields $0 = b$. Substituting 25 for a and 0 for b in the equation $t = ad + b$ yields $t = 25d + 0$, or $t = 25d$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6f02a16a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 6f02a16a

The linear function g is defined by $g(x) = b - 15x$, where b is a constant. If $g(c + 7) = \frac{c}{4}$, where c is a constant, which of the following expressions represents the value of b ?

- A. $\frac{15c}{4}$
- B. $\frac{19c}{4} + 7$
- C. $\frac{61c}{4} + 105$
- D. $15c + 105$

ID: 6f02a16a Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $g(c + 7) = \frac{c}{4}$. Therefore, for the given linear function g , when $x = c + 7$, $g(x) = \frac{c}{4}$. Substituting $c + 7$ for x and $\frac{c}{4}$ for $g(x)$ in $g(x) = b - 15x$ yields $\frac{c}{4} = b - 15(c + 7)$. Applying the distributive property to the right-hand side of this equation yields $\frac{c}{4} = b - 15c - 105$. Adding $15c$ to both sides of this equation yields $\frac{c}{4} + 15c = b - 105$. Adding 105 to both sides of this equation yields $\frac{c}{4} + 15c + 105 = b$, or $\frac{61c}{4} + 105 = b$. Therefore, the expression that represents the value of b is $\frac{61c}{4} + 105$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 10898fb9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 10898fb9

The cost of renting a backhoe for up to **10** days is **\$270** for the first day and **\$135** for each additional day. Which of the following equations gives the cost y , in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

- A. $y = 270x - 135$
- B. $y = 270x + 135$
- C. $y = 135x + 270$
- D. $y = 135x + 135$

ID: 10898fb9 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the cost of renting a backhoe for up to **10** days is **\$270** for the first day and **\$135** for each additional day. Therefore, the cost y , in dollars, for x days, where $x \leq 10$, is the sum of the cost for the first day, **\$270**, and the cost for the additional $x - 1$ days, **\$135($x - 1$)**. It follows that $y = 270 + 135(x - 1)$, which is equivalent to $y = 270 + 135x - 135$, or $y = 135x + 135$.

Choice A is incorrect. This equation represents a situation where the cost of renting a backhoe is **\$135** for the first day and **\$270** for each additional day.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 95023651

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 95023651

A bowl contains **20** ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by **1** ounce every **4** days. If **9** ounces of water remain in this bowl, for how many days has it been uncovered?

- A. **3**
- B. **7**
- C. **36**
- D. **44**

ID: 95023651 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the bowl starts with **20** ounces of water and has **9** ounces of water remaining after a period of time has passed. The amount of water the bowl has lost during the time period can be found by subtracting the remaining amount of water from the amount of water the bowl starts with, which yields **20** – **9** ounces, or **11** ounces. This means the bowl loses **11** ounces of water during that period of time. It's given that the amount of water decreases by **1** ounce every **4** days. Letting ***t*** represent the number of days the bowl has been uncovered, it follows that $\frac{1}{4} = \frac{11}{t}$. Multiplying both sides of this equation by **4*t*** yields ***t* = 44**. Therefore, the bowl has been uncovered for **44** days.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of ***t*** for the equation $\frac{1}{4} = \frac{9}{t}$, not $\frac{1}{4} = \frac{11}{t}$.

Question Difficulty: Medium

Question ID 6f9dabea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 6f9dabea

$$\begin{aligned}24x + y &= 48 \\ 6x + y &= 72\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 6f9dabea Answer

Correct Answer: 80

Rationale

The correct answer is **80**. Subtracting the second equation in the given system from the first equation yields $(24x + y) - (6x + y) = 48 - 72$, which is equivalent to $24x - 6x + y - y = -24$, or $18x = -24$. Dividing each side of this equation by **3** yields $6x = -8$. Substituting -8 for $6x$ in the second equation yields $-8 + y = 72$. Adding **8** to both sides of this equation yields $y = 80$.

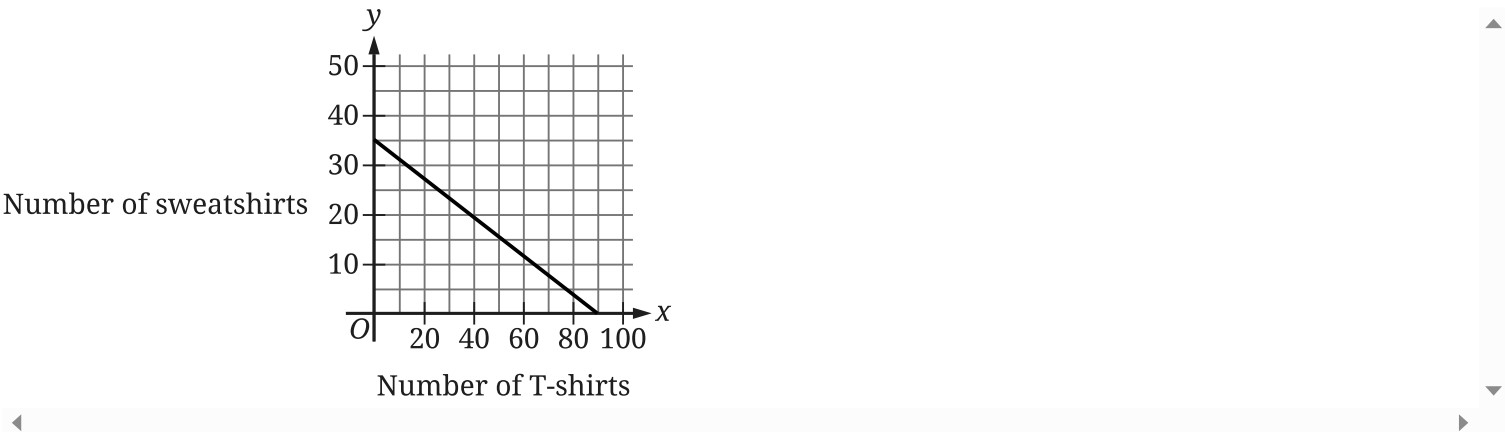
Alternate approach: Multiplying each side of the second equation in the given system by **4** yields $24x + 4y = 288$. Subtracting the first equation in the given system from this equation yields $(24x + 4y) - (24x + y) = 288 - 48$, which is equivalent to $24x - 24x + 4y - y = 240$, or $3y = 240$. Dividing each side of this equation by **3** yields $y = 80$.

Question Difficulty: Hard

Question ID b4c22438

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: b4c22438



The graph models the relationship between the number of T-shirts, x , and the number of sweatshirts, y , that Kira can purchase for a school fundraiser. Which equation could represent this relationship?

- A. $y = 7x + 18$
- B. $7x + 18y = 630$
- C. $y = 18x + 7$
- D. $18x + 7y = 630$

ID: b4c22438 Answer

Correct Answer: B

Rationale

Choice B is correct. A line in the xy -plane can be written as $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept. The graph shown is a line passing through the points $(0, 35)$ and $(90, 0)$. Substituting 0 for x and 35 for y in the equation $y = mx + b$ yields $35 = m(0) + b$, or $35 = b$. Substituting 35 for b , 90 for x , and 0 for y in the equation $y = mx + b$ yields $0 = 90m + 35$. Subtracting 35 from both sides of this equation yields $-35 = 90m$. Dividing both sides of this equation by 90 yields $-\frac{35}{90} = m$, or $-\frac{7}{18} = m$. Substituting $-\frac{7}{18}$ for m and 35 for b in the equation $y = mx + b$ yields $y = -\frac{7}{18}x + 35$. Multiplying both sides of this equation by 18 yields $18y = -7x + 35(18)$, or $18y = -7x + 630$. Adding $7x$ to both sides of this equation yields $7x + 18y = 630$. Therefore, the equation $7x + 18y = 630$ represents the relationship between x and y modeled by the graph.

Choice A is incorrect. The point $(0, 35)$ is not on the graph of this equation, since $7(0) + 18 = 18$, not 35 .

Choice C is incorrect. The point $(0, 35)$ is not on the graph of this equation, since $18(0) + 7 = 7$, not 35 .

Choice D is incorrect. The point $(90, 0)$ is not on the graph of this equation, since $18(90) + 7(0) = 1,620$, not 630 .

Question Difficulty: Medium

Question ID 85059bec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 85059bec

$$\begin{aligned}6 + 7r &= pw \\ 7r - 5w &= 5w + 11\end{aligned}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

ID: 85059bec Answer

Correct Answer: 10

Rationale

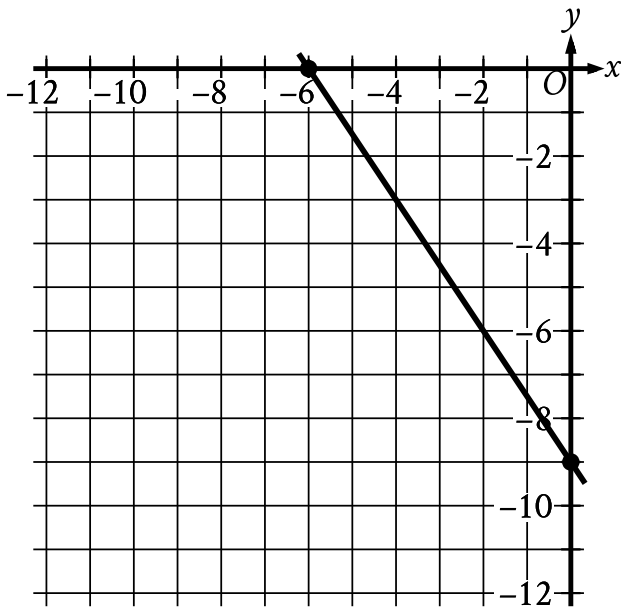
The correct answer is **10**. Solving by substitution, the given system of equations, where p is a constant, can be written so that the left-hand side of each equation is equal to $7r$. Subtracting **6** from each side of the first equation in the given system, $6 + 7r = pw$, yields $7r = pw - 6$. Adding $5w$ to each side of the second equation in the given system, $7r - 5w = 5w + 11$, yields $7r = 10w + 11$. Since the left-hand side of each equation is equal to $7r$, setting the the right-hand side of the equations equal to each other yields $pw - 6 = 10w + 11$. A linear equation in one variable, w , has no solution if and only if the equation is false; that is, when there's no value of w that produces a true statement. For the equation $pw - 6 = 10w + 11$, there's no value of w that produces a true statement when $pw = 10w$. Therefore, for the equation $pw - 6 = 10w + 11$, there's no value of w that produces a true statement when the value of p is **10**. It follows that in the given system of equations, the system has no solution when the value of p is **10**.

Question Difficulty: Hard

Question ID 097a4f83

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 097a4f83



The graph of line h is shown in the xy -plane. Line k (not shown) is defined by $sx + 48y = t$, where s and t are constants. If line k is graphed in this xy -plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is **NOT** a possible value of t ?

- A. -432
- B. -9
- C. 72
- D. 288

ID: 097a4f83 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that if line k is graphed in the xy -plane with line h , the result is the graph of a system of two linear equations that has no solution. It follows that line h and line k don't intersect. This means that line k and line h have the same slope and different y -intercepts. The graph shows that the y -intercept of line h is $(0, -9)$. The y -coordinate of the y -intercept of line k is the value of y when $x = 0$ in the equation defining line k . Substituting 0 for x in the equation $sx + 48y = t$ yields $s(0) + 48y = t$, or $48y = t$. Dividing both sides of this equation by 48 yields $y = \frac{t}{48}$. Therefore, the y -coordinate of the y -intercept of line k is $\frac{t}{48}$, so the y -intercept of line k is $(0, \frac{t}{48})$. Since line k and line h have different y -intercepts, $\frac{t}{48}$ can't be equal to -9 . It follows that t can't be equal to $-9(48)$, or -432 . Thus, -432 isn't a possible value of t .

Choice B is incorrect. This isn't a possible value of the y -coordinate of the y -intercept of line k , but it is a possible value of t .

Choice C is incorrect. This is the value of s , not a value of t that isn't possible.

Choice D is incorrect. If $t = 288$, then the y -intercept of line k is $(0, \frac{288}{48})$, or $(0, 6)$, which is different from the y -intercept of line h , $(0, -9)$. Therefore, 288 is a possible value of t .

Question Difficulty: Hard

Question ID 86e98b7d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 86e98b7d

If $2(3t - 10) + t = 40 + 4t$, what is the value of $3t$?

ID: 86e98b7d Answer

Correct Answer: 60

Rationale

The correct answer is **60**. Subtracting t from both sides of the given equation yields $2(3t - 10) = 40 + 3t$. Applying the distributive property to the left-hand side of this equation yields $6t - 20 = 40 + 3t$. Adding **20** to both sides of this equation yields $6t = 60 + 3t$. Subtracting $3t$ from both sides of this equation yields $3t = 60$. Therefore, the value of $3t$ is **60**.

Question Difficulty: Hard

Question ID ece4e9e6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: ece4e9e6

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under **220** miles. Rhett drove at an average speed of **35 miles per hour (mph)**, and Jessica drove at an average speed of **40 mph**. Which of the following inequalities represents this situation, where ***r*** is the number of hours Rhett drove and ***j*** is the number of hours Jessica drove?

- A. $35r + 40j > 220$
- B. $35r + 40j < 220$
- C. $40r + 35j > 220$
- D. $40r + 35j < 220$

ID: ece4e9e6 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that Rhett drove at an average speed of **35** miles per hour and that he drove for ***r*** hours. Multiplying **35** miles per hour by ***r*** hours yields **35*r*** miles, or the distance that Rhett drove. It’s also given that Jessica drove at an average speed of **40** miles per hour and that she drove for ***j*** hours. Multiplying **40** miles per hour by ***j*** hours yields **40*j*** miles, or the distance that Jessica drove. The total distance, in miles, that Rhett and Jessica drove can be represented by the expression **35*r* + 40*j***. It’s given that the total distance they drove was under **220** miles. Therefore, the inequality **35*r* + 40*j* < 220** represents this situation.

Choice A is incorrect. This inequality represents a situation in which the total distance Rhett and Jessica drove was over, rather than under, **220** miles.

Choice C is incorrect. This inequality represents a situation in which Rhett drove at an average speed of **40**, rather than **35**, miles per hour, Jessica drove at an average speed of **35**, rather than **40**, miles per hour, and the total distance they drove was over, rather than under, **220** miles.

Choice D is incorrect. This inequality represents a situation in which Rhett drove at an average speed of **40**, rather than **35**, miles per hour, and Jessica drove at an average speed of **35**, rather than **40**, miles per hour.

Question Difficulty: Medium